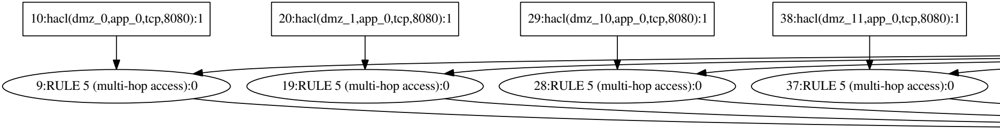
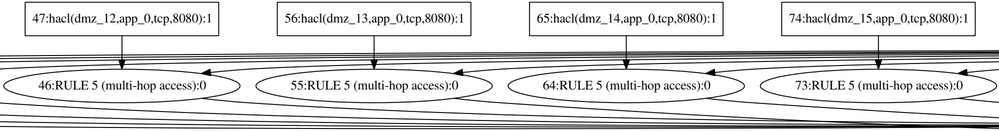
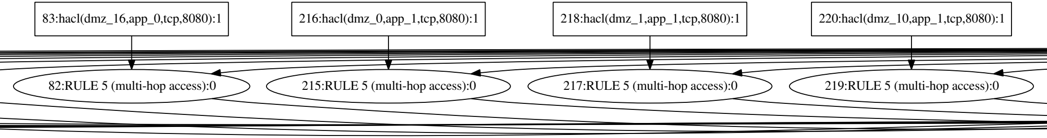
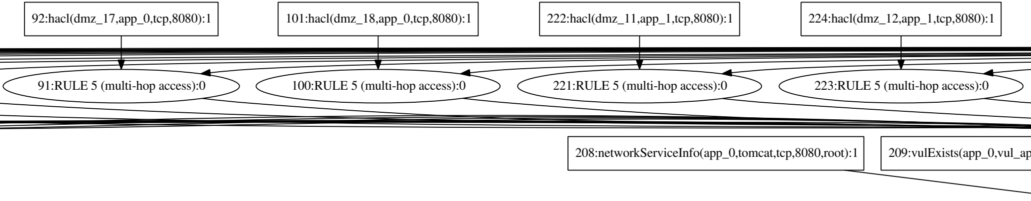
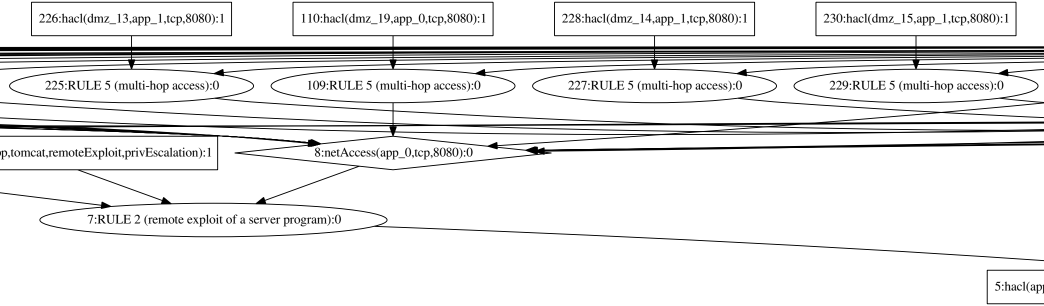
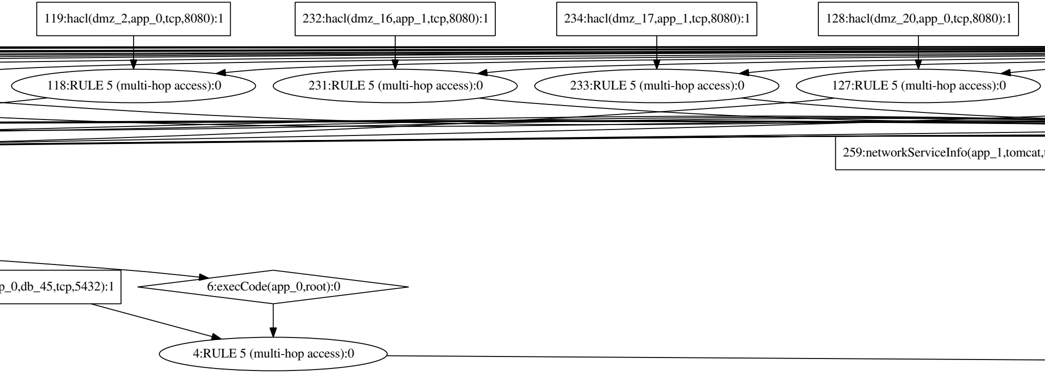

10:hacl(dmz_0,app_0,tcp,8080):1
20:hacl(dmz_1,app_0,tcp,8080):1
29:hacl(dmz_10,app_0,tcp,8080):1
38:hacl(dmz_11,app_0,tcp,8080):1
9:RULE 5 (multi-hop access):0
19:RULE 5 (multi-hop access):0
28:RULE 5 (multi-hop access):0
37:RULE 5 (multi-hop access):0

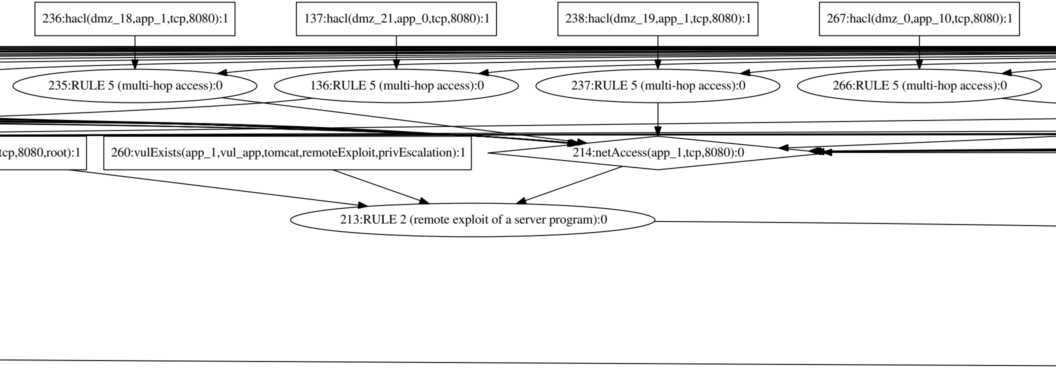

47:hacl(dmz_12,app_0,tcp,8080):1
56:hacl(dmz_13,app_0,tcp,8080):1
65:hacl(dmz_14,app_0,tcp,8080):1
74:hacl(dmz_15,app_0,tcp,8080):1
46:RULE 5 (multi-hop access):0
55:RULE 5 (multi-hop access):0
64:RULE 5 (multi-hop access):0
73:RULE 5 (multi-hop access):0

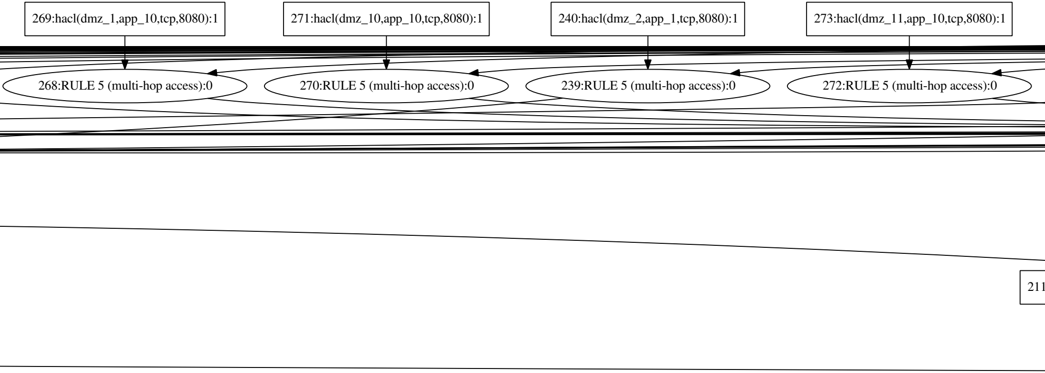

83:hacl(dmz_16,app_0,tcp,8080):1
216:hacl(dmz_0,app_1,tcp,8080):1
218:hacl(dmz_1,app_1,tcp,8080):1
220:hacl(dmz_10,app_1,tcp,8080):1
82:RULE 5 (multi-hop access):0
215:RULE 5 (multi-hop access):0
217:RULE 5 (multi-hop access):0
219:RULE 5 (multi-hop access):0

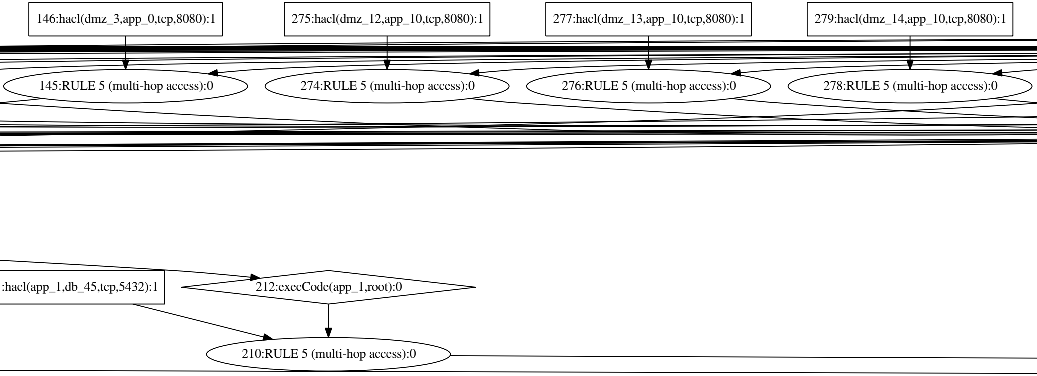

92:hacl(dmz_17,app_0,tcp,8080):1
101:hacl(dmz_18,app_0,tcp,8080):1
222:hacl(dmz_11,app_1,tcp,8080):1
224:hacl(dmz_12,app_1,tcp,8080):1
91:RULE 5 (multi-hop access):0
100:RULE 5 (multi-hop access):0
221:RULE 5 (multi-hop access):0
223:RULE 5 (multi-hop access):0
208:networkServiceInfo(app_0,tomcat,tcp,8080,root):1
209:vulExists(app_0,vul_ap

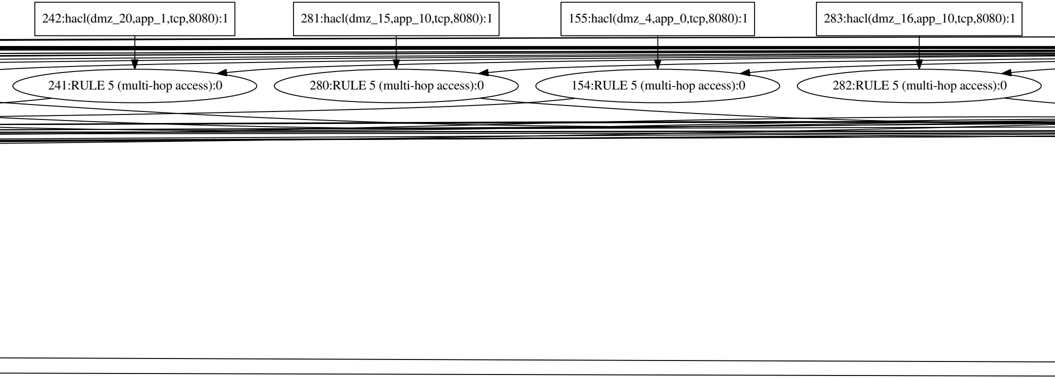

226:hacl(dmz_13,app_1,tcp,8080):1
110:hacl(dmz_19,app_0,tcp,8080):1
228:hacl(dmz_14,app_1,tcp,8080):1
230:hacl(dmz_15,app_1,tcp,8080):1
225:RULE 5 (multi-hop access):0
109:RULE 5 (multi-hop access):0
227:RULE 5 (multi-hop access):0
229:RULE 5 (multi-hop access):0
p,tomcat,remoteExploit,privEscalation):1
8:netAccess(app_0,tcp,8080):0
7:RULE 2 (remote exploit of a server program):0
5:hacl(app

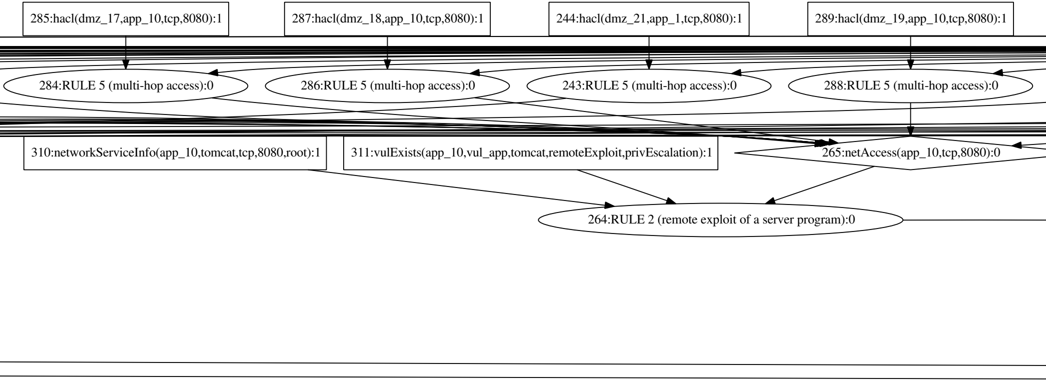

119:hacl(dmz_2,app_0,tcp,8080):1
232:hacl(dmz_16,app_1,tcp,8080):1
234:hacl(dmz_17,app_1,tcp,8080):1
128:hacl(dmz_20,app_0,tcp,8080):1
118:RULE 5 (multi-hop access):0
231:RULE 5 (multi-hop access):0
233:RULE 5 (multi-hop access):0
127:RULE 5 (multi-hop access):0
259:networkServiceInfo(app_1,tomcat,t
p_0,db_45,tcp,5432):1
6:execCode(app_0,root):0
4:RULE 5 (multi-hop access):0

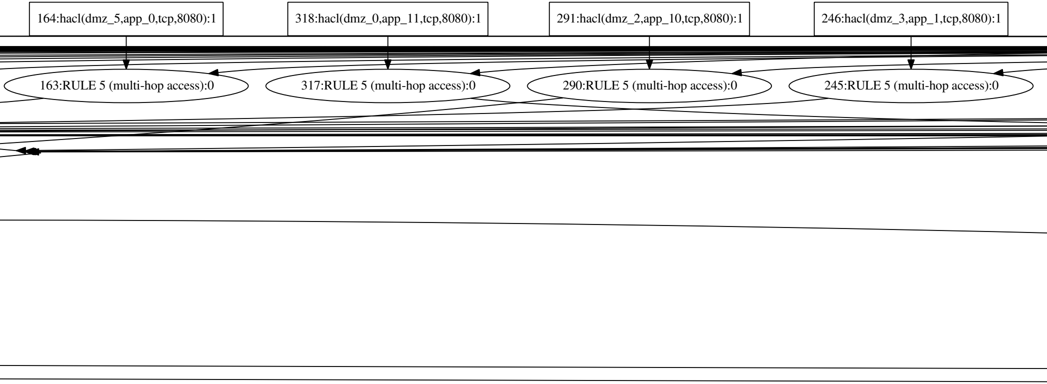

236:hacl(dmz_18,app_1,tcp,8080):1
137:hacl(dmz_21,app_0,tcp,8080):1
238:hacl(dmz_19,app_1,tcp,8080):1
267:hacl(dmz_0,app_10,tcp,8080):1
235:RULE 5 (multi-hop access):0
136:RULE 5 (multi-hop access):0
237:RULE 5 (multi-hop access):0
266:RULE 5 (multi-hop access):0
tcp,8080,root):1
260:vulExists(app_1,vul_app,tomcat,remoteExploit,privEscalation):1
214:netAccess(app_1,tcp,8080):0
213:RULE 2 (remote exploit of a server program):0

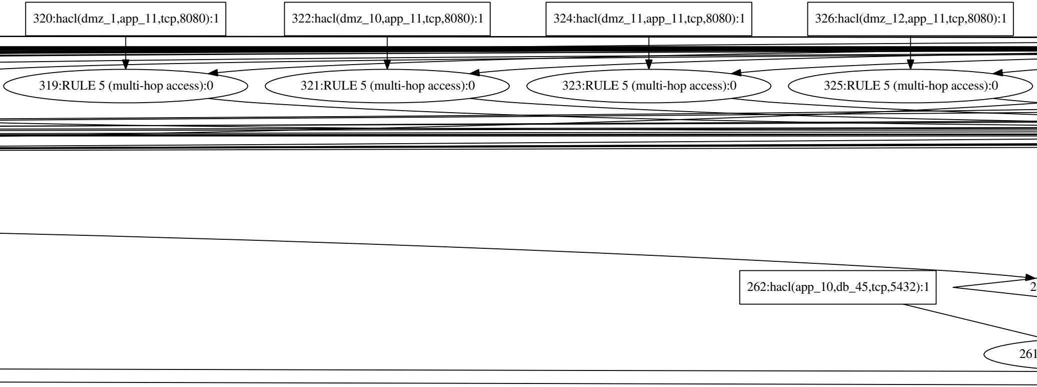

269:hacl(dmz_1,app_10,tcp,8080):1
271:hacl(dmz_10,app_10,tcp,8080):1
240:hacl(dmz_2,app_1,tcp,8080):1
273:hacl(dmz_11,app_10,tcp,8080):1
268:RULE 5 (multi-hop access):0
270:RULE 5 (multi-hop access):0
239:RULE 5 (multi-hop access):0
272:RULE 5 (multi-hop access):0
211

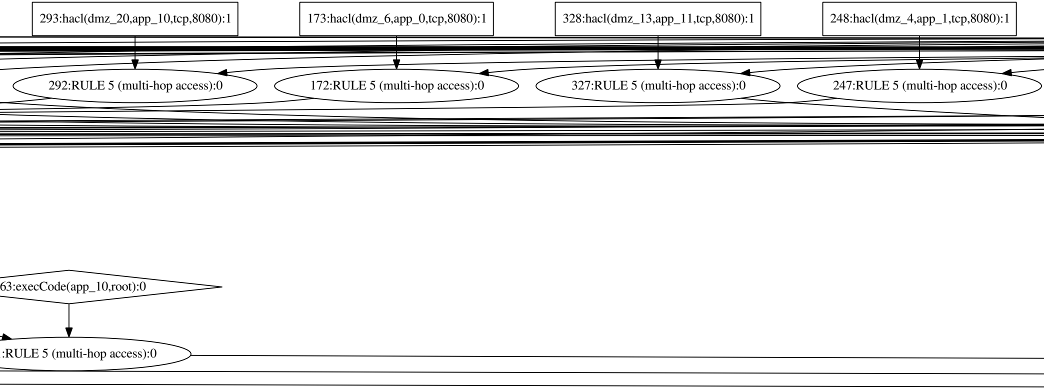

146:hacl(dmz_3,app_0,tcp,8080):1
275:hacl(dmz_12,app_10,tcp,8080):1
277:hacl(dmz_13,app_10,tcp,8080):1
279:hacl(dmz_14,app_10,tcp,8080):1
145:RULE 5 (multi-hop access):0
274:RULE 5 (multi-hop access):0
276:RULE 5 (multi-hop access):0
278:RULE 5 (multi-hop access):0
:hacl(app_1,db_45,tcp,5432):1
212:execCode(app_1,root):0
210:RULE 5 (multi-hop access):0

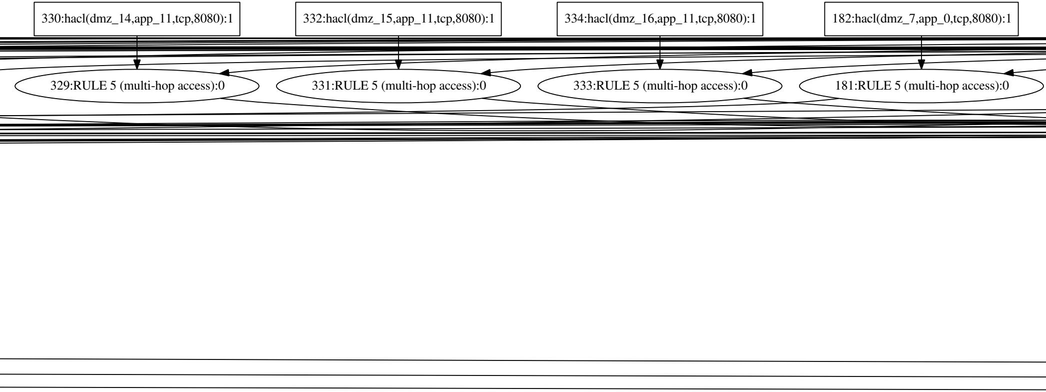

242:hacl(dmz_20,app_1,tcp,8080):1
281:hacl(dmz_15,app_10,tcp,8080):1
155:hacl(dmz_4,app_0,tcp,8080):1
283:hacl(dmz_16,app_10,tcp,8080):1
241:RULE 5 (multi-hop access):0
280:RULE 5 (multi-hop access):0
154:RULE 5 (multi-hop access):0
282:RULE 5 (multi-hop access):0

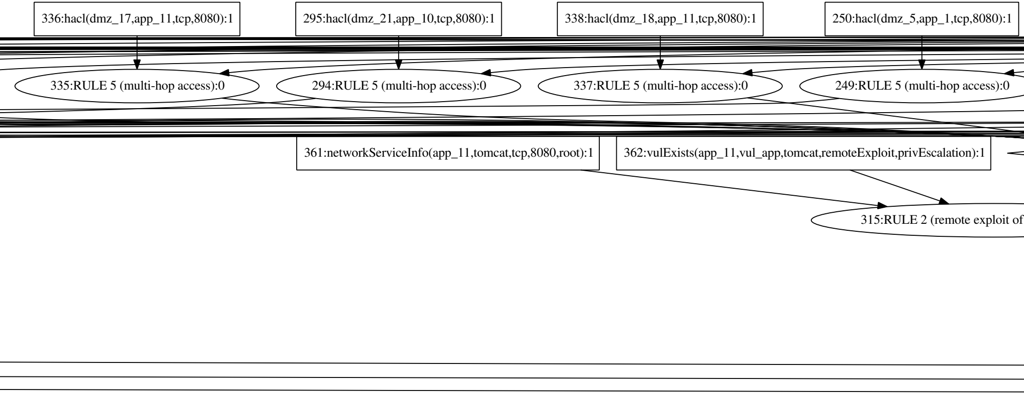

285:hacl(dmz_17,app_10,tcp,8080):1
287:hacl(dmz_18,app_10,tcp,8080):1
244:hacl(dmz_21,app_1,tcp,8080):1
289:hacl(dmz_19,app_10,tcp,8080):1
284:RULE 5 (multi-hop access):0
286:RULE 5 (multi-hop access):0
243:RULE 5 (multi-hop access):0
288:RULE 5 (multi-hop access):0
310:networkServiceInfo(app_10,tomcat,tcp,8080,root):1
311:vulExists(app_10,vul_app,tomcat,remoteExploit,privEscalation):1
265:netAccess(app_10,tcp,8080):0
264:RULE 2 (remote exploit of a server program):0

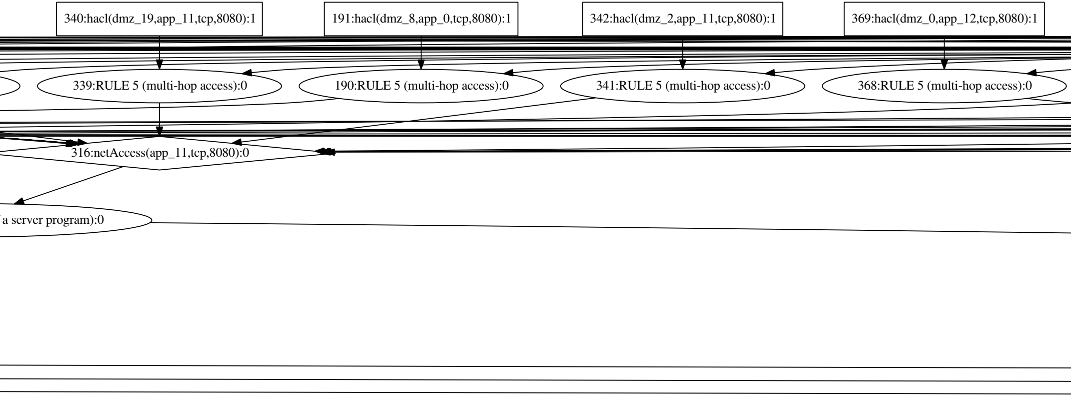

164:hacl(dmz_5,app_0,tcp,8080):1
318:hacl(dmz_0,app_11,tcp,8080):1
291:hacl(dmz_2,app_10,tcp,8080):1
246:hacl(dmz_3,app_1,tcp,8080):1
163:RULE 5 (multi-hop access):0
317:RULE 5 (multi-hop access):0
290:RULE 5 (multi-hop access):0
245:RULE 5 (multi-hop access):0

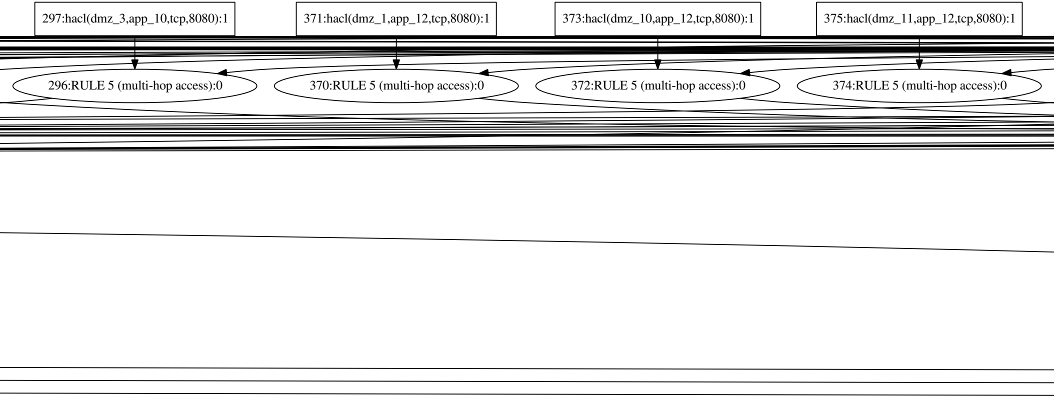

320:hacl(dmz_1,app_11,tcp,8080):1
322:hacl(dmz_10,app_11,tcp,8080):1
324:hacl(dmz_11,app_11,tcp,8080):1
326:hacl(dmz_12,app_11,tcp,8080):1
319:RULE 5 (multi-hop access):0
321:RULE 5 (multi-hop access):0
323:RULE 5 (multi-hop access):0
325:RULE 5 (multi-hop access):0
262:hacl(app_10,db_45,tcp,5432):1
261

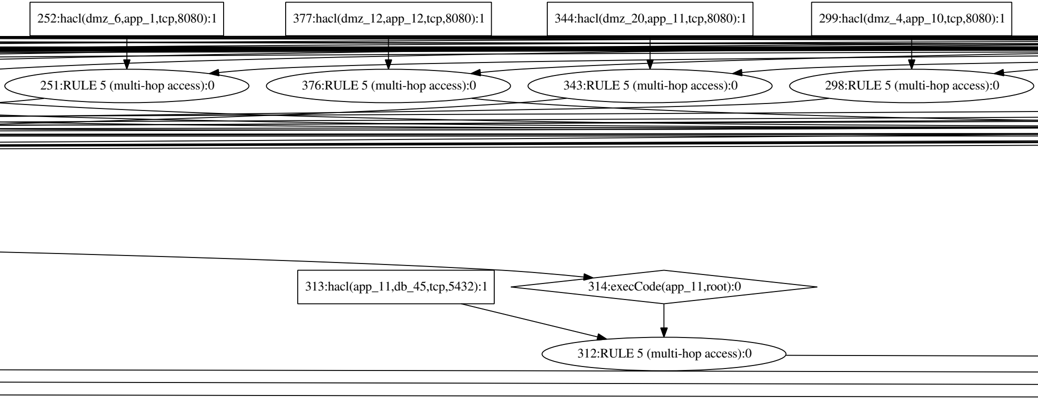

293:hacl(dmz_20,app_10,tcp,8080):1
173:hacl(dmz_6,app_0,tcp,8080):1
328:hacl(dmz_13,app_11,tcp,8080):1
248:hacl(dmz_4,app_1,tcp,8080):1
292:RULE 5 (multi-hop access):0
172:RULE 5 (multi-hop access):0
327:RULE 5 (multi-hop access):0
247:RULE 5 (multi-hop access):0
63:execCode(app_10,root):0
:RULE 5 (multi-hop access):0

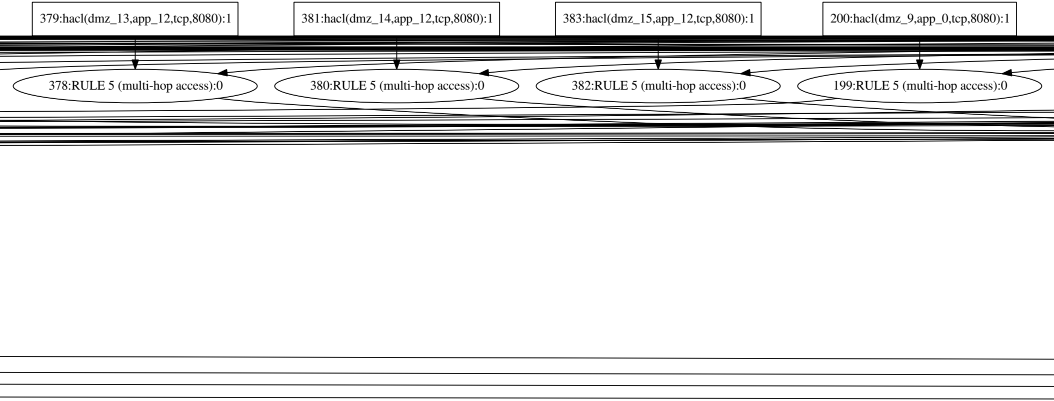

330:hacl(dmz_14,app_11,tcp,8080):1
332:hacl(dmz_15,app_11,tcp,8080):1
334:hacl(dmz_16,app_11,tcp,8080):1
182:hacl(dmz_7,app_0,tcp,8080):1
329:RULE 5 (multi-hop access):0
331:RULE 5 (multi-hop access):0
333:RULE 5 (multi-hop access):0
181:RULE 5 (multi-hop access):0

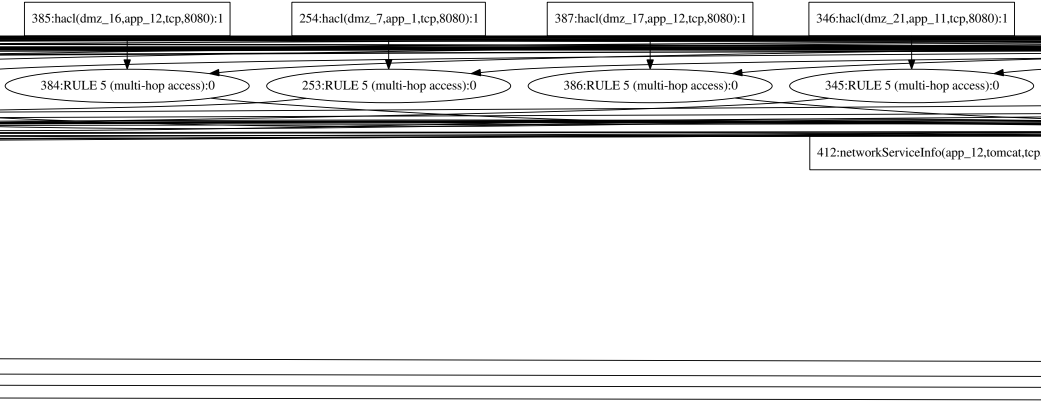

336:hacl(dmz_17,app_11,tcp,8080):1
295:hacl(dmz_21,app_10,tcp,8080):1
338:hacl(dmz_18,app_11,tcp,8080):1
250:hacl(dmz_5,app_1,tcp,8080):1
335:RULE 5 (multi-hop access):0
294:RULE 5 (multi-hop access):0
337:RULE 5 (multi-hop access):0
249:RULE 5 (multi-hop access):0
361:networkServiceInfo(app_11,tomcat,tcp,8080,root):1
362:vulExists(app_11,vul_app,tomcat,remoteExploit,privEscalation):1
315:RULE 2 (remote exploit of

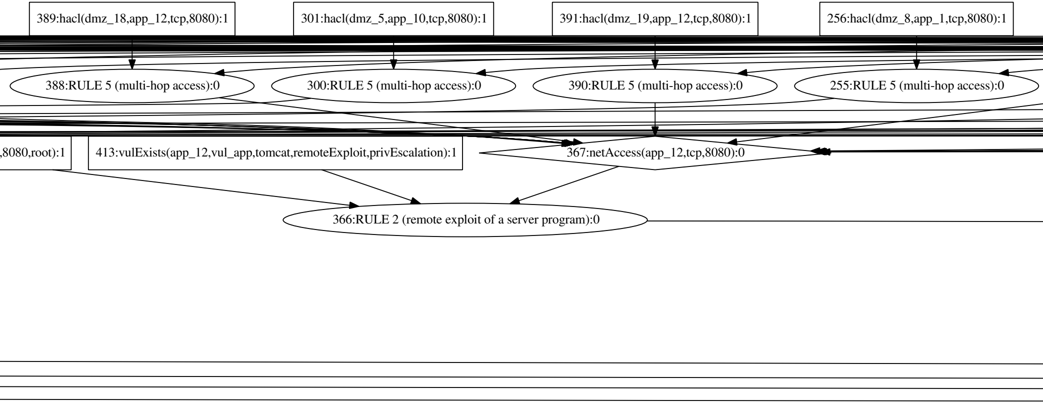

340:hacl(dmz_19,app_11,tcp,8080):1
191:hacl(dmz_8,app_0,tcp,8080):1
342:hacl(dmz_2,app_11,tcp,8080):1
369:hacl(dmz_0,app_12,tcp,8080):1
339:RULE 5 (multi-hop access):0
190:RULE 5 (multi-hop access):0
341:RULE 5 (multi-hop access):0
368:RULE 5 (multi-hop access):0
316:netAccess(app_11,tcp,8080):0
a server program):0

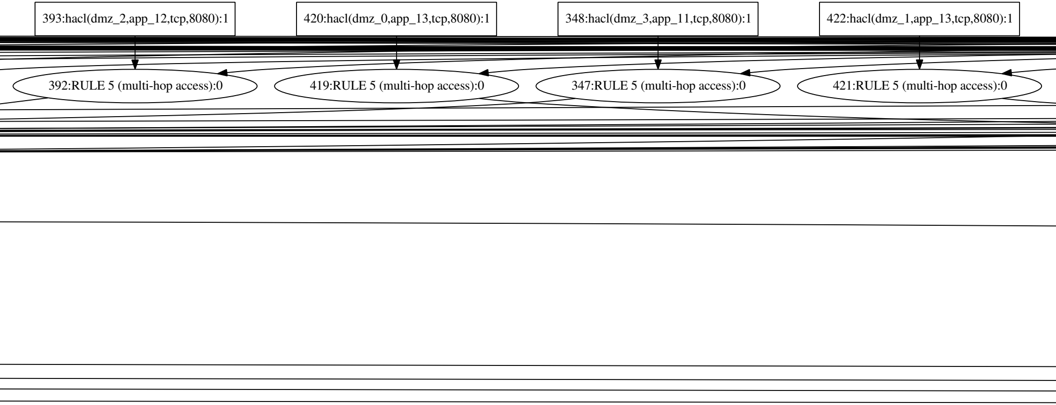

297:hacl(dmz_3,app_10,tcp,8080):1
371:hacl(dmz_1,app_12,tcp,8080):1
373:hacl(dmz_10,app_12,tcp,8080):1
375:hacl(dmz_11,app_12,tcp,8080):1
296:RULE 5 (multi-hop access):0
370:RULE 5 (multi-hop access):0
372:RULE 5 (multi-hop access):0
374:RULE 5 (multi-hop access):0

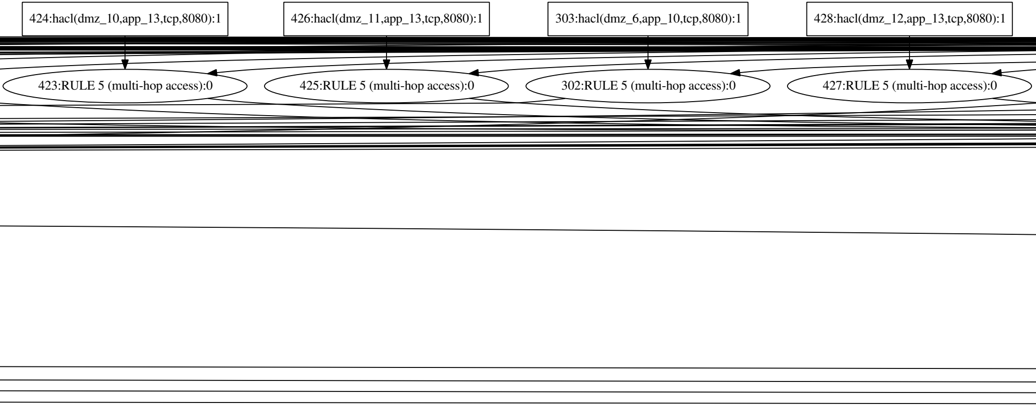

252:hacl(dmz_6,app_1,tcp,8080):1
377:hacl(dmz_12,app_12,tcp,8080):1
344:hacl(dmz_20,app_11,tcp,8080):1
299:hacl(dmz_4,app_10,tcp,8080):1
251:RULE 5 (multi-hop access):0
376:RULE 5 (multi-hop access):0
343:RULE 5 (multi-hop access):0
298:RULE 5 (multi-hop access):0
313:hacl(app_11,db_45,tcp,5432):1
314:execCode(app_11,root):0
312:RULE 5 (multi-hop access):0

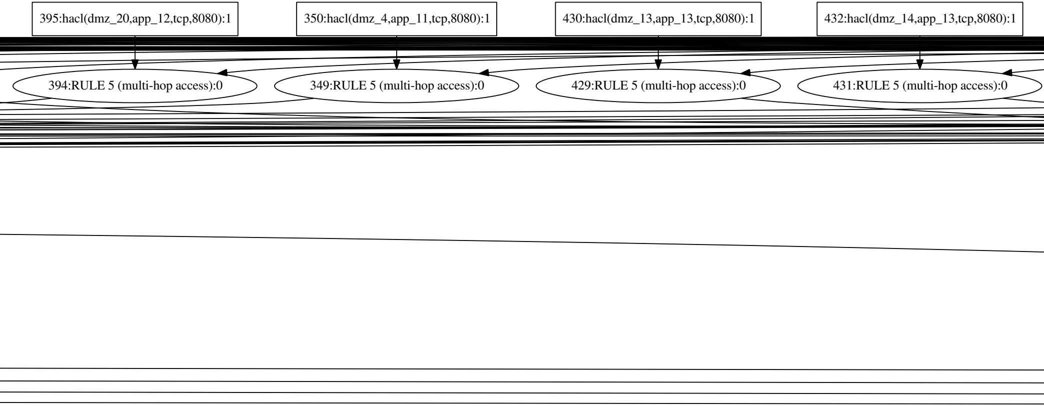

379:hacl(dmz_13,app_12,tcp,8080):1
381:hacl(dmz_14,app_12,tcp,8080):1
383:hacl(dmz_15,app_12,tcp,8080):1
200:hacl(dmz_9,app_0,tcp,8080):1
378:RULE 5 (multi-hop access):0
380:RULE 5 (multi-hop access):0
382:RULE 5 (multi-hop access):0
199:RULE 5 (multi-hop access):0

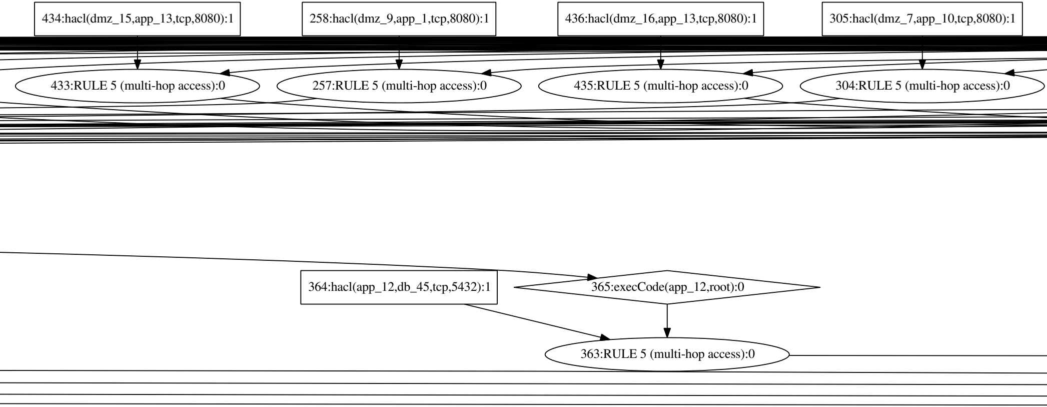

385:hacl(dmz_16,app_12,tcp,8080):1
254:hacl(dmz_7,app_1,tcp,8080):1
387:hacl(dmz_17,app_12,tcp,8080):1
346:hacl(dmz_21,app_11,tcp,8080):1
384:RULE 5 (multi-hop access):0
253:RULE 5 (multi-hop access):0
386:RULE 5 (multi-hop access):0
345:RULE 5 (multi-hop access):0
412:networkServiceInfo(app_12,tomcat,tcp

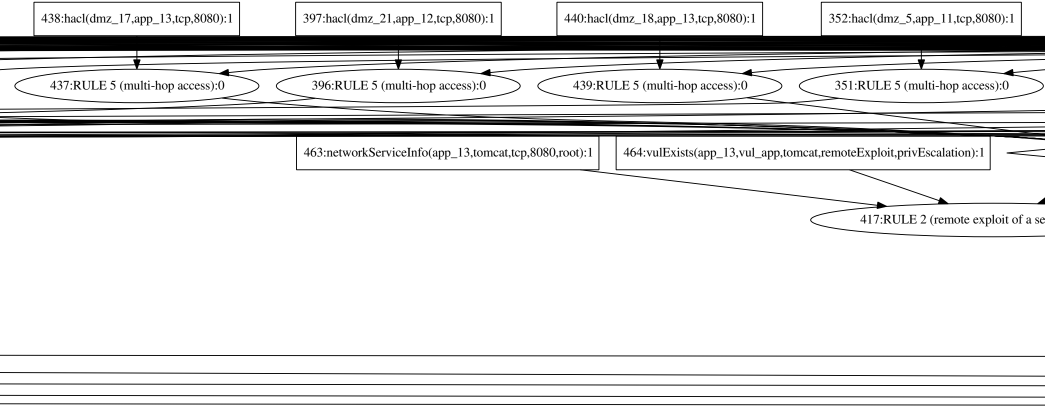

389:hacl(dmz_18,app_12,tcp,8080):1
301:hacl(dmz_5,app_10,tcp,8080):1
391:hacl(dmz_19,app_12,tcp,8080):1
256:hacl(dmz_8,app_1,tcp,8080):1
388:RULE 5 (multi-hop access):0
300:RULE 5 (multi-hop access):0
390:RULE 5 (multi-hop access):0
255:RULE 5 (multi-hop access):0
,8080,root):1
413:vulExists(app_12,vul_app,tomcat,remoteExploit,privEscalation):1
367:netAccess(app_12,tcp,8080):0
366:RULE 2 (remote exploit of a server program):0

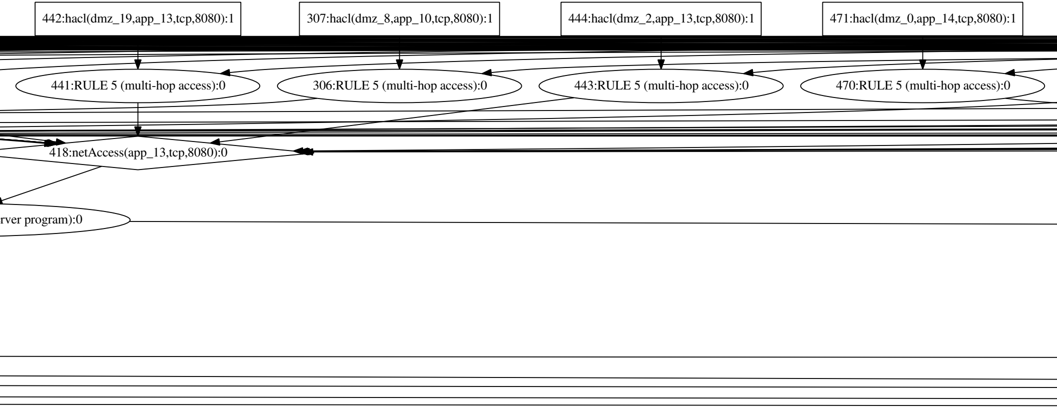

393:hacl(dmz_2,app_12,tcp,8080):1
420:hacl(dmz_0,app_13,tcp,8080):1
348:hacl(dmz_3,app_11,tcp,8080):1
422:hacl(dmz_1,app_13,tcp,8080):1
392:RULE 5 (multi-hop access):0
419:RULE 5 (multi-hop access):0
347:RULE 5 (multi-hop access):0
421:RULE 5 (multi-hop access):0

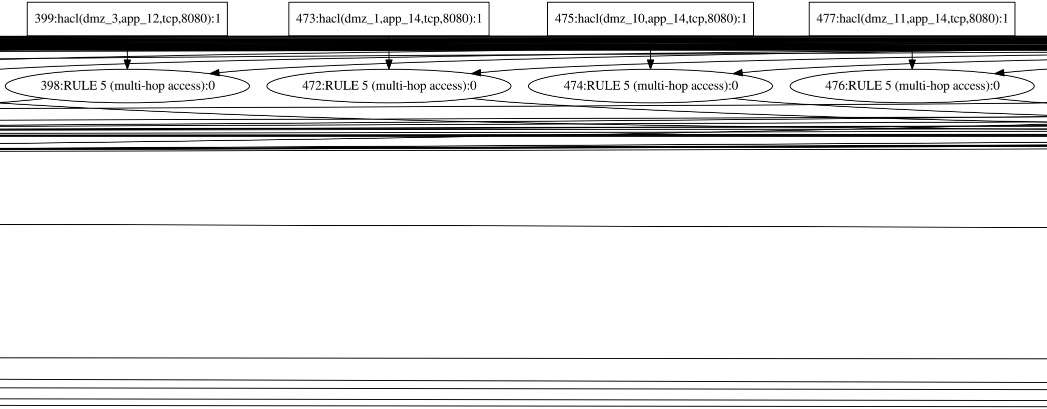

424:hacl(dmz_10,app_13,tcp,8080):1
426:hacl(dmz_11,app_13,tcp,8080):1
303:hacl(dmz_6,app_10,tcp,8080):1
428:hacl(dmz_12,app_13,tcp,8080):1
423:RULE 5 (multi-hop access):0
425:RULE 5 (multi-hop access):0
302:RULE 5 (multi-hop access):0
427:RULE 5 (multi-hop access):0

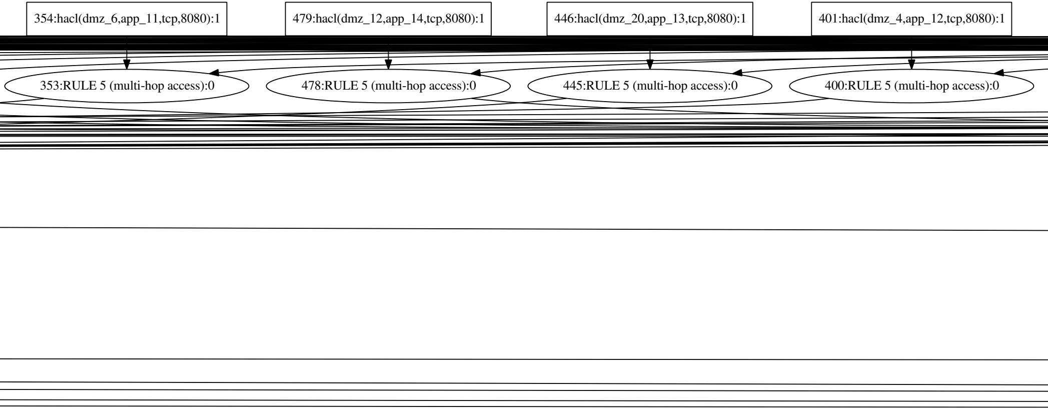

395:hacl(dmz_20,app_12,tcp,8080):1
350:hacl(dmz_4,app_11,tcp,8080):1
430:hacl(dmz_13,app_13,tcp,8080):1
432:hacl(dmz_14,app_13,tcp,8080):1
394:RULE 5 (multi-hop access):0
349:RULE 5 (multi-hop access):0
429:RULE 5 (multi-hop access):0
431:RULE 5 (multi-hop access):0

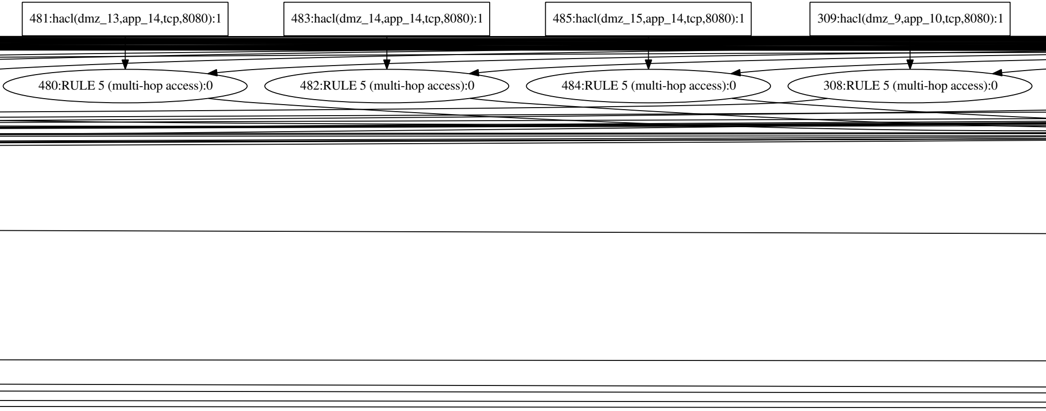

434:hacl(dmz_15,app_13,tcp,8080):1
258:hacl(dmz_9,app_1,tcp,8080):1
436:hacl(dmz_16,app_13,tcp,8080):1
305:hacl(dmz_7,app_10,tcp,8080):1
433:RULE 5 (multi-hop access):0
257:RULE 5 (multi-hop access):0
435:RULE 5 (multi-hop access):0
304:RULE 5 (multi-hop access):0
364:hacl(app_12,db_45,tcp,5432):1
365:execCode(app_12,root):0
363:RULE 5 (multi-hop access):0

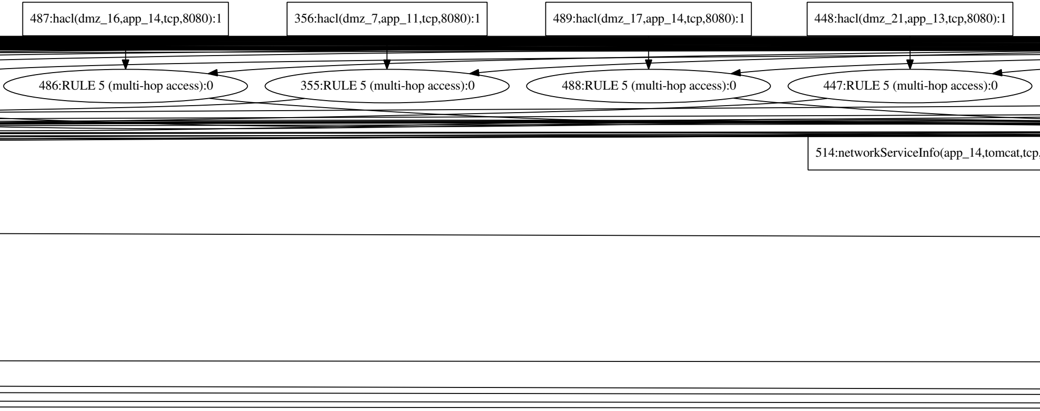

438:hacl(dmz_17,app_13,tcp,8080):1
397:hacl(dmz_21,app_12,tcp,8080):1
440:hacl(dmz_18,app_13,tcp,8080):1
352:hacl(dmz_5,app_11,tcp,8080):1
437:RULE 5 (multi-hop access):0
396:RULE 5 (multi-hop access):0
439:RULE 5 (multi-hop access):0
351:RULE 5 (multi-hop access):0
463:networkServiceInfo(app_13,tomcat,tcp,8080,root):1
464:vulExists(app_13,vul_app,tomcat,remoteExploit,privEscalation):1
417:RULE 2 (remote exploit of a se

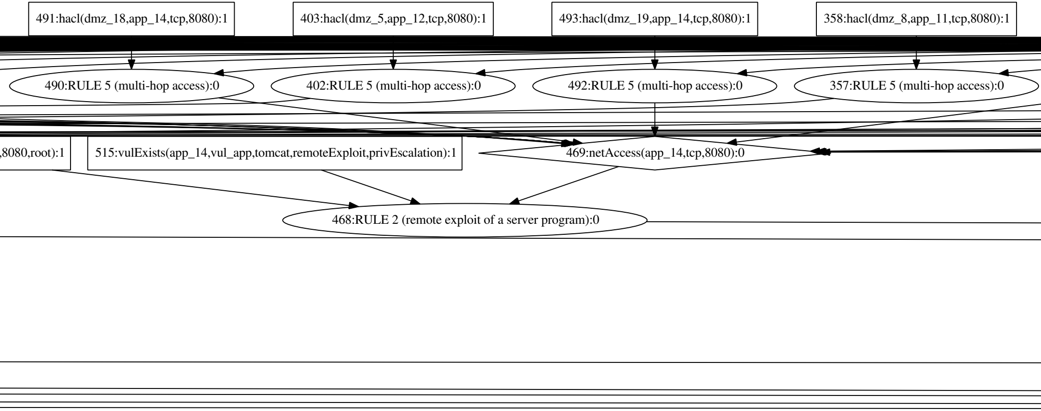

442:hacl(dmz_19,app_13,tcp,8080):1
307:hacl(dmz_8,app_10,tcp,8080):1
444:hacl(dmz_2,app_13,tcp,8080):1
471:hacl(dmz_0,app_14,tcp,8080):1
441:RULE 5 (multi-hop access):0
306:RULE 5 (multi-hop access):0
443:RULE 5 (multi-hop access):0
470:RULE 5 (multi-hop access):0
418:netAccess(app_13,tcp,8080):0
rver program):0

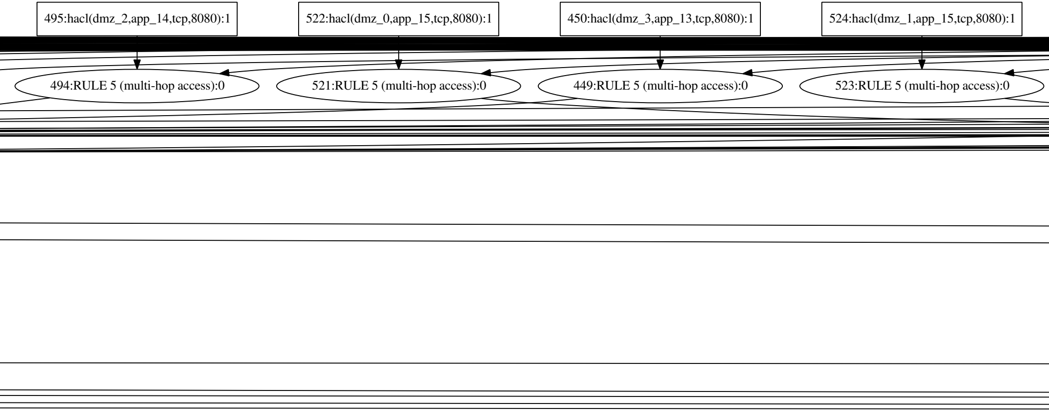

399:hacl(dmz_3,app_12,tcp,8080):1
473:hacl(dmz_1,app_14,tcp,8080):1
475:hacl(dmz_10,app_14,tcp,8080):1
477:hacl(dmz_11,app_14,tcp,8080):1
398:RULE 5 (multi-hop access):0
472:RULE 5 (multi-hop access):0
474:RULE 5 (multi-hop access):0
476:RULE 5 (multi-hop access):0

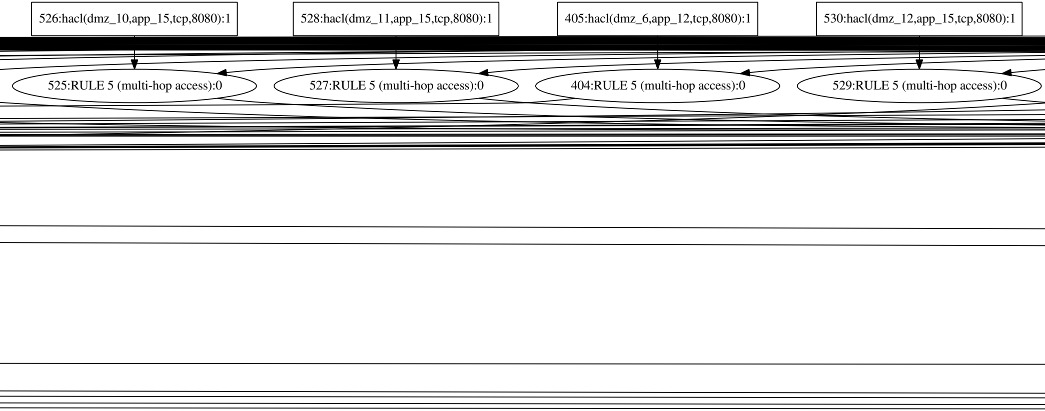

354:hacl(dmz_6,app_11,tcp,8080):1
479:hacl(dmz_12,app_14,tcp,8080):1
446:hacl(dmz_20,app_13,tcp,8080):1
401:hacl(dmz_4,app_12,tcp,8080):1
353:RULE 5 (multi-hop access):0
478:RULE 5 (multi-hop access):0
445:RULE 5 (multi-hop access):0
400:RULE 5 (multi-hop access):0

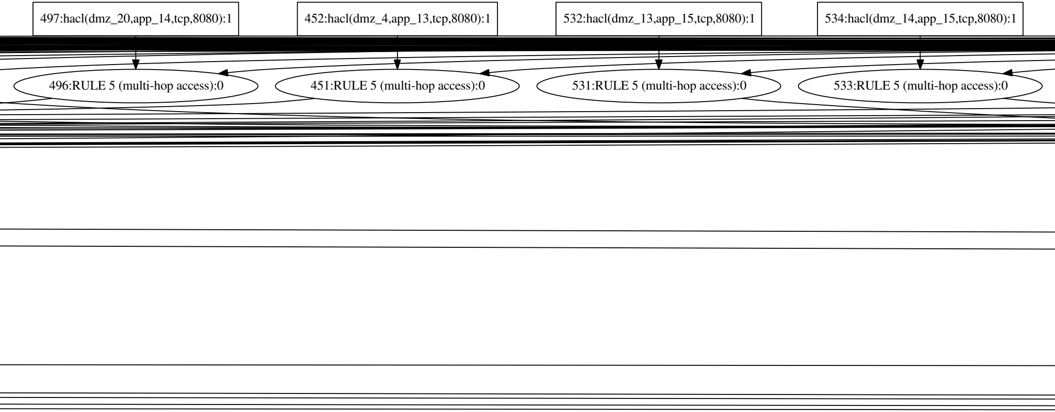

481:hacl(dmz_13,app_14,tcp,8080):1
483:hacl(dmz_14,app_14,tcp,8080):1
485:hacl(dmz_15,app_14,tcp,8080):1
309:hacl(dmz_9,app_10,tcp,8080):1
480:RULE 5 (multi-hop access):0
482:RULE 5 (multi-hop access):0
484:RULE 5 (multi-hop access):0
308:RULE 5 (multi-hop access):0

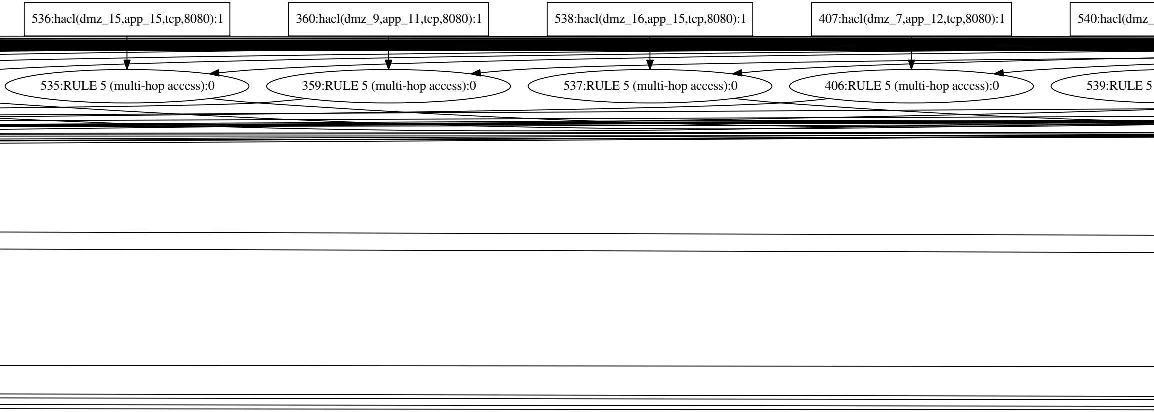

487:hacl(dmz_16,app_14,tcp,8080):1
356:hacl(dmz_7,app_11,tcp,8080):1
489:hacl(dmz_17,app_14,tcp,8080):1
448:hacl(dmz_21,app_13,tcp,8080):1
486:RULE 5 (multi-hop access):0
355:RULE 5 (multi-hop access):0
488:RULE 5 (multi-hop access):0
447:RULE 5 (multi-hop access):0
514:networkServiceInfo(app_14,tomcat,tcp,

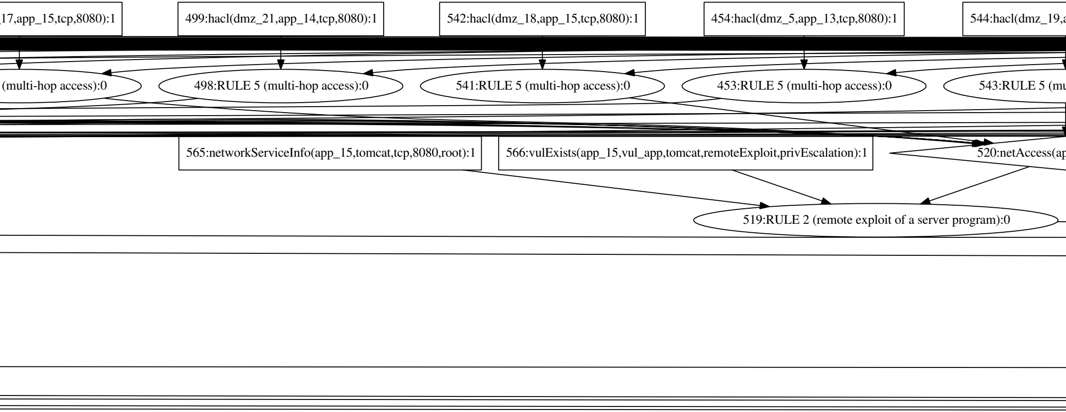

491:hacl(dmz_18,app_14,tcp,8080):1
403:hacl(dmz_5,app_12,tcp,8080):1
493:hacl(dmz_19,app_14,tcp,8080):1
358:hacl(dmz_8,app_11,tcp,8080):1
490:RULE 5 (multi-hop access):0
402:RULE 5 (multi-hop access):0
492:RULE 5 (multi-hop access):0
357:RULE 5 (multi-hop access):0
,8080,root):1
515:vulExists(app_14,vul_app,tomcat,remoteExploit,privEscalation):1
469:netAccess(app_14,tcp,8080):0
468:RULE 2 (remote exploit of a server program):0

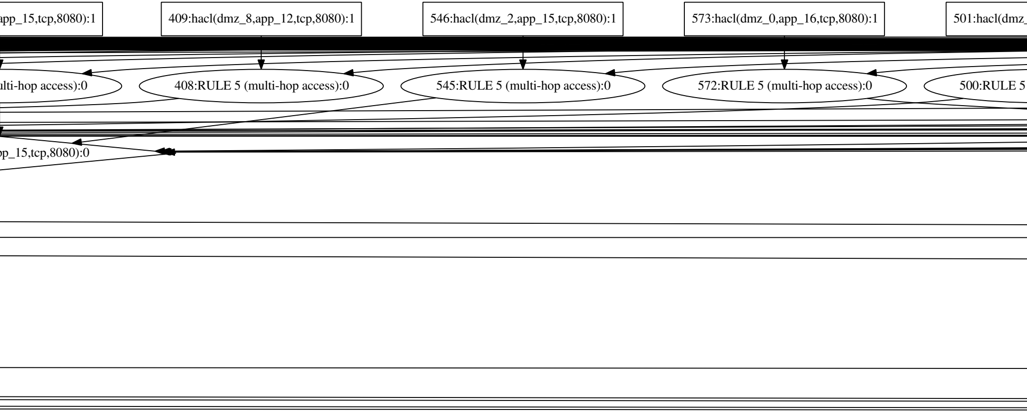

495:hacl(dmz_2,app_14,tcp,8080):1
522:hacl(dmz_0,app_15,tcp,8080):1
450:hacl(dmz_3,app_13,tcp,8080):1
524:hacl(dmz_1,app_15,tcp,8080):1
494:RULE 5 (multi-hop access):0
521:RULE 5 (multi-hop access):0
449:RULE 5 (multi-hop access):0
523:RULE 5 (multi-hop access):0

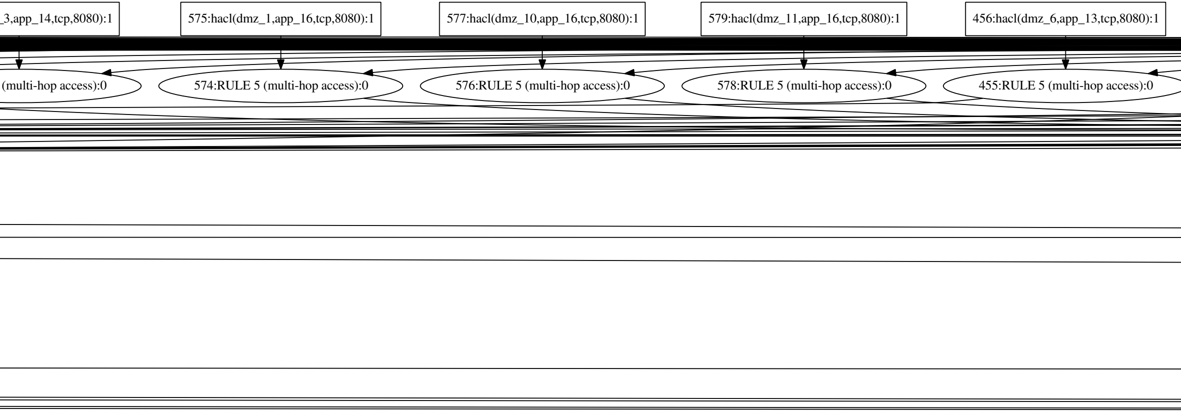

526:hacl(dmz_10,app_15,tcp,8080):1
528:hacl(dmz_11,app_15,tcp,8080):1
405:hacl(dmz_6,app_12,tcp,8080):1
530:hacl(dmz_12,app_15,tcp,8080):1
525:RULE 5 (multi-hop access):0
527:RULE 5 (multi-hop access):0
404:RULE 5 (multi-hop access):0
529:RULE 5 (multi-hop access):0

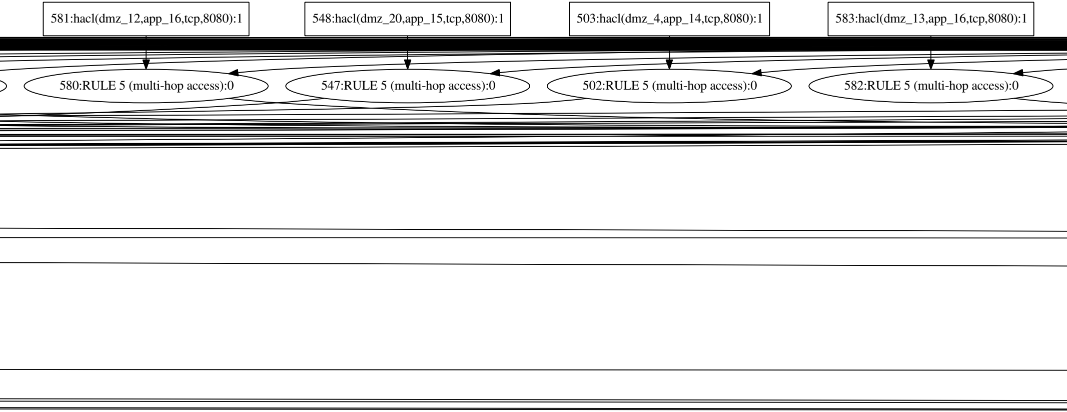

497:hacl(dmz_20,app_14,tcp,8080):1
452:hacl(dmz_4,app_13,tcp,8080):1
532:hacl(dmz_13,app_15,tcp,8080):1
534:hacl(dmz_14,app_15,tcp,8080):1
496:RULE 5 (multi-hop access):0
451:RULE 5 (multi-hop access):0
531:RULE 5 (multi-hop access):0
533:RULE 5 (multi-hop access):0

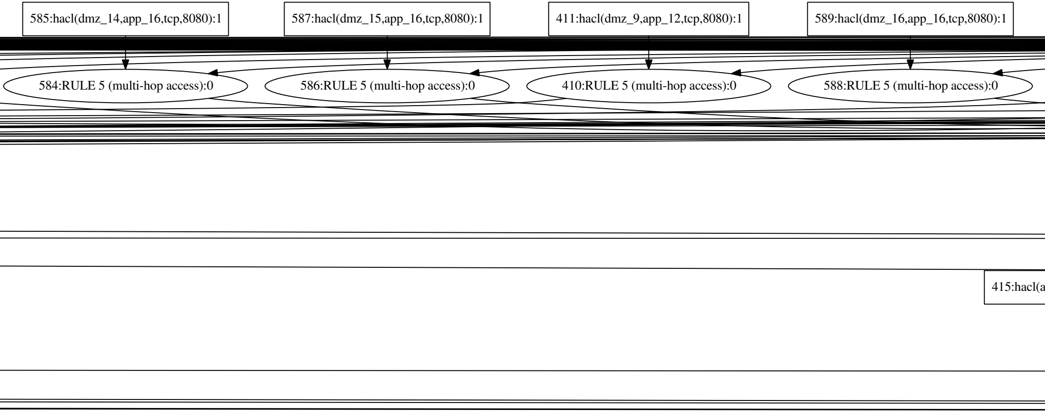

536:hacl(dmz_15,app_15,tcp,8080):1
360:hacl(dmz_9,app_11,tcp,8080):1
538:hacl(dmz_16,app_15,tcp,8080):1
407:hacl(dmz_7,app_12,tcp,8080):1
540:hacl(dmz_
535:RULE 5 (multi-hop access):0
359:RULE 5 (multi-hop access):0
537:RULE 5 (multi-hop access):0
406:RULE 5 (multi-hop access):0
539:RULE 5

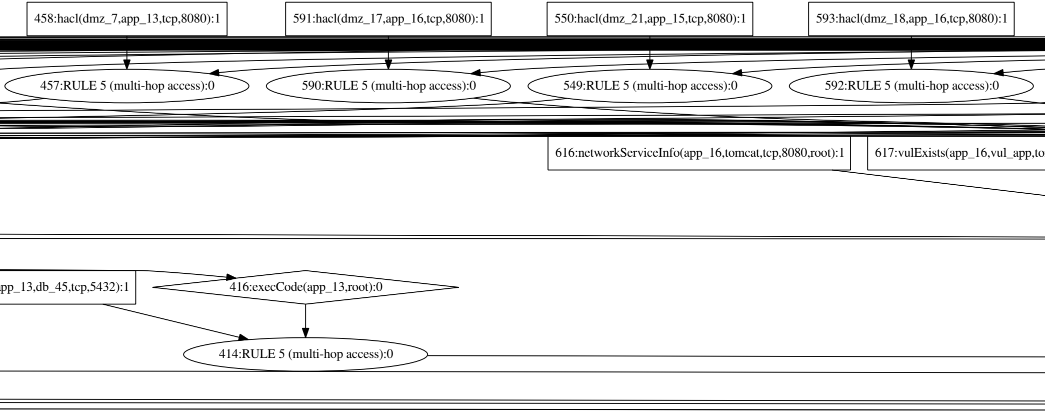

17,app_15,tcp,8080):1
499:hacl(dmz_21,app_14,tcp,8080):1
542:hacl(dmz_18,app_15,tcp,8080):1
454:hacl(dmz_5,app_13,tcp,8080):1
544:hacl(dmz_19,a
(multi-hop access):0
498:RULE 5 (multi-hop access):0
541:RULE 5 (multi-hop access):0
453:RULE 5 (multi-hop access):0
543:RULE 5 (mu
565:networkServiceInfo(app_15,tomcat,tcp,8080,root):1
566:vulExists(app_15,vul_app,tomcat,remoteExploit,privEscalation):1
520:netAccess(ap
519:RULE 2 (remote exploit of a server program):0

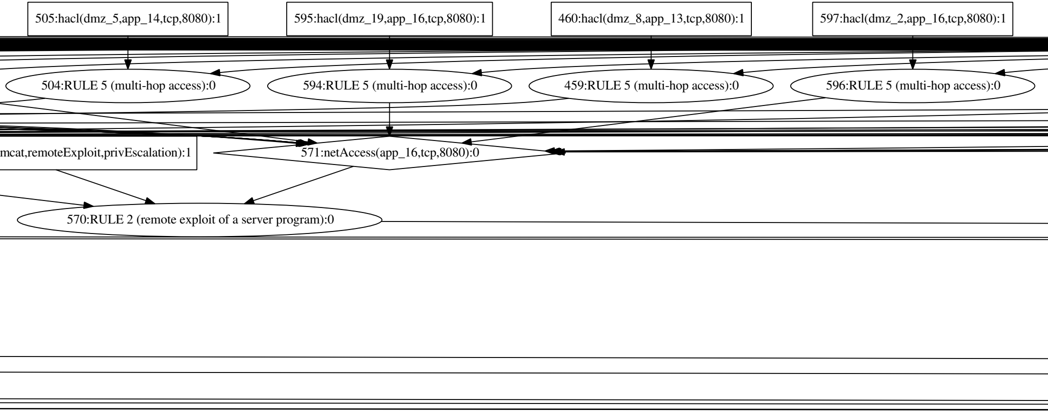

app_15,tcp,8080):1
409:hacl(dmz_8,app_12,tcp,8080):1
546:hacl(dmz_2,app_15,tcp,8080):1
573:hacl(dmz_0,app_16,tcp,8080):1
501:hacl(dmz_
lti-hop access):0
408:RULE 5 (multi-hop access):0
545:RULE 5 (multi-hop access):0
572:RULE 5 (multi-hop access):0
500:RULE 5
p_15,tcp,8080):0

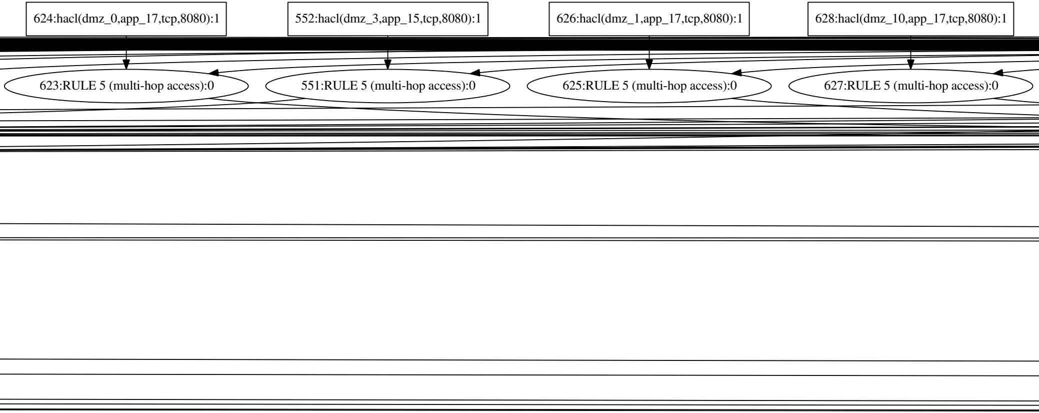

_3,app_14,tcp,8080):1
575:hacl(dmz_1,app_16,tcp,8080):1
577:hacl(dmz_10,app_16,tcp,8080):1
579:hacl(dmz_11,app_16,tcp,8080):1
456:hacl(dmz_6,app_13,tcp,8080):1
(multi-hop access):0
574:RULE 5 (multi-hop access):0
576:RULE 5 (multi-hop access):0
578:RULE 5 (multi-hop access):0
455:RULE 5 (multi-hop access):0

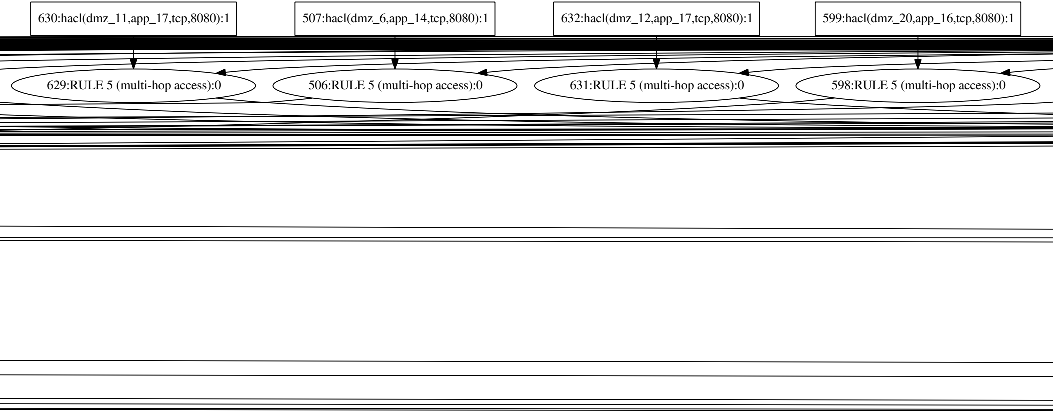

581:hacl(dmz_12,app_16,tcp,8080):1
548:hacl(dmz_20,app_15,tcp,8080):1
503:hacl(dmz_4,app_14,tcp,8080):1
583:hacl(dmz_13,app_16,tcp,8080):1
580:RULE 5 (multi-hop access):0
547:RULE 5 (multi-hop access):0
502:RULE 5 (multi-hop access):0
582:RULE 5 (multi-hop access):0

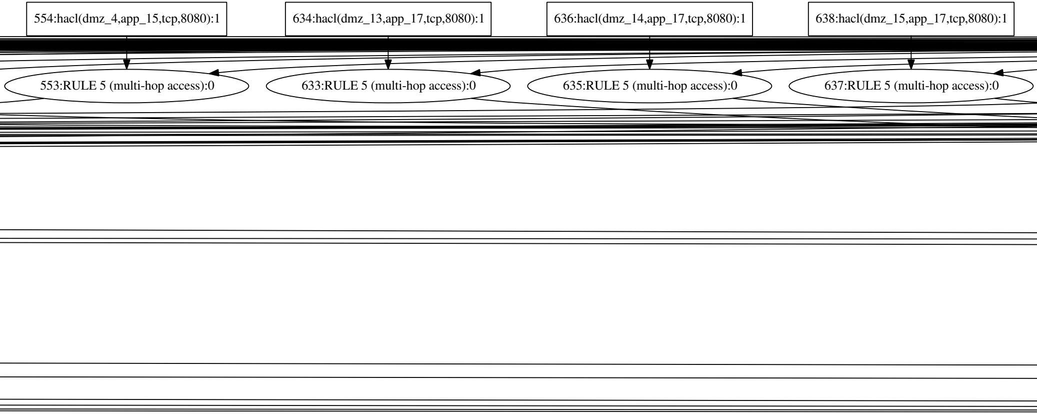

585:hacl(dmz_14,app_16,tcp,8080):1
587:hacl(dmz_15,app_16,tcp,8080):1
411:hacl(dmz_9,app_12,tcp,8080):1
589:hacl(dmz_16,app_16,tcp,8080):1
584:RULE 5 (multi-hop access):0
586:RULE 5 (multi-hop access):0
410:RULE 5 (multi-hop access):0
588:RULE 5 (multi-hop access):0
415:hacl(a

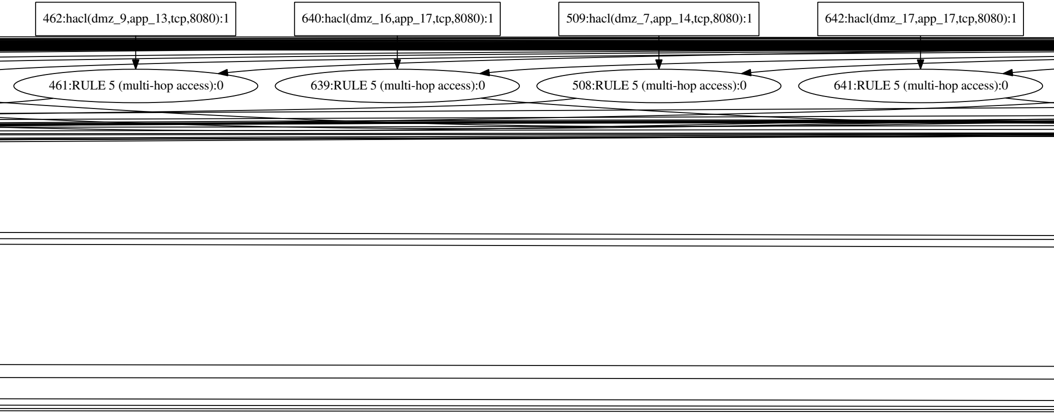

458:hacl(dmz_7,app_13,tcp,8080):1
591:hacl(dmz_17,app_16,tcp,8080):1
550:hacl(dmz_21,app_15,tcp,8080):1
593:hacl(dmz_18,app_16,tcp,8080):1
457:RULE 5 (multi-hop access):0
590:RULE 5 (multi-hop access):0
549:RULE 5 (multi-hop access):0
592:RULE 5 (multi-hop access):0
616:networkServiceInfo(app_16,tomcat,tcp,8080,root):1
617:vulExists(app_16,vul_app,to
pp_13,db_45,tcp,5432):1
416:execCode(app_13,root):0
414:RULE 5 (multi-hop access):0

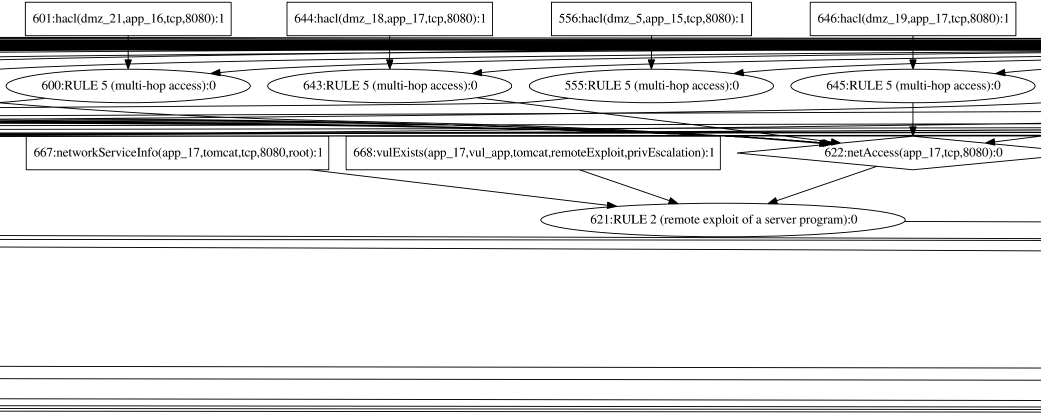

505:hacl(dmz_5,app_14,tcp,8080):1
595:hacl(dmz_19,app_16,tcp,8080):1
460:hacl(dmz_8,app_13,tcp,8080):1
597:hacl(dmz_2,app_16,tcp,8080):1
504:RULE 5 (multi-hop access):0
594:RULE 5 (multi-hop access):0
459:RULE 5 (multi-hop access):0
596:RULE 5 (multi-hop access):0
mcat,remoteExploit,privEscalation):1
571:netAccess(app_16,tcp,8080):0
570:RULE 2 (remote exploit of a server program):0

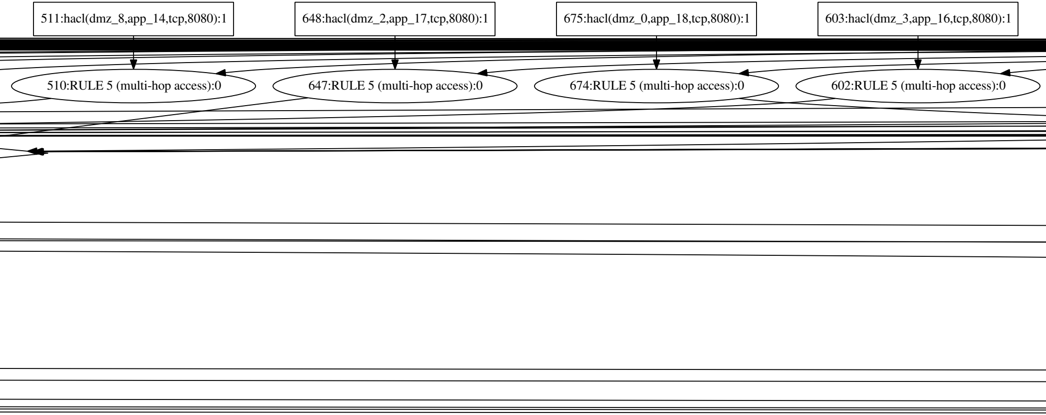

624:hacl(dmz_0,app_17,tcp,8080):1
552:hacl(dmz_3,app_15,tcp,8080):1
626:hacl(dmz_1,app_17,tcp,8080):1
628:hacl(dmz_10,app_17,tcp,8080):1
623:RULE 5 (multi-hop access):0
551:RULE 5 (multi-hop access):0
625:RULE 5 (multi-hop access):0
627:RULE 5 (multi-hop access):0

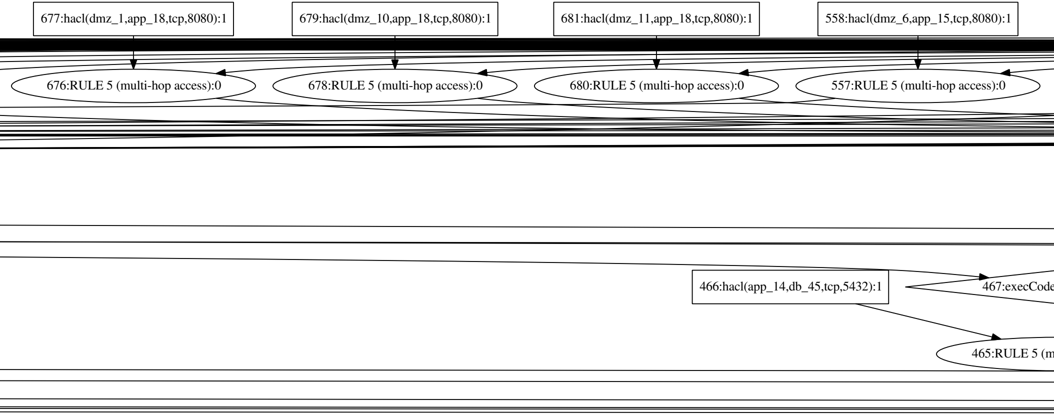

630:hacl(dmz_11,app_17,tcp,8080):1
507:hacl(dmz_6,app_14,tcp,8080):1
632:hacl(dmz_12,app_17,tcp,8080):1
599:hacl(dmz_20,app_16,tcp,8080):1
629:RULE 5 (multi-hop access):0
506:RULE 5 (multi-hop access):0
631:RULE 5 (multi-hop access):0
598:RULE 5 (multi-hop access):0

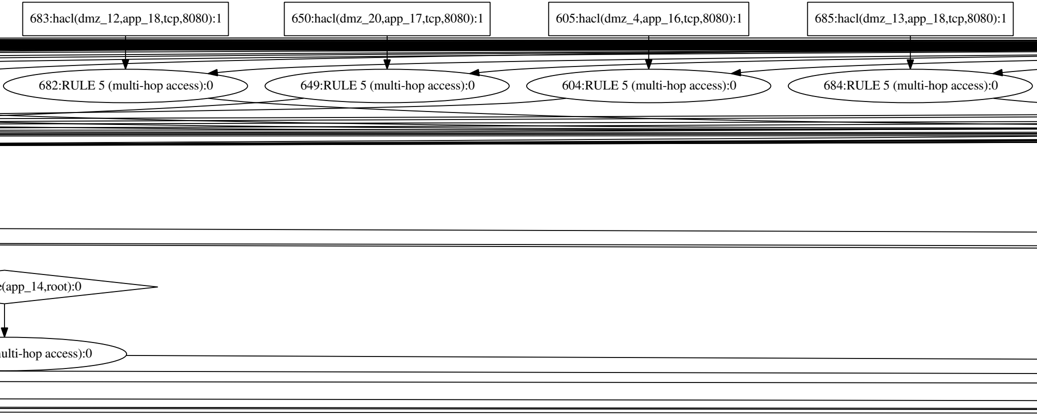

554:hacl(dmz_4,app_15,tcp,8080):1
634:hacl(dmz_13,app_17,tcp,8080):1
636:hacl(dmz_14,app_17,tcp,8080):1
638:hacl(dmz_15,app_17,tcp,8080):1
553:RULE 5 (multi-hop access):0
633:RULE 5 (multi-hop access):0
635:RULE 5 (multi-hop access):0
637:RULE 5 (multi-hop access):0

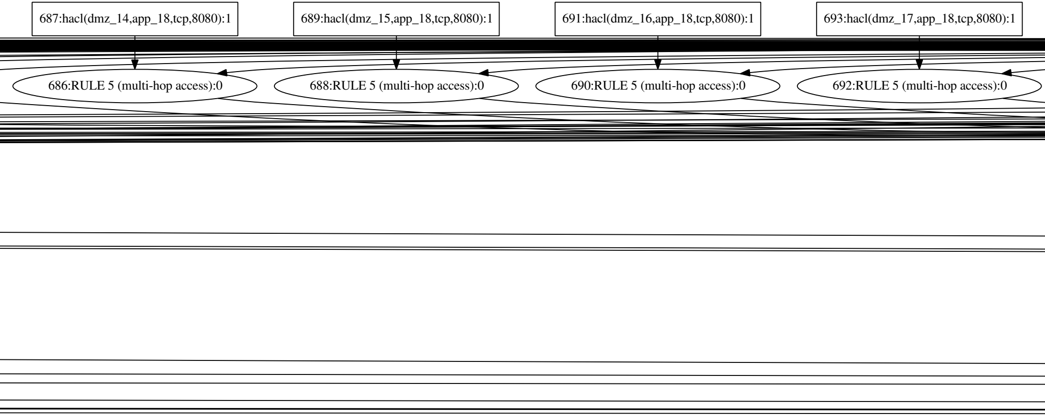

462:hacl(dmz_9,app_13,tcp,8080):1
640:hacl(dmz_16,app_17,tcp,8080):1
509:hacl(dmz_7,app_14,tcp,8080):1
642:hacl(dmz_17,app_17,tcp,8080):1
461:RULE 5 (multi-hop access):0
639:RULE 5 (multi-hop access):0
508:RULE 5 (multi-hop access):0
641:RULE 5 (multi-hop access):0

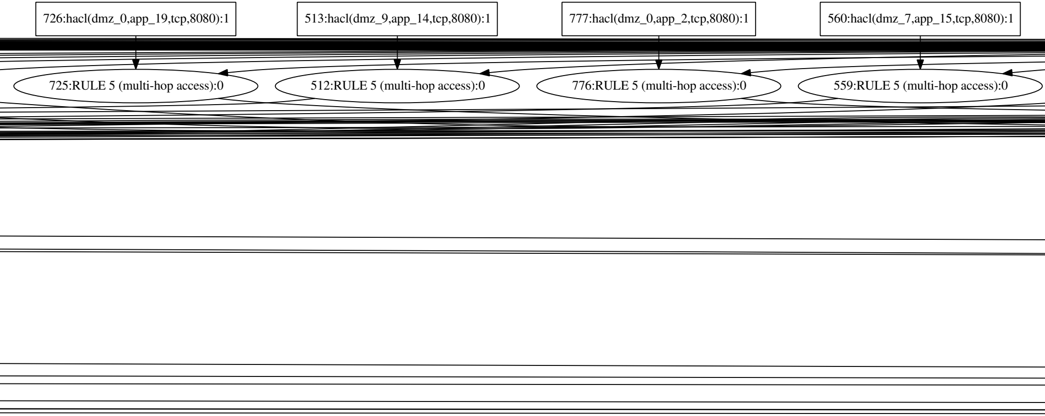

601:hacl(dmz_21,app_16,tcp,8080):1
644:hacl(dmz_18,app_17,tcp,8080):1
556:hacl(dmz_5,app_15,tcp,8080):1
646:hacl(dmz_19,app_17,tcp,8080):1
600:RULE 5 (multi-hop access):0
643:RULE 5 (multi-hop access):0
555:RULE 5 (multi-hop access):0
645:RULE 5 (multi-hop access):0
667:networkServiceInfo(app_17,tomcat,tcp,8080,root):1
668:vulExists(app_17,vul_app,tomcat,remoteExploit,privEscalation):1
622:netAccess(app_17,tcp,8080):0
621:RULE 2 (remote exploit of a server program):0

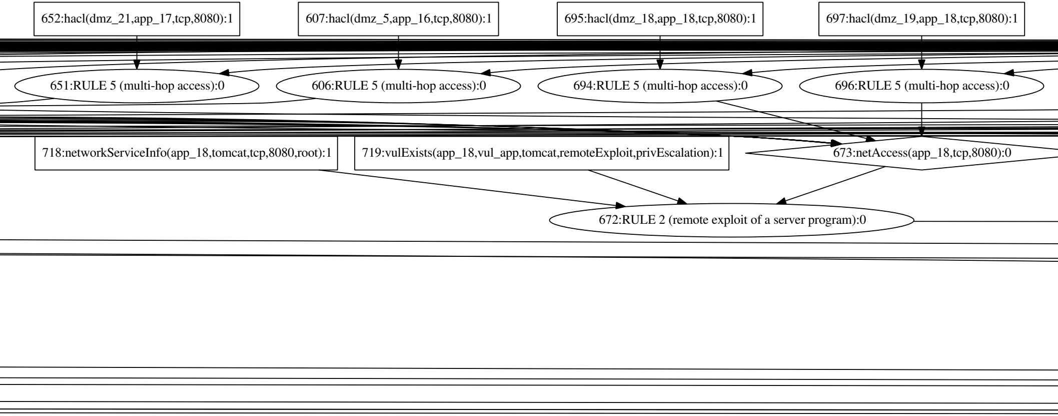

511:hacl(dmz_8,app_14,tcp,8080):1
648:hacl(dmz_2,app_17,tcp,8080):1
675:hacl(dmz_0,app_18,tcp,8080):1
603:hacl(dmz_3,app_16,tcp,8080):1
510:RULE 5 (multi-hop access):0
647:RULE 5 (multi-hop access):0
674:RULE 5 (multi-hop access):0
602:RULE 5 (multi-hop access):0

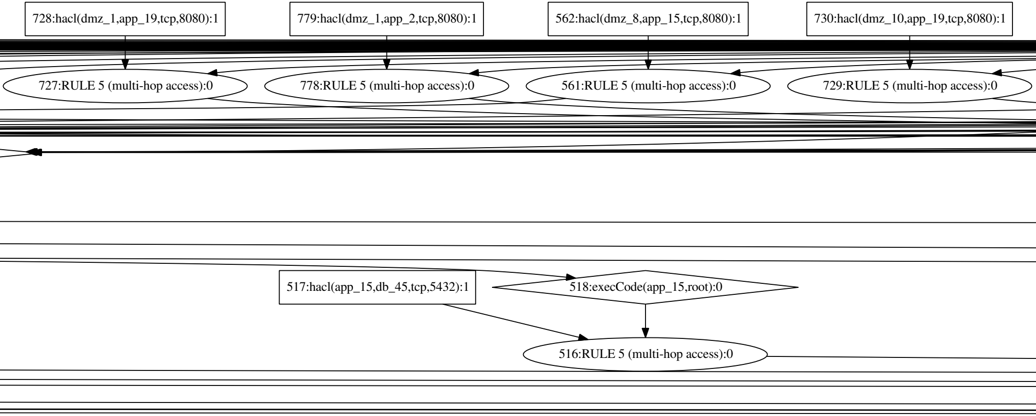

677:hacl(dmz_1,app_18,tcp,8080):1
679:hacl(dmz_10,app_18,tcp,8080):1
681:hacl(dmz_11,app_18,tcp,8080):1
558:hacl(dmz_6,app_15,tcp,8080):1
676:RULE 5 (multi-hop access):0
678:RULE 5 (multi-hop access):0
680:RULE 5 (multi-hop access):0
557:RULE 5 (multi-hop access):0
466:hacl(app_14,db_45,tcp,5432):1
467:execCode
465:RULE 5 (m

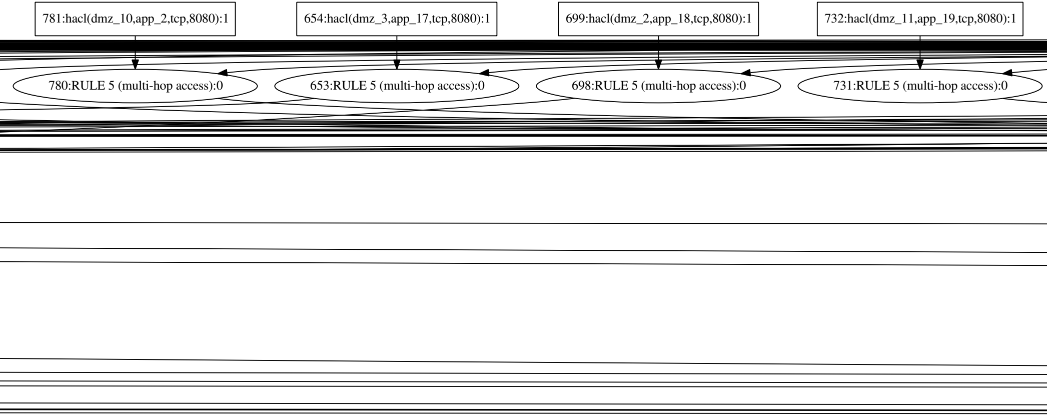

683:hacl(dmz_12,app_18,tcp,8080):1
650:hacl(dmz_20,app_17,tcp,8080):1
605:hacl(dmz_4,app_16,tcp,8080):1
685:hacl(dmz_13,app_18,tcp,8080):1
682:RULE 5 (multi-hop access):0
649:RULE 5 (multi-hop access):0
604:RULE 5 (multi-hop access):0
684:RULE 5 (multi-hop access):0
e(app_14,root):0
ulti-hop access):0

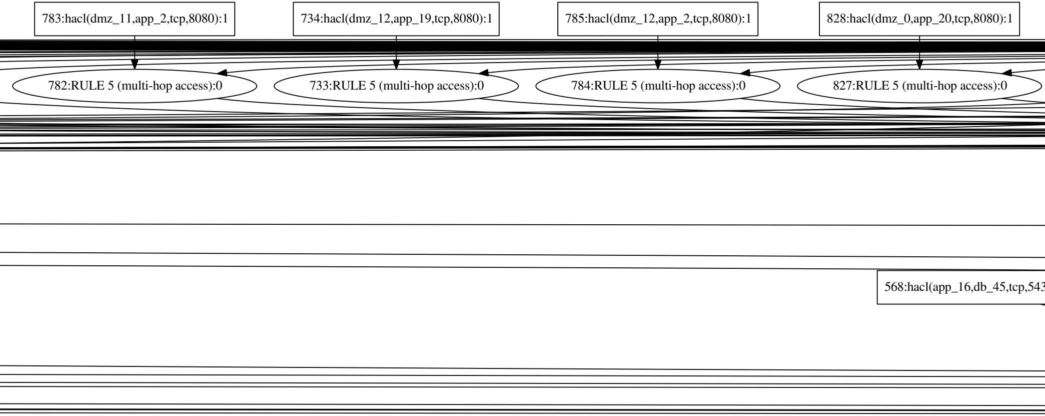

687:hacl(dmz_14,app_18,tcp,8080):1
689:hacl(dmz_15,app_18,tcp,8080):1
691:hacl(dmz_16,app_18,tcp,8080):1
693:hacl(dmz_17,app_18,tcp,8080):1
686:RULE 5 (multi-hop access):0
688:RULE 5 (multi-hop access):0
690:RULE 5 (multi-hop access):0
692:RULE 5 (multi-hop access):0

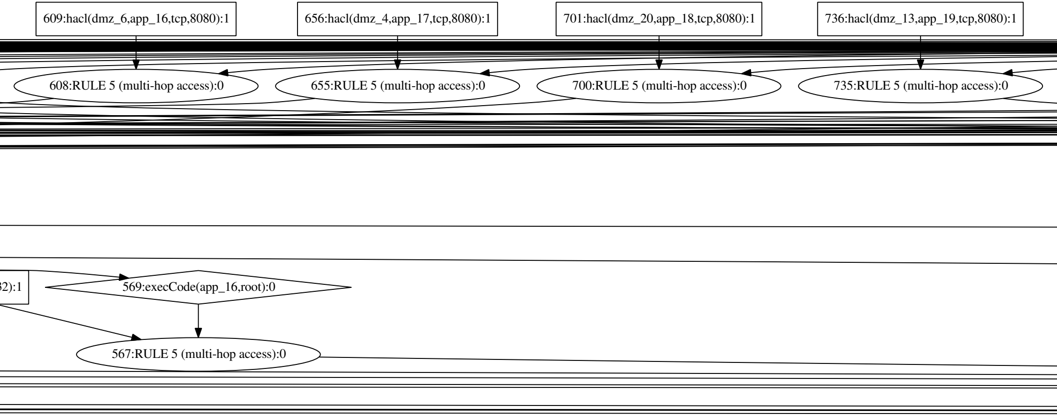

726:hacl(dmz_0,app_19,tcp,8080):1
513:hacl(dmz_9,app_14,tcp,8080):1
777:hacl(dmz_0,app_2,tcp,8080):1
560:hacl(dmz_7,app_15,tcp,8080):1
725:RULE 5 (multi-hop access):0
512:RULE 5 (multi-hop access):0
776:RULE 5 (multi-hop access):0
559:RULE 5 (multi-hop access):0

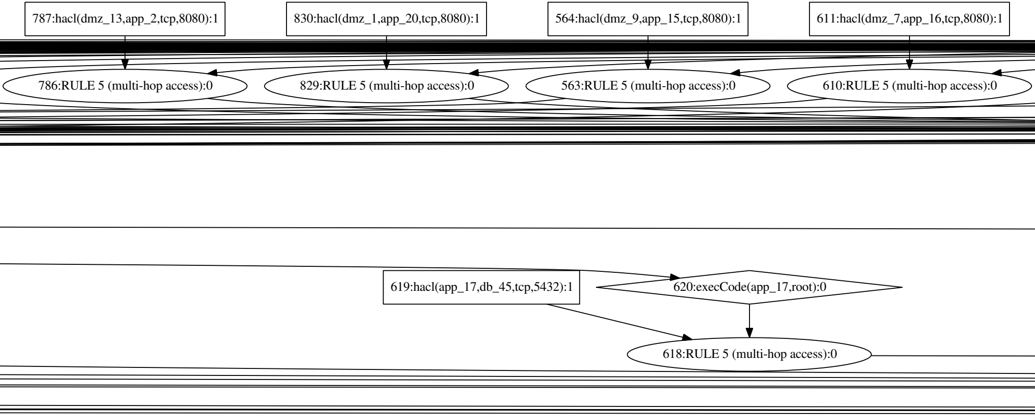

652:hacl(dmz_21,app_17,tcp,8080):1
607:hacl(dmz_5,app_16,tcp,8080):1
695:hacl(dmz_18,app_18,tcp,8080):1
697:hacl(dmz_19,app_18,tcp,8080):1
651:RULE 5 (multi-hop access):0
606:RULE 5 (multi-hop access):0
694:RULE 5 (multi-hop access):0
696:RULE 5 (multi-hop access):0
718:networkServiceInfo(app_18,tomcat,tcp,8080,root):1
719:vulExists(app_18,vul_app,tomcat,remoteExploit,privEscalation):1
673:netAccess(app_18,tcp,8080):0
672:RULE 2 (remote exploit of a server program):0

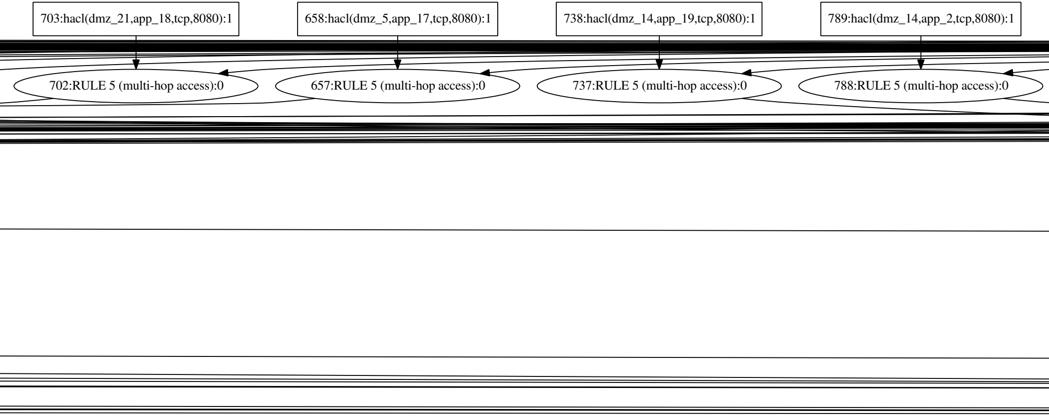

728:hacl(dmz_1,app_19,tcp,8080):1
779:hacl(dmz_1,app_2,tcp,8080):1
562:hacl(dmz_8,app_15,tcp,8080):1
730:hacl(dmz_10,app_19,tcp,8080):1
727:RULE 5 (multi-hop access):0
778:RULE 5 (multi-hop access):0
561:RULE 5 (multi-hop access):0
729:RULE 5 (multi-hop access):0
517:hacl(app_15,db_45,tcp,5432):1
518:execCode(app_15,root):0
516:RULE 5 (multi-hop access):0

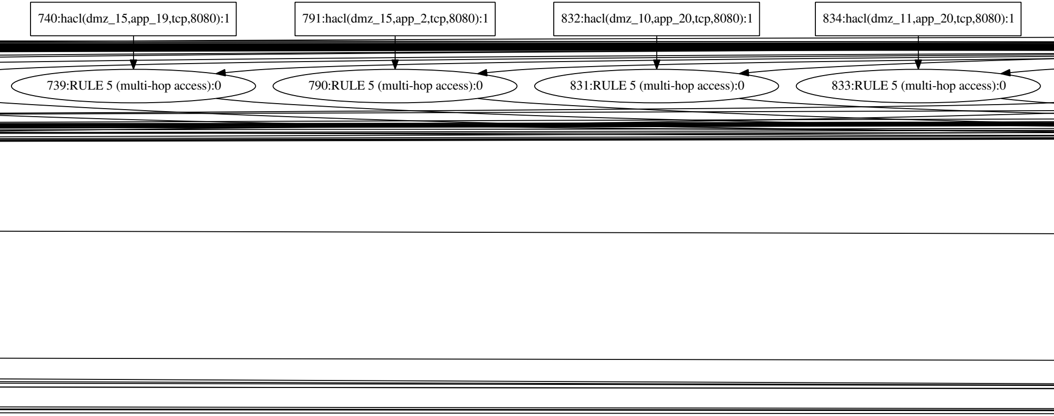

781:hacl(dmz_10,app_2,tcp,8080):1
654:hacl(dmz_3,app_17,tcp,8080):1
699:hacl(dmz_2,app_18,tcp,8080):1
732:hacl(dmz_11,app_19,tcp,8080):1
780:RULE 5 (multi-hop access):0
653:RULE 5 (multi-hop access):0
698:RULE 5 (multi-hop access):0
731:RULE 5 (multi-hop access):0

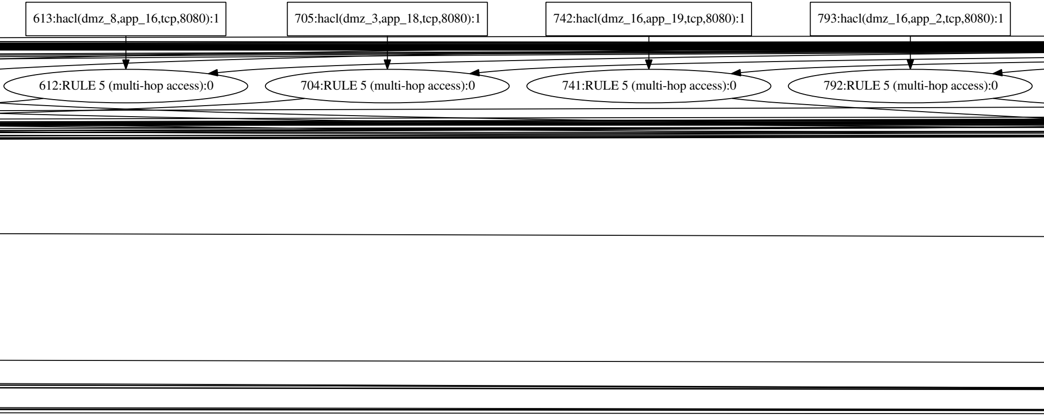

783:hacl(dmz_11,app_2,tcp,8080):1
734:hacl(dmz_12,app_19,tcp,8080):1
785:hacl(dmz_12,app_2,tcp,8080):1
828:hacl(dmz_0,app_20,tcp,8080):1
782:RULE 5 (multi-hop access):0
733:RULE 5 (multi-hop access):0
784:RULE 5 (multi-hop access):0
827:RULE 5 (multi-hop access):0
568:hacl(app_16,db_45,tcp,543

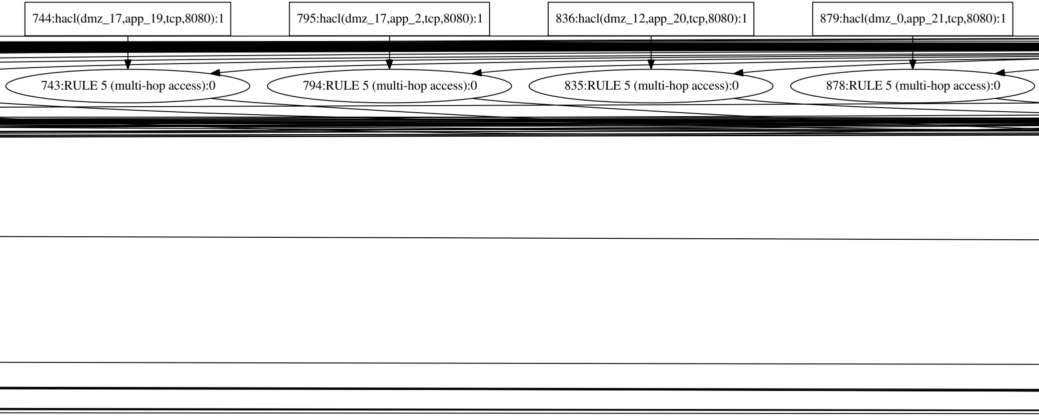

609:hacl(dmz_6,app_16,tcp,8080):1
656:hacl(dmz_4,app_17,tcp,8080):1
701:hacl(dmz_20,app_18,tcp,8080):1
736:hacl(dmz_13,app_19,tcp,8080):1
608:RULE 5 (multi-hop access):0
655:RULE 5 (multi-hop access):0
700:RULE 5 (multi-hop access):0
735:RULE 5 (multi-hop access):0
2):1
569:execCode(app_16,root):0
567:RULE 5 (multi-hop access):0

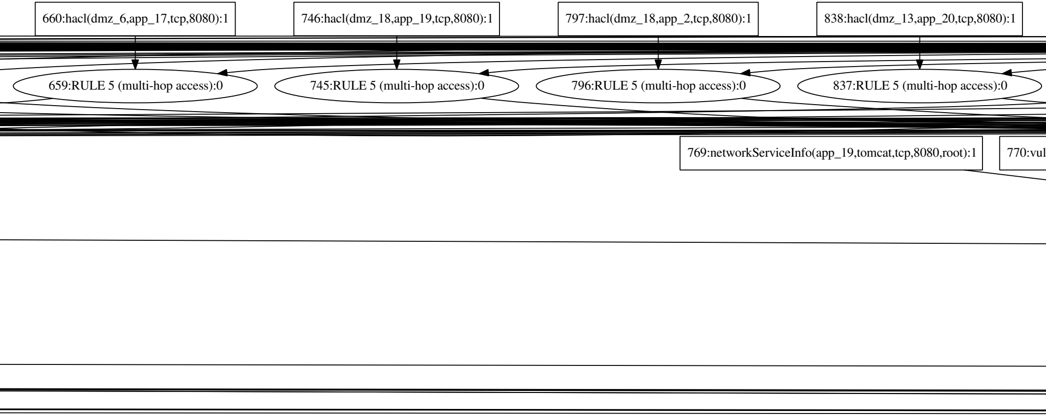

787:hacl(dmz_13,app_2,tcp,8080):1
830:hacl(dmz_1,app_20,tcp,8080):1
564:hacl(dmz_9,app_15,tcp,8080):1
611:hacl(dmz_7,app_16,tcp,8080):1
786:RULE 5 (multi-hop access):0
829:RULE 5 (multi-hop access):0
563:RULE 5 (multi-hop access):0
610:RULE 5 (multi-hop access):0
619:hacl(app_17,db_45,tcp,5432):1
620:execCode(app_17,root):0
618:RULE 5 (multi-hop access):0

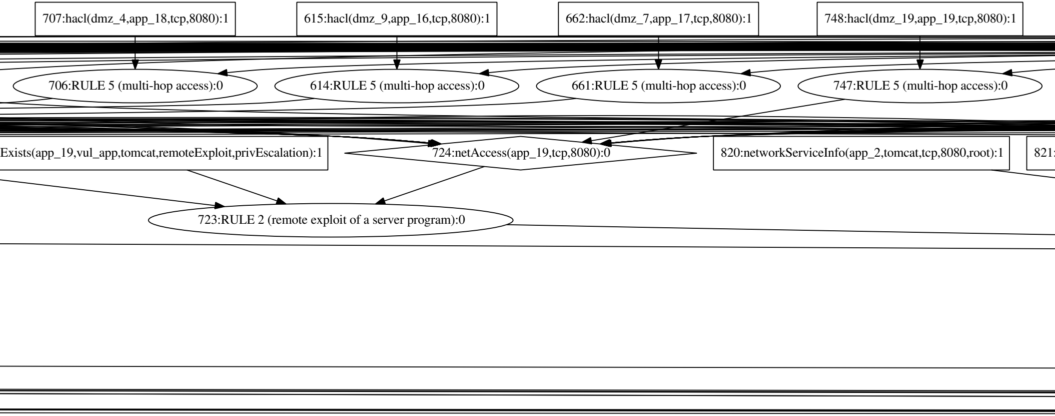

703:hacl(dmz_21,app_18,tcp,8080):1
658:hacl(dmz_5,app_17,tcp,8080):1
738:hacl(dmz_14,app_19,tcp,8080):1
789:hacl(dmz_14,app_2,tcp,8080):1
702:RULE 5 (multi-hop access):0
657:RULE 5 (multi-hop access):0
737:RULE 5 (multi-hop access):0
788:RULE 5 (multi-hop access):0

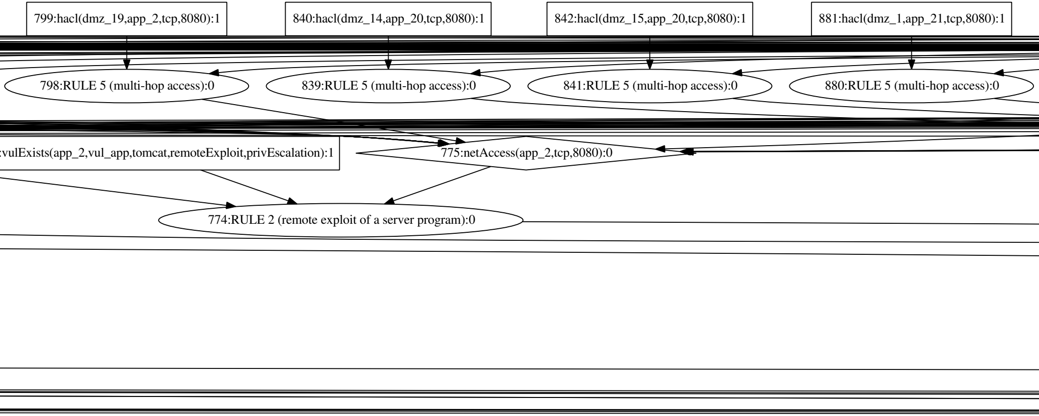

740:hacl(dmz_15,app_19,tcp,8080):1
791:hacl(dmz_15,app_2,tcp,8080):1
832:hacl(dmz_10,app_20,tcp,8080):1
834:hacl(dmz_11,app_20,tcp,8080):1
739:RULE 5 (multi-hop access):0
790:RULE 5 (multi-hop access):0
831:RULE 5 (multi-hop access):0
833:RULE 5 (multi-hop access):0

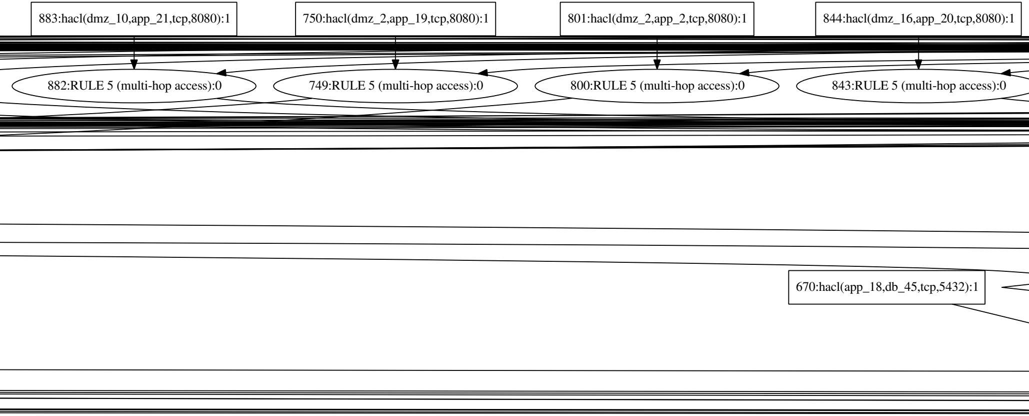

613:hacl(dmz_8,app_16,tcp,8080):1
705:hacl(dmz_3,app_18,tcp,8080):1
742:hacl(dmz_16,app_19,tcp,8080):1
793:hacl(dmz_16,app_2,tcp,8080):1
612:RULE 5 (multi-hop access):0
704:RULE 5 (multi-hop access):0
741:RULE 5 (multi-hop access):0
792:RULE 5 (multi-hop access):0

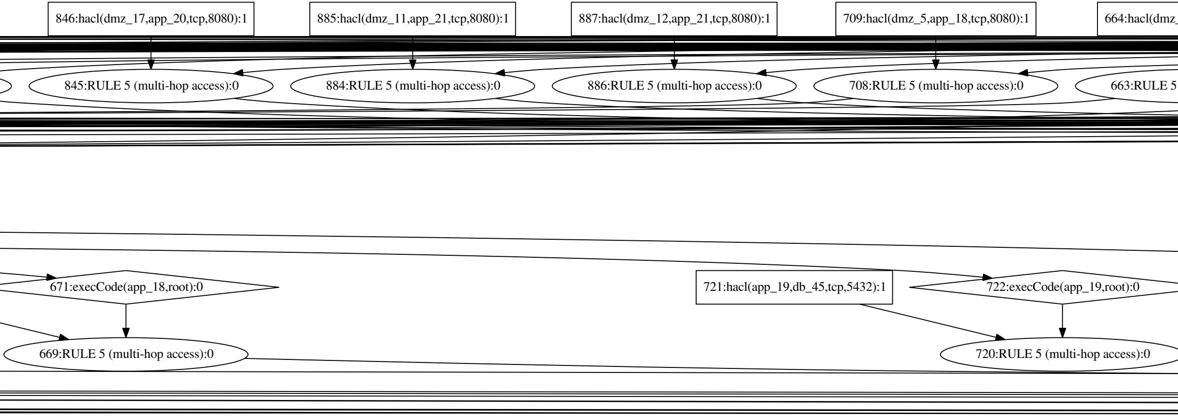

744:hacl(dmz_17,app_19,tcp,8080):1
795:hacl(dmz_17,app_2,tcp,8080):1
836:hacl(dmz_12,app_20,tcp,8080):1
879:hacl(dmz_0,app_21,tcp,8080):1
743:RULE 5 (multi-hop access):0
794:RULE 5 (multi-hop access):0
835:RULE 5 (multi-hop access):0
878:RULE 5 (multi-hop access):0

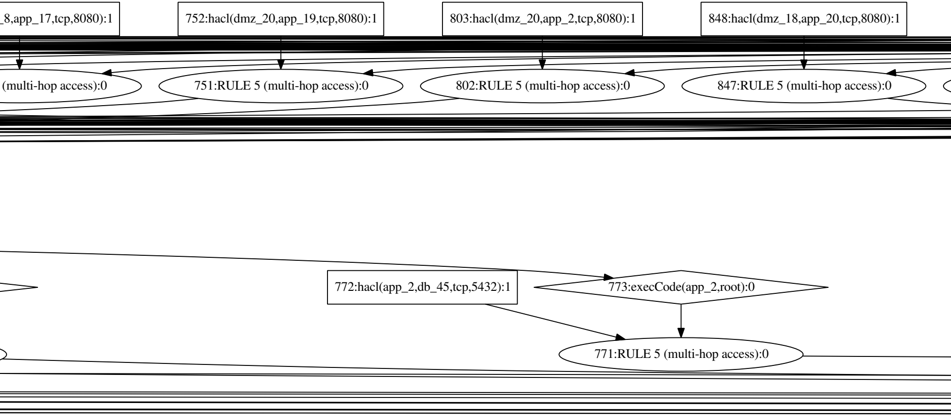

660:hacl(dmz_6,app_17,tcp,8080):1
746:hacl(dmz_18,app_19,tcp,8080):1
797:hacl(dmz_18,app_2,tcp,8080):1
838:hacl(dmz_13,app_20,tcp,8080):1
659:RULE 5 (multi-hop access):0
745:RULE 5 (multi-hop access):0
796:RULE 5 (multi-hop access):0
837:RULE 5 (multi-hop access):0
769:networkServiceInfo(app_19,tomcat,tcp,8080,root):1
770:vul

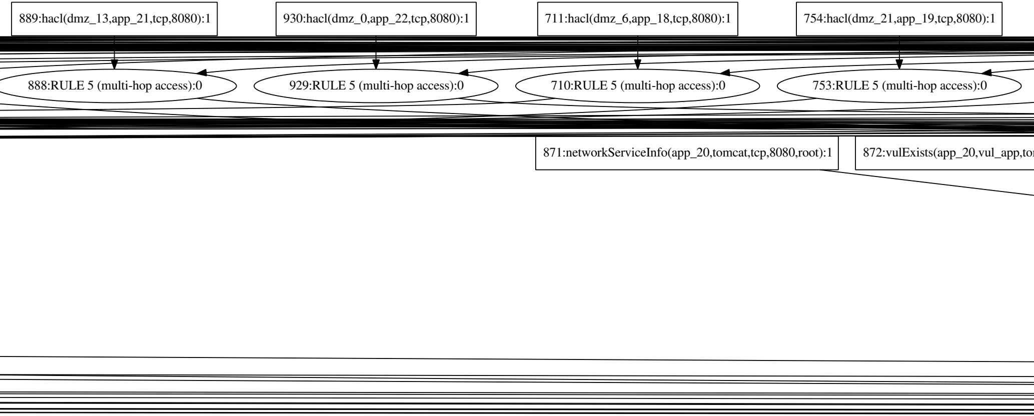

707:hacl(dmz_4,app_18,tcp,8080):1
615:hacl(dmz_9,app_16,tcp,8080):1
662:hacl(dmz_7,app_17,tcp,8080):1
748:hacl(dmz_19,app_19,tcp,8080):1
706:RULE 5 (multi-hop access):0
614:RULE 5 (multi-hop access):0
661:RULE 5 (multi-hop access):0
747:RULE 5 (multi-hop access):0
Exists(app_19,vul_app,tomcat,remoteExploit,privEscalation):1
724:netAccess(app_19,tcp,8080):0
820:networkServiceInfo(app_2,tomcat,tcp,8080,root):1
821:
723:RULE 2 (remote exploit of a server program):0

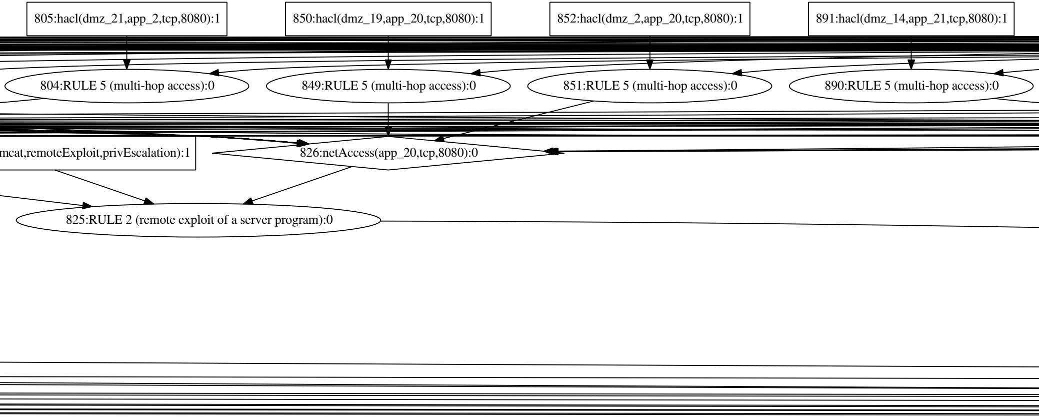

799:hacl(dmz_19,app_2,tcp,8080):1
840:hacl(dmz_14,app_20,tcp,8080):1
842:hacl(dmz_15,app_20,tcp,8080):1
881:hacl(dmz_1,app_21,tcp,8080):1
798:RULE 5 (multi-hop access):0
839:RULE 5 (multi-hop access):0
841:RULE 5 (multi-hop access):0
880:RULE 5 (multi-hop access):0
:vulExists(app_2,vul_app,tomcat,remoteExploit,privEscalation):1
775:netAccess(app_2,tcp,8080):0
774:RULE 2 (remote exploit of a server program):0

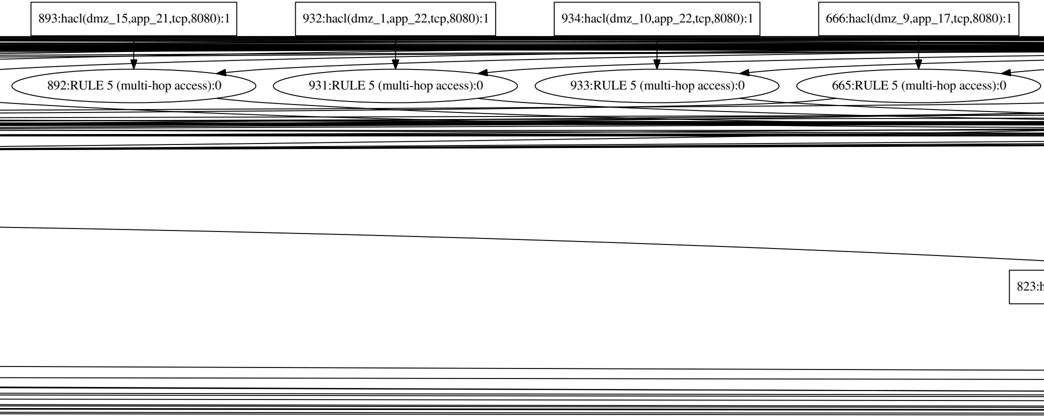

883:hacl(dmz_10,app_21,tcp,8080):1
750:hacl(dmz_2,app_19,tcp,8080):1
801:hacl(dmz_2,app_2,tcp,8080):1
844:hacl(dmz_16,app_20,tcp,8080):1
882:RULE 5 (multi-hop access):0
749:RULE 5 (multi-hop access):0
800:RULE 5 (multi-hop access):0
843:RULE 5 (multi-hop access):0
670:hacl(app_18,db_45,tcp,5432):1

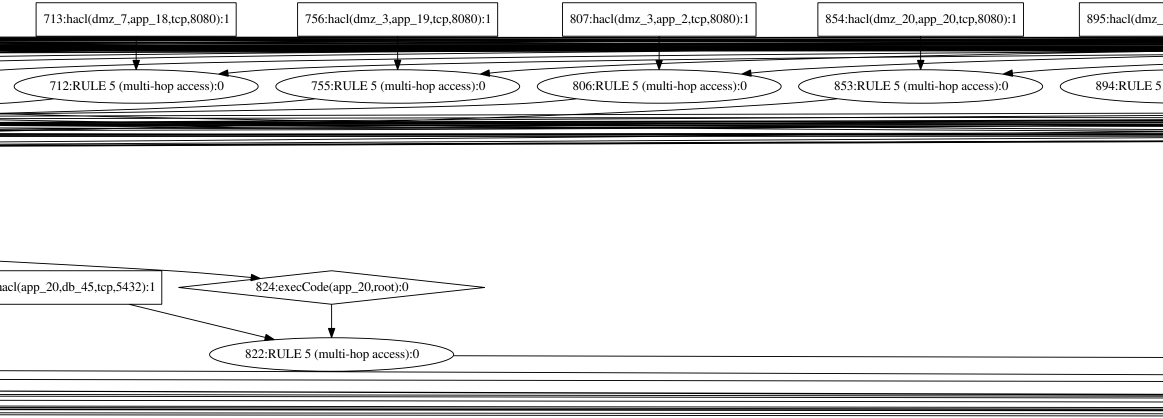

846:hacl(dmz_17,app_20,tcp,8080):1
885:hacl(dmz_11,app_21,tcp,8080):1
887:hacl(dmz_12,app_21,tcp,8080):1
709:hacl(dmz_5,app_18,tcp,8080):1
664:hacl(dmz_
845:RULE 5 (multi-hop access):0
884:RULE 5 (multi-hop access):0
886:RULE 5 (multi-hop access):0
708:RULE 5 (multi-hop access):0
663:RULE 5
671:execCode(app_18,root):0
721:hacl(app_19,db_45,tcp,5432):1
722:execCode(app_19,root):0
669:RULE 5 (multi-hop access):0
720:RULE 5 (multi-hop access):0

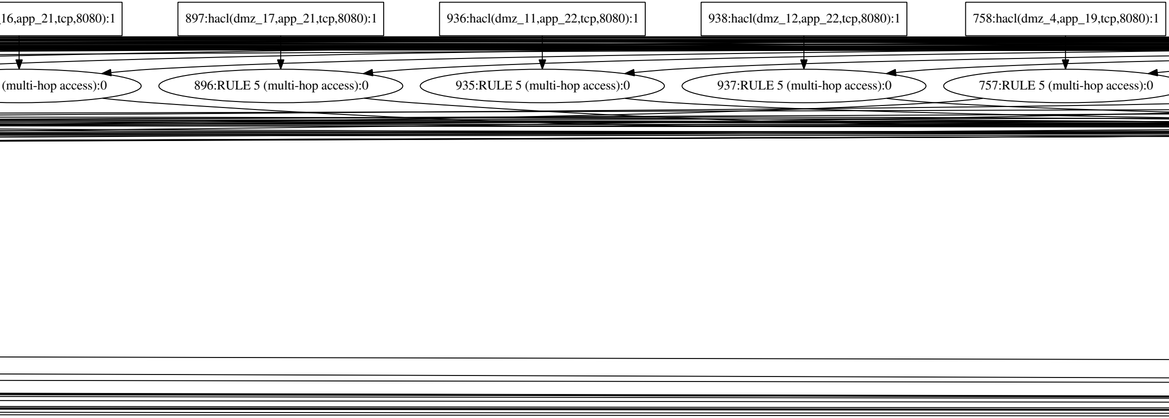

_8,app_17,tcp,8080):1
752:hacl(dmz_20,app_19,tcp,8080):1
803:hacl(dmz_20,app_2,tcp,8080):1
848:hacl(dmz_18,app_20,tcp,8080):1
(multi-hop access):0
751:RULE 5 (multi-hop access):0
802:RULE 5 (multi-hop access):0
847:RULE 5 (multi-hop access):0
772:hacl(app_2,db_45,tcp,5432):1
773:execCode(app_2,root):0
771:RULE 5 (multi-hop access):0

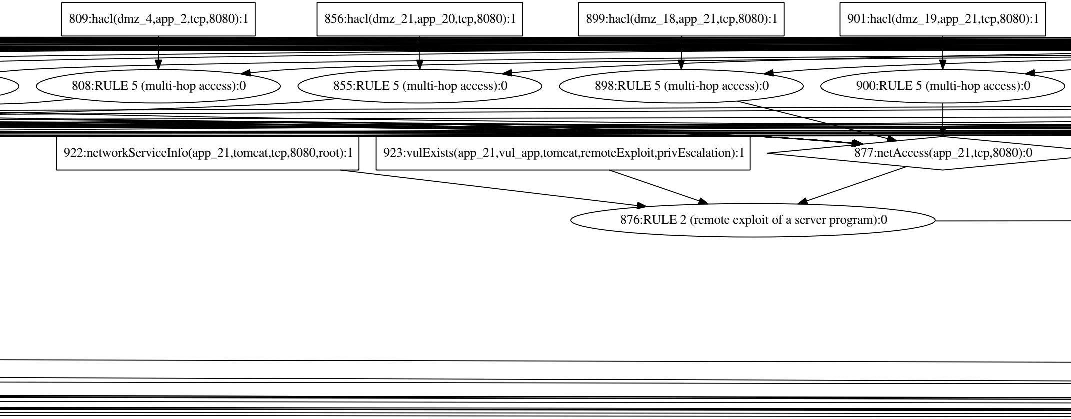

889:hacl(dmz_13,app_21,tcp,8080):1
930:hacl(dmz_0,app_22,tcp,8080):1
711:hacl(dmz_6,app_18,tcp,8080):1
754:hacl(dmz_21,app_19,tcp,8080):1
888:RULE 5 (multi-hop access):0
929:RULE 5 (multi-hop access):0
710:RULE 5 (multi-hop access):0
753:RULE 5 (multi-hop access):0
871:networkServiceInfo(app_20,tomcat,tcp,8080,root):1
872:vulExists(app_20,vul_app,to

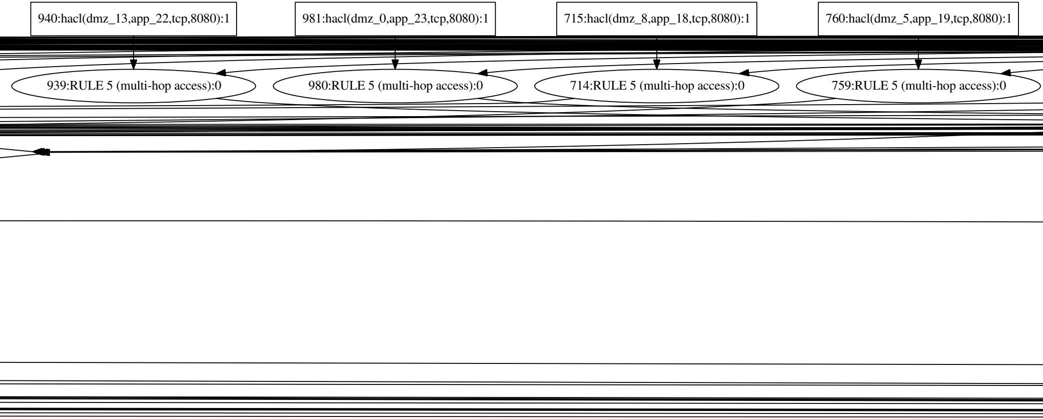

805:hacl(dmz_21,app_2,tcp,8080):1
850:hacl(dmz_19,app_20,tcp,8080):1
852:hacl(dmz_2,app_20,tcp,8080):1
891:hacl(dmz_14,app_21,tcp,8080):1
804:RULE 5 (multi-hop access):0
849:RULE 5 (multi-hop access):0
851:RULE 5 (multi-hop access):0
890:RULE 5 (multi-hop access):0
mcat,remoteExploit,privEscalation):1
826:netAccess(app_20,tcp,8080):0
825:RULE 2 (remote exploit of a server program):0

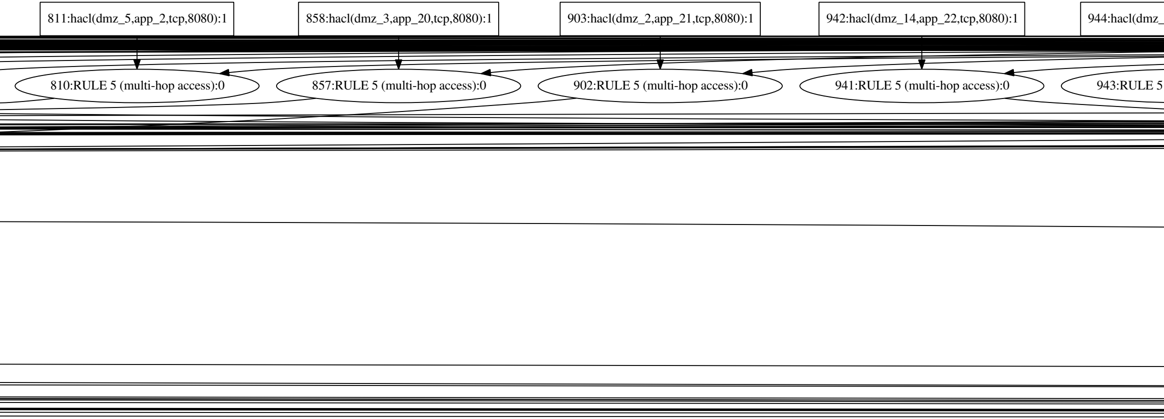

893:hacl(dmz_15,app_21,tcp,8080):1
932:hacl(dmz_1,app_22,tcp,8080):1
934:hacl(dmz_10,app_22,tcp,8080):1
666:hacl(dmz_9,app_17,tcp,8080):1
892:RULE 5 (multi-hop access):0
931:RULE 5 (multi-hop access):0
933:RULE 5 (multi-hop access):0
665:RULE 5 (multi-hop access):0
823:h

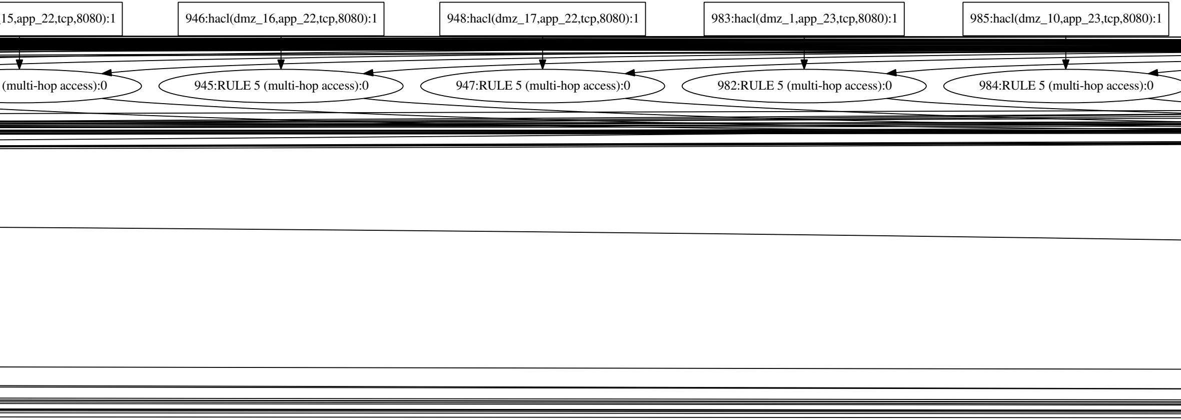

713:hacl(dmz_7,app_18,tcp,8080):1
756:hacl(dmz_3,app_19,tcp,8080):1
807:hacl(dmz_3,app_2,tcp,8080):1
854:hacl(dmz_20,app_20,tcp,8080):1
895:hacl(dmz_
712:RULE 5 (multi-hop access):0
755:RULE 5 (multi-hop access):0
806:RULE 5 (multi-hop access):0
853:RULE 5 (multi-hop access):0
894:RULE 5
hacl(app_20,db_45,tcp,5432):1
824:execCode(app_20,root):0
822:RULE 5 (multi-hop access):0

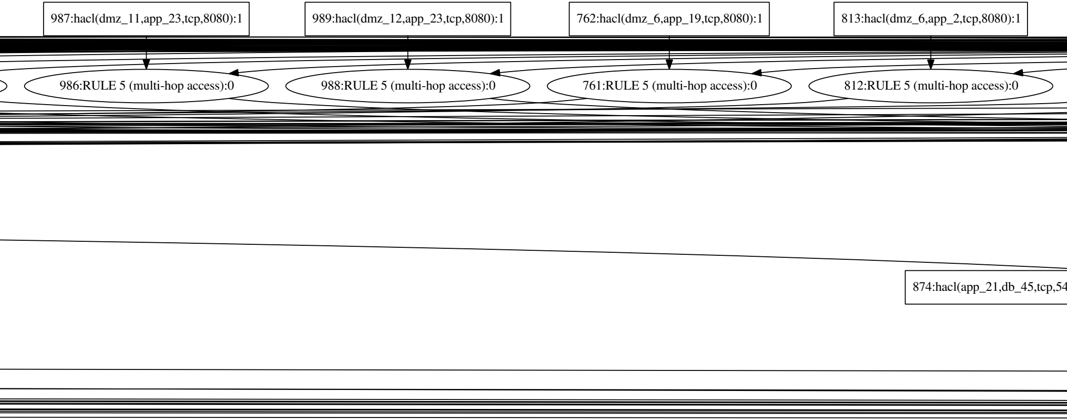

16,app_21,tcp,8080):1
897:hacl(dmz_17,app_21,tcp,8080):1
936:hacl(dmz_11,app_22,tcp,8080):1
938:hacl(dmz_12,app_22,tcp,8080):1
758:hacl(dmz_4,app_19,tcp,8080):1
(multi-hop access):0
896:RULE 5 (multi-hop access):0
935:RULE 5 (multi-hop access):0
937:RULE 5 (multi-hop access):0
757:RULE 5 (multi-hop access):0

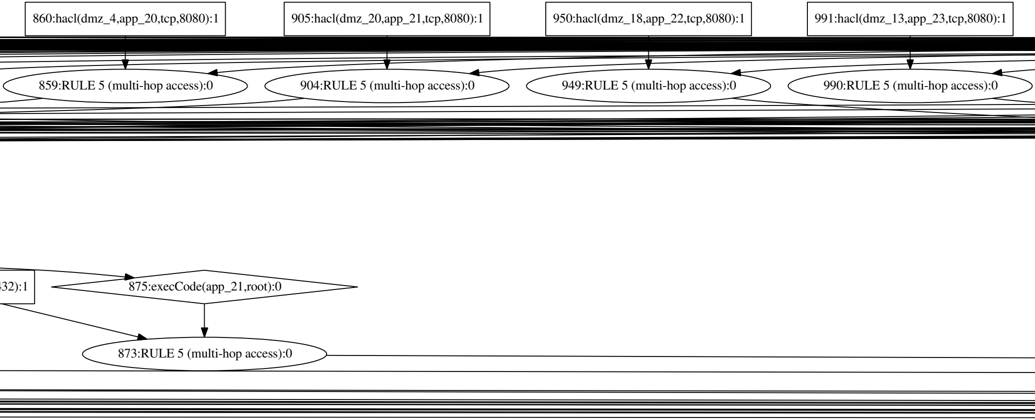

809:hacl(dmz_4,app_2,tcp,8080):1
856:hacl(dmz_21,app_20,tcp,8080):1
899:hacl(dmz_18,app_21,tcp,8080):1
901:hacl(dmz_19,app_21,tcp,8080):1
808:RULE 5 (multi-hop access):0
855:RULE 5 (multi-hop access):0
898:RULE 5 (multi-hop access):0
900:RULE 5 (multi-hop access):0
922:networkServiceInfo(app_21,tomcat,tcp,8080,root):1
923:vulExists(app_21,vul_app,tomcat,remoteExploit,privEscalation):1
877:netAccess(app_21,tcp,8080):0
876:RULE 2 (remote exploit of a server program):0

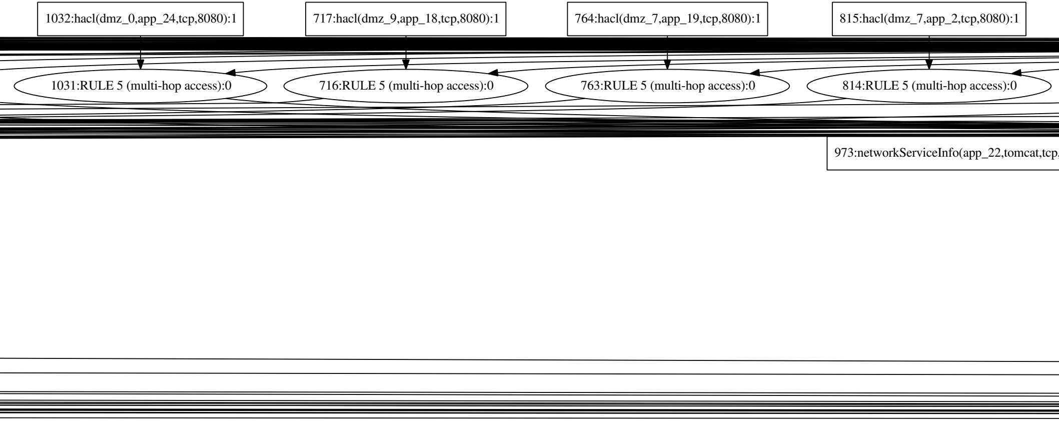

940:hacl(dmz_13,app_22,tcp,8080):1
981:hacl(dmz_0,app_23,tcp,8080):1
715:hacl(dmz_8,app_18,tcp,8080):1
760:hacl(dmz_5,app_19,tcp,8080):1
939:RULE 5 (multi-hop access):0
980:RULE 5 (multi-hop access):0
714:RULE 5 (multi-hop access):0
759:RULE 5 (multi-hop access):0

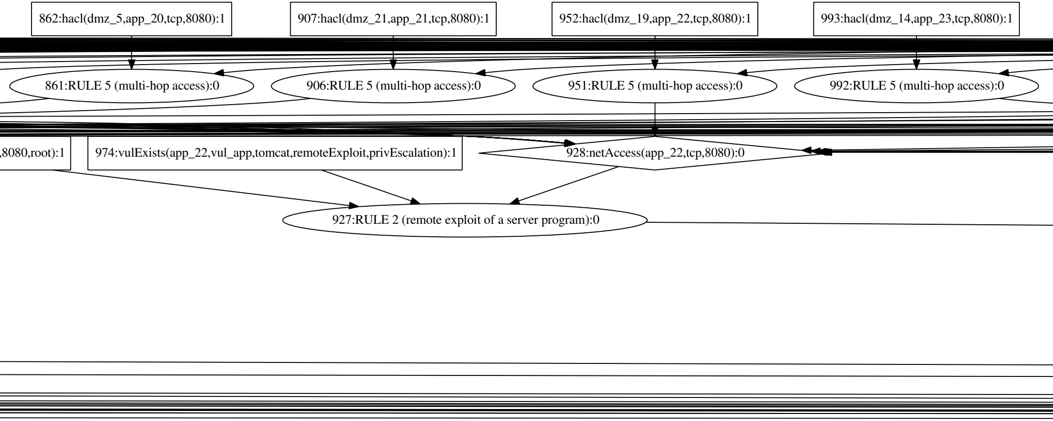

811:hacl(dmz_5,app_2,tcp,8080):1
858:hacl(dmz_3,app_20,tcp,8080):1
903:hacl(dmz_2,app_21,tcp,8080):1
942:hacl(dmz_14,app_22,tcp,8080):1
944:hacl(dmz_
810:RULE 5 (multi-hop access):0
857:RULE 5 (multi-hop access):0
902:RULE 5 (multi-hop access):0
941:RULE 5 (multi-hop access):0
943:RULE 5

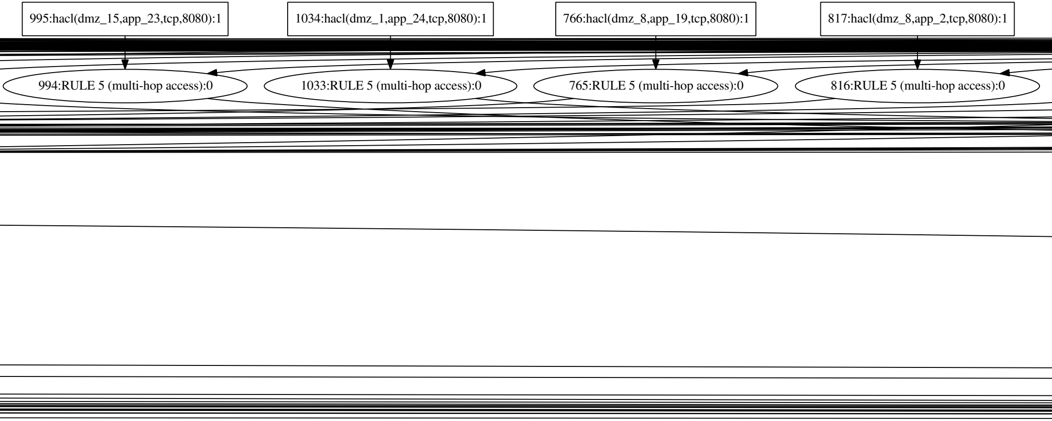

15,app_22,tcp,8080):1
946:hacl(dmz_16,app_22,tcp,8080):1
948:hacl(dmz_17,app_22,tcp,8080):1
983:hacl(dmz_1,app_23,tcp,8080):1
985:hacl(dmz_10,app_23,tcp,8080):1
(multi-hop access):0
945:RULE 5 (multi-hop access):0
947:RULE 5 (multi-hop access):0
982:RULE 5 (multi-hop access):0
984:RULE 5 (multi-hop access):0

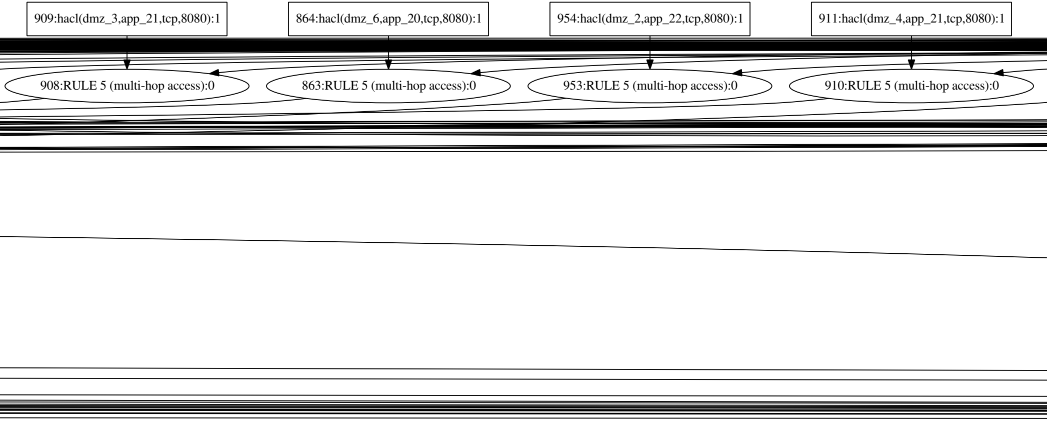

987:hacl(dmz_11,app_23,tcp,8080):1
989:hacl(dmz_12,app_23,tcp,8080):1
762:hacl(dmz_6,app_19,tcp,8080):1
813:hacl(dmz_6,app_2,tcp,8080):1
986:RULE 5 (multi-hop access):0
988:RULE 5 (multi-hop access):0
761:RULE 5 (multi-hop access):0
812:RULE 5 (multi-hop access):0
874:hacl(app_21,db_45,tcp,54

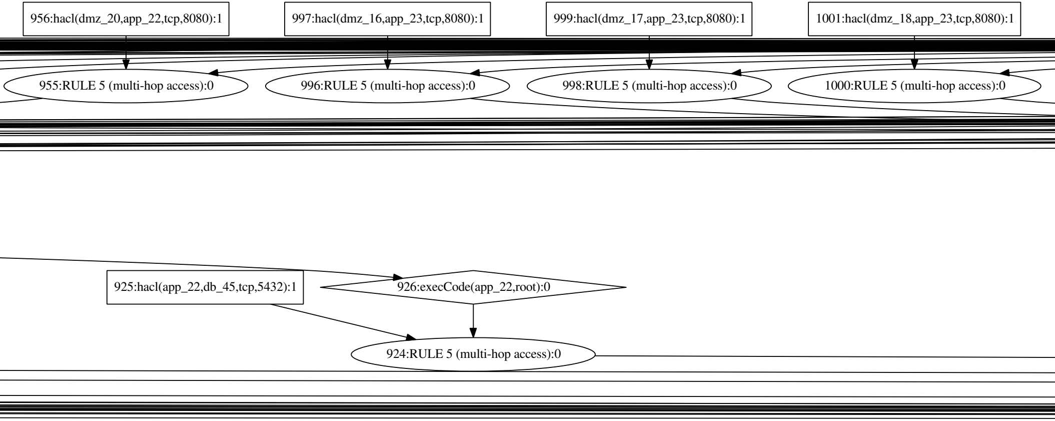

860:hacl(dmz_4,app_20,tcp,8080):1
905:hacl(dmz_20,app_21,tcp,8080):1
950:hacl(dmz_18,app_22,tcp,8080):1
991:hacl(dmz_13,app_23,tcp,8080):1
859:RULE 5 (multi-hop access):0
904:RULE 5 (multi-hop access):0
949:RULE 5 (multi-hop access):0
990:RULE 5 (multi-hop access):0
432):1
875:execCode(app_21,root):0
873:RULE 5 (multi-hop access):0

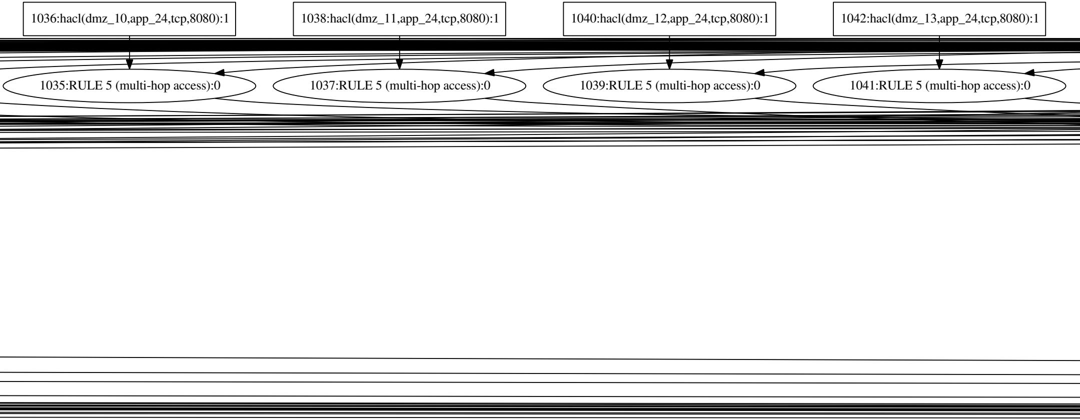

1032:hacl(dmz_0,app_24,tcp,8080):1
717:hacl(dmz_9,app_18,tcp,8080):1
764:hacl(dmz_7,app_19,tcp,8080):1
815:hacl(dmz_7,app_2,tcp,8080):1
1031:RULE 5 (multi-hop access):0
716:RULE 5 (multi-hop access):0
763:RULE 5 (multi-hop access):0
814:RULE 5 (multi-hop access):0
973:networkServiceInfo(app_22,tomcat,tcp,

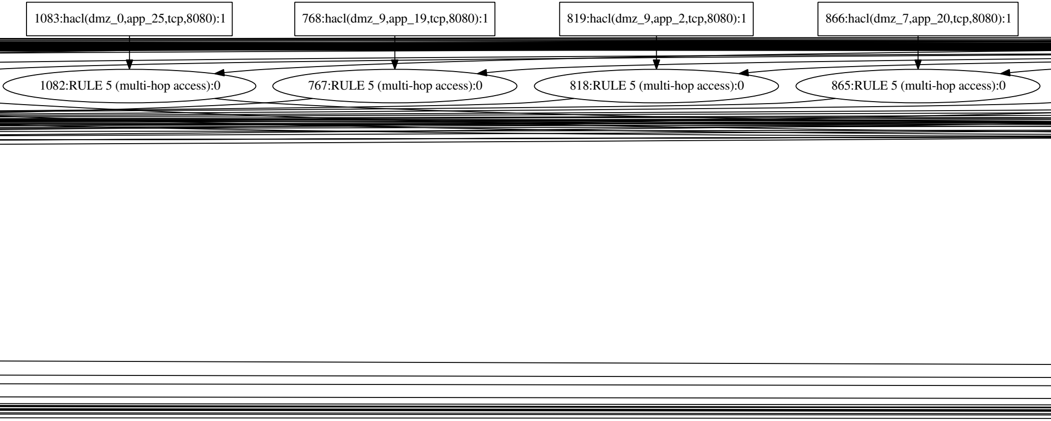

862:hacl(dmz_5,app_20,tcp,8080):1
907:hacl(dmz_21,app_21,tcp,8080):1
952:hacl(dmz_19,app_22,tcp,8080):1
993:hacl(dmz_14,app_23,tcp,8080):1
861:RULE 5 (multi-hop access):0
906:RULE 5 (multi-hop access):0
951:RULE 5 (multi-hop access):0
992:RULE 5 (multi-hop access):0
,8080,root):1
974:vulExists(app_22,vul_app,tomcat,remoteExploit,privEscalation):1
928:netAccess(app_22,tcp,8080):0
927:RULE 2 (remote exploit of a server program):0

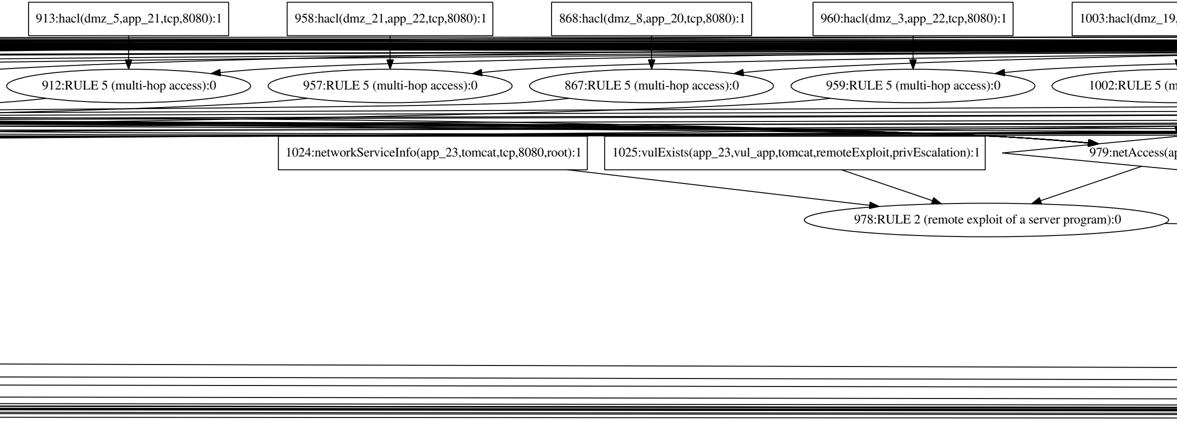

995:hacl(dmz_15,app_23,tcp,8080):1
1034:hacl(dmz_1,app_24,tcp,8080):1
766:hacl(dmz_8,app_19,tcp,8080):1
817:hacl(dmz_8,app_2,tcp,8080):1
994:RULE 5 (multi-hop access):0
1033:RULE 5 (multi-hop access):0
765:RULE 5 (multi-hop access):0
816:RULE 5 (multi-hop access):0

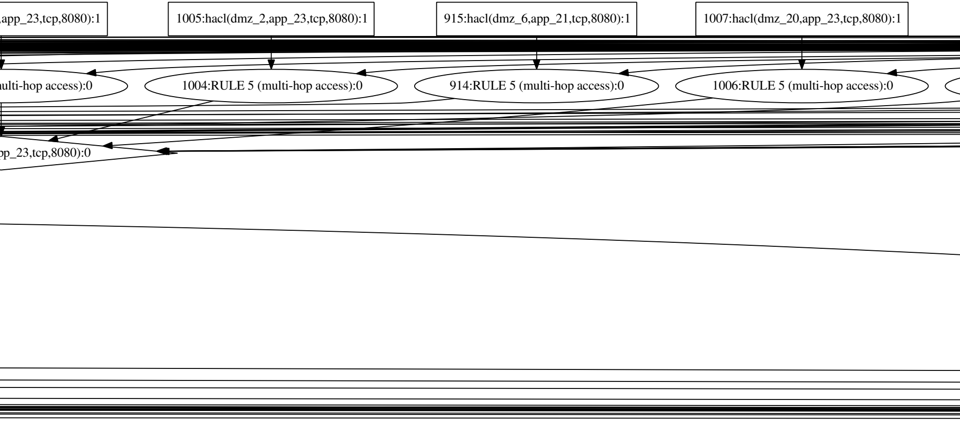

909:hacl(dmz_3,app_21,tcp,8080):1
864:hacl(dmz_6,app_20,tcp,8080):1
954:hacl(dmz_2,app_22,tcp,8080):1
911:hacl(dmz_4,app_21,tcp,8080):1
908:RULE 5 (multi-hop access):0
863:RULE 5 (multi-hop access):0
953:RULE 5 (multi-hop access):0
910:RULE 5 (multi-hop access):0

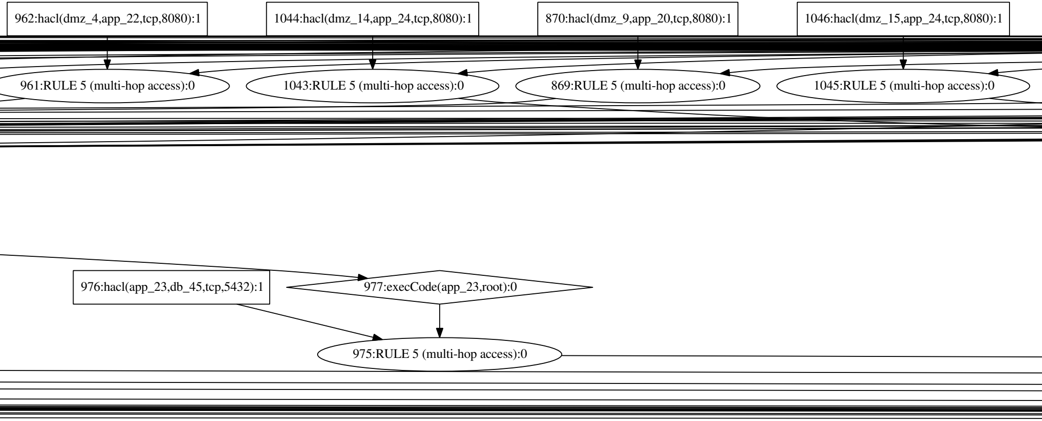

956:hacl(dmz_20,app_22,tcp,8080):1
997:hacl(dmz_16,app_23,tcp,8080):1
999:hacl(dmz_17,app_23,tcp,8080):1
1001:hacl(dmz_18,app_23,tcp,8080):1
955:RULE 5 (multi-hop access):0
996:RULE 5 (multi-hop access):0
998:RULE 5 (multi-hop access):0
1000:RULE 5 (multi-hop access):0
925:hacl(app_22,db_45,tcp,5432):1
926:execCode(app_22,root):0
924:RULE 5 (multi-hop access):0

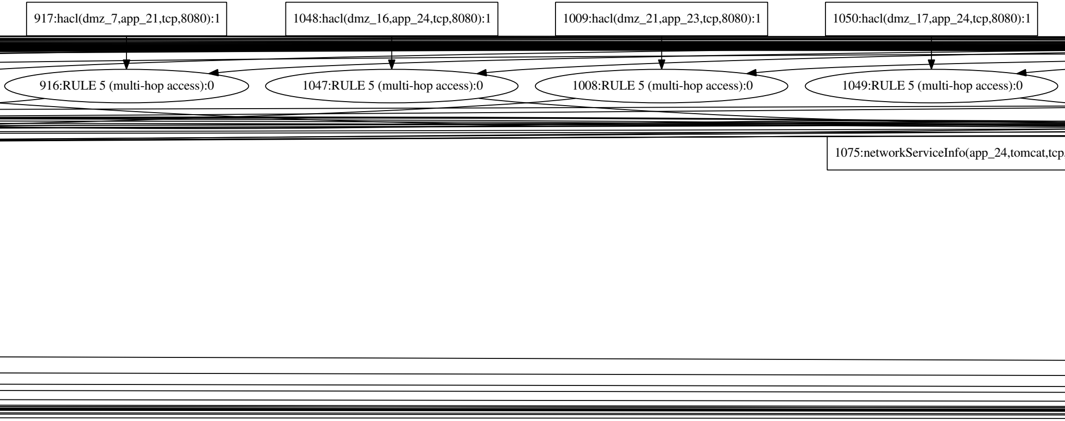

1036:hacl(dmz_10,app_24,tcp,8080):1
1038:hacl(dmz_11,app_24,tcp,8080):1
1040:hacl(dmz_12,app_24,tcp,8080):1
1042:hacl(dmz_13,app_24,tcp,8080):1
1035:RULE 5 (multi-hop access):0
1037:RULE 5 (multi-hop access):0
1039:RULE 5 (multi-hop access):0
1041:RULE 5 (multi-hop access):0

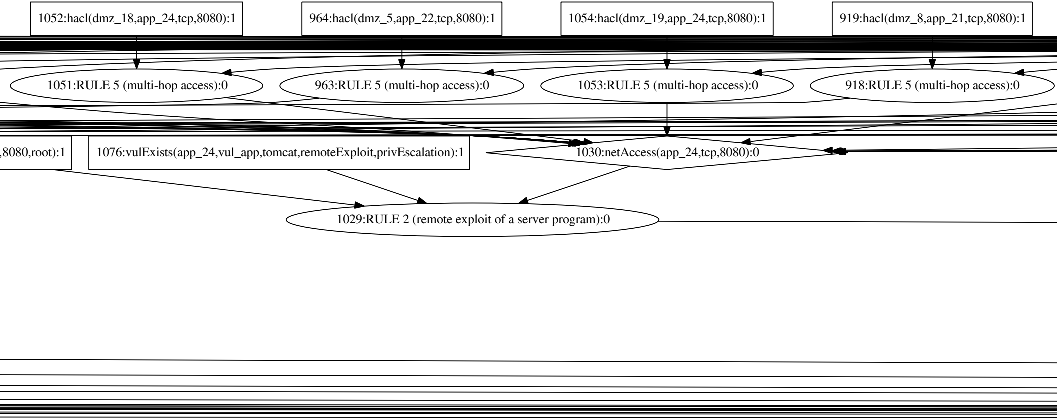

1083:hacl(dmz_0,app_25,tcp,8080):1
768:hacl(dmz_9,app_19,tcp,8080):1
819:hacl(dmz_9,app_2,tcp,8080):1
866:hacl(dmz_7,app_20,tcp,8080):1
1082:RULE 5 (multi-hop access):0
767:RULE 5 (multi-hop access):0
818:RULE 5 (multi-hop access):0
865:RULE 5 (multi-hop access):0

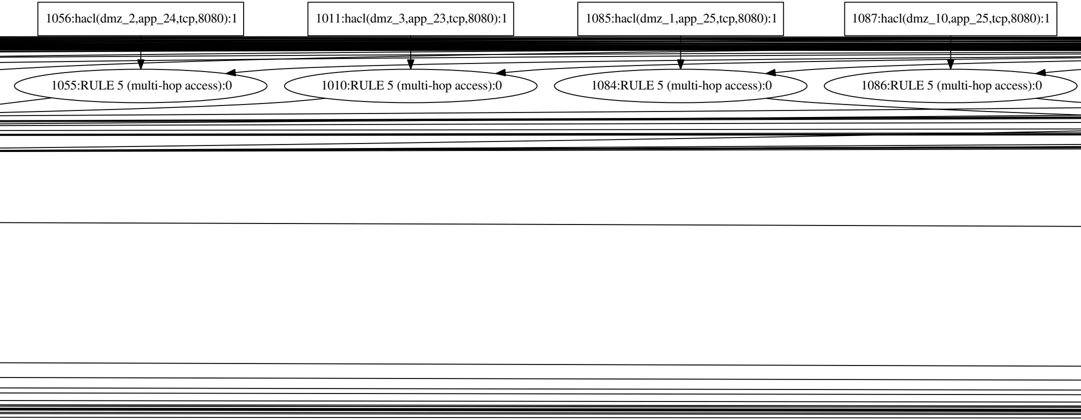

913:hacl(dmz_5,app_21,tcp,8080):1
958:hacl(dmz_21,app_22,tcp,8080):1
868:hacl(dmz_8,app_20,tcp,8080):1
960:hacl(dmz_3,app_22,tcp,8080):1
1003:hacl(dmz_19,
912:RULE 5 (multi-hop access):0
957:RULE 5 (multi-hop access):0
867:RULE 5 (multi-hop access):0
959:RULE 5 (multi-hop access):0
1002:RULE 5 (m
1024:networkServiceInfo(app_23,tomcat,tcp,8080,root):1
1025:vulExists(app_23,vul_app,tomcat,remoteExploit,privEscalation):1
979:netAccess(ap
978:RULE 2 (remote exploit of a server program):0

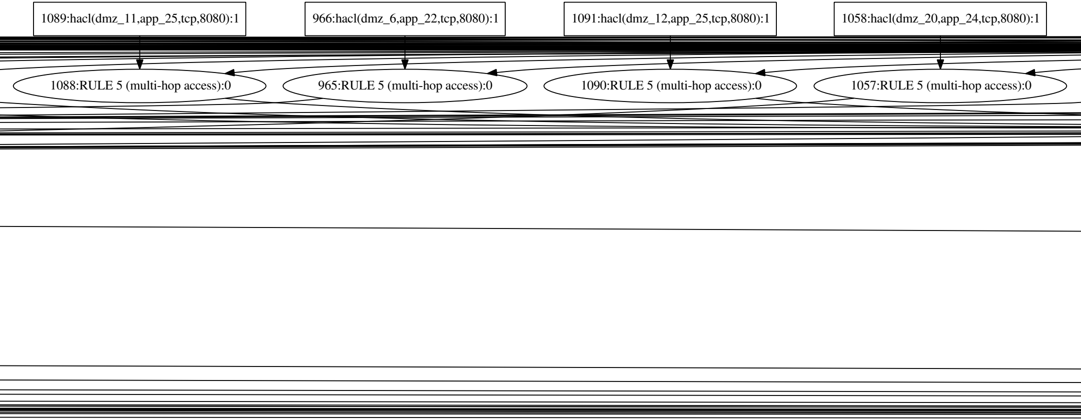

,app_23,tcp,8080):1
1005:hacl(dmz_2,app_23,tcp,8080):1
915:hacl(dmz_6,app_21,tcp,8080):1
1007:hacl(dmz_20,app_23,tcp,8080):1
ulti-hop access):0
1004:RULE 5 (multi-hop access):0
914:RULE 5 (multi-hop access):0
1006:RULE 5 (multi-hop access):0
pp_23,tcp,8080):0

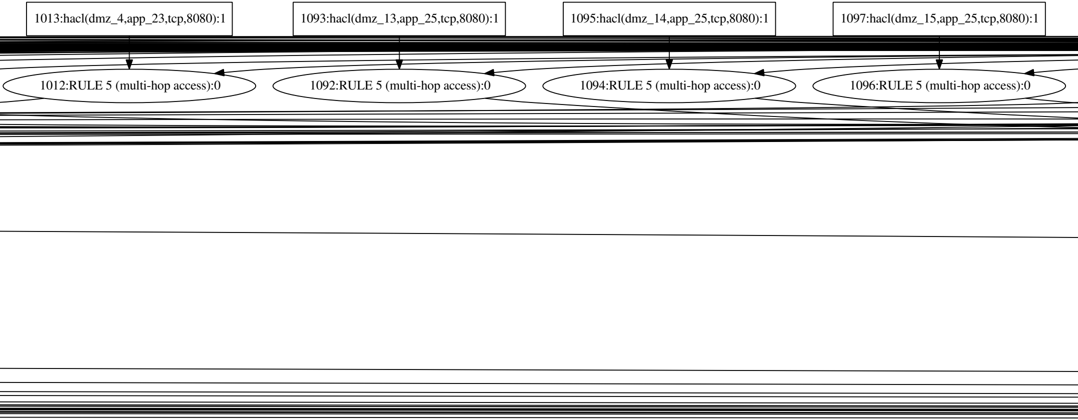

962:hacl(dmz_4,app_22,tcp,8080):1
1044:hacl(dmz_14,app_24,tcp,8080):1
870:hacl(dmz_9,app_20,tcp,8080):1
1046:hacl(dmz_15,app_24,tcp,8080):1
961:RULE 5 (multi-hop access):0
1043:RULE 5 (multi-hop access):0
869:RULE 5 (multi-hop access):0
1045:RULE 5 (multi-hop access):0
976:hacl(app_23,db_45,tcp,5432):1
977:execCode(app_23,root):0
975:RULE 5 (multi-hop access):0

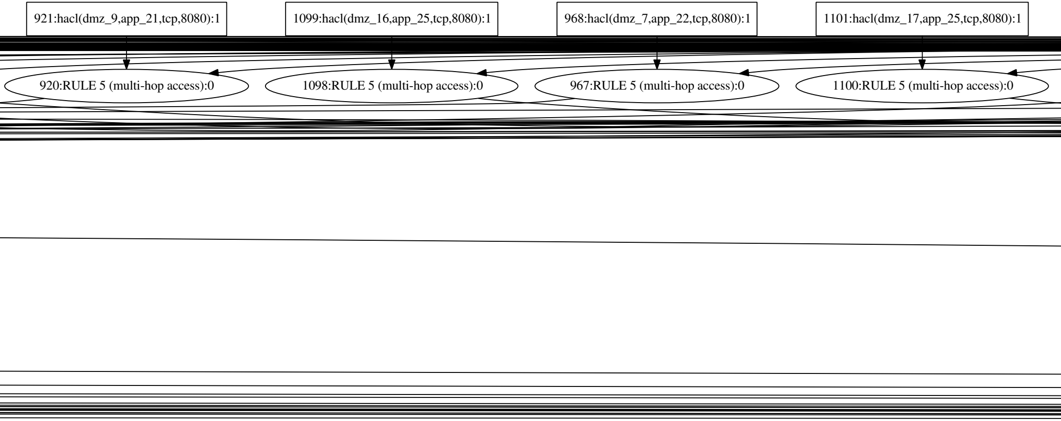

917:hacl(dmz_7,app_21,tcp,8080):1
1048:hacl(dmz_16,app_24,tcp,8080):1
1009:hacl(dmz_21,app_23,tcp,8080):1
1050:hacl(dmz_17,app_24,tcp,8080):1
916:RULE 5 (multi-hop access):0
1047:RULE 5 (multi-hop access):0
1008:RULE 5 (multi-hop access):0
1049:RULE 5 (multi-hop access):0
1075:networkServiceInfo(app_24,tomcat,tcp,

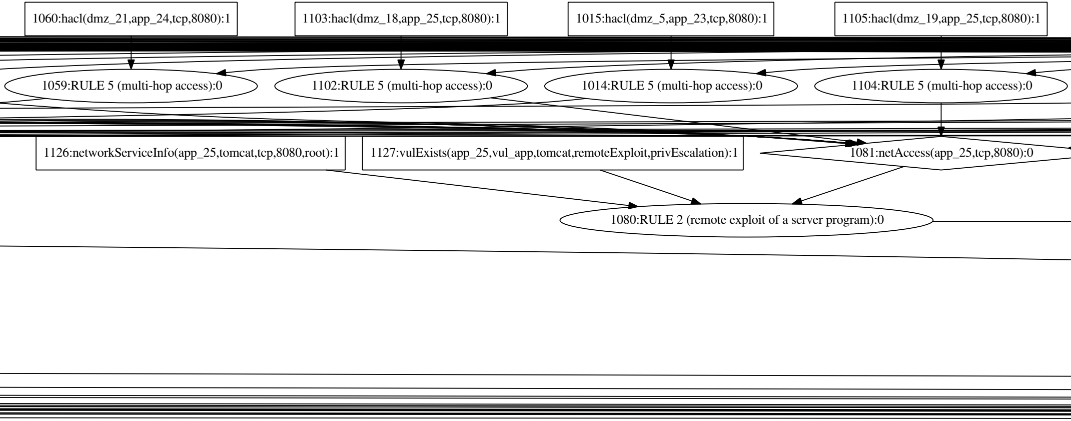

1052:hacl(dmz_18,app_24,tcp,8080):1
964:hacl(dmz_5,app_22,tcp,8080):1
1054:hacl(dmz_19,app_24,tcp,8080):1
919:hacl(dmz_8,app_21,tcp,8080):1
1051:RULE 5 (multi-hop access):0
963:RULE 5 (multi-hop access):0
1053:RULE 5 (multi-hop access):0
918:RULE 5 (multi-hop access):0
,8080,root):1
1076:vulExists(app_24,vul_app,tomcat,remoteExploit,privEscalation):1
1030:netAccess(app_24,tcp,8080):0
1029:RULE 2 (remote exploit of a server program):0

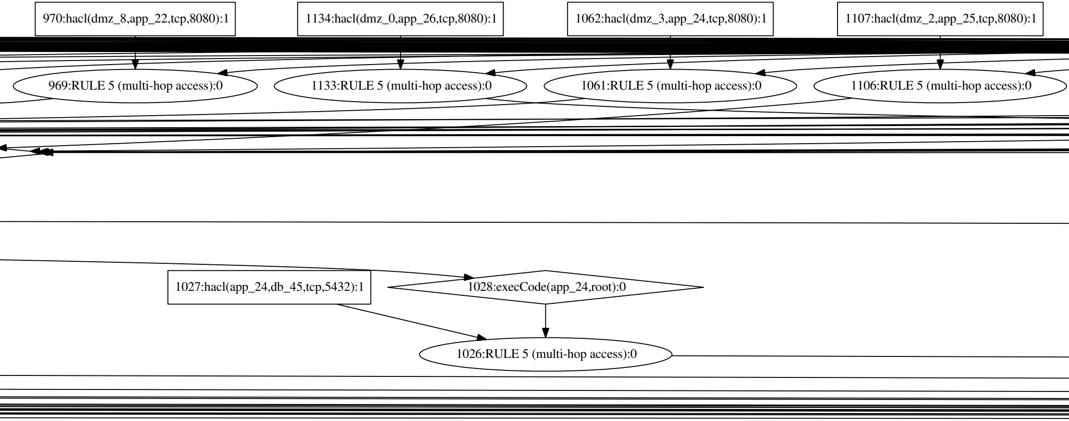

1056:hacl(dmz_2,app_24,tcp,8080):1
1011:hacl(dmz_3,app_23,tcp,8080):1
1085:hacl(dmz_1,app_25,tcp,8080):1
1087:hacl(dmz_10,app_25,tcp,8080):1
1055:RULE 5 (multi-hop access):0
1010:RULE 5 (multi-hop access):0
1084:RULE 5 (multi-hop access):0
1086:RULE 5 (multi-hop access):0

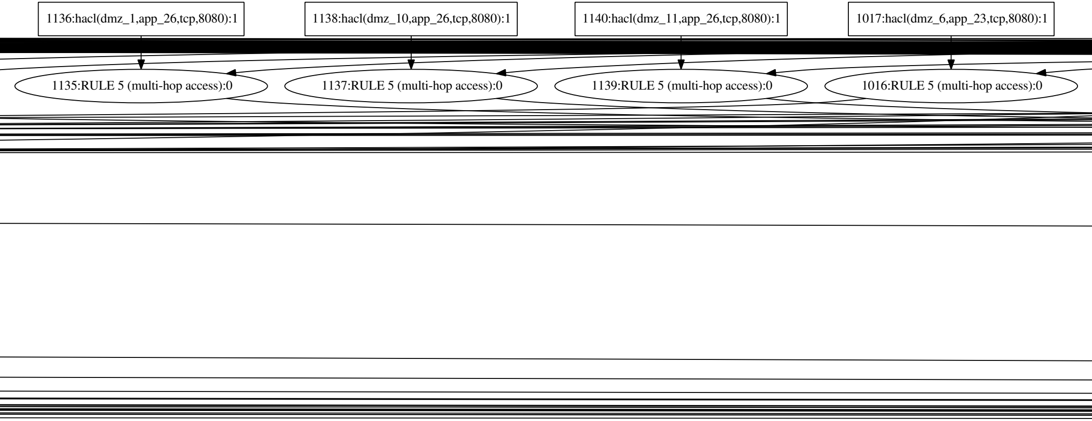

1089:hacl(dmz_11,app_25,tcp,8080):1
966:hacl(dmz_6,app_22,tcp,8080):1
1091:hacl(dmz_12,app_25,tcp,8080):1
1058:hacl(dmz_20,app_24,tcp,8080):1
1088:RULE 5 (multi-hop access):0
965:RULE 5 (multi-hop access):0
1090:RULE 5 (multi-hop access):0
1057:RULE 5 (multi-hop access):0

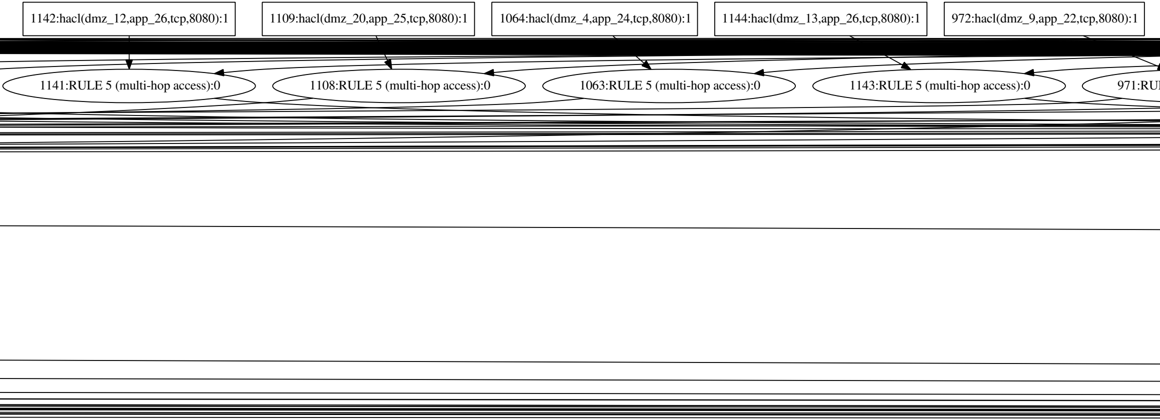

1013:hacl(dmz_4,app_23,tcp,8080):1
1093:hacl(dmz_13,app_25,tcp,8080):1
1095:hacl(dmz_14,app_25,tcp,8080):1
1097:hacl(dmz_15,app_25,tcp,8080):1
1012:RULE 5 (multi-hop access):0
1092:RULE 5 (multi-hop access):0
1094:RULE 5 (multi-hop access):0
1096:RULE 5 (multi-hop access):0

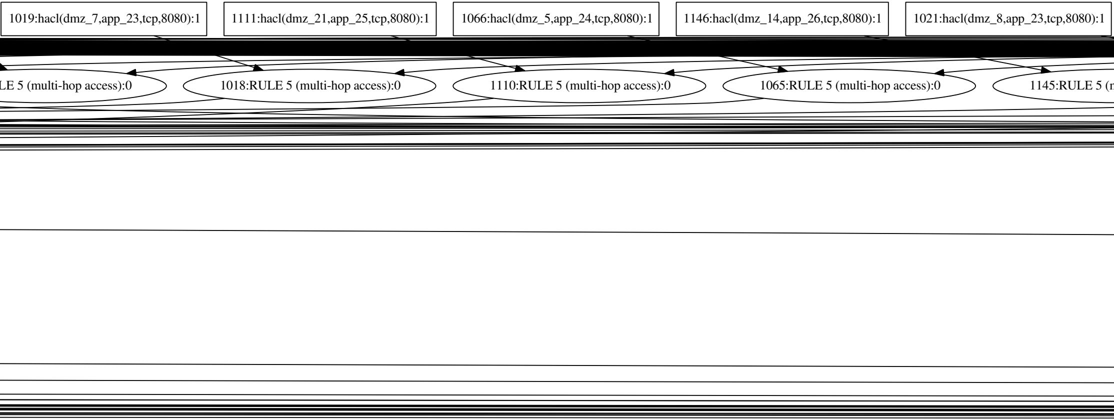

921:hacl(dmz_9,app_21,tcp,8080):1
1099:hacl(dmz_16,app_25,tcp,8080):1
968:hacl(dmz_7,app_22,tcp,8080):1
1101:hacl(dmz_17,app_25,tcp,8080):1
920:RULE 5 (multi-hop access):0
1098:RULE 5 (multi-hop access):0
967:RULE 5 (multi-hop access):0
1100:RULE 5 (multi-hop access):0

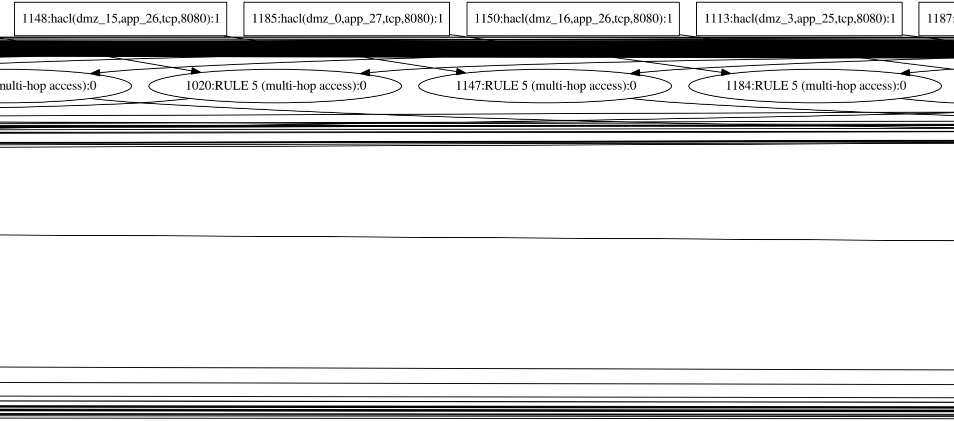

1060:hacl(dmz_21,app_24,tcp,8080):1
1103:hacl(dmz_18,app_25,tcp,8080):1
1015:hacl(dmz_5,app_23,tcp,8080):1
1105:hacl(dmz_19,app_25,tcp,8080):1
1059:RULE 5 (multi-hop access):0
1102:RULE 5 (multi-hop access):0
1014:RULE 5 (multi-hop access):0
1104:RULE 5 (multi-hop access):0
1126:networkServiceInfo(app_25,tomcat,tcp,8080,root):1
1127:vulExists(app_25,vul_app,tomcat,remoteExploit,privEscalation):1
1081:netAccess(app_25,tcp,8080):0
1080:RULE 2 (remote exploit of a server program):0

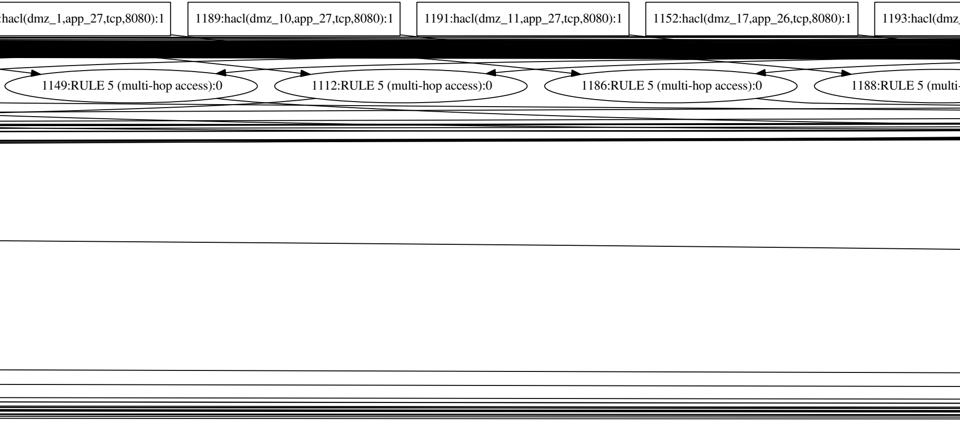

970:hacl(dmz_8,app_22,tcp,8080):1
1134:hacl(dmz_0,app_26,tcp,8080):1
1062:hacl(dmz_3,app_24,tcp,8080):1
1107:hacl(dmz_2,app_25,tcp,8080):1
969:RULE 5 (multi-hop access):0
1133:RULE 5 (multi-hop access):0
1061:RULE 5 (multi-hop access):0
1106:RULE 5 (multi-hop access):0
1027:hacl(app_24,db_45,tcp,5432):1
1028:execCode(app_24,root):0
1026:RULE 5 (multi-hop access):0

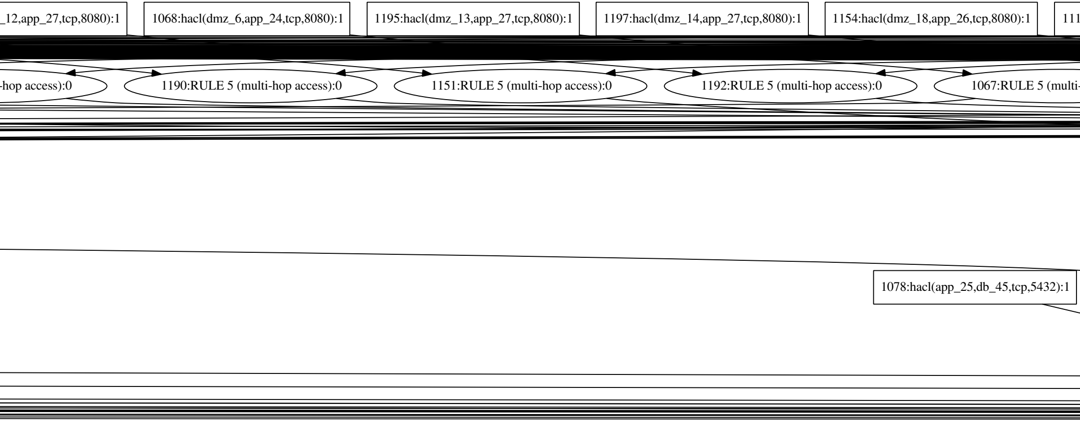

1136:hacl(dmz_1,app_26,tcp,8080):1
1138:hacl(dmz_10,app_26,tcp,8080):1
1140:hacl(dmz_11,app_26,tcp,8080):1
1017:hacl(dmz_6,app_23,tcp,8080):1
1135:RULE 5 (multi-hop access):0
1137:RULE 5 (multi-hop access):0
1139:RULE 5 (multi-hop access):0
1016:RULE 5 (multi-hop access):0

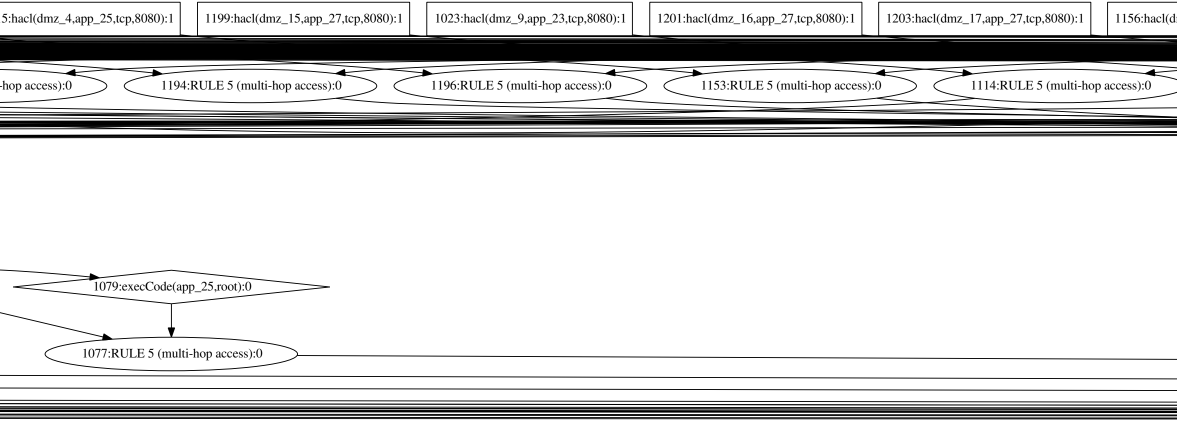

1142:hacl(dmz_12,app_26,tcp,8080):1
1109:hacl(dmz_20,app_25,tcp,8080):1
1064:hacl(dmz_4,app_24,tcp,8080):1
1144:hacl(dmz_13,app_26,tcp,8080):1
972:hacl(dmz_9,app_22,tcp,8080):1
1141:RULE 5 (multi-hop access):0
1108:RULE 5 (multi-hop access):0
1063:RULE 5 (multi-hop access):0
1143:RULE 5 (multi-hop access):0
971:RU

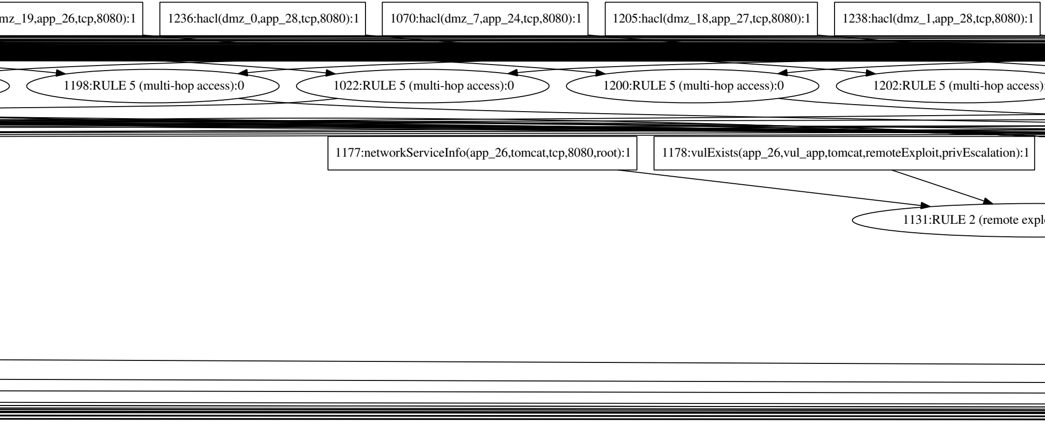

1019:hacl(dmz_7,app_23,tcp,8080):1
1111:hacl(dmz_21,app_25,tcp,8080):1
1066:hacl(dmz_5,app_24,tcp,8080):1
1146:hacl(dmz_14,app_26,tcp,8080):1
1021:hacl(dmz_8,app_23,tcp,8080):1
LE 5 (multi-hop access):0
1018:RULE 5 (multi-hop access):0
1110:RULE 5 (multi-hop access):0
1065:RULE 5 (multi-hop access):0
1145:RULE 5 (m

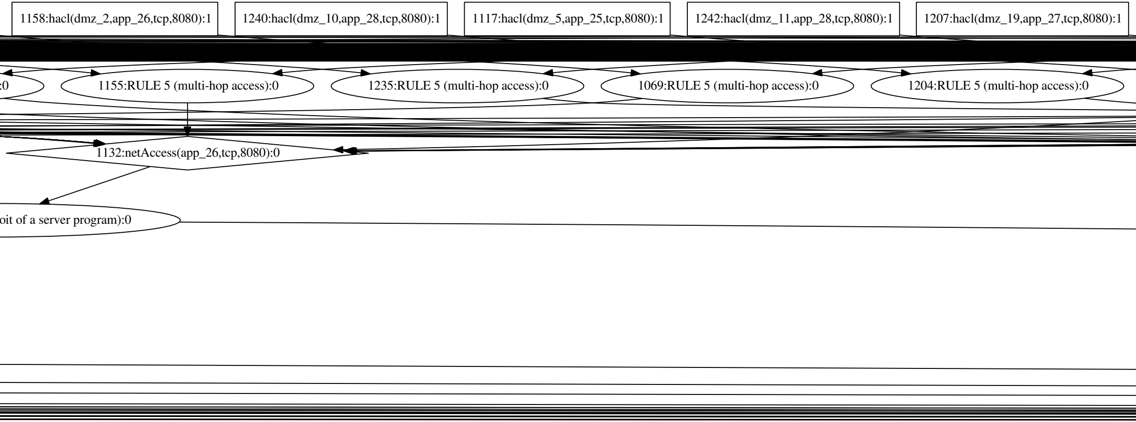

1148:hacl(dmz_15,app_26,tcp,8080):1
1185:hacl(dmz_0,app_27,tcp,8080):1
1150:hacl(dmz_16,app_26,tcp,8080):1
1113:hacl(dmz_3,app_25,tcp,8080):1
1187:
ulti-hop access):0
1020:RULE 5 (multi-hop access):0
1147:RULE 5 (multi-hop access):0
1184:RULE 5 (multi-hop access):0

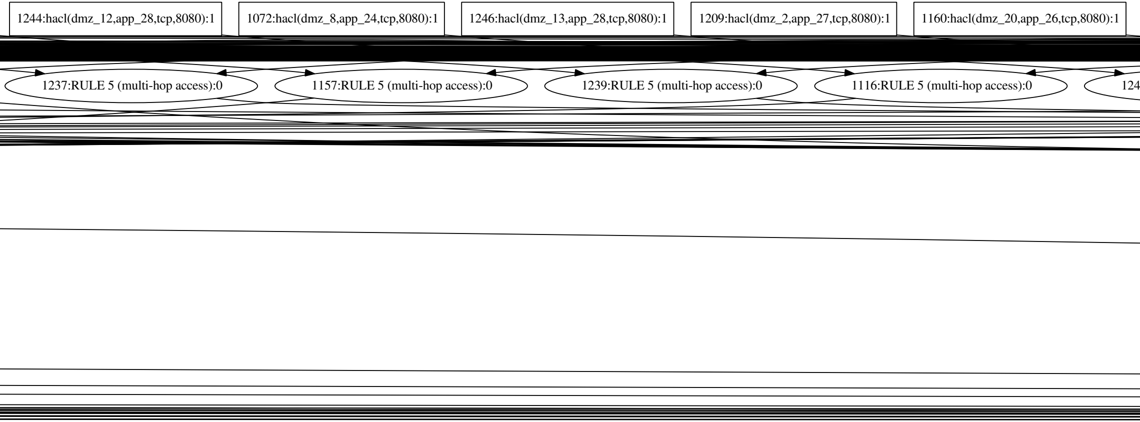

:hacl(dmz_1,app_27,tcp,8080):1
1189:hacl(dmz_10,app_27,tcp,8080):1
1191:hacl(dmz_11,app_27,tcp,8080):1
1152:hacl(dmz_17,app_26,tcp,8080):1
1193:hacl(dmz_
1149:RULE 5 (multi-hop access):0
1112:RULE 5 (multi-hop access):0
1186:RULE 5 (multi-hop access):0
1188:RULE 5 (multi-

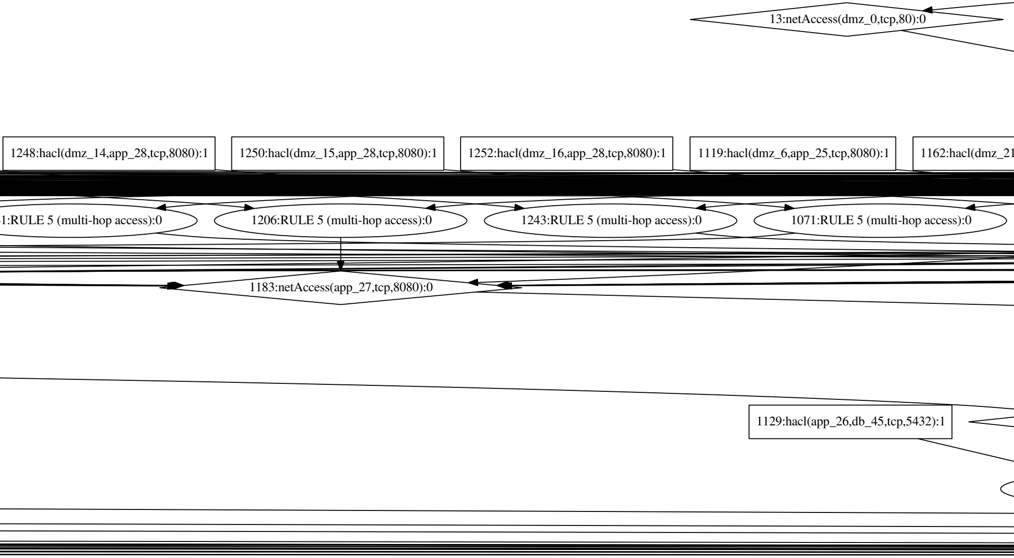

_12,app_27,tcp,8080):1
1068:hacl(dmz_6,app_24,tcp,8080):1
1195:hacl(dmz_13,app_27,tcp,8080):1
1197:hacl(dmz_14,app_27,tcp,8080):1
1154:hacl(dmz_18,app_26,tcp,8080):1
111
-hop access):0
1190:RULE 5 (multi-hop access):0
1151:RULE 5 (multi-hop access):0
1192:RULE 5 (multi-hop access):0
1067:RULE 5 (multi-
1078:hacl(app_25,db_45,tcp,5432):1

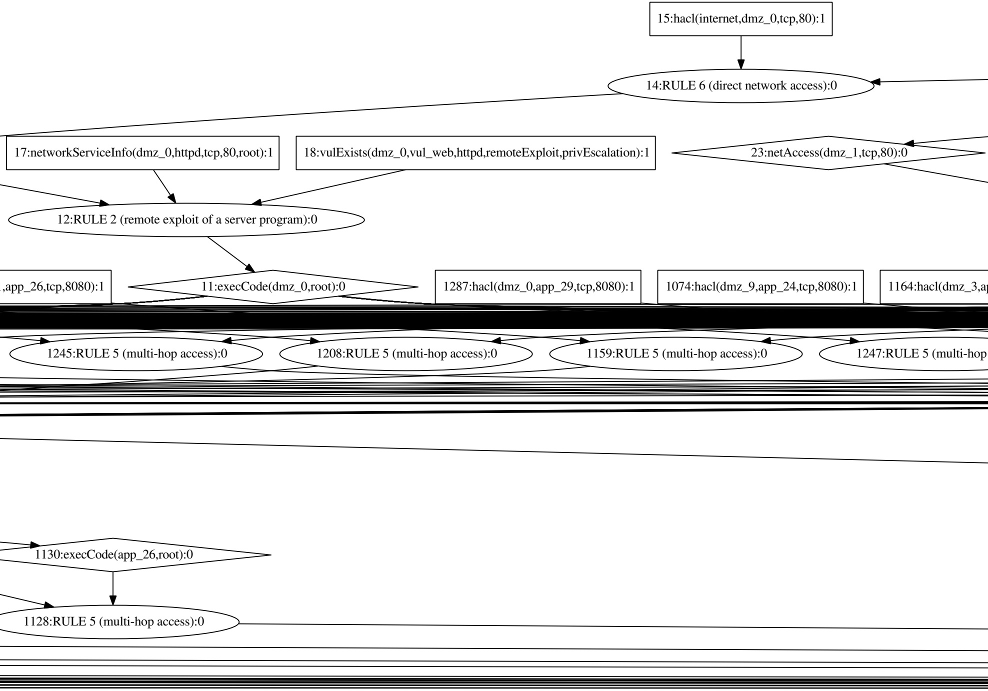

5:hacl(dmz_4,app_25,tcp,8080):1
1199:hacl(dmz_15,app_27,tcp,8080):1
1023:hacl(dmz_9,app_23,tcp,8080):1
1201:hacl(dmz_16,app_27,tcp,8080):1
1203:hacl(dmz_17,app_27,tcp,8080):1
1156:hacl(d
-hop access):0
1194:RULE 5 (multi-hop access):0
1196:RULE 5 (multi-hop access):0
1153:RULE 5 (multi-hop access):0
1114:RULE 5 (multi-hop access):0
1079:execCode(app_25,root):0
1077:RULE 5 (multi-hop access):0

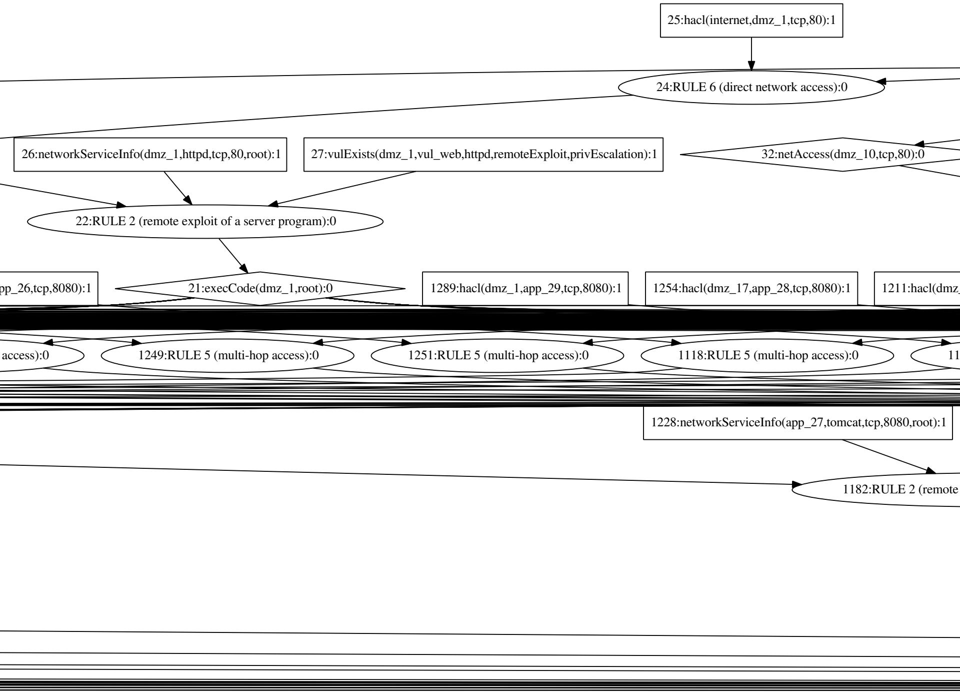

mz_19,app_26,tcp,8080):1
1236:hacl(dmz_0,app_28,tcp,8080):1
1070:hacl(dmz_7,app_24,tcp,8080):1
1205:hacl(dmz_18,app_27,tcp,8080):1
1238:hacl(dmz_1,app_28,tcp,8080):1
1198:RULE 5 (multi-hop access):0
1022:RULE 5 (multi-hop access):0
1200:RULE 5 (multi-hop access):0
1202:RULE 5 (multi-hop access)
1177:networkServiceInfo(app_26,tomcat,tcp,8080,root):1
1178:vulExists(app_26,vul_app,tomcat,remoteExploit,privEscalation):1
1131:RULE 2 (remote expl

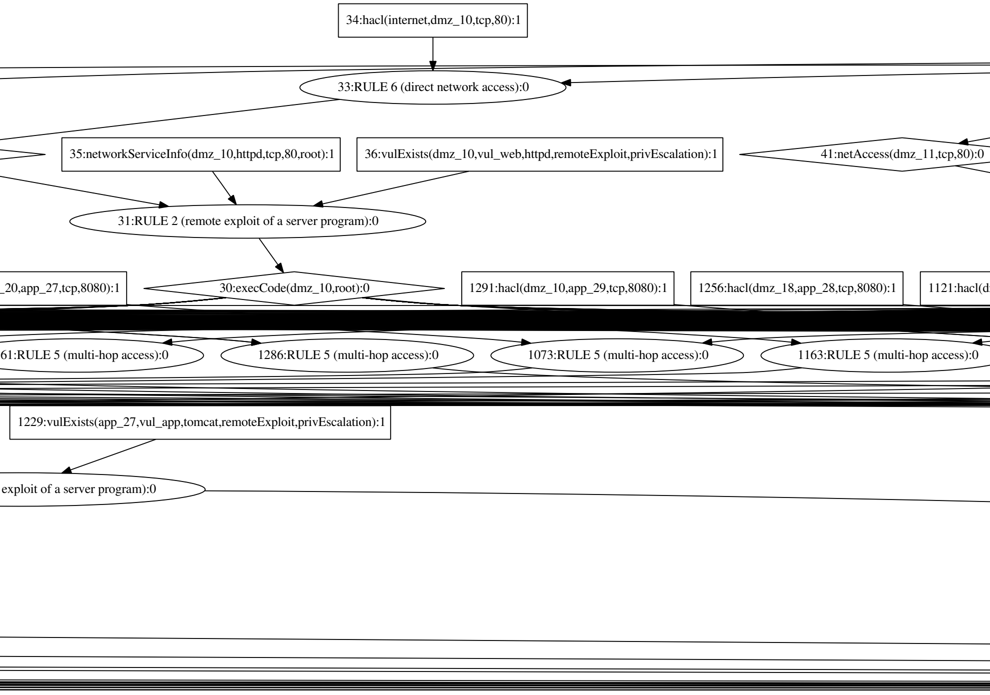

1158:hacl(dmz_2,app_26,tcp,8080):1
1240:hacl(dmz_10,app_28,tcp,8080):1
1117:hacl(dmz_5,app_25,tcp,8080):1
1242:hacl(dmz_11,app_28,tcp,8080):1
1207:hacl(dmz_19,app_27,tcp,8080):1
:0
1155:RULE 5 (multi-hop access):0
1235:RULE 5 (multi-hop access):0
1069:RULE 5 (multi-hop access):0
1204:RULE 5 (multi-hop access):0
1132:netAccess(app_26,tcp,8080):0
oit of a server program):0

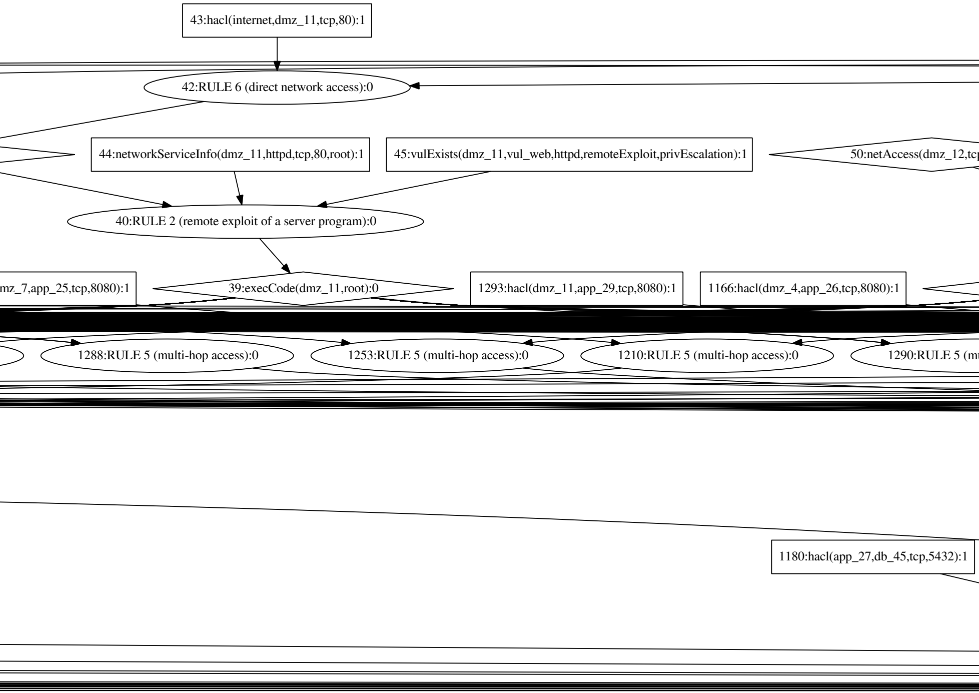

1244:hacl(dmz_12,app_28,tcp,8080):1
1072:hacl(dmz_8,app_24,tcp,8080):1
1246:hacl(dmz_13,app_28,tcp,8080):1
1209:hacl(dmz_2,app_27,tcp,8080):1
1160:hacl(dmz_20,app_26,tcp,8080):1
1237:RULE 5 (multi-hop access):0
1157:RULE 5 (multi-hop access):0
1239:RULE 5 (multi-hop access):0
1116:RULE 5 (multi-hop access):0
124

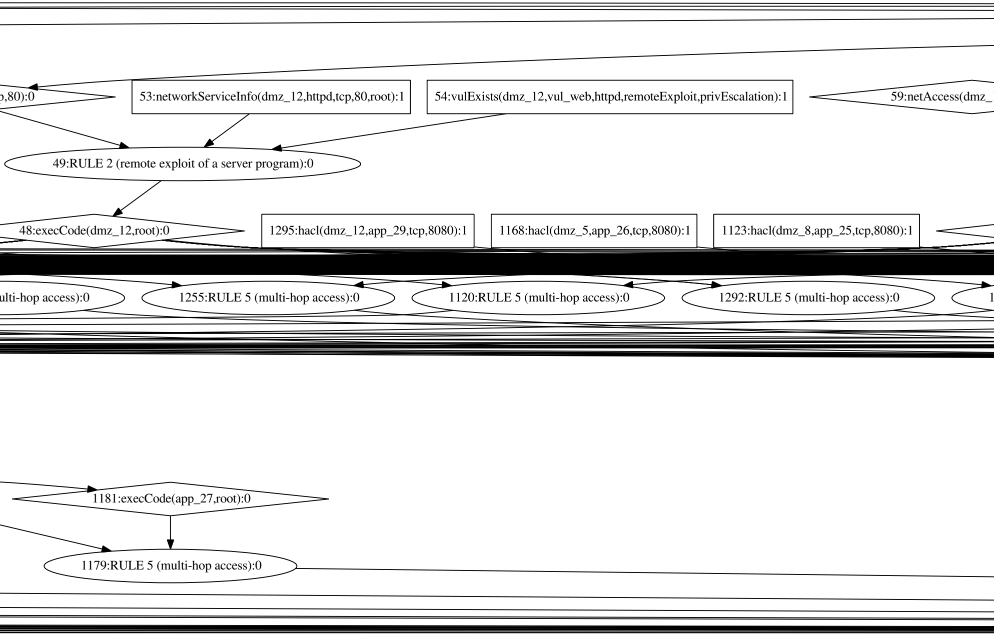

13:netAccess(dmz_0,tcp,80):0
1248:hacl(dmz_14,app_28,tcp,8080):1
1250:hacl(dmz_15,app_28,tcp,8080):1
1252:hacl(dmz_16,app_28,tcp,8080):1
1119:hacl(dmz_6,app_25,tcp,8080):1
1162:hacl(dmz_21
1:RULE 5 (multi-hop access):0
1206:RULE 5 (multi-hop access):0
1243:RULE 5 (multi-hop access):0
1071:RULE 5 (multi-hop access):0
1183:netAccess(app_27,tcp,8080):0
1129:hacl(app_26,db_45,tcp,5432):1

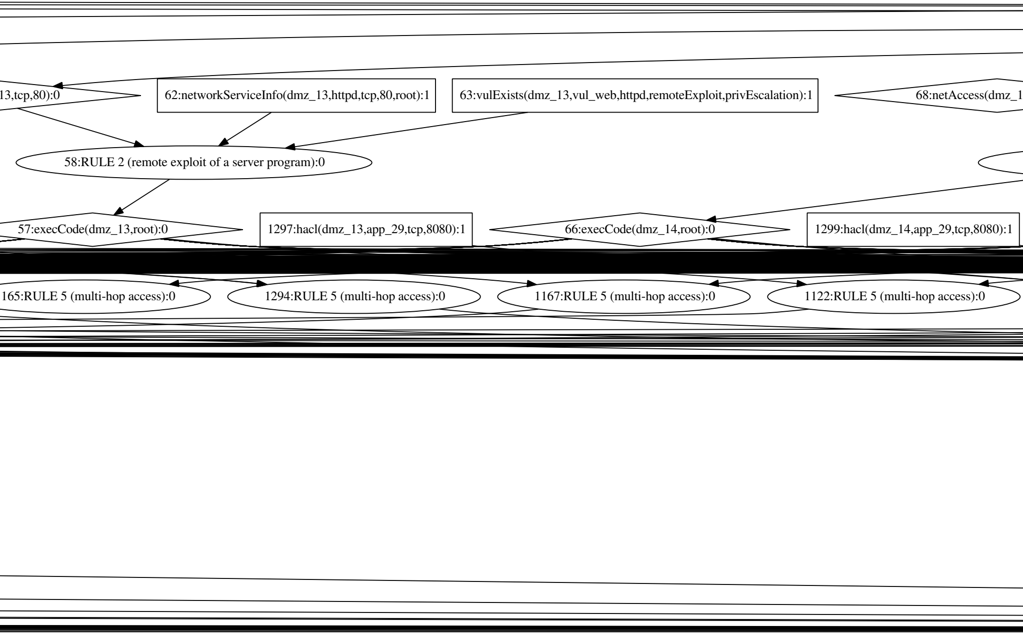

15:hacl(internet,dmz_0,tcp,80):1
14:RULE 6 (direct network access):0
17:networkServiceInfo(dmz_0,httpd,tcp,80,root):1
18:vulExists(dmz_0,vul_web,httpd,remoteExploit,privEscalation):1
23:netAccess(dmz_1,tcp,80):0
12:RULE 2 (remote exploit of a server program):0
,app_26,tcp,8080):1
11:execCode(dmz_0,root):0
1287:hacl(dmz_0,app_29,tcp,8080):1
1074:hacl(dmz_9,app_24,tcp,8080):1
1164:hacl(dmz_3,ap
1245:RULE 5 (multi-hop access):0
1208:RULE 5 (multi-hop access):0
1159:RULE 5 (multi-hop access):0
1247:RULE 5 (multi-hop
1130:execCode(app_26,root):0
1128:RULE 5 (multi-hop access):0

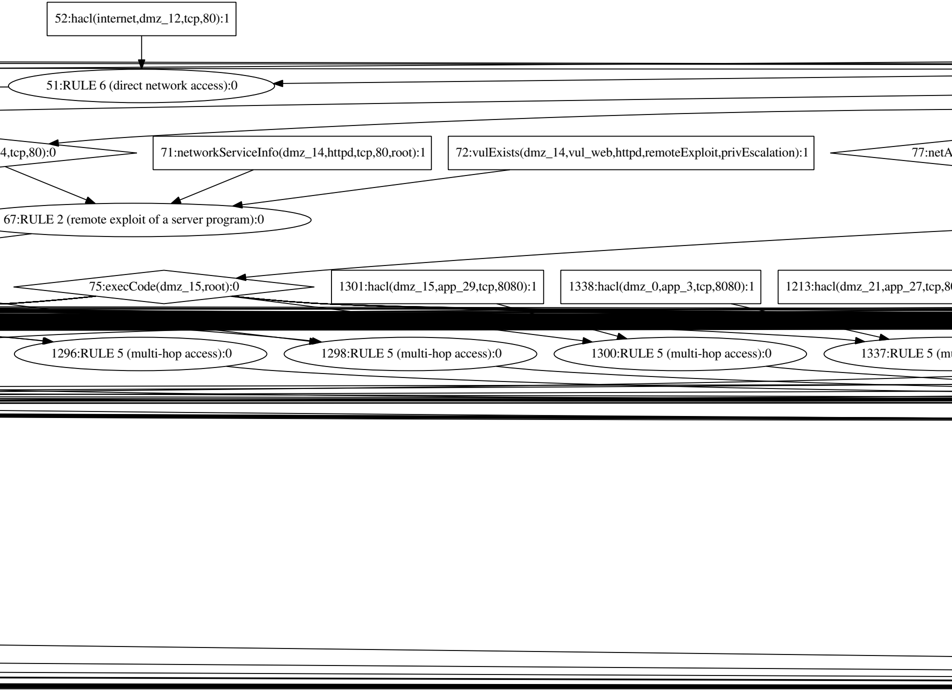

25:hacl(internet,dmz_1,tcp,80):1
24:RULE 6 (direct network access):0
26:networkServiceInfo(dmz_1,httpd,tcp,80,root):1
27:vulExists(dmz_1,vul_web,httpd,remoteExploit,privEscalation):1
32:netAccess(dmz_10,tcp,80):0
22:RULE 2 (remote exploit of a server program):0
pp_26,tcp,8080):1
21:execCode(dmz_1,root):0
1289:hacl(dmz_1,app_29,tcp,8080):1
1254:hacl(dmz_17,app_28,tcp,8080):1
1211:hacl(dmz
access):0
1249:RULE 5 (multi-hop access):0
1251:RULE 5 (multi-hop access):0
1118:RULE 5 (multi-hop access):0
11
1228:networkServiceInfo(app_27,tomcat,tcp,8080,root):1
1182:RULE 2 (remote

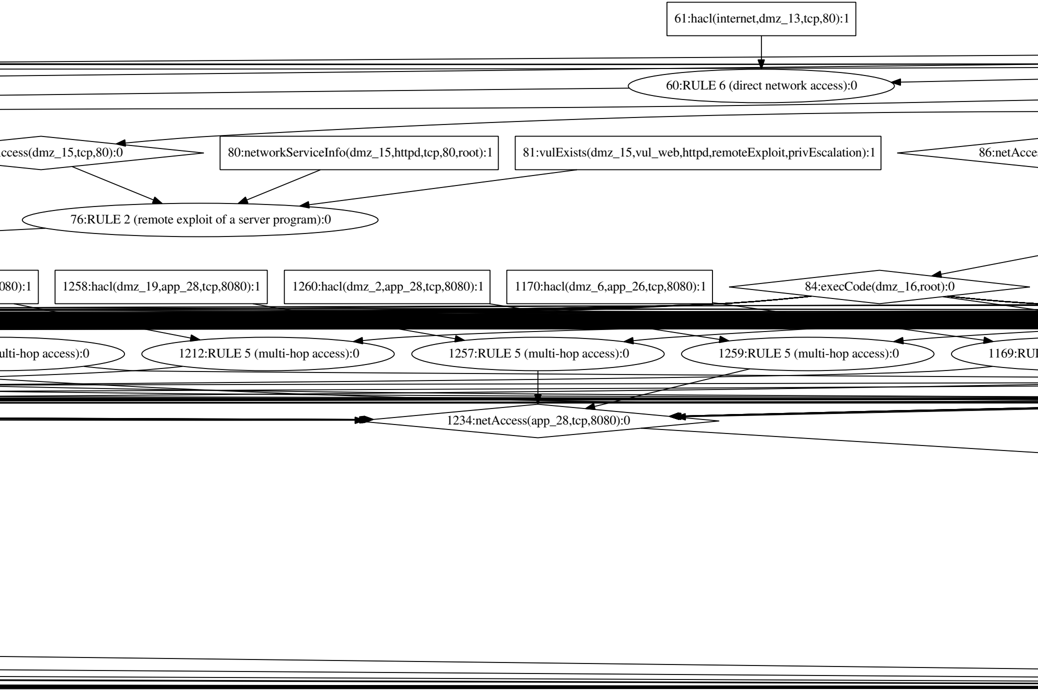

34:hacl(internet,dmz_10,tcp,80):1
33:RULE 6 (direct network access):0
35:networkServiceInfo(dmz_10,httpd,tcp,80,root):1
36:vulExists(dmz_10,vul_web,httpd,remoteExploit,privEscalation):1
41:netAccess(dmz_11,tcp,80):0
31:RULE 2 (remote exploit of a server program):0
_20,app_27,tcp,8080):1
30:execCode(dmz_10,root):0
1291:hacl(dmz_10,app_29,tcp,8080):1
1256:hacl(dmz_18,app_28,tcp,8080):1
1121:hacl(d
61:RULE 5 (multi-hop access):0
1286:RULE 5 (multi-hop access):0
1073:RULE 5 (multi-hop access):0
1163:RULE 5 (multi-hop access):0
1229:vulExists(app_27,vul_app,tomcat,remoteExploit,privEscalation):1
exploit of a server program):0

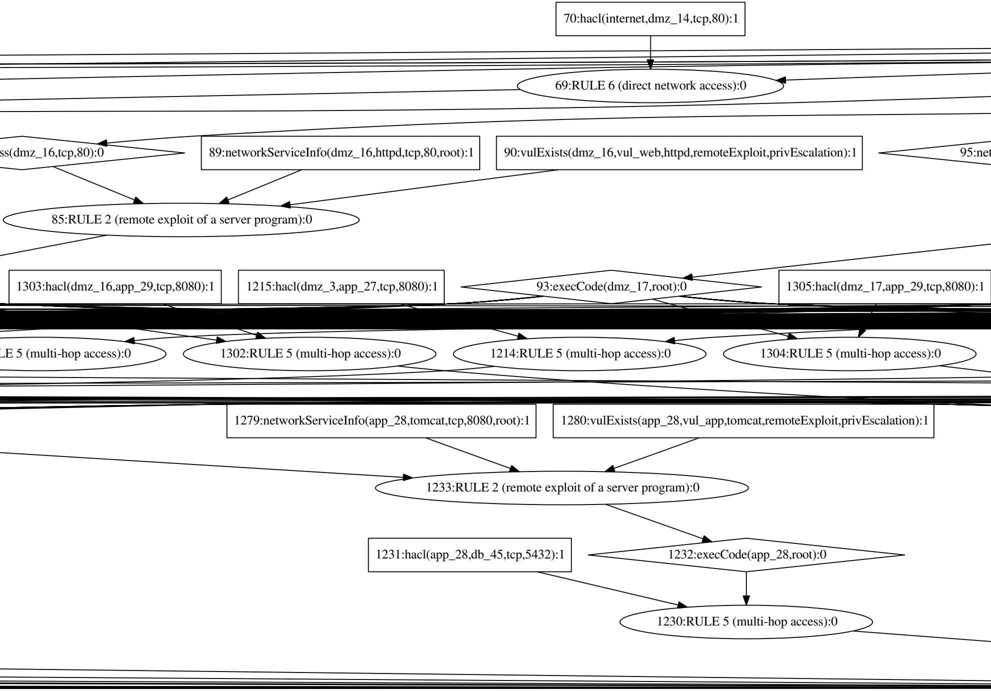

43:hacl(internet,dmz_11,tcp,80):1
42:RULE 6 (direct network access):0
44:networkServiceInfo(dmz_11,httpd,tcp,80,root):1
45:vulExists(dmz_11,vul_web,httpd,remoteExploit,privEscalation):1
50:netAccess(dmz_12,tc
40:RULE 2 (remote exploit of a server program):0
mz_7,app_25,tcp,8080):1
39:execCode(dmz_11,root):0
1293:hacl(dmz_11,app_29,tcp,8080):1
1166:hacl(dmz_4,app_26,tcp,8080):1
1288:RULE 5 (multi-hop access):0
1253:RULE 5 (multi-hop access):0
1210:RULE 5 (multi-hop access):0
1290:RULE 5 (m
1180:hacl(app_27,db_45,tcp,5432):1

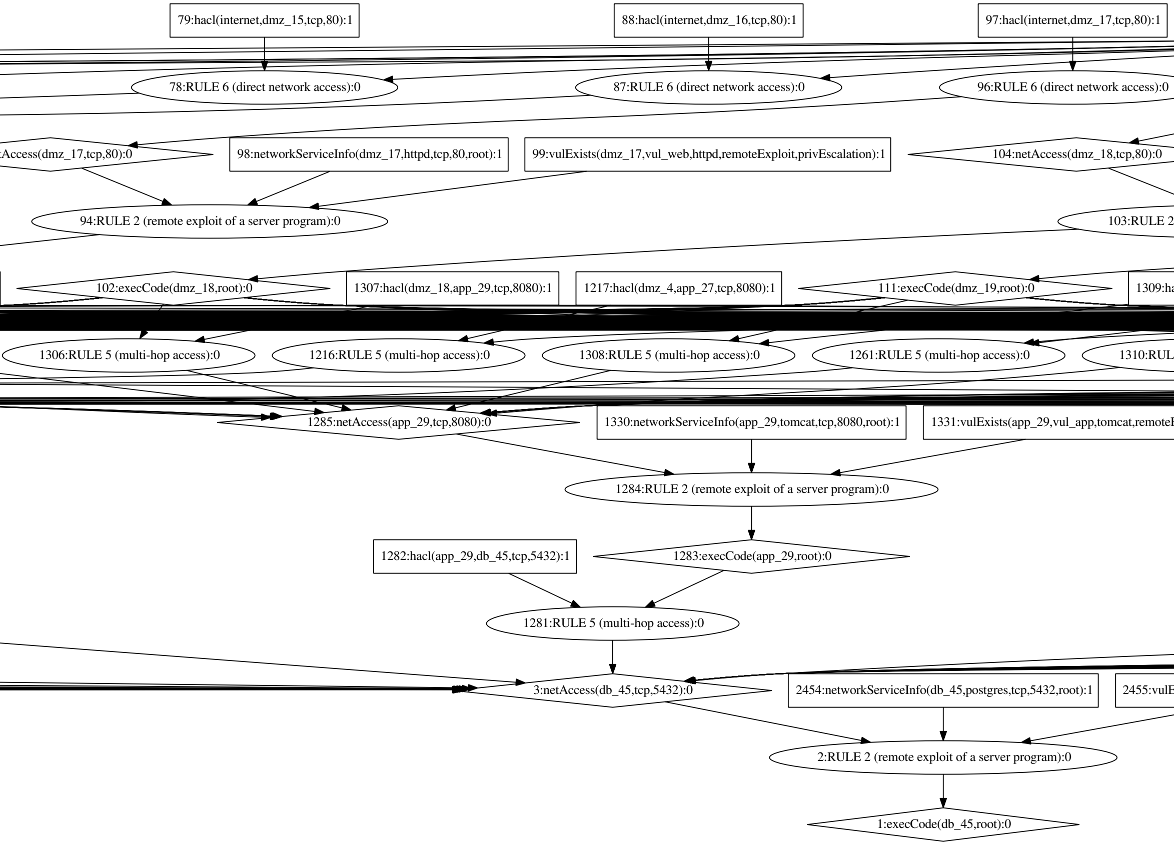

p,80):0
53:networkServiceInfo(dmz_12,httpd,tcp,80,root):1
54:vulExists(dmz_12,vul_web,httpd,remoteExploit,privEscalation):1
59:netAccess(dmz_
49:RULE 2 (remote exploit of a server program):0
48:execCode(dmz_12,root):0
1295:hacl(dmz_12,app_29,tcp,8080):1
1168:hacl(dmz_5,app_26,tcp,8080):1
1123:hacl(dmz_8,app_25,tcp,8080):1
ulti-hop access):0
1255:RULE 5 (multi-hop access):0
1120:RULE 5 (multi-hop access):0
1292:RULE 5 (multi-hop access):0
1
1181:execCode(app_27,root):0
1179:RULE 5 (multi-hop access):0

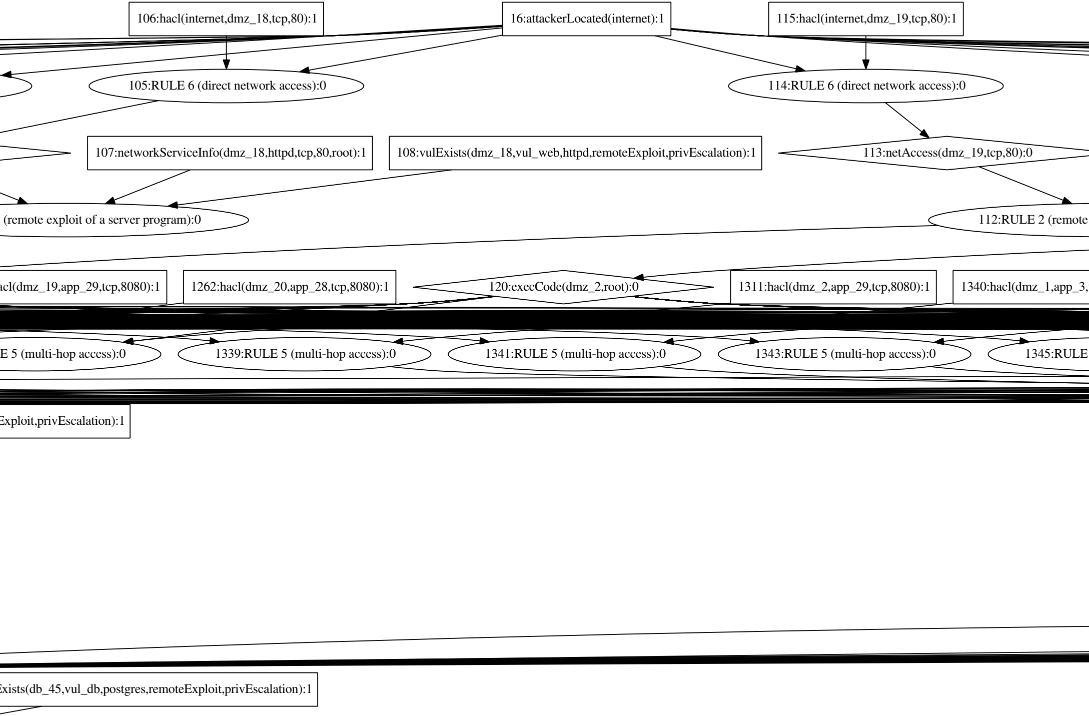

13,tcp,80):0
62:networkServiceInfo(dmz_13,httpd,tcp,80,root):1
63:vulExists(dmz_13,vul_web,httpd,remoteExploit,privEscalation):1
68:netAccess(dmz_1
58:RULE 2 (remote exploit of a server program):0
57:execCode(dmz_13,root):0
1297:hacl(dmz_13,app_29,tcp,8080):1
66:execCode(dmz_14,root):0
1299:hacl(dmz_14,app_29,tcp,8080):1
165:RULE 5 (multi-hop access):0
1294:RULE 5 (multi-hop access):0
1167:RULE 5 (multi-hop access):0
1122:RULE 5 (multi-hop access):0

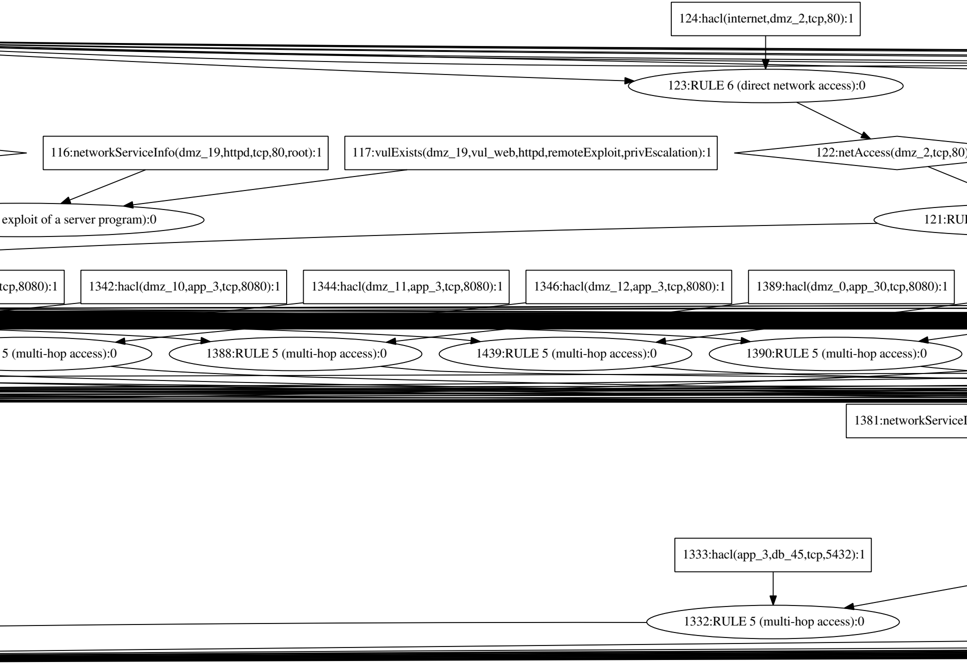

52:hacl(internet,dmz_12,tcp,80):1
51:RULE 6 (direct network access):0
4,tcp,80):0
71:networkServiceInfo(dmz_14,httpd,tcp,80,root):1
72:vulExists(dmz_14,vul_web,httpd,remoteExploit,privEscalation):1
77:netA
67:RULE 2 (remote exploit of a server program):0
75:execCode(dmz_15,root):0
1301:hacl(dmz_15,app_29,tcp,8080):1
1338:hacl(dmz_0,app_3,tcp,8080):1
1213:hacl(dmz_21,app_27,tcp,80
1296:RULE 5 (multi-hop access):0
1298:RULE 5 (multi-hop access):0
1300:RULE 5 (multi-hop access):0
1337:RULE 5 (mu

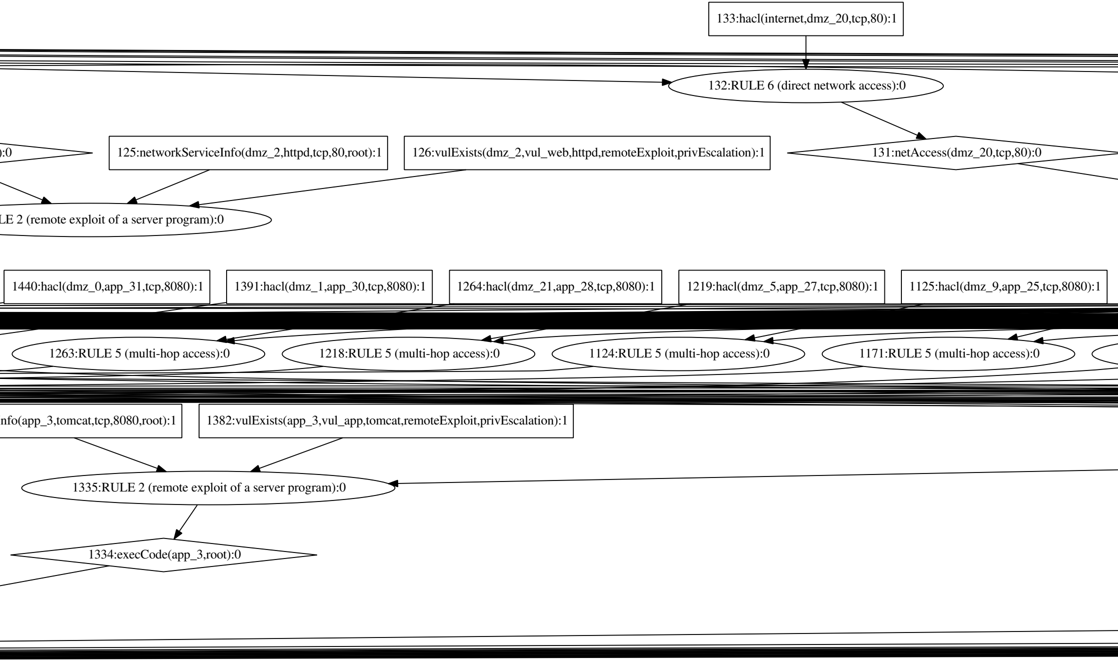

61:hacl(internet,dmz_13,tcp,80):1
60:RULE 6 (direct network access):0
ccess(dmz_15,tcp,80):0
80:networkServiceInfo(dmz_15,httpd,tcp,80,root):1
81:vulExists(dmz_15,vul_web,httpd,remoteExploit,privEscalation):1
86:netAcce
76:RULE 2 (remote exploit of a server program):0
080):1
1258:hacl(dmz_19,app_28,tcp,8080):1
1260:hacl(dmz_2,app_28,tcp,8080):1
1170:hacl(dmz_6,app_26,tcp,8080):1
84:execCode(dmz_16,root):0
lti-hop access):0
1212:RULE 5 (multi-hop access):0
1257:RULE 5 (multi-hop access):0
1259:RULE 5 (multi-hop access):0
1169:RUL
1234:netAccess(app_28,tcp,8080):0

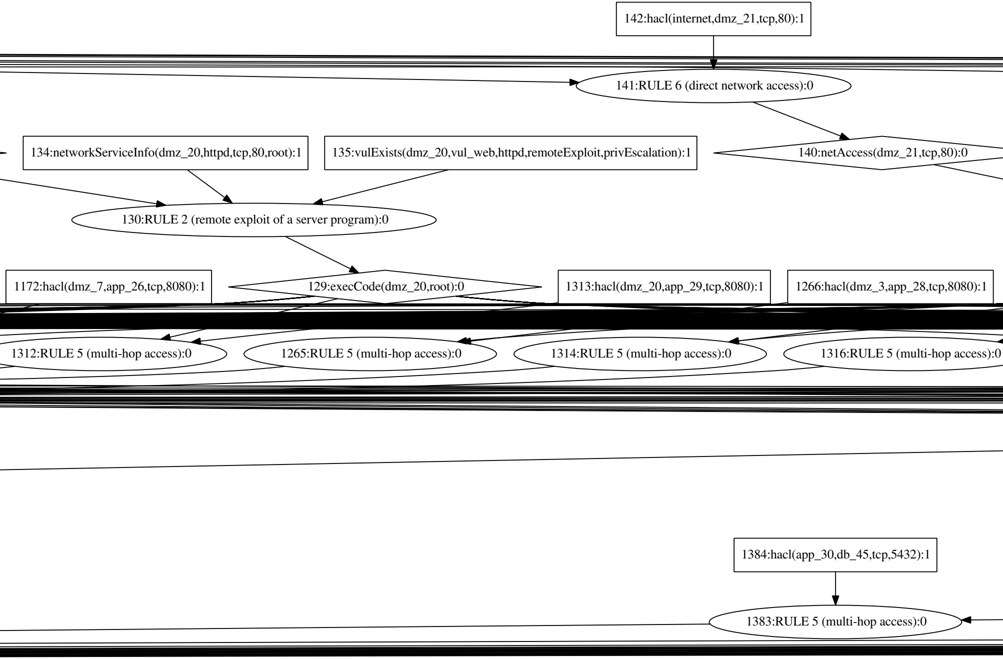

70:hacl(internet,dmz_14,tcp,80):1
69:RULE 6 (direct network access):0
ss(dmz_16,tcp,80):0
89:networkServiceInfo(dmz_16,httpd,tcp,80,root):1
90:vulExists(dmz_16,vul_web,httpd,remoteExploit,privEscalation):1
95:net
85:RULE 2 (remote exploit of a server program):0
1303:hacl(dmz_16,app_29,tcp,8080):1
1215:hacl(dmz_3,app_27,tcp,8080):1
93:execCode(dmz_17,root):0
1305:hacl(dmz_17,app_29,tcp,8080):1
LE 5 (multi-hop access):0
1302:RULE 5 (multi-hop access):0
1214:RULE 5 (multi-hop access):0
1304:RULE 5 (multi-hop access):0
1279:networkServiceInfo(app_28,tomcat,tcp,8080,root):1
1280:vulExists(app_28,vul_app,tomcat,remoteExploit,privEscalation):1
1233:RULE 2 (remote exploit of a server program):0
1231:hacl(app_28,db_45,tcp,5432):1
1232:execCode(app_28,root):0
1230:RULE 5 (multi-hop access):0

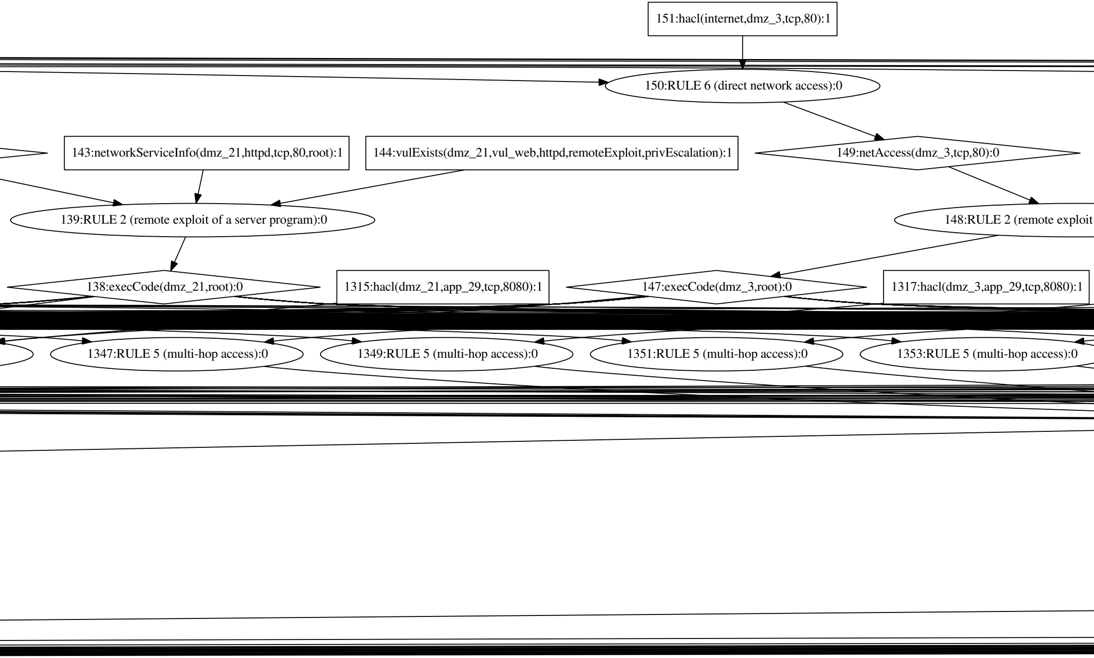

79:hacl(internet,dmz_15,tcp,80):1
88:hacl(internet,dmz_16,tcp,80):1
97:hacl(internet,dmz_17,tcp,80):1
78:RULE 6 (direct network access):0
87:RULE 6 (direct network access):0
96:RULE 6 (direct network access):0
:Access(dmz_17,tcp,80):0
98:networkServiceInfo(dmz_17,httpd,tcp,80,root):1
99:vulExists(dmz_17,vul_web,httpd,remoteExploit,privEscalation):1
104:netAccess(dmz_18,tcp,80):0
94:RULE 2 (remote exploit of a server program):0
103:RULE 2
102:execCode(dmz_18,root):0
1307:hacl(dmz_18,app_29,tcp,8080):1
1217:hacl(dmz_4,app_27,tcp,8080):1
111:execCode(dmz_19,root):0
1309:ha
1306:RULE 5 (multi-hop access):0
1216:RULE 5 (multi-hop access):0
1308:RULE 5 (multi-hop access):0
1261:RULE 5 (multi-hop access):0
1310:RUL
1285:netAccess(app_29,tcp,8080):0
1330:networkServiceInfo(app_29,tomcat,tcp,8080,root):1
1331:vulExists(app_29,vul_app,tomcat,remoteE
1284:RULE 2 (remote exploit of a server program):0
1282:hacl(app_29,db_45,tcp,5432):1
1283:execCode(app_29,root):0
1281:RULE 5 (multi-hop access):0
3:netAccess(db_45,tcp,5432):0
2454:networkServiceInfo(db_45,postgres,tcp,5432,root):1
2455:vulE
2:RULE 2 (remote exploit of a server program):0
1:execCode(db_45,root):0

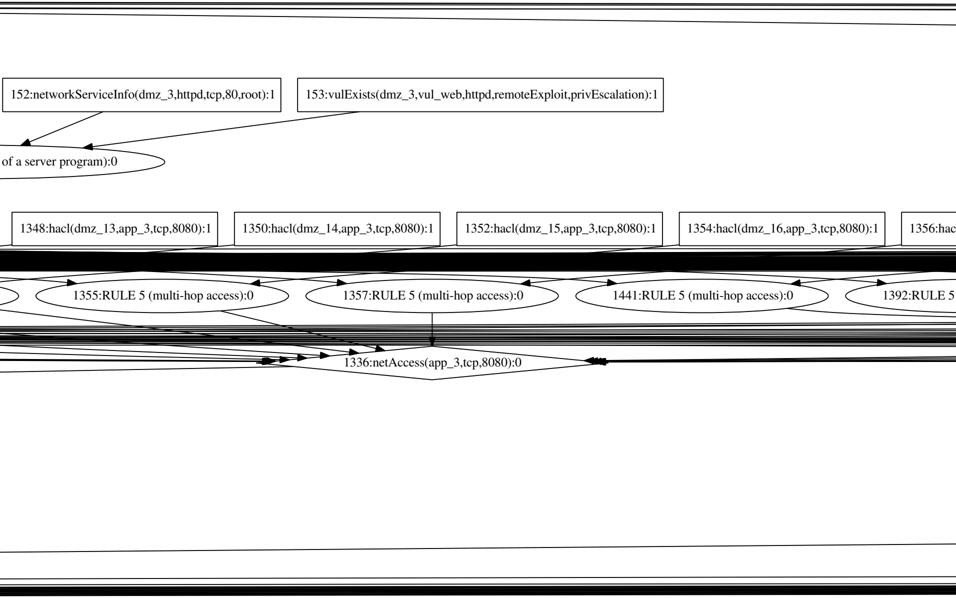

106:hacl(internet,dmz_18,tcp,80):1
16:attackerLocated(internet):1
115:hacl(internet,dmz_19,tcp,80):1
105:RULE 6 (direct network access):0
114:RULE 6 (direct network access):0
107:networkServiceInfo(dmz_18,httpd,tcp,80,root):1
108:vulExists(dmz_18,vul_web,httpd,remoteExploit,privEscalation):1
113:netAccess(dmz_19,tcp,80):0
(remote exploit of a server program):0
112:RULE 2 (remote
acl(dmz_19,app_29,tcp,8080):1
1262:hacl(dmz_20,app_28,tcp,8080):1
120:execCode(dmz_2,root):0
1311:hacl(dmz_2,app_29,tcp,8080):1
1340:hacl(dmz_1,app_3,
E 5 (multi-hop access):0
1339:RULE 5 (multi-hop access):0
1341:RULE 5 (multi-hop access):0
1343:RULE 5 (multi-hop access):0
1345:RULE
Exploit,privEscalation):1
Exists(db_45,vul_db,postgres,remoteExploit,privEscalation):1

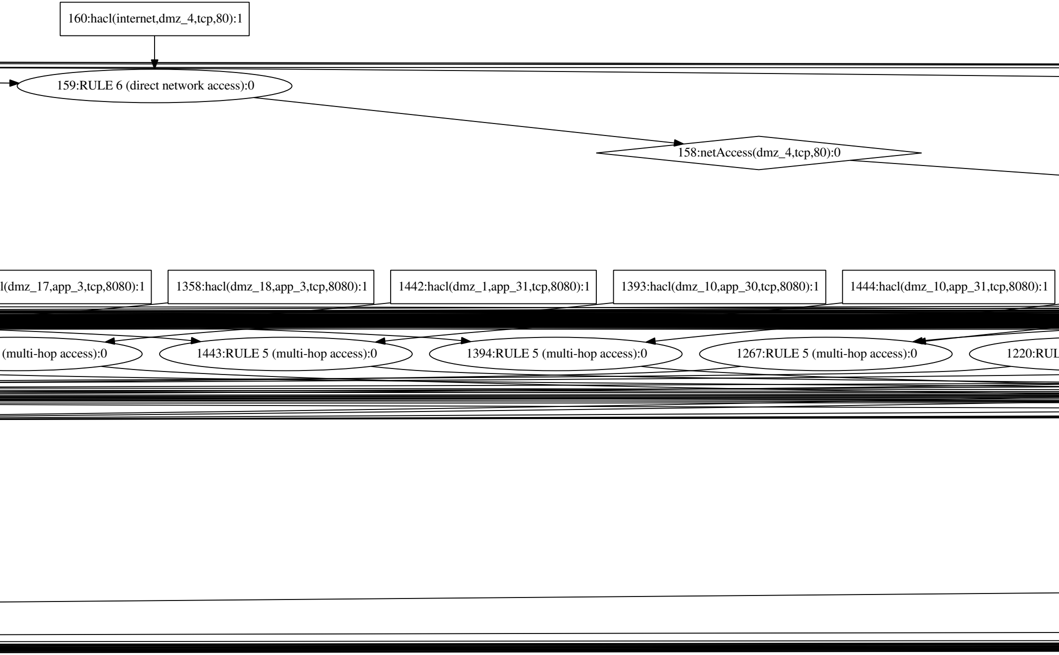

124:hacl(internet,dmz_2,tcp,80):1
123:RULE 6 (direct network access):0
116:networkServiceInfo(dmz_19,httpd,tcp,80,root):1
117:vulExists(dmz_19,vul_web,httpd,remoteExploit,privEscalation):1
122:netAccess(dmz_2,tcp,80)
exploit of a server program):0
121:RU
tcp,8080):1
1342:hacl(dmz_10,app_3,tcp,8080):1
1344:hacl(dmz_11,app_3,tcp,8080):1
1346:hacl(dmz_12,app_3,tcp,8080):1
1389:hacl(dmz_0,app_30,tcp,8080):1
5 (multi-hop access):0
1388:RULE 5 (multi-hop access):0
1439:RULE 5 (multi-hop access):0
1390:RULE 5 (multi-hop access):0
1381:networkService
1333:hacl(app_3,db_45,tcp,5432):1
1332:RULE 5 (multi-hop access):0

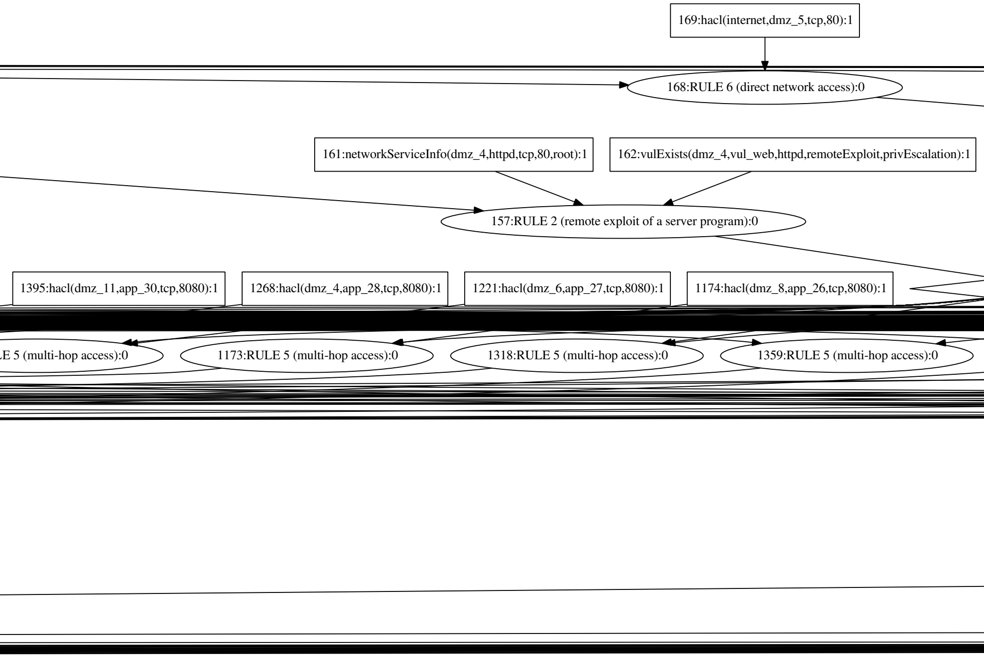

133:hacl(internet,dmz_20,tcp,80):1
132:RULE 6 (direct network access):0
):0
125:networkServiceInfo(dmz_2,httpd,tcp,80,root):1
126:vulExists(dmz_2,vul_web,httpd,remoteExploit,privEscalation):1
131:netAccess(dmz_20,tcp,80):0
LE 2 (remote exploit of a server program):0
1440:hacl(dmz_0,app_31,tcp,8080):1
1391:hacl(dmz_1,app_30,tcp,8080):1
1264:hacl(dmz_21,app_28,tcp,8080):1
1219:hacl(dmz_5,app_27,tcp,8080):1
1125:hacl(dmz_9,app_25,tcp,8080):1
1263:RULE 5 (multi-hop access):0
1218:RULE 5 (multi-hop access):0
1124:RULE 5 (multi-hop access):0
1171:RULE 5 (multi-hop access):0
nfo(app_3,tomcat,tcp,8080,root):1
1382:vulExists(app_3,vul_app,tomcat,remoteExploit,privEscalation):1
1335:RULE 2 (remote exploit of a server program):0
1334:execCode(app_3,root):0

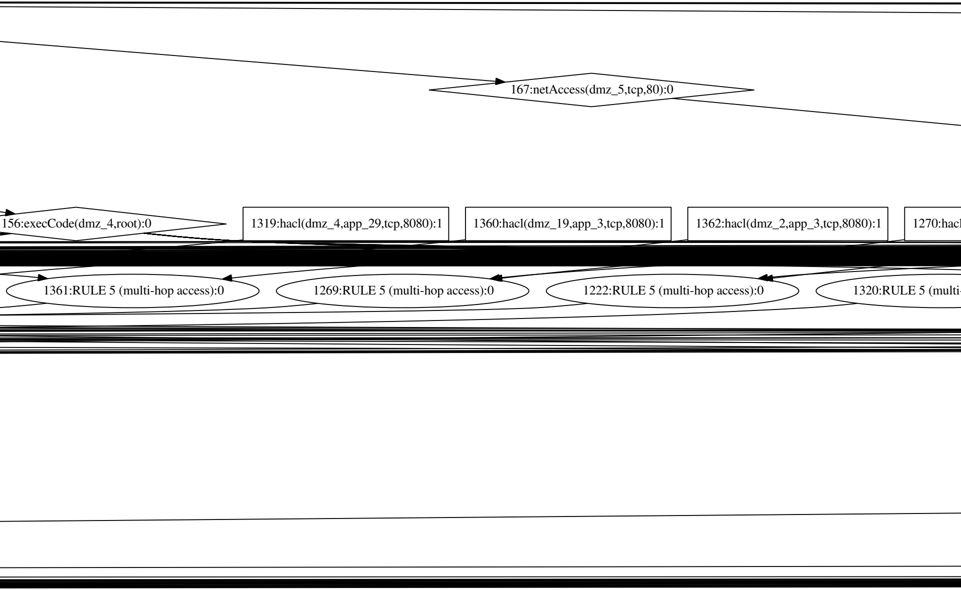

142:hacl(internet,dmz_21,tcp,80):1
141:RULE 6 (direct network access):0
134:networkServiceInfo(dmz_20,httpd,tcp,80,root):1
135:vulExists(dmz_20,vul_web,httpd,remoteExploit,privEscalation):1
140:netAccess(dmz_21,tcp,80):0
130:RULE 2 (remote exploit of a server program):0
1172:hacl(dmz_7,app_26,tcp,8080):1
129:execCode(dmz_20,root):0
1313:hacl(dmz_20,app_29,tcp,8080):1
1266:hacl(dmz_3,app_28,tcp,8080):1
1312:RULE 5 (multi-hop access):0
1265:RULE 5 (multi-hop access):0
1314:RULE 5 (multi-hop access):0
1316:RULE 5 (multi-hop access):0
1384:hacl(app_30,db_45,tcp,5432):1
1383:RULE 5 (multi-hop access):0

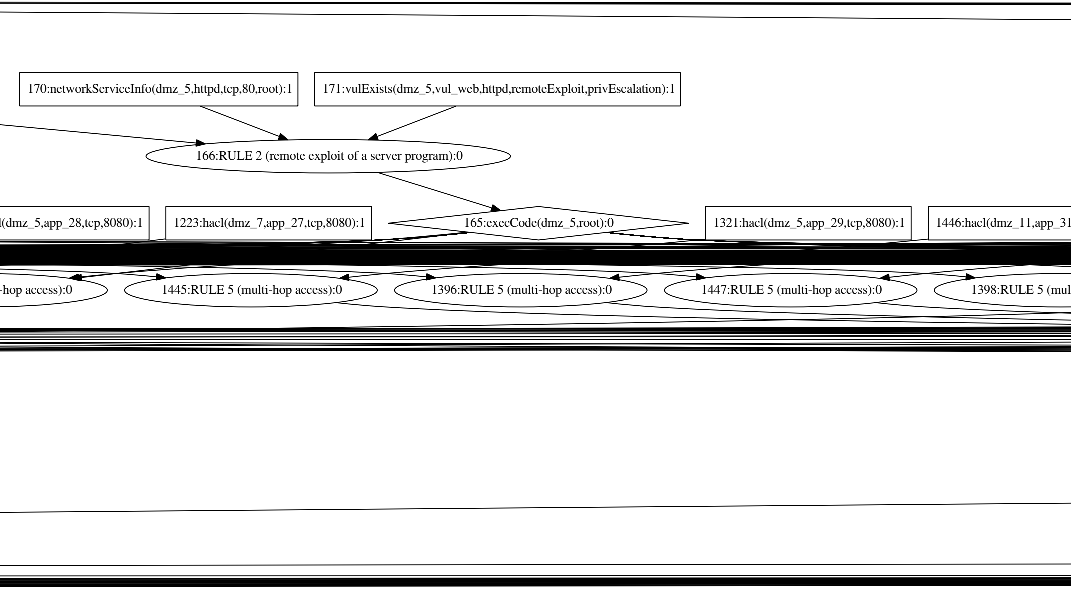

151:hacl(internet,dmz_3,tcp,80):1
150:RULE 6 (direct network access):0
143:networkServiceInfo(dmz_21,httpd,tcp,80,root):1
144:vulExists(dmz_21,vul_web,httpd,remoteExploit,privEscalation):1
149:netAccess(dmz_3,tcp,80):0
139:RULE 2 (remote exploit of a server program):0
148:RULE 2 (remote exploit
138:execCode(dmz_21,root):0
1315:hacl(dmz_21,app_29,tcp,8080):1
147:execCode(dmz_3,root):0
1317:hacl(dmz_3,app_29,tcp,8080):1
1347:RULE 5 (multi-hop access):0
1349:RULE 5 (multi-hop access):0
1351:RULE 5 (multi-hop access):0
1353:RULE 5 (multi-hop access):0

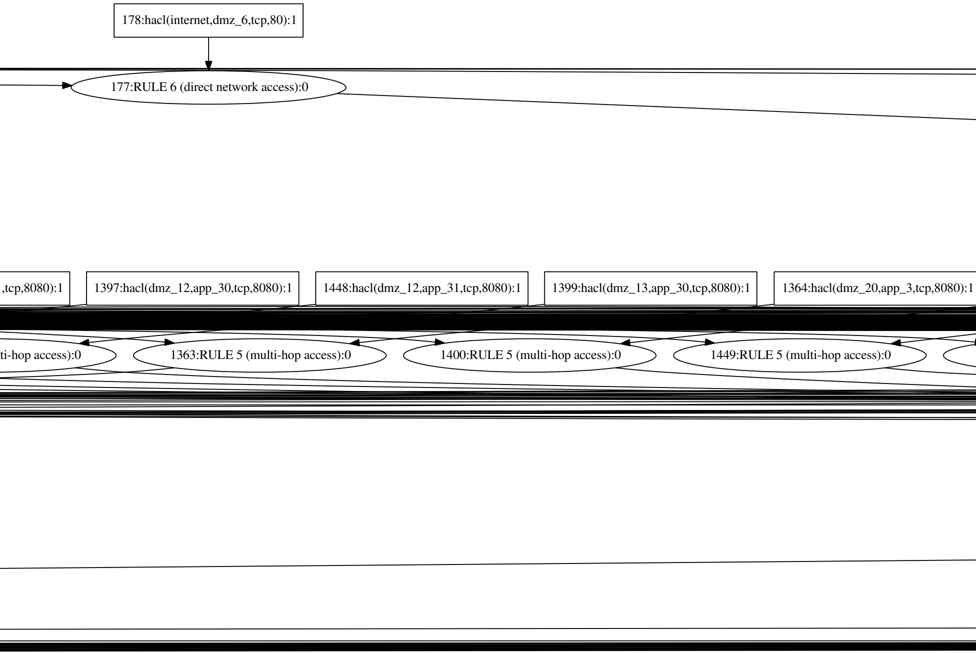

152:networkServiceInfo(dmz_3,httpd,tcp,80,root):1
153:vulExists(dmz_3,vul_web,httpd,remoteExploit,privEscalation):1
of a server program):0
1348:hacl(dmz_13,app_3,tcp,8080):1
1350:hacl(dmz_14,app_3,tcp,8080):1
1352:hacl(dmz_15,app_3,tcp,8080):1
1354:hacl(dmz_16,app_3,tcp,8080):1
1356:hac
1355:RULE 5 (multi-hop access):0
1357:RULE 5 (multi-hop access):0
1441:RULE 5 (multi-hop access):0
1392:RULE 5
1336:netAccess(app_3,tcp,8080):0

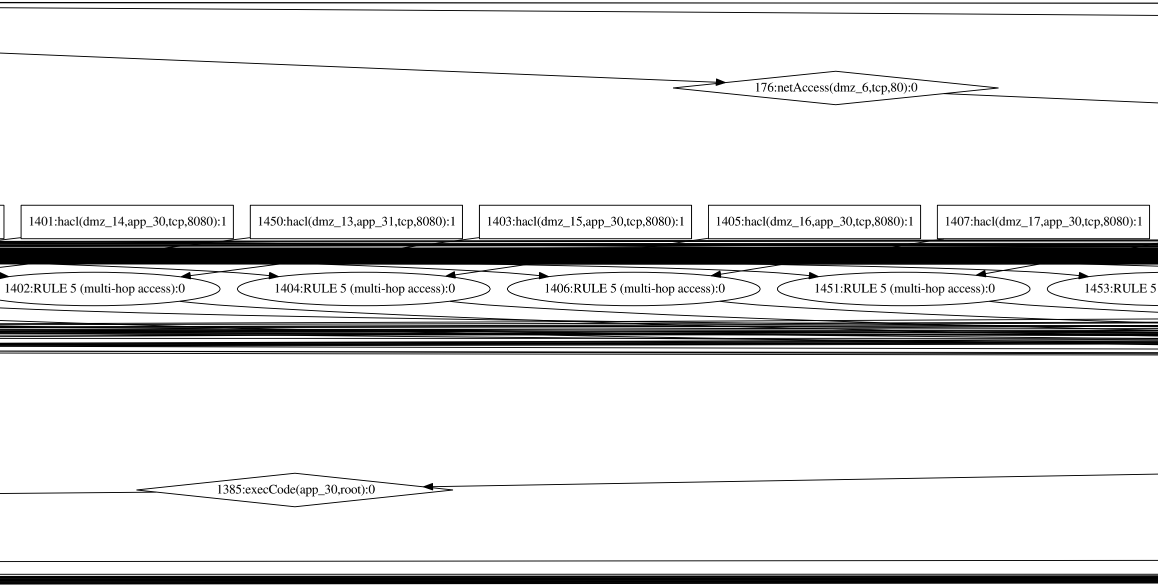

160:hacl(internet,dmz_4,tcp,80):1
159:RULE 6 (direct network access):0
158:netAccess(dmz_4,tcp,80):0
l(dmz_17,app_3,tcp,8080):1
1358:hacl(dmz_18,app_3,tcp,8080):1
1442:hacl(dmz_1,app_31,tcp,8080):1
1393:hacl(dmz_10,app_30,tcp,8080):1
1444:hacl(dmz_10,app_31,tcp,8080):1
(multi-hop access):0
1443:RULE 5 (multi-hop access):0
1394:RULE 5 (multi-hop access):0
1267:RULE 5 (multi-hop access):0
1220:RUL

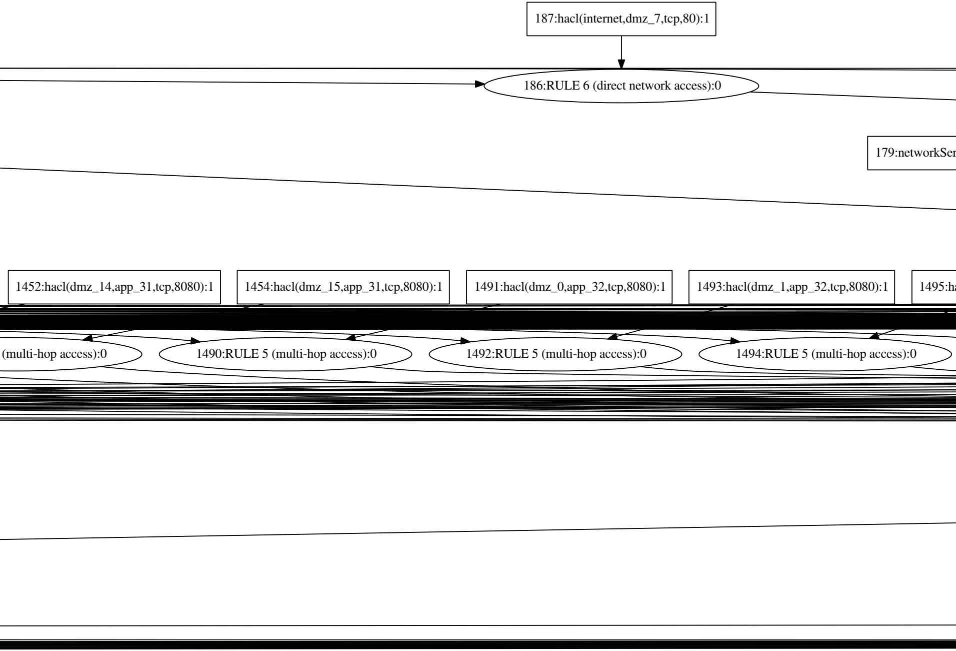

169:hacl(internet,dmz_5,tcp,80):1
168:RULE 6 (direct network access):0
161:networkServiceInfo(dmz_4,httpd,tcp,80,root):1
162:vulExists(dmz_4,vul_web,httpd,remoteExploit,privEscalation):1
157:RULE 2 (remote exploit of a server program):0
1395:hacl(dmz_11,app_30,tcp,8080):1
1268:hacl(dmz_4,app_28,tcp,8080):1
1221:hacl(dmz_6,app_27,tcp,8080):1
1174:hacl(dmz_8,app_26,tcp,8080):1
LE 5 (multi-hop access):0
1173:RULE 5 (multi-hop access):0
1318:RULE 5 (multi-hop access):0
1359:RULE 5 (multi-hop access):0

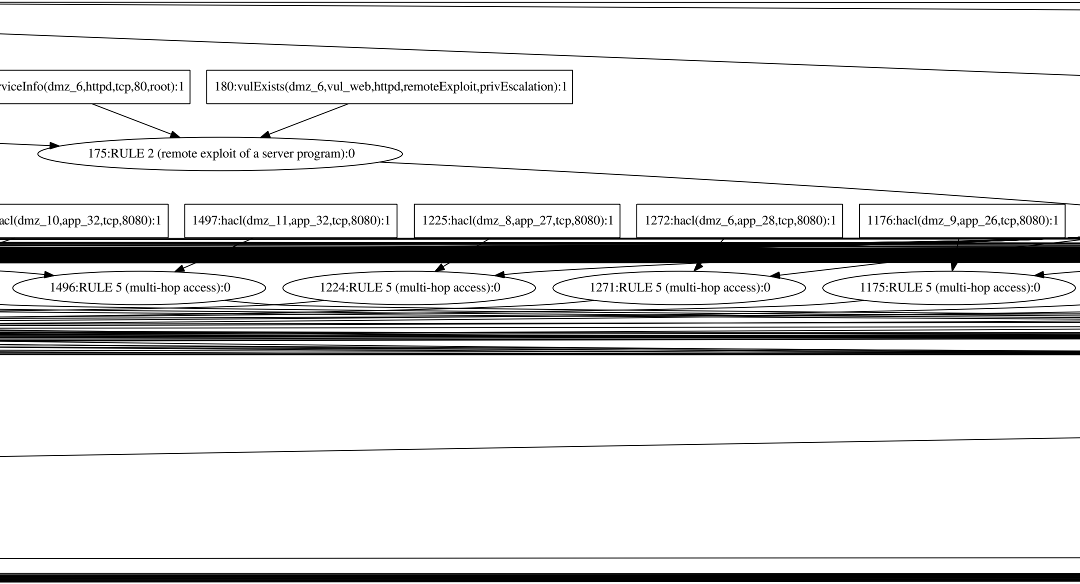

167:netAccess(dmz_5,tcp,80):0
156:execCode(dmz_4,root):0
1319:hacl(dmz_4,app_29,tcp,8080):1
1360:hacl(dmz_19,app_3,tcp,8080):1
1362:hacl(dmz_2,app_3,tcp,8080):1
1270:hacl
1361:RULE 5 (multi-hop access):0
1269:RULE 5 (multi-hop access):0
1222:RULE 5 (multi-hop access):0
1320:RULE 5 (multi-

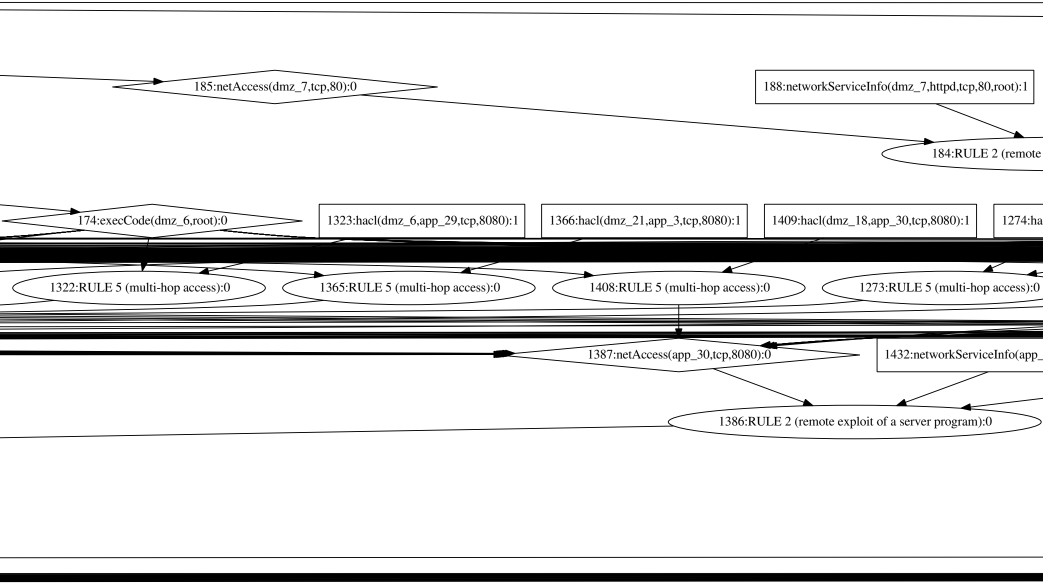

170:networkServiceInfo(dmz_5,httpd,tcp,80,root):1
171:vulExists(dmz_5,vul_web,httpd,remoteExploit,privEscalation):1
166:RULE 2 (remote exploit of a server program):0
l(dmz_5,app_28,tcp,8080):1
1223:hacl(dmz_7,app_27,tcp,8080):1
165:execCode(dmz_5,root):0
1321:hacl(dmz_5,app_29,tcp,8080):1
1446:hacl(dmz_11,app_31
-hop access):0
1445:RULE 5 (multi-hop access):0
1396:RULE 5 (multi-hop access):0
1447:RULE 5 (multi-hop access):0
1398:RULE 5 (mul

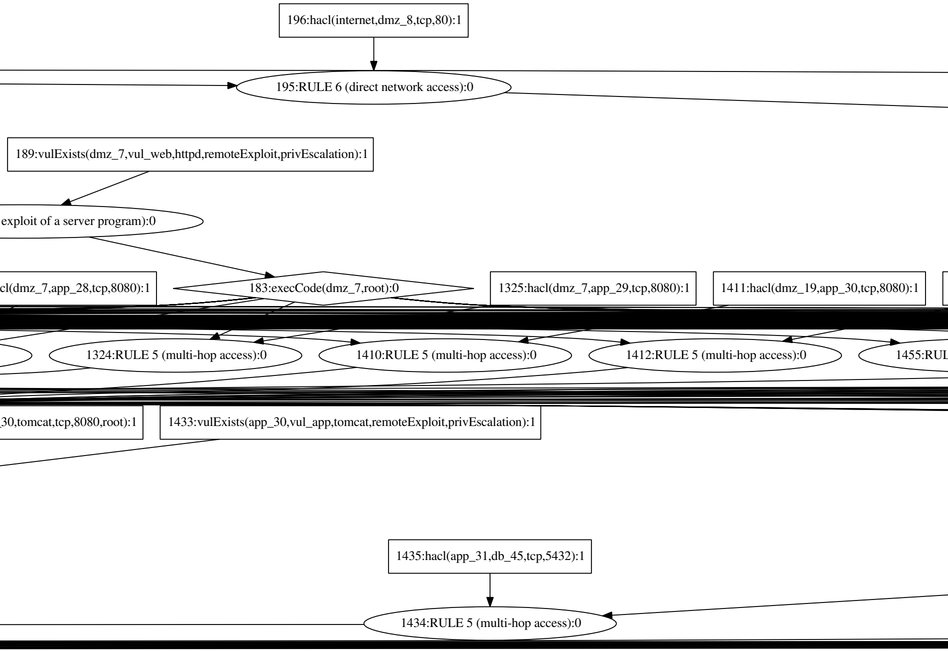

178:hacl(internet,dmz_6,tcp,80):1
177:RULE 6 (direct network access):0
,tcp,8080):1
1397:hacl(dmz_12,app_30,tcp,8080):1
1448:hacl(dmz_12,app_31,tcp,8080):1
1399:hacl(dmz_13,app_30,tcp,8080):1
1364:hacl(dmz_20,app_3,tcp,8080):1
ti-hop access):0
1363:RULE 5 (multi-hop access):0
1400:RULE 5 (multi-hop access):0
1449:RULE 5 (multi-hop access):0

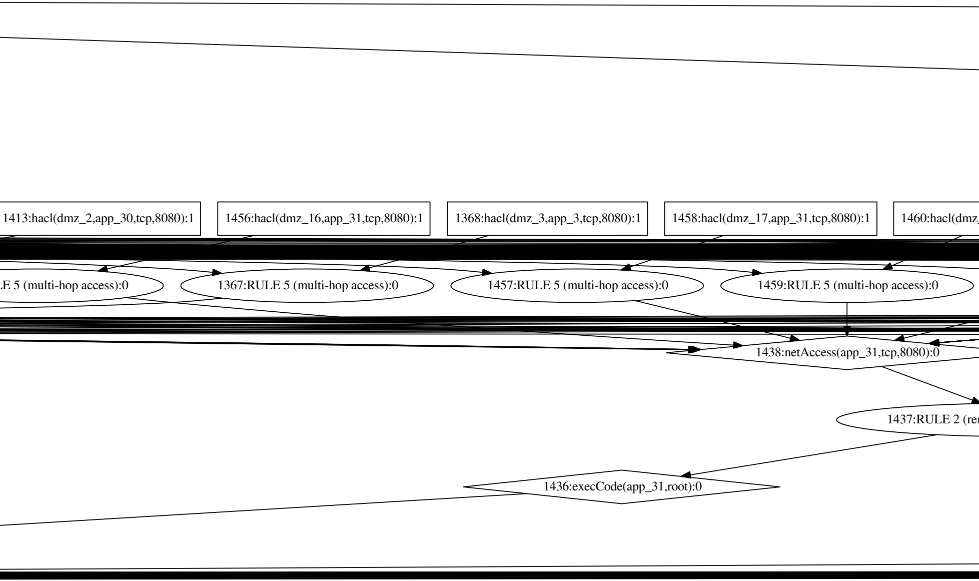

176:netAccess(dmz_6,tcp,80):0
1401:hacl(dmz_14,app_30,tcp,8080):1
1450:hacl(dmz_13,app_31,tcp,8080):1
1403:hacl(dmz_15,app_30,tcp,8080):1
1405:hacl(dmz_16,app_30,tcp,8080):1
1407:hacl(dmz_17,app_30,tcp,8080):1
1402:RULE 5 (multi-hop access):0
1404:RULE 5 (multi-hop access):0
1406:RULE 5 (multi-hop access):0
1451:RULE 5 (multi-hop access):0
1453:RULE 5
1385:execCode(app_30,root):0

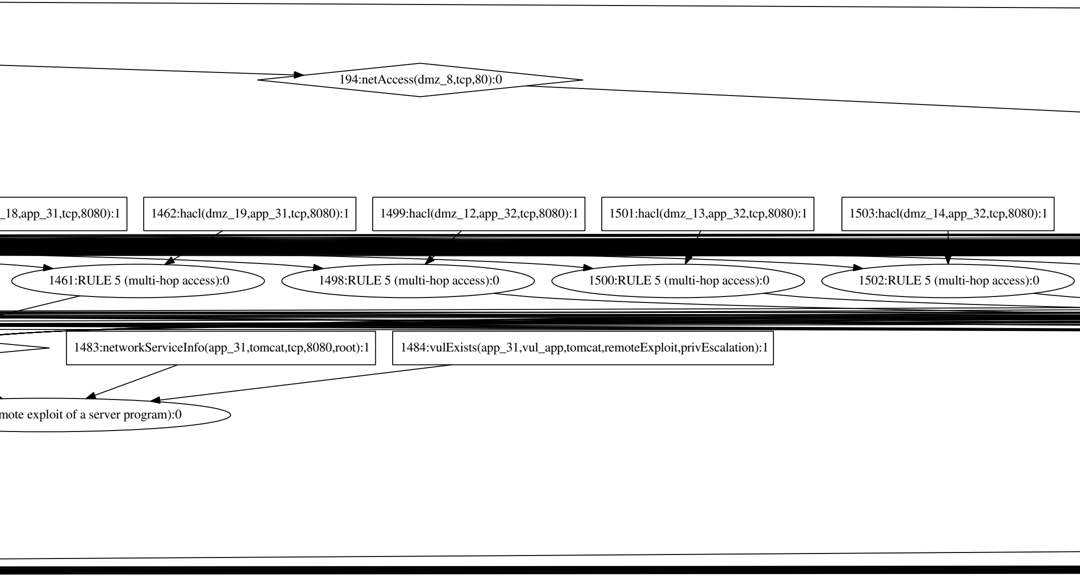

187:hacl(internet,dmz_7,tcp,80):1
186:RULE 6 (direct network access):0
179:networkSer
1452:hacl(dmz_14,app_31,tcp,8080):1
1454:hacl(dmz_15,app_31,tcp,8080):1
1491:hacl(dmz_0,app_32,tcp,8080):1
1493:hacl(dmz_1,app_32,tcp,8080):1
1495:ha
(multi-hop access):0
1490:RULE 5 (multi-hop access):0
1492:RULE 5 (multi-hop access):0
1494:RULE 5 (multi-hop access):0

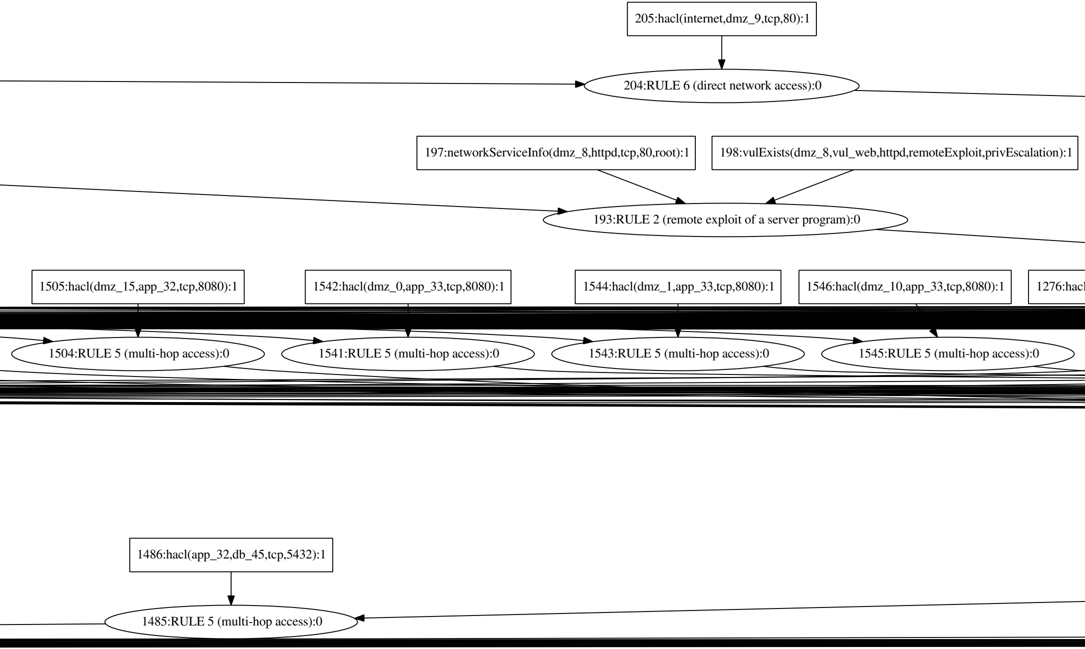

viceInfo(dmz_6,httpd,tcp,80,root):1
180:vulExists(dmz_6,vul_web,httpd,remoteExploit,privEscalation):1
175:RULE 2 (remote exploit of a server program):0
acl(dmz_10,app_32,tcp,8080):1
1497:hacl(dmz_11,app_32,tcp,8080):1
1225:hacl(dmz_8,app_27,tcp,8080):1
1272:hacl(dmz_6,app_28,tcp,8080):1
1176:hacl(dmz_9,app_26,tcp,8080):1
1496:RULE 5 (multi-hop access):0
1224:RULE 5 (multi-hop access):0
1271:RULE 5 (multi-hop access):0
1175:RULE 5 (multi-hop access):0

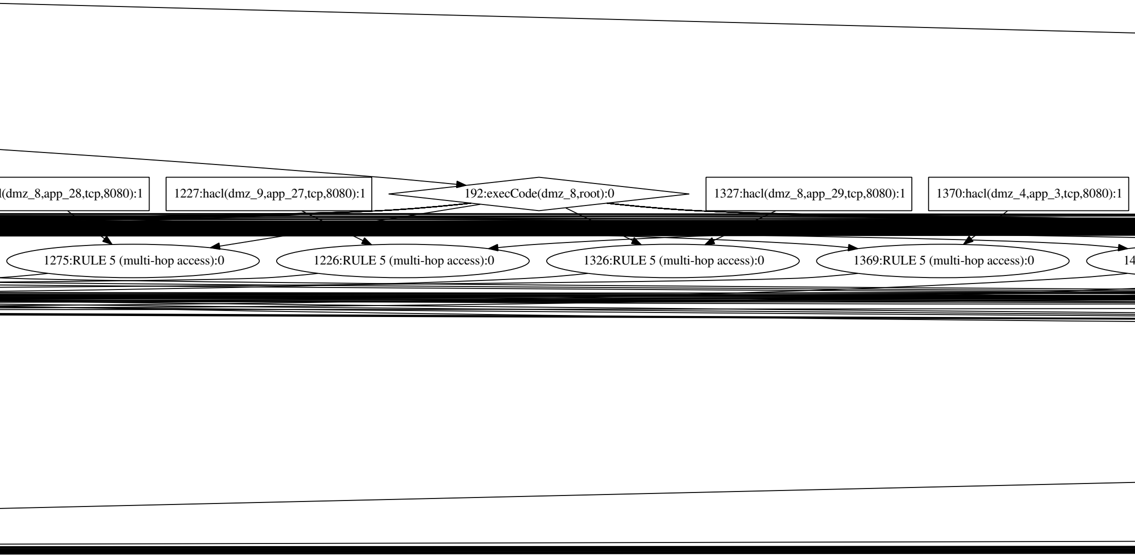

185:netAccess(dmz_7,tcp,80):0
188:networkServiceInfo(dmz_7,httpd,tcp,80,root):1
184:RULE 2 (remote
174:execCode(dmz_6,root):0
1323:hacl(dmz_6,app_29,tcp,8080):1
1366:hacl(dmz_21,app_3,tcp,8080):1
1409:hacl(dmz_18,app_30,tcp,8080):1
1274:ha
1322:RULE 5 (multi-hop access):0
1365:RULE 5 (multi-hop access):0
1408:RULE 5 (multi-hop access):0
1273:RULE 5 (multi-hop access):0
1387:netAccess(app_30,tcp,8080):0
1432:networkServiceInfo(app_
1386:RULE 2 (remote exploit of a server program):0

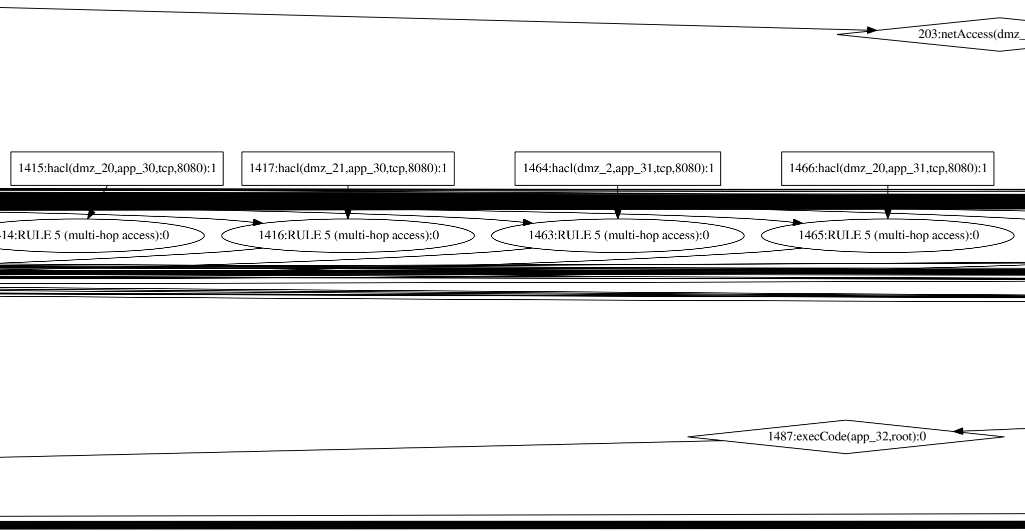

196:hacl(internet,dmz_8,tcp,80):1
195:RULE 6 (direct network access):0
189:vulExists(dmz_7,vul_web,httpd,remoteExploit,privEscalation):1
exploit of a server program):0
cl(dmz_7,app_28,tcp,8080):1
183:execCode(dmz_7,root):0
1325:hacl(dmz_7,app_29,tcp,8080):1
1411:hacl(dmz_19,app_30,tcp,8080):1
1324:RULE 5 (multi-hop access):0
1410:RULE 5 (multi-hop access):0
1412:RULE 5 (multi-hop access):0
1455:RUL
30,tomcat,tcp,8080,root):1
1433:vulExists(app_30,vul_app,tomcat,remoteExploit,privEscalation):1
1435:hacl(app_31,db_45,tcp,5432):1
1434:RULE 5 (multi-hop access):0

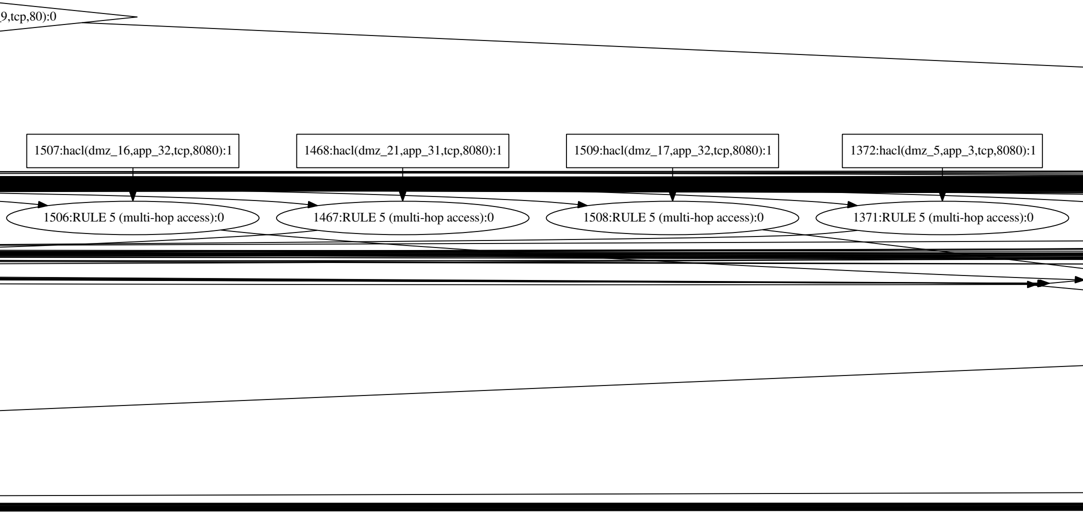

1413:hacl(dmz_2,app_30,tcp,8080):1
1456:hacl(dmz_16,app_31,tcp,8080):1
1368:hacl(dmz_3,app_3,tcp,8080):1
1458:hacl(dmz_17,app_31,tcp,8080):1
1460:hacl(dmz_
LE 5 (multi-hop access):0
1367:RULE 5 (multi-hop access):0
1457:RULE 5 (multi-hop access):0
1459:RULE 5 (multi-hop access):0
1438:netAccess(app_31,tcp,8080):0
1437:RULE 2 (re
1436:execCode(app_31,root):0

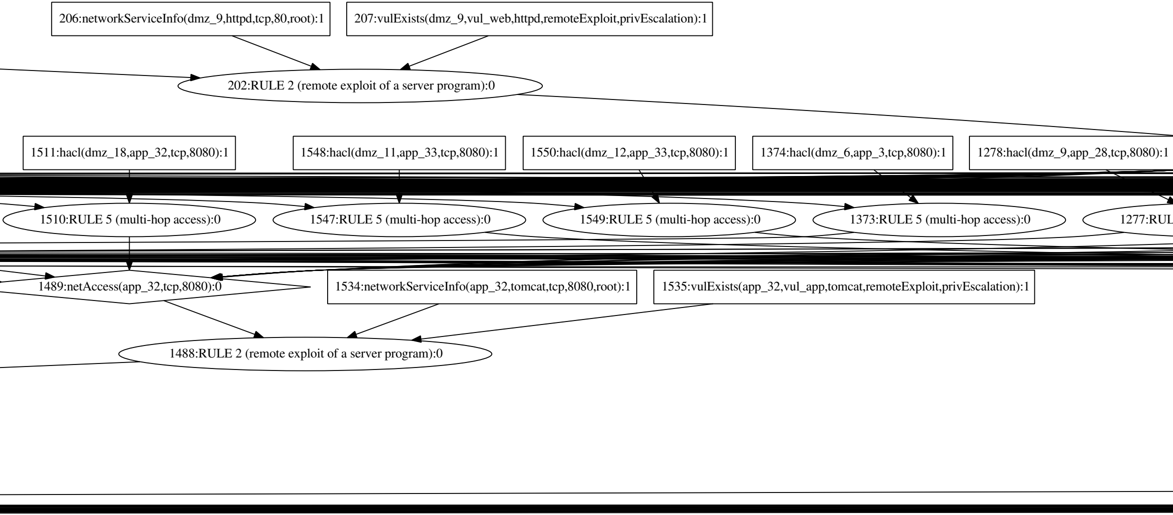

194:netAccess(dmz_8,tcp,80):0
_18,app_31,tcp,8080):1
1462:hacl(dmz_19,app_31,tcp,8080):1
1499:hacl(dmz_12,app_32,tcp,8080):1
1501:hacl(dmz_13,app_32,tcp,8080):1
1503:hacl(dmz_14,app_32,tcp,8080):1
1461:RULE 5 (multi-hop access):0
1498:RULE 5 (multi-hop access):0
1500:RULE 5 (multi-hop access):0
1502:RULE 5 (multi-hop access):0
1483:networkServiceInfo(app_31,tomcat,tcp,8080,root):1
1484:vulExists(app_31,vul_app,tomcat,remoteExploit,privEscalation):1
mote exploit of a server program):0

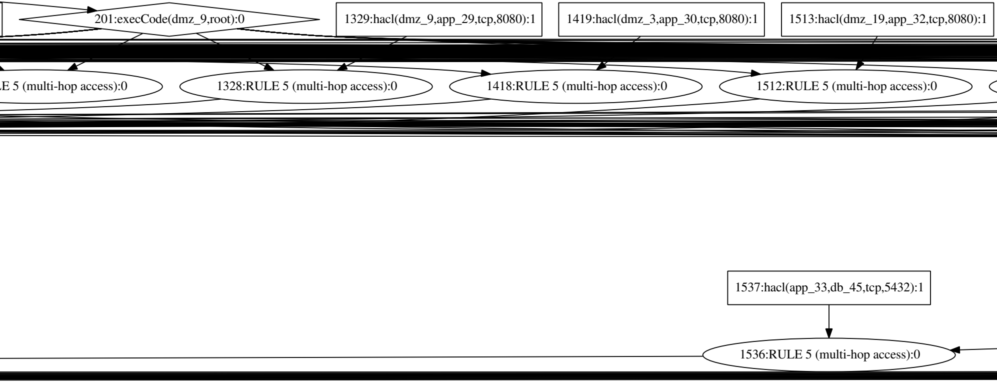

205:hacl(internet,dmz_9,tcp,80):1
204:RULE 6 (direct network access):0
197:networkServiceInfo(dmz_8,httpd,tcp,80,root):1
198:vulExists(dmz_8,vul_web,httpd,remoteExploit,privEscalation):1
193:RULE 2 (remote exploit of a server program):0
1505:hacl(dmz_15,app_32,tcp,8080):1
1542:hacl(dmz_0,app_33,tcp,8080):1
1544:hacl(dmz_1,app_33,tcp,8080):1
1546:hacl(dmz_10,app_33,tcp,8080):1
1276:hac
1504:RULE 5 (multi-hop access):0
1541:RULE 5 (multi-hop access):0
1543:RULE 5 (multi-hop access):0
1545:RULE 5 (multi-hop access):0
1486:hacl(app_32,db_45,tcp,5432):1
1485:RULE 5 (multi-hop access):0

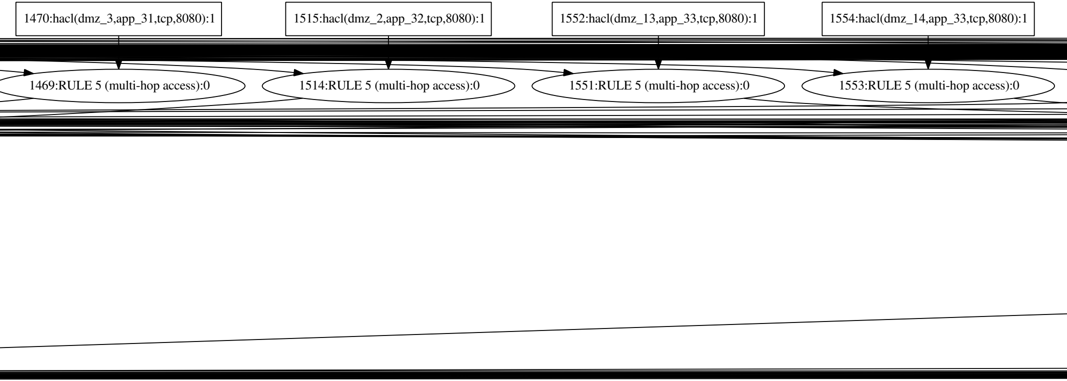

l(dmz_8,app_28,tcp,8080):1
1227:hacl(dmz_9,app_27,tcp,8080):1
192:execCode(dmz_8,root):0
1327:hacl(dmz_8,app_29,tcp,8080):1
1370:hacl(dmz_4,app_3,tcp,8080):1
1275:RULE 5 (multi-hop access):0
1226:RULE 5 (multi-hop access):0
1326:RULE 5 (multi-hop access):0
1369:RULE 5 (multi-hop access):0
14

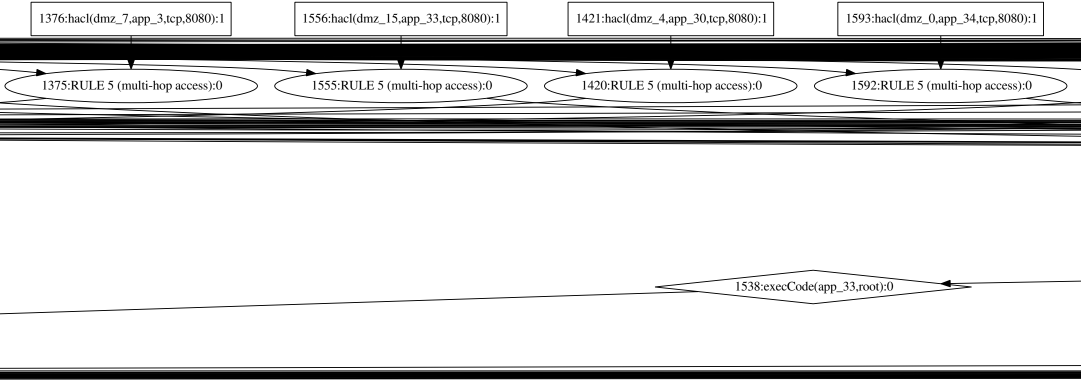

203:netAccess(dmz_
1415:hacl(dmz_20,app_30,tcp,8080):1
1417:hacl(dmz_21,app_30,tcp,8080):1
1464:hacl(dmz_2,app_31,tcp,8080):1
1466:hacl(dmz_20,app_31,tcp,8080):1
14:RULE 5 (multi-hop access):0
1416:RULE 5 (multi-hop access):0
1463:RULE 5 (multi-hop access):0
1465:RULE 5 (multi-hop access):0
1487:execCode(app_32,root):0

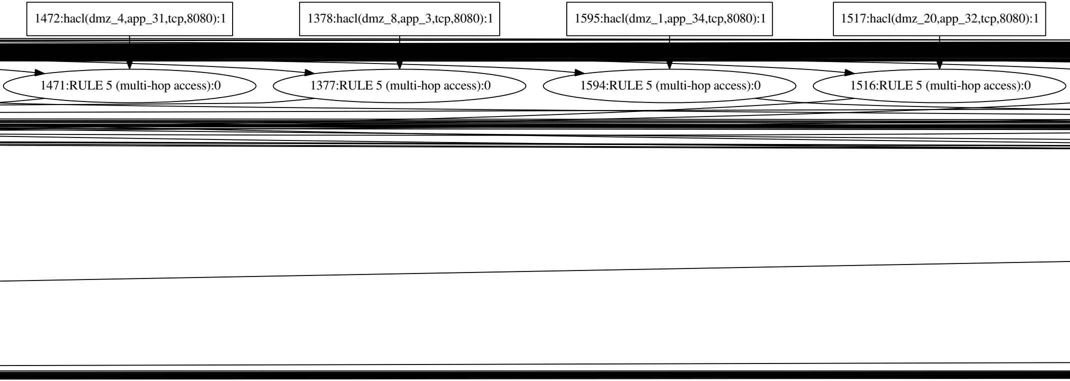

9,tcp,80):0
1507:hacl(dmz_16,app_32,tcp,8080):1
1468:hacl(dmz_21,app_31,tcp,8080):1
1509:hacl(dmz_17,app_32,tcp,8080):1
1372:hacl(dmz_5,app_3,tcp,8080):1
1506:RULE 5 (multi-hop access):0
1467:RULE 5 (multi-hop access):0
1508:RULE 5 (multi-hop access):0
1371:RULE 5 (multi-hop access):0

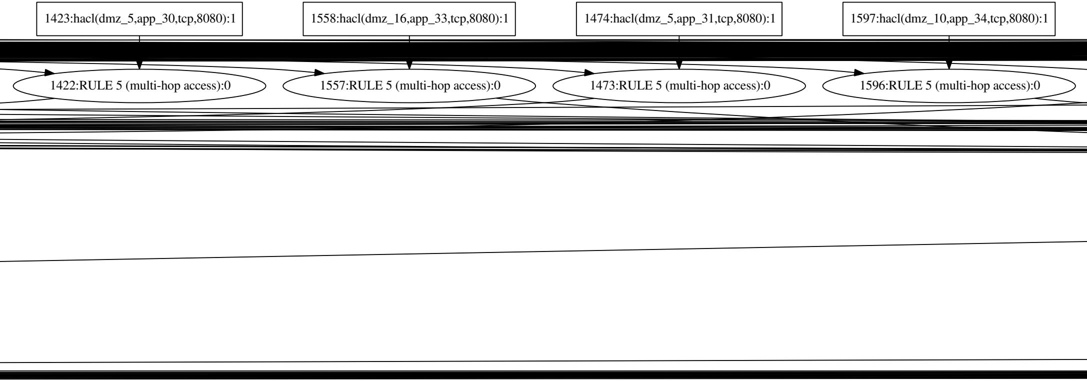

206:networkServiceInfo(dmz_9,httpd,tcp,80,root):1
207:vulExists(dmz_9,vul_web,httpd,remoteExploit,privEscalation):1
202:RULE 2 (remote exploit of a server program):0
1511:hacl(dmz_18,app_32,tcp,8080):1
1548:hacl(dmz_11,app_33,tcp,8080):1
1550:hacl(dmz_12,app_33,tcp,8080):1
1374:hacl(dmz_6,app_3,tcp,8080):1
1278:hacl(dmz_9,app_28,tcp,8080):1
1510:RULE 5 (multi-hop access):0
1547:RULE 5 (multi-hop access):0
1549:RULE 5 (multi-hop access):0
1373:RULE 5 (multi-hop access):0
1277:RUL
1489:netAccess(app_32,tcp,8080):0
1534:networkServiceInfo(app_32,tomcat,tcp,8080,root):1
1535:vulExists(app_32,vul_app,tomcat,remoteExploit,privEscalation):1
1488:RULE 2 (remote exploit of a server program):0

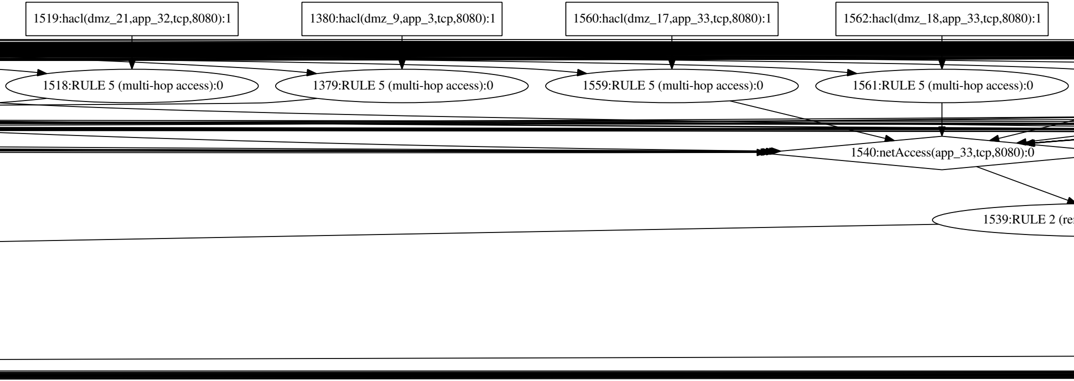

201:execCode(dmz_9,root):0
1329:hacl(dmz_9,app_29,tcp,8080):1
1419:hacl(dmz_3,app_30,tcp,8080):1
1513:hacl(dmz_19,app_32,tcp,8080):1
E 5 (multi-hop access):0
1328:RULE 5 (multi-hop access):0
1418:RULE 5 (multi-hop access):0
1512:RULE 5 (multi-hop access):0
1537:hacl(app_33,db_45,tcp,5432):1
1536:RULE 5 (multi-hop access):0

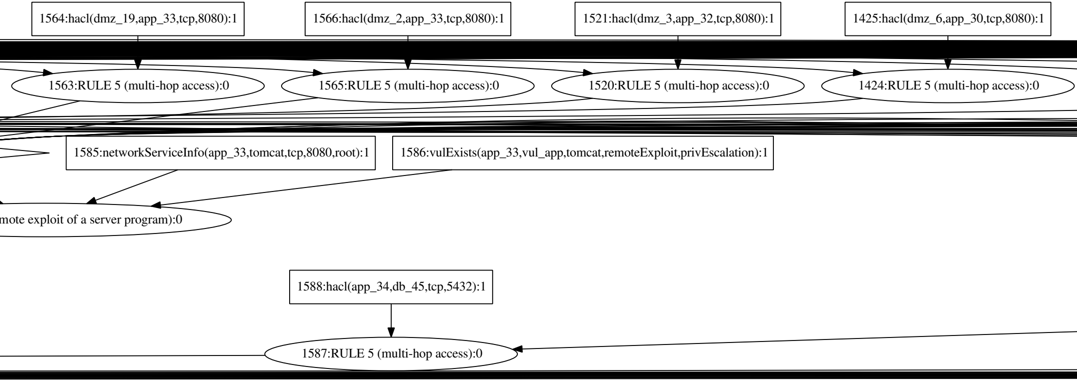

1470:hacl(dmz_3,app_31,tcp,8080):1
1515:hacl(dmz_2,app_32,tcp,8080):1
1552:hacl(dmz_13,app_33,tcp,8080):1
1554:hacl(dmz_14,app_33,tcp,8080):1
1469:RULE 5 (multi-hop access):0
1514:RULE 5 (multi-hop access):0
1551:RULE 5 (multi-hop access):0
1553:RULE 5 (multi-hop access):0

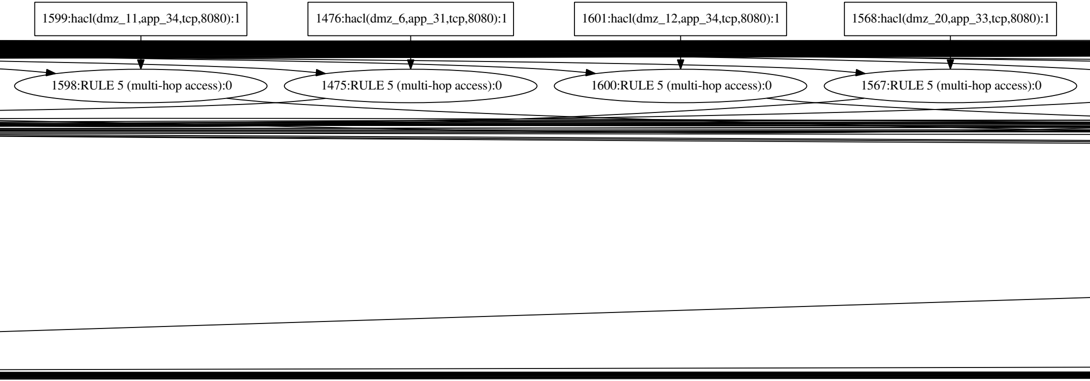

1376:hacl(dmz_7,app_3,tcp,8080):1
1556:hacl(dmz_15,app_33,tcp,8080):1
1421:hacl(dmz_4,app_30,tcp,8080):1
1593:hacl(dmz_0,app_34,tcp,8080):1
1375:RULE 5 (multi-hop access):0
1555:RULE 5 (multi-hop access):0
1420:RULE 5 (multi-hop access):0
1592:RULE 5 (multi-hop access):0
1538:execCode(app_33,root):0

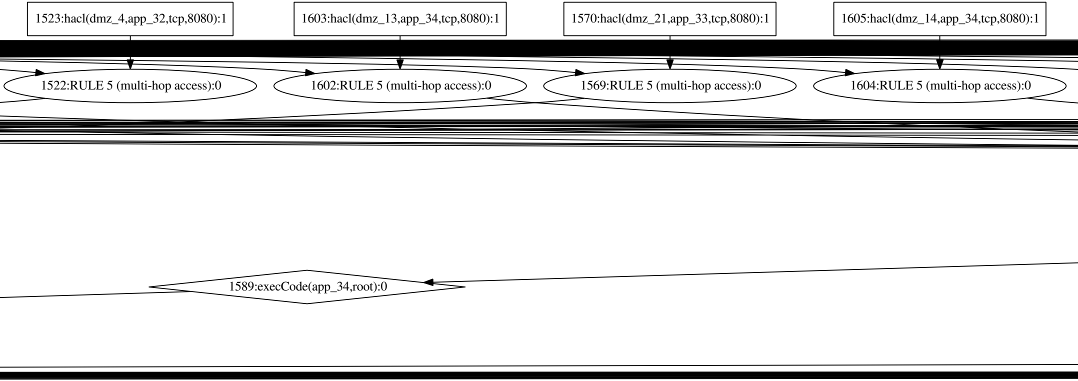

1472:hacl(dmz_4,app_31,tcp,8080):1
1378:hacl(dmz_8,app_3,tcp,8080):1
1595:hacl(dmz_1,app_34,tcp,8080):1
1517:hacl(dmz_20,app_32,tcp,8080):1
1471:RULE 5 (multi-hop access):0
1377:RULE 5 (multi-hop access):0
1594:RULE 5 (multi-hop access):0
1516:RULE 5 (multi-hop access):0

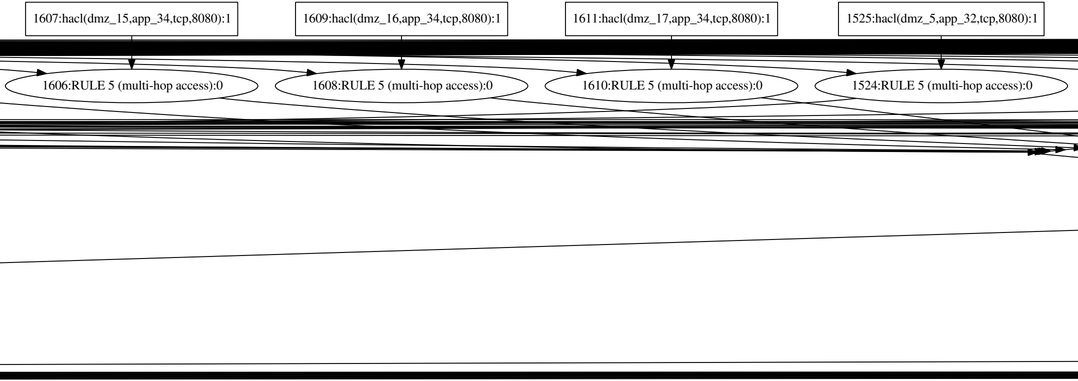

1423:hacl(dmz_5,app_30,tcp,8080):1
1558:hacl(dmz_16,app_33,tcp,8080):1
1474:hacl(dmz_5,app_31,tcp,8080):1
1597:hacl(dmz_10,app_34,tcp,8080):1
1422:RULE 5 (multi-hop access):0
1557:RULE 5 (multi-hop access):0
1473:RULE 5 (multi-hop access):0
1596:RULE 5 (multi-hop access):0

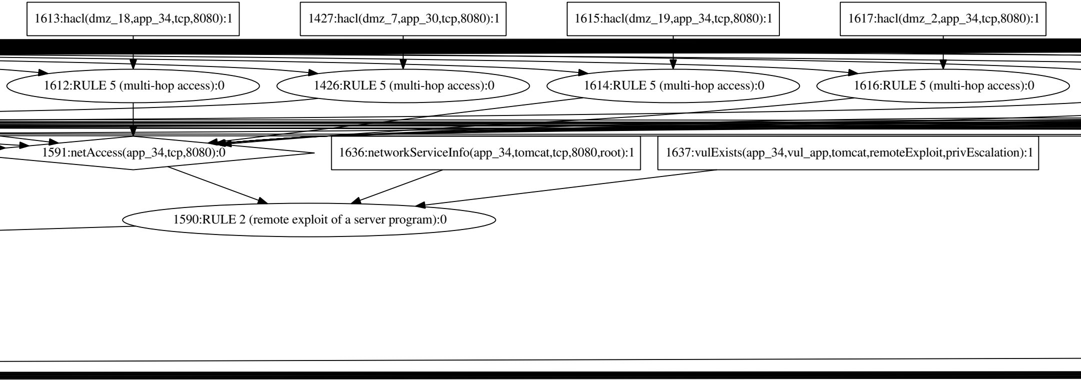

1519:hacl(dmz_21,app_32,tcp,8080):1
1380:hacl(dmz_9,app_3,tcp,8080):1
1560:hacl(dmz_17,app_33,tcp,8080):1
1562:hacl(dmz_18,app_33,tcp,8080):1
1518:RULE 5 (multi-hop access):0
1379:RULE 5 (multi-hop access):0
1559:RULE 5 (multi-hop access):0
1561:RULE 5 (multi-hop access):0
1540:netAccess(app_33,tcp,8080):0
1539:RULE 2 (re

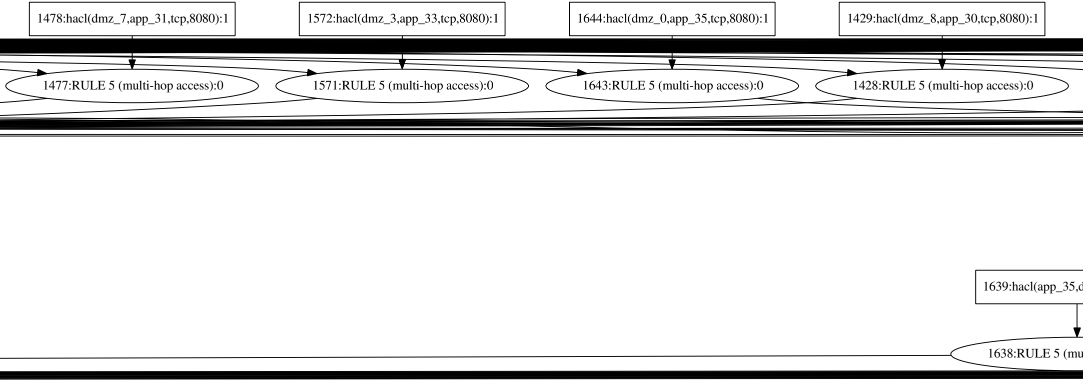

1564:hacl(dmz_19,app_33,tcp,8080):1
1566:hacl(dmz_2,app_33,tcp,8080):1
1521:hacl(dmz_3,app_32,tcp,8080):1
1425:hacl(dmz_6,app_30,tcp,8080):1
1563:RULE 5 (multi-hop access):0
1565:RULE 5 (multi-hop access):0
1520:RULE 5 (multi-hop access):0
1424:RULE 5 (multi-hop access):0
1585:networkServiceInfo(app_33,tomcat,tcp,8080,root):1
1586:vulExists(app_33,vul_app,tomcat,remoteExploit,privEscalation):1
mote exploit of a server program):0
1588:hacl(app_34,db_45,tcp,5432):1
1587:RULE 5 (multi-hop access):0

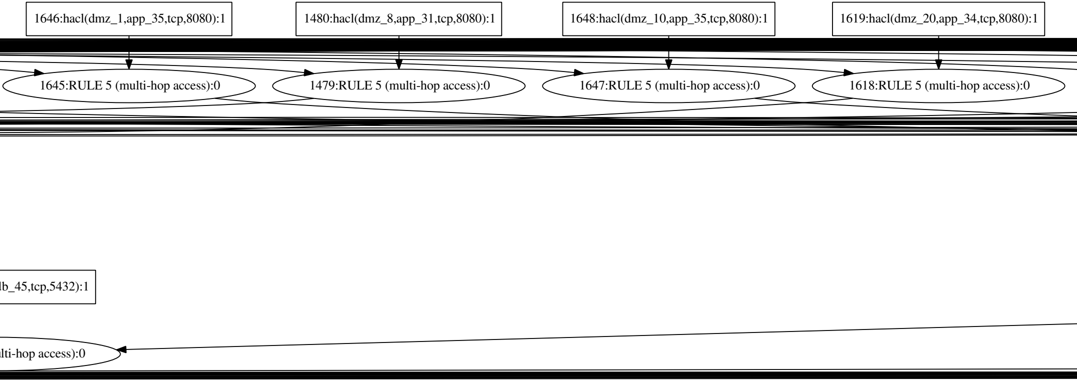

1599:hacl(dmz_11,app_34,tcp,8080):1
1476:hacl(dmz_6,app_31,tcp,8080):1
1601:hacl(dmz_12,app_34,tcp,8080):1
1568:hacl(dmz_20,app_33,tcp,8080):1
1598:RULE 5 (multi-hop access):0
1475:RULE 5 (multi-hop access):0
1600:RULE 5 (multi-hop access):0
1567:RULE 5 (multi-hop access):0

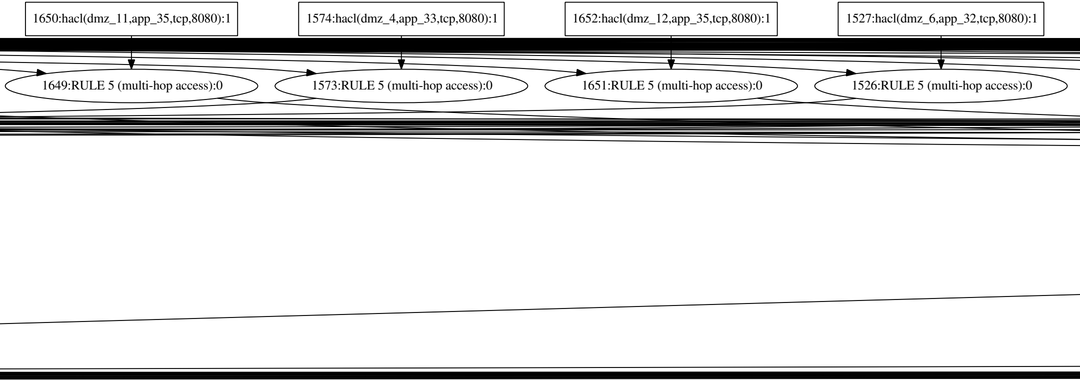

1523:hacl(dmz_4,app_32,tcp,8080):1
1603:hacl(dmz_13,app_34,tcp,8080):1
1570:hacl(dmz_21,app_33,tcp,8080):1
1605:hacl(dmz_14,app_34,tcp,8080):1
1522:RULE 5 (multi-hop access):0
1602:RULE 5 (multi-hop access):0
1569:RULE 5 (multi-hop access):0
1604:RULE 5 (multi-hop access):0
1589:execCode(app_34,root):0

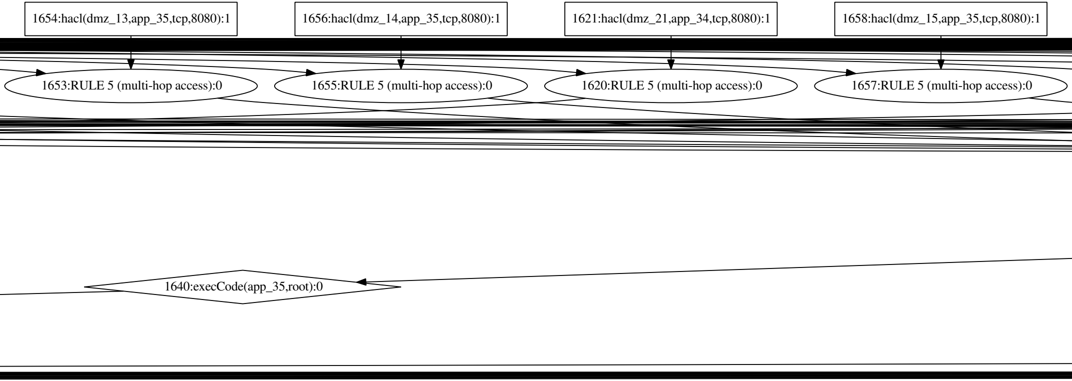

1607:hacl(dmz_15,app_34,tcp,8080):1
1609:hacl(dmz_16,app_34,tcp,8080):1
1611:hacl(dmz_17,app_34,tcp,8080):1
1525:hacl(dmz_5,app_32,tcp,8080):1
1606:RULE 5 (multi-hop access):0
1608:RULE 5 (multi-hop access):0
1610:RULE 5 (multi-hop access):0
1524:RULE 5 (multi-hop access):0

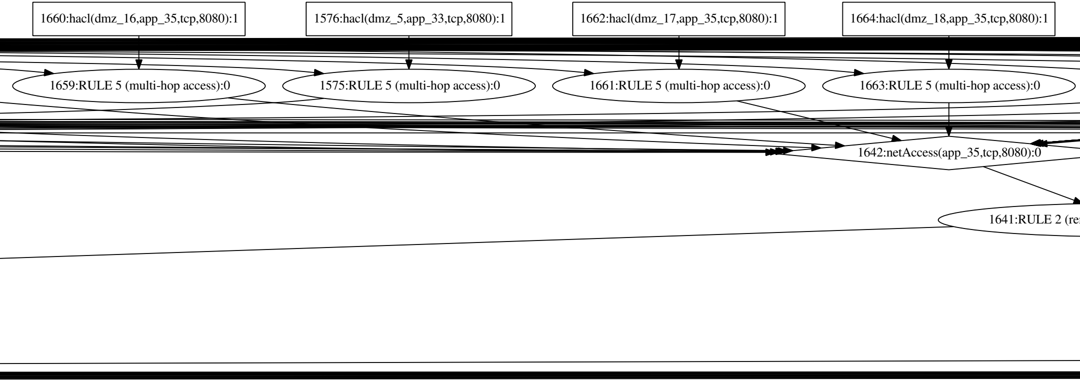

1613:hacl(dmz_18,app_34,tcp,8080):1
1427:hacl(dmz_7,app_30,tcp,8080):1
1615:hacl(dmz_19,app_34,tcp,8080):1
1617:hacl(dmz_2,app_34,tcp,8080):1
1612:RULE 5 (multi-hop access):0
1426:RULE 5 (multi-hop access):0
1614:RULE 5 (multi-hop access):0
1616:RULE 5 (multi-hop access):0
1591:netAccess(app_34,tcp,8080):0
1636:networkServiceInfo(app_34,tomcat,tcp,8080,root):1
1637:vulExists(app_34,vul_app,tomcat,remoteExploit,privEscalation):1
1590:RULE 2 (remote exploit of a server program):0

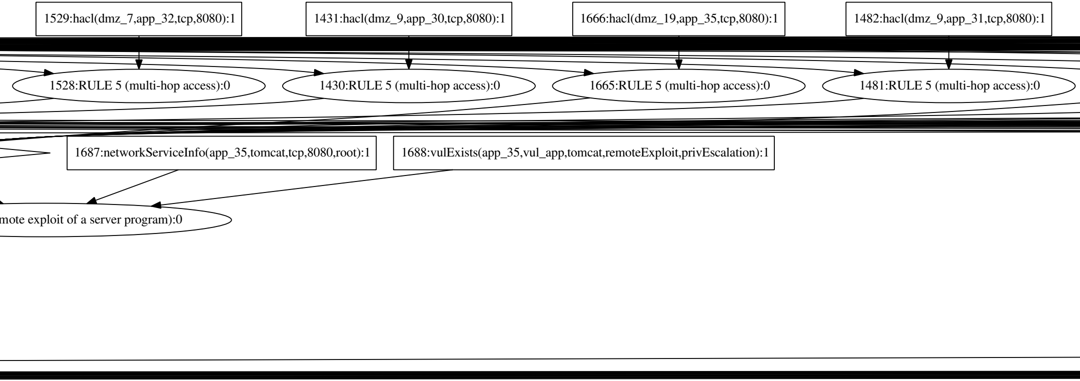

1478:hacl(dmz_7,app_31,tcp,8080):1
1572:hacl(dmz_3,app_33,tcp,8080):1
1644:hacl(dmz_0,app_35,tcp,8080):1
1429:hacl(dmz_8,app_30,tcp,8080):1
1477:RULE 5 (multi-hop access):0
1571:RULE 5 (multi-hop access):0
1643:RULE 5 (multi-hop access):0
1428:RULE 5 (multi-hop access):0
1639:hacl(app_35,d
1638:RULE 5 (mu

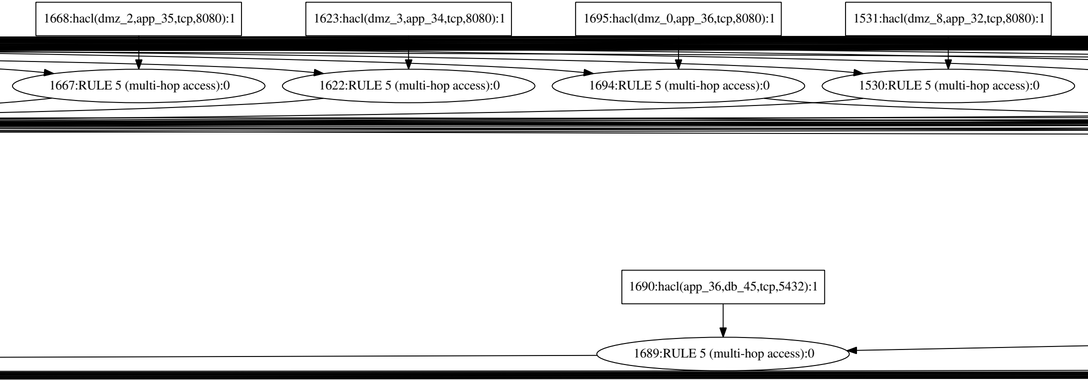

1646:hacl(dmz_1,app_35,tcp,8080):1
1480:hacl(dmz_8,app_31,tcp,8080):1
1648:hacl(dmz_10,app_35,tcp,8080):1
1619:hacl(dmz_20,app_34,tcp,8080):1
1645:RULE 5 (multi-hop access):0
1479:RULE 5 (multi-hop access):0
1647:RULE 5 (multi-hop access):0
1618:RULE 5 (multi-hop access):0
lb_45,tcp,5432):1
lti-hop access):0

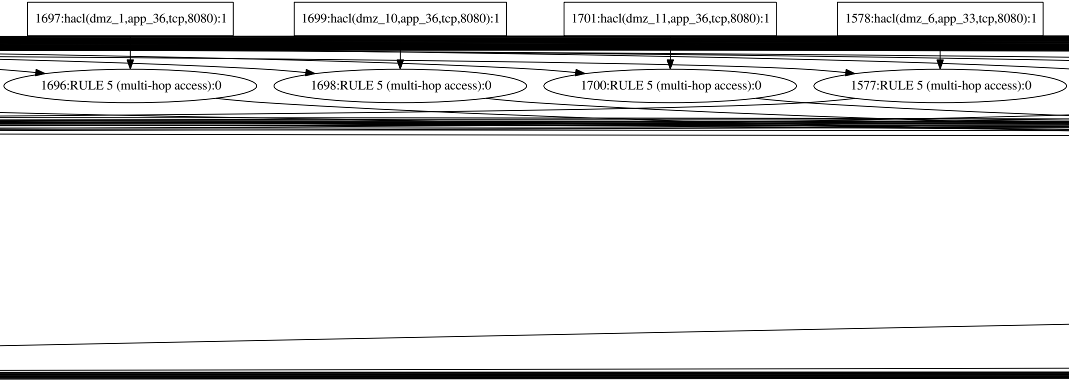

1650:hacl(dmz_11,app_35,tcp,8080):1
1574:hacl(dmz_4,app_33,tcp,8080):1
1652:hacl(dmz_12,app_35,tcp,8080):1
1527:hacl(dmz_6,app_32,tcp,8080):1
1649:RULE 5 (multi-hop access):0
1573:RULE 5 (multi-hop access):0
1651:RULE 5 (multi-hop access):0
1526:RULE 5 (multi-hop access):0

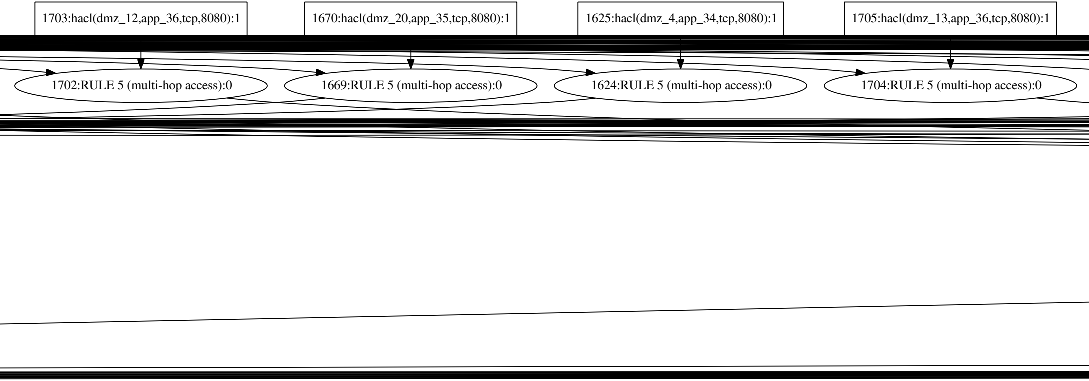

1654:hacl(dmz_13,app_35,tcp,8080):1
1656:hacl(dmz_14,app_35,tcp,8080):1
1621:hacl(dmz_21,app_34,tcp,8080):1
1658:hacl(dmz_15,app_35,tcp,8080):1
1653:RULE 5 (multi-hop access):0
1655:RULE 5 (multi-hop access):0
1620:RULE 5 (multi-hop access):0
1657:RULE 5 (multi-hop access):0
1640:execCode(app_35,root):0

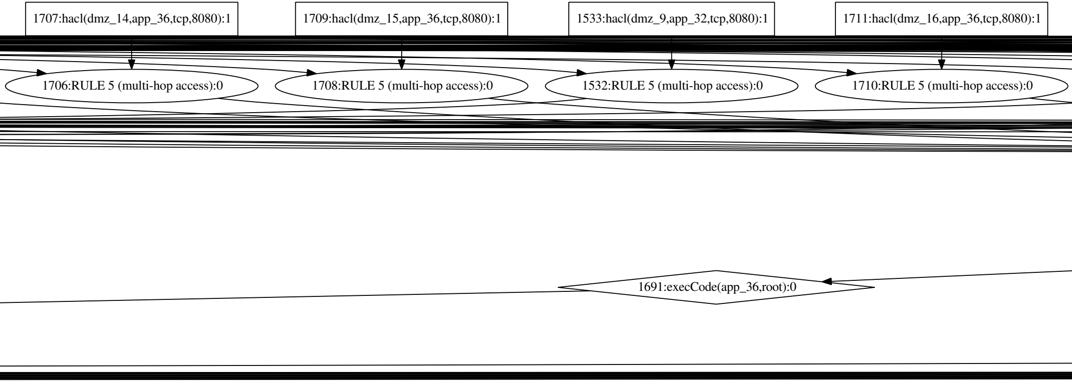

1660:hacl(dmz_16,app_35,tcp,8080):1
1576:hacl(dmz_5,app_33,tcp,8080):1
1662:hacl(dmz_17,app_35,tcp,8080):1
1664:hacl(dmz_18,app_35,tcp,8080):1
1659:RULE 5 (multi-hop access):0
1575:RULE 5 (multi-hop access):0
1661:RULE 5 (multi-hop access):0
1663:RULE 5 (multi-hop access):0
1642:netAccess(app_35,tcp,8080):0
1641:RULE 2 (re

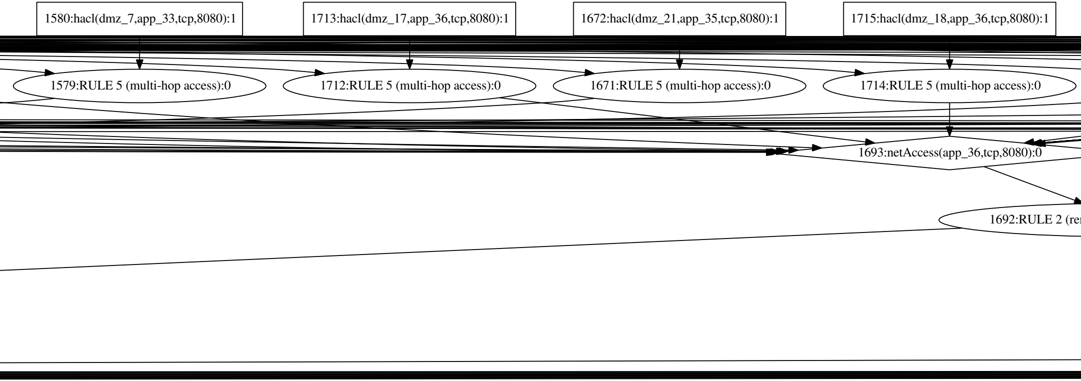

1529:hacl(dmz_7,app_32,tcp,8080):1
1431:hacl(dmz_9,app_30,tcp,8080):1
1666:hacl(dmz_19,app_35,tcp,8080):1
1482:hacl(dmz_9,app_31,tcp,8080):1
1528:RULE 5 (multi-hop access):0
1430:RULE 5 (multi-hop access):0
1665:RULE 5 (multi-hop access):0
1481:RULE 5 (multi-hop access):0
1687:networkServiceInfo(app_35,tomcat,tcp,8080,root):1
1688:vulExists(app_35,vul_app,tomcat,remoteExploit,privEscalation):1
mote exploit of a server program):0

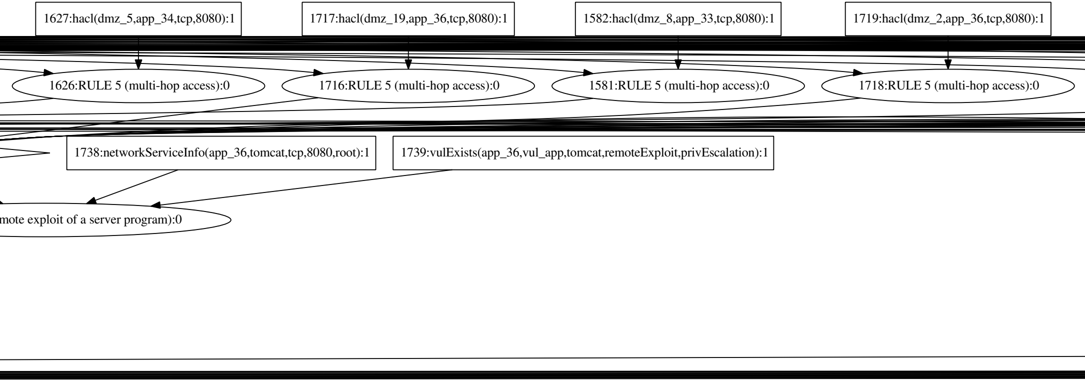

1668:hacl(dmz_2,app_35,tcp,8080):1
1623:hacl(dmz_3,app_34,tcp,8080):1
1695:hacl(dmz_0,app_36,tcp,8080):1
1531:hacl(dmz_8,app_32,tcp,8080):1
1667:RULE 5 (multi-hop access):0
1622:RULE 5 (multi-hop access):0
1694:RULE 5 (multi-hop access):0
1530:RULE 5 (multi-hop access):0
1690:hacl(app_36,db_45,tcp,5432):1
1689:RULE 5 (multi-hop access):0

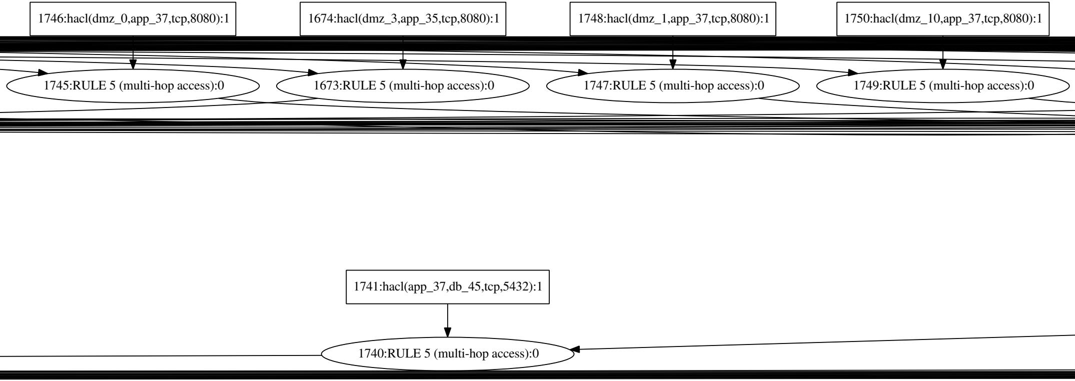

1697:hacl(dmz_1,app_36,tcp,8080):1
1699:hacl(dmz_10,app_36,tcp,8080):1
1701:hacl(dmz_11,app_36,tcp,8080):1
1578:hacl(dmz_6,app_33,tcp,8080):1
1696:RULE 5 (multi-hop access):0
1698:RULE 5 (multi-hop access):0
1700:RULE 5 (multi-hop access):0
1577:RULE 5 (multi-hop access):0

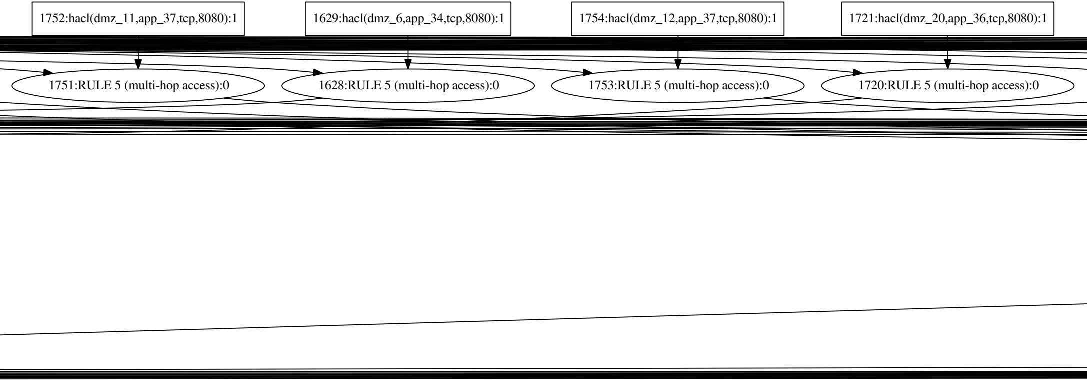

1703:hacl(dmz_12,app_36,tcp,8080):1
1670:hacl(dmz_20,app_35,tcp,8080):1
1625:hacl(dmz_4,app_34,tcp,8080):1
1705:hacl(dmz_13,app_36,tcp,8080):1
1702:RULE 5 (multi-hop access):0
1669:RULE 5 (multi-hop access):0
1624:RULE 5 (multi-hop access):0
1704:RULE 5 (multi-hop access):0

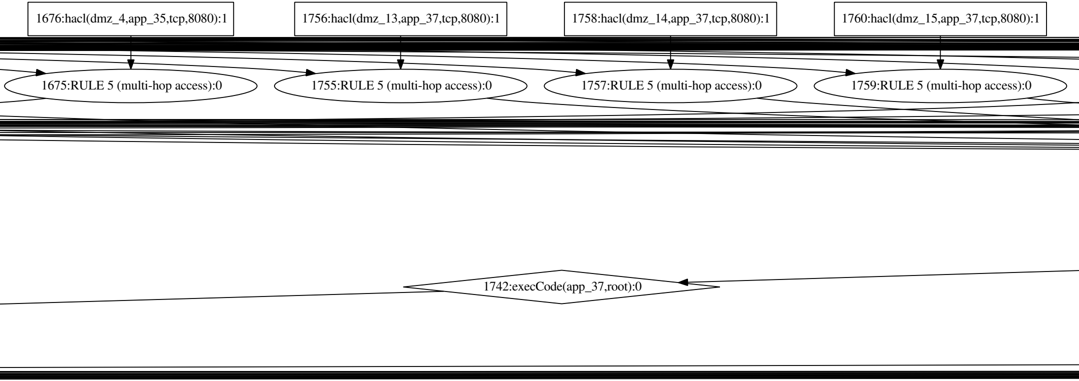

1707:hacl(dmz_14,app_36,tcp,8080):1
1709:hacl(dmz_15,app_36,tcp,8080):1
1533:hacl(dmz_9,app_32,tcp,8080):1
1711:hacl(dmz_16,app_36,tcp,8080):1
1706:RULE 5 (multi-hop access):0
1708:RULE 5 (multi-hop access):0
1532:RULE 5 (multi-hop access):0
1710:RULE 5 (multi-hop access):0
1691:execCode(app_36,root):0

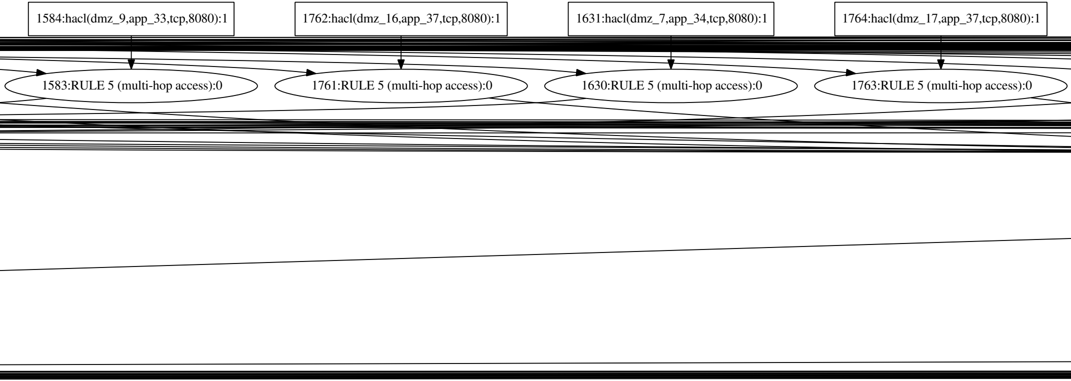

1580:hacl(dmz_7,app_33,tcp,8080):1
1713:hacl(dmz_17,app_36,tcp,8080):1
1672:hacl(dmz_21,app_35,tcp,8080):1
1715:hacl(dmz_18,app_36,tcp,8080):1
1579:RULE 5 (multi-hop access):0
1712:RULE 5 (multi-hop access):0
1671:RULE 5 (multi-hop access):0
1714:RULE 5 (multi-hop access):0
1693:netAccess(app_36,tcp,8080):0
1692:RULE 2 (re

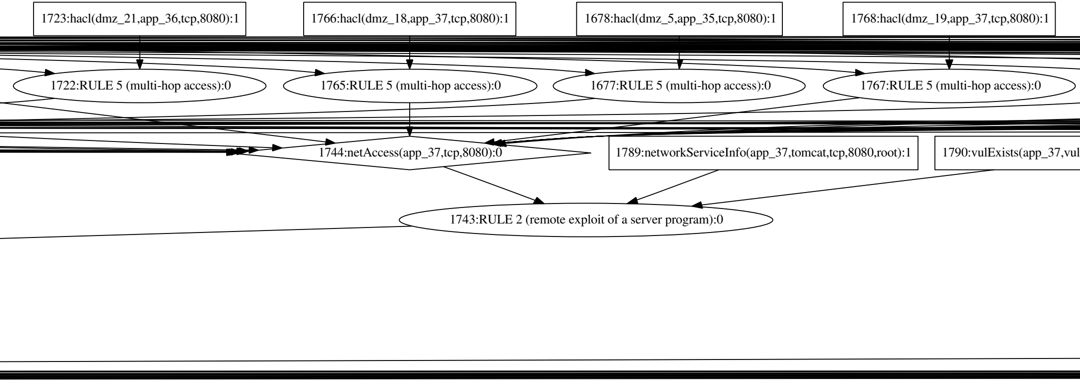

1627:hacl(dmz_5,app_34,tcp,8080):1
1717:hacl(dmz_19,app_36,tcp,8080):1
1582:hacl(dmz_8,app_33,tcp,8080):1
1719:hacl(dmz_2,app_36,tcp,8080):1
1626:RULE 5 (multi-hop access):0
1716:RULE 5 (multi-hop access):0
1581:RULE 5 (multi-hop access):0
1718:RULE 5 (multi-hop access):0
1738:networkServiceInfo(app_36,tomcat,tcp,8080,root):1
1739:vulExists(app_36,vul_app,tomcat,remoteExploit,privEscalation):1
mote exploit of a server program):0

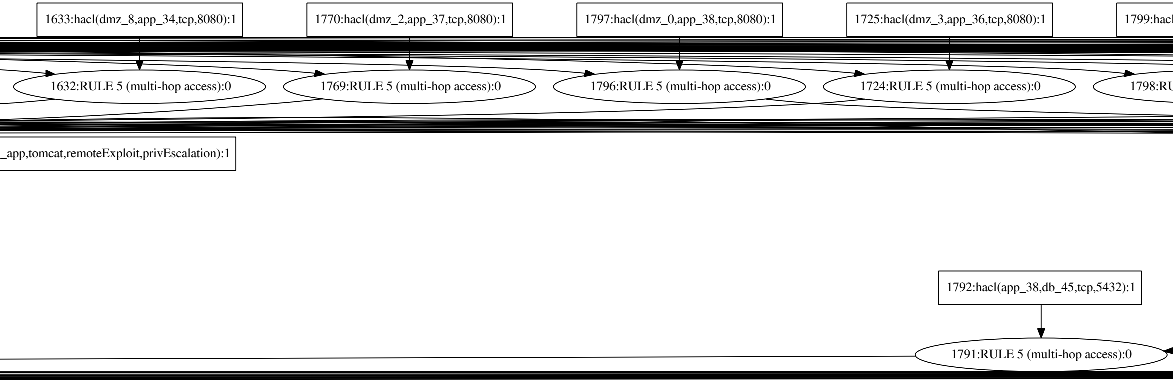

1746:hacl(dmz_0,app_37,tcp,8080):1
1674:hacl(dmz_3,app_35,tcp,8080):1
1748:hacl(dmz_1,app_37,tcp,8080):1
1750:hacl(dmz_10,app_37,tcp,8080):1
1745:RULE 5 (multi-hop access):0
1673:RULE 5 (multi-hop access):0
1747:RULE 5 (multi-hop access):0
1749:RULE 5 (multi-hop access):0
1741:hacl(app_37,db_45,tcp,5432):1
1740:RULE 5 (multi-hop access):0

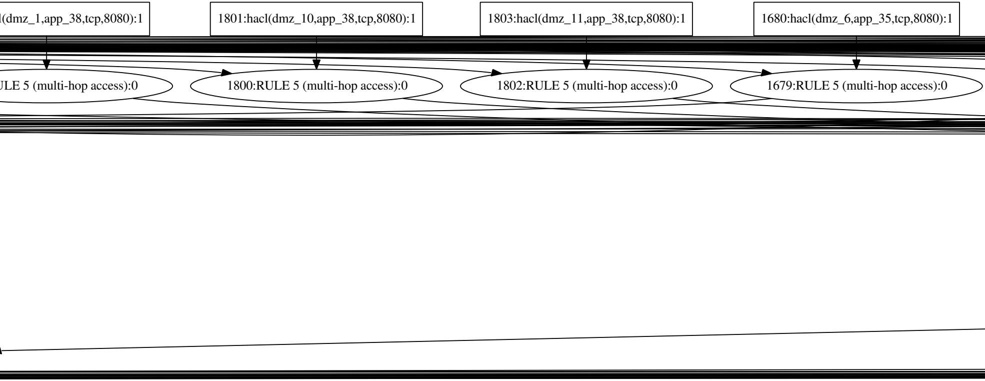

1752:hacl(dmz_11,app_37,tcp,8080):1
1629:hacl(dmz_6,app_34,tcp,8080):1
1754:hacl(dmz_12,app_37,tcp,8080):1
1721:hacl(dmz_20,app_36,tcp,8080):1
1751:RULE 5 (multi-hop access):0
1628:RULE 5 (multi-hop access):0
1753:RULE 5 (multi-hop access):0
1720:RULE 5 (multi-hop access):0

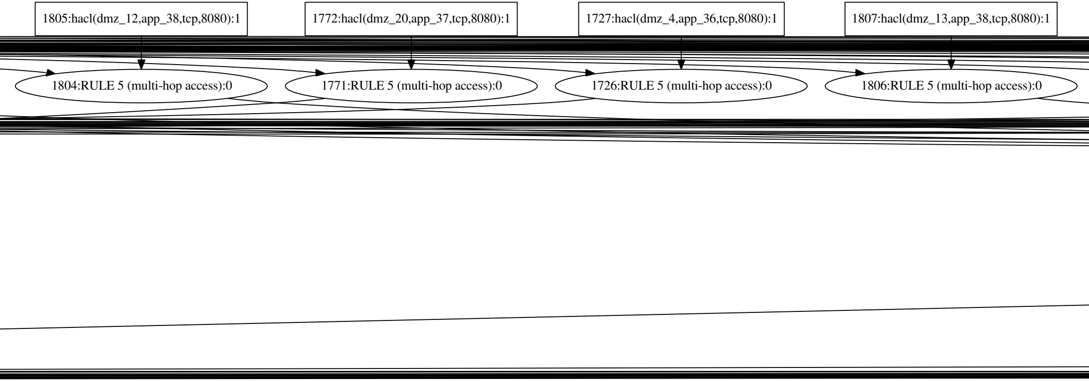

1676:hacl(dmz_4,app_35,tcp,8080):1
1756:hacl(dmz_13,app_37,tcp,8080):1
1758:hacl(dmz_14,app_37,tcp,8080):1
1760:hacl(dmz_15,app_37,tcp,8080):1
1675:RULE 5 (multi-hop access):0
1755:RULE 5 (multi-hop access):0
1757:RULE 5 (multi-hop access):0
1759:RULE 5 (multi-hop access):0
1742:execCode(app_37,root):0

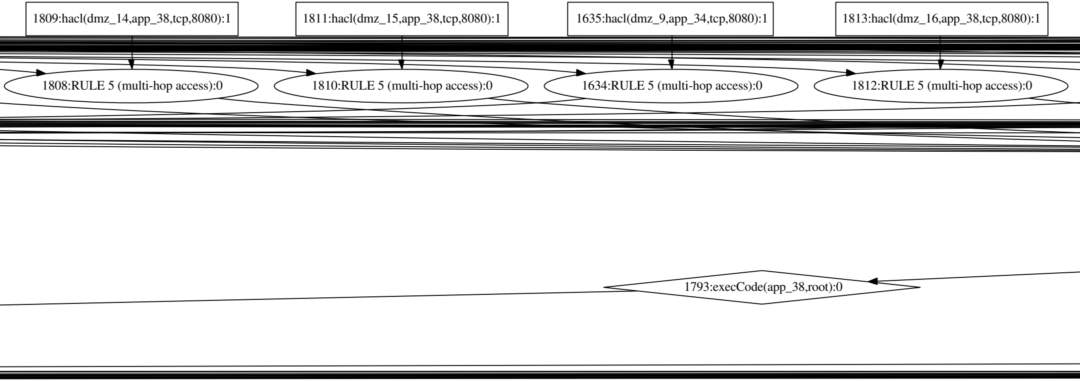

1584:hacl(dmz_9,app_33,tcp,8080):1
1762:hacl(dmz_16,app_37,tcp,8080):1
1631:hacl(dmz_7,app_34,tcp,8080):1
1764:hacl(dmz_17,app_37,tcp,8080):1
1583:RULE 5 (multi-hop access):0
1761:RULE 5 (multi-hop access):0
1630:RULE 5 (multi-hop access):0
1763:RULE 5 (multi-hop access):0

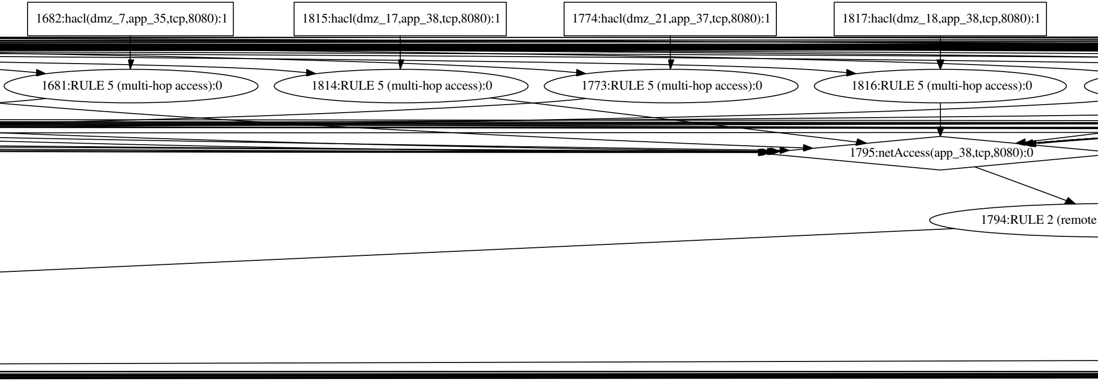

1723:hacl(dmz_21,app_36,tcp,8080):1
1766:hacl(dmz_18,app_37,tcp,8080):1
1678:hacl(dmz_5,app_35,tcp,8080):1
1768:hacl(dmz_19,app_37,tcp,8080):1
1722:RULE 5 (multi-hop access):0
1765:RULE 5 (multi-hop access):0
1677:RULE 5 (multi-hop access):0
1767:RULE 5 (multi-hop access):0
1744:netAccess(app_37,tcp,8080):0
1789:networkServiceInfo(app_37,tomcat,tcp,8080,root):1
1790:vulExists(app_37,vul
1743:RULE 2 (remote exploit of a server program):0

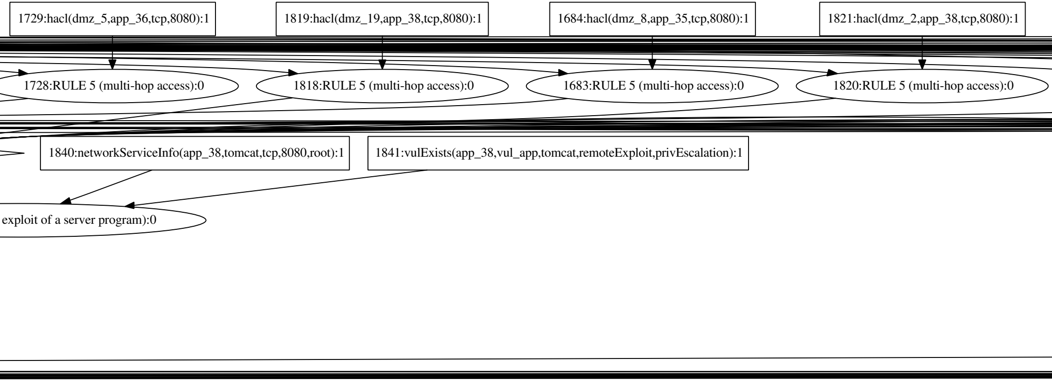

1633:hacl(dmz_8,app_34,tcp,8080):1
1770:hacl(dmz_2,app_37,tcp,8080):1
1797:hacl(dmz_0,app_38,tcp,8080):1
1725:hacl(dmz_3,app_36,tcp,8080):1
1799:hacl
1632:RULE 5 (multi-hop access):0
1769:RULE 5 (multi-hop access):0
1796:RULE 5 (multi-hop access):0
1724:RULE 5 (multi-hop access):0
1798:RU
_app,tomcat,remoteExploit,privEscalation):1
1792:hacl(app_38,db_45,tcp,5432):1
1791:RULE 5 (multi-hop access):0

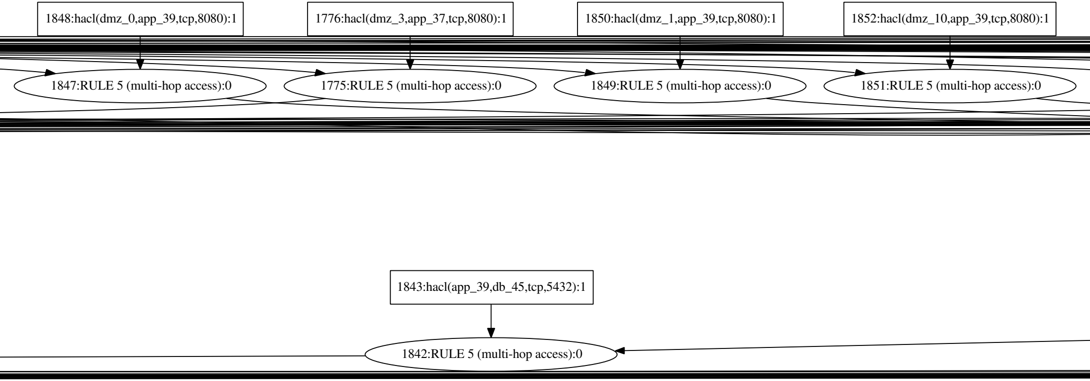

l(dmz_1,app_38,tcp,8080):1
1801:hacl(dmz_10,app_38,tcp,8080):1
1803:hacl(dmz_11,app_38,tcp,8080):1
1680:hacl(dmz_6,app_35,tcp,8080):1
ULE 5 (multi-hop access):0
1800:RULE 5 (multi-hop access):0
1802:RULE 5 (multi-hop access):0
1679:RULE 5 (multi-hop access):0

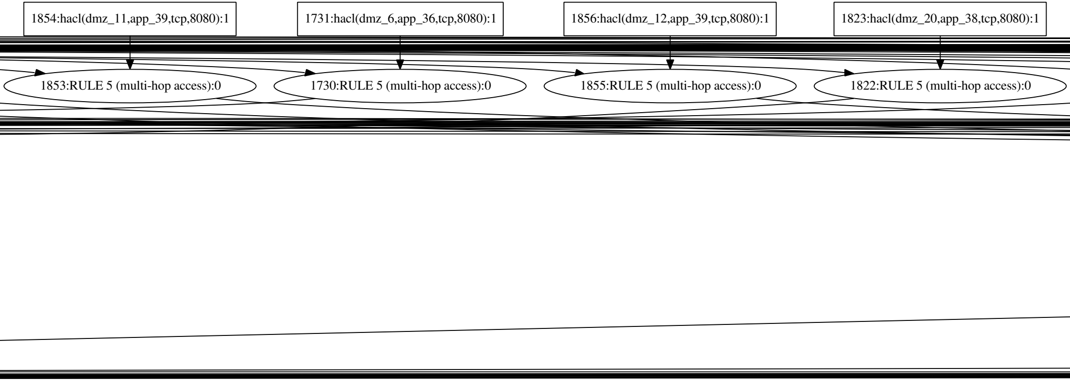

1805:hacl(dmz_12,app_38,tcp,8080):1
1772:hacl(dmz_20,app_37,tcp,8080):1
1727:hacl(dmz_4,app_36,tcp,8080):1
1807:hacl(dmz_13,app_38,tcp,8080):1
1804:RULE 5 (multi-hop access):0
1771:RULE 5 (multi-hop access):0
1726:RULE 5 (multi-hop access):0
1806:RULE 5 (multi-hop access):0

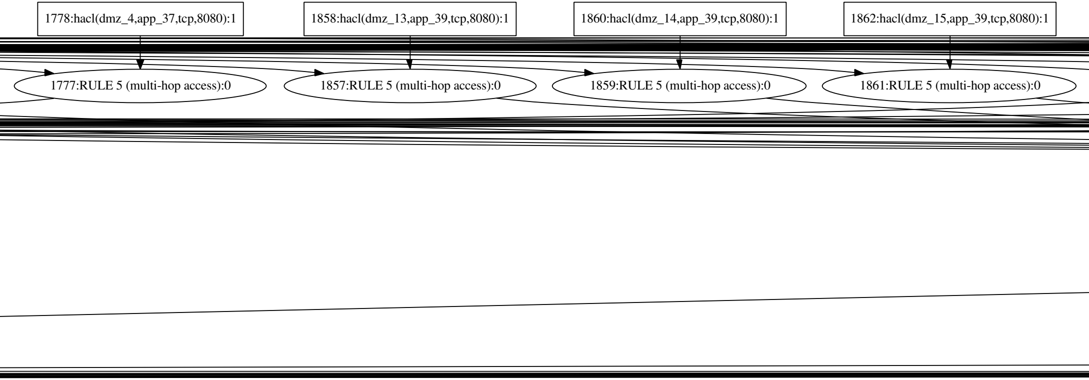

1809:hacl(dmz_14,app_38,tcp,8080):1
1811:hacl(dmz_15,app_38,tcp,8080):1
1635:hacl(dmz_9,app_34,tcp,8080):1
1813:hacl(dmz_16,app_38,tcp,8080):1
1808:RULE 5 (multi-hop access):0
1810:RULE 5 (multi-hop access):0
1634:RULE 5 (multi-hop access):0
1812:RULE 5 (multi-hop access):0
1793:execCode(app_38,root):0

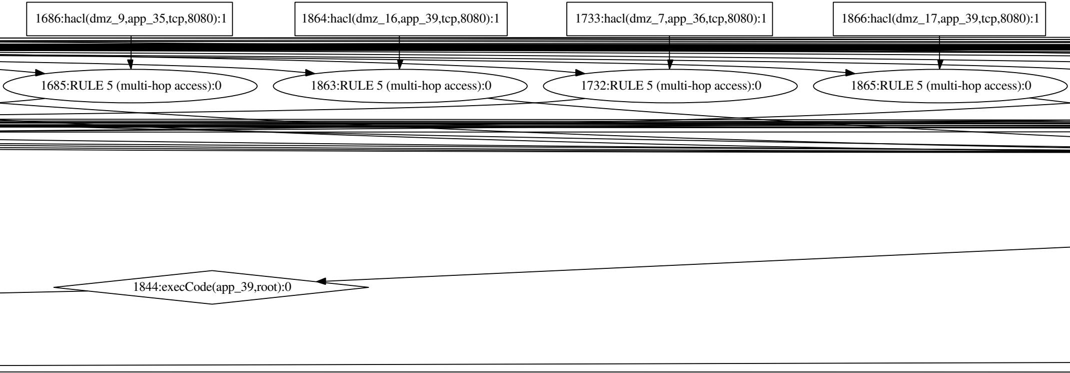

1682:hacl(dmz_7,app_35,tcp,8080):1
1815:hacl(dmz_17,app_38,tcp,8080):1
1774:hacl(dmz_21,app_37,tcp,8080):1
1817:hacl(dmz_18,app_38,tcp,8080):1
1681:RULE 5 (multi-hop access):0
1814:RULE 5 (multi-hop access):0
1773:RULE 5 (multi-hop access):0
1816:RULE 5 (multi-hop access):0
1795:netAccess(app_38,tcp,8080):0
1794:RULE 2 (remote

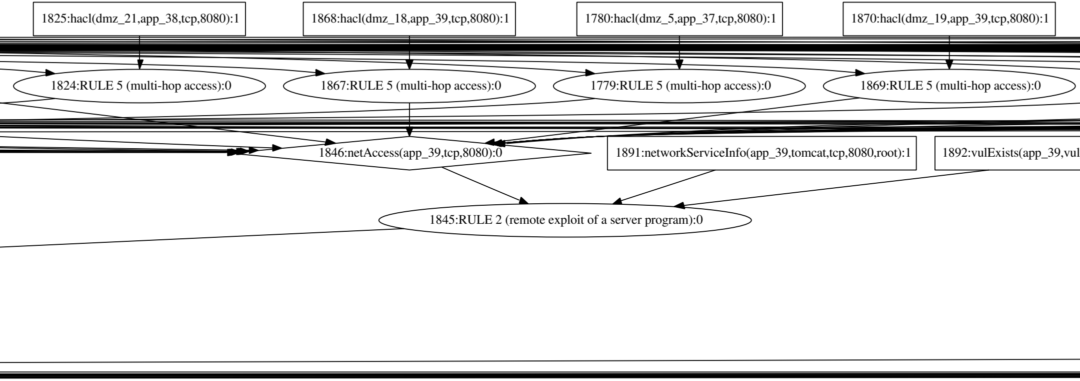

1729:hacl(dmz_5,app_36,tcp,8080):1
1819:hacl(dmz_19,app_38,tcp,8080):1
1684:hacl(dmz_8,app_35,tcp,8080):1
1821:hacl(dmz_2,app_38,tcp,8080):1
1728:RULE 5 (multi-hop access):0
1818:RULE 5 (multi-hop access):0
1683:RULE 5 (multi-hop access):0
1820:RULE 5 (multi-hop access):0
1840:networkServiceInfo(app_38,tomcat,tcp,8080,root):1
1841:vulExists(app_38,vul_app,tomcat,remoteExploit,privEscalation):1
exploit of a server program):0

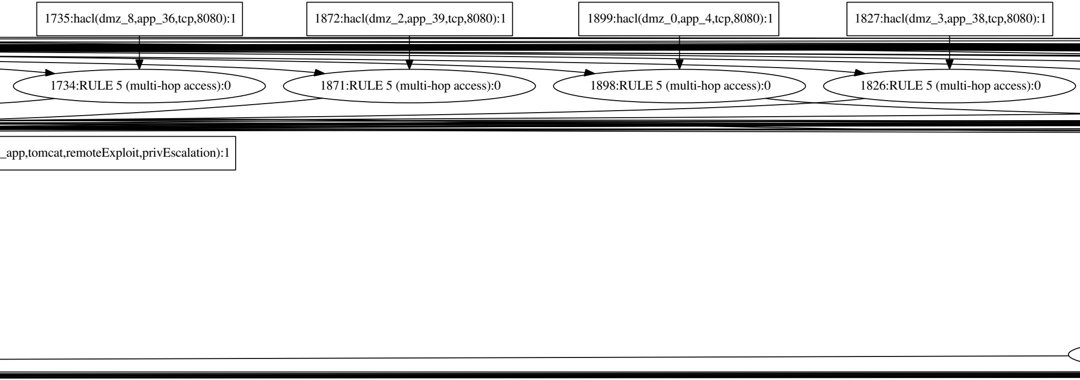

1848:hacl(dmz_0,app_39,tcp,8080):1
1776:hacl(dmz_3,app_37,tcp,8080):1
1850:hacl(dmz_1,app_39,tcp,8080):1
1852:hacl(dmz_10,app_39,tcp,8080):1
1847:RULE 5 (multi-hop access):0
1775:RULE 5 (multi-hop access):0
1849:RULE 5 (multi-hop access):0
1851:RULE 5 (multi-hop access):0
1843:hacl(app_39,db_45,tcp,5432):1
1842:RULE 5 (multi-hop access):0

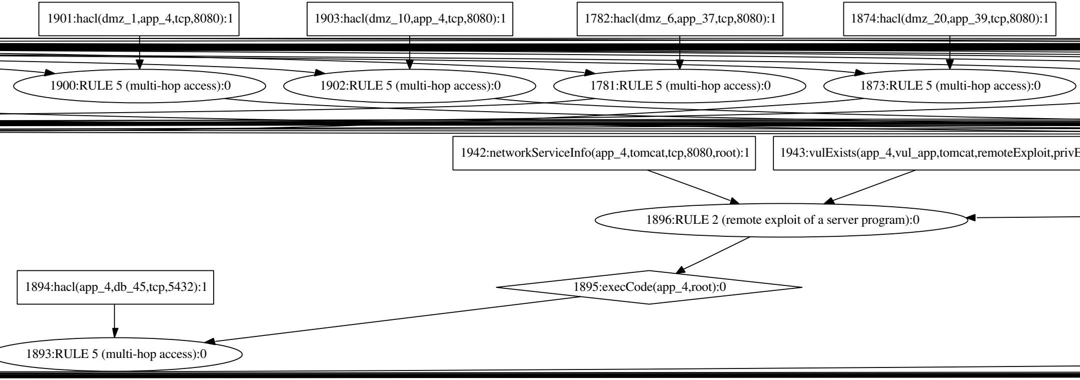

1854:hacl(dmz_11,app_39,tcp,8080):1
1731:hacl(dmz_6,app_36,tcp,8080):1
1856:hacl(dmz_12,app_39,tcp,8080):1
1823:hacl(dmz_20,app_38,tcp,8080):1
1853:RULE 5 (multi-hop access):0
1730:RULE 5 (multi-hop access):0
1855:RULE 5 (multi-hop access):0
1822:RULE 5 (multi-hop access):0

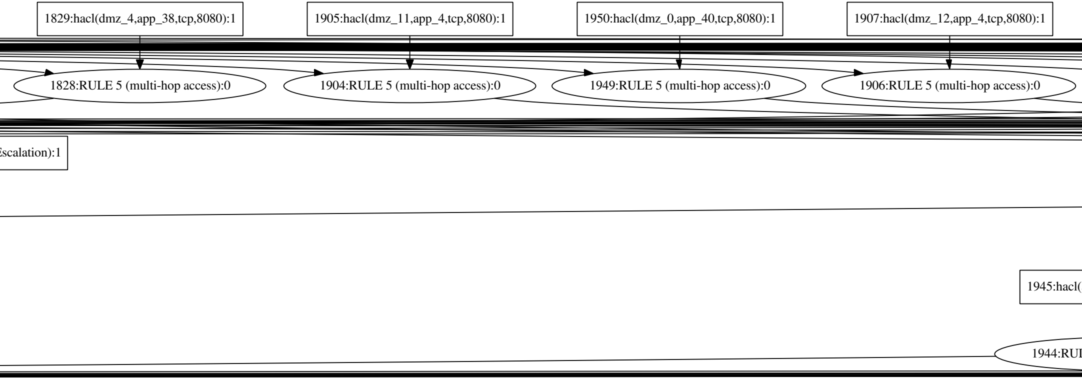

1778:hacl(dmz_4,app_37,tcp,8080):1
1858:hacl(dmz_13,app_39,tcp,8080):1
1860:hacl(dmz_14,app_39,tcp,8080):1
1862:hacl(dmz_15,app_39,tcp,8080):1
1777:RULE 5 (multi-hop access):0
1857:RULE 5 (multi-hop access):0
1859:RULE 5 (multi-hop access):0
1861:RULE 5 (multi-hop access):0

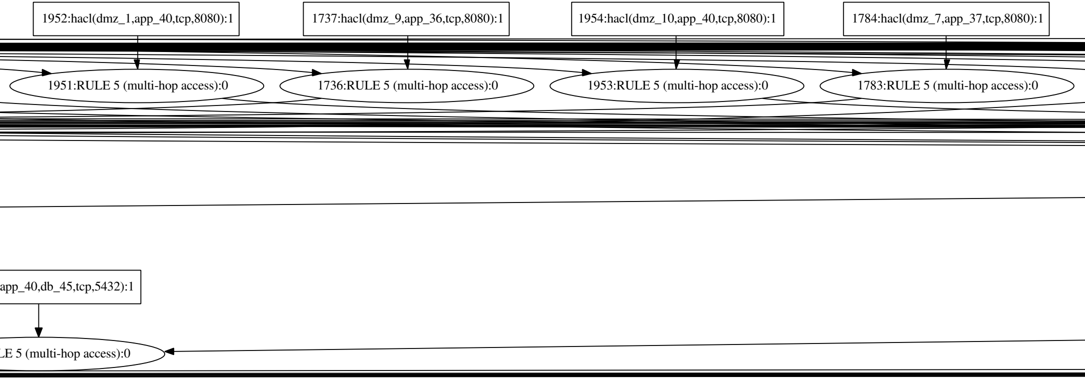

1686:hacl(dmz_9,app_35,tcp,8080):1
1864:hacl(dmz_16,app_39,tcp,8080):1
1733:hacl(dmz_7,app_36,tcp,8080):1
1866:hacl(dmz_17,app_39,tcp,8080):1
1685:RULE 5 (multi-hop access):0
1863:RULE 5 (multi-hop access):0
1732:RULE 5 (multi-hop access):0
1865:RULE 5 (multi-hop access):0
1844:execCode(app_39,root):0

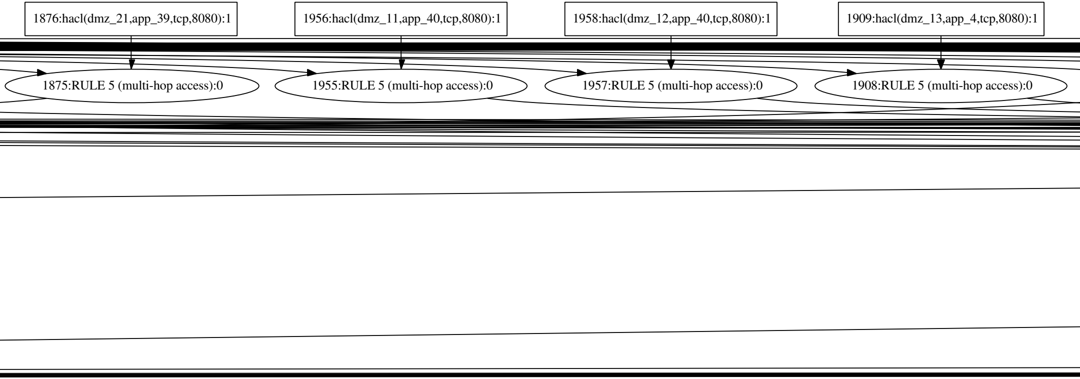

1825:hacl(dmz_21,app_38,tcp,8080):1
1868:hacl(dmz_18,app_39,tcp,8080):1
1780:hacl(dmz_5,app_37,tcp,8080):1
1870:hacl(dmz_19,app_39,tcp,8080):1
1824:RULE 5 (multi-hop access):0
1867:RULE 5 (multi-hop access):0
1779:RULE 5 (multi-hop access):0
1869:RULE 5 (multi-hop access):0
1846:netAccess(app_39,tcp,8080):0
1891:networkServiceInfo(app_39,tomcat,tcp,8080,root):1
1892:vulExists(app_39,vul
1845:RULE 2 (remote exploit of a server program):0

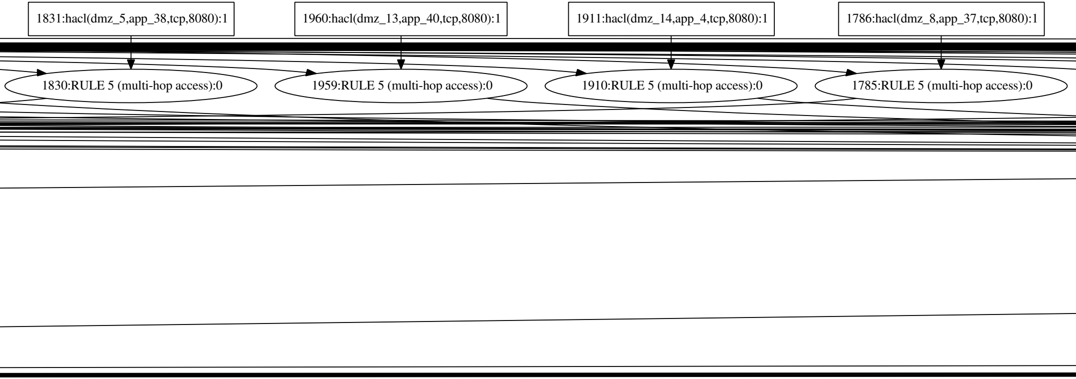

1735:hacl(dmz_8,app_36,tcp,8080):1
1872:hacl(dmz_2,app_39,tcp,8080):1
1899:hacl(dmz_0,app_4,tcp,8080):1
1827:hacl(dmz_3,app_38,tcp,8080):1
1734:RULE 5 (multi-hop access):0
1871:RULE 5 (multi-hop access):0
1898:RULE 5 (multi-hop access):0
1826:RULE 5 (multi-hop access):0
_app,tomcat,remoteExploit,privEscalation):1

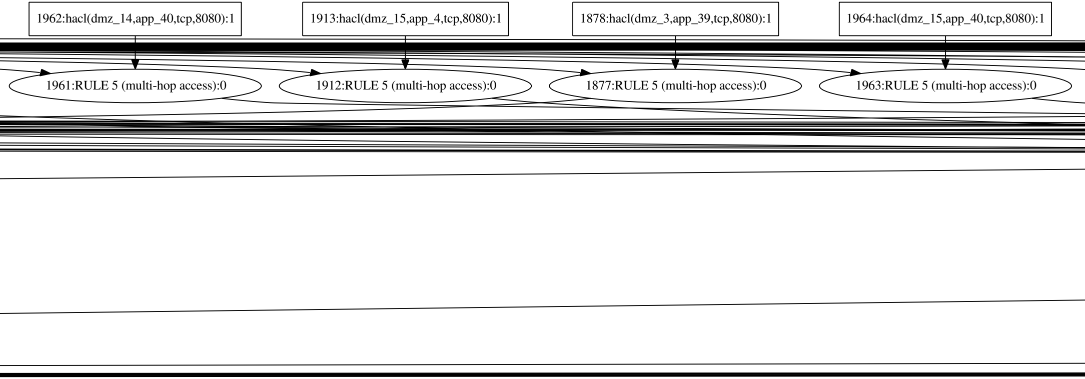

1901:hacl(dmz_1,app_4,tcp,8080):1
1903:hacl(dmz_10,app_4,tcp,8080):1
1782:hacl(dmz_6,app_37,tcp,8080):1
1874:hacl(dmz_20,app_39,tcp,8080):1
1900:RULE 5 (multi-hop access):0
1902:RULE 5 (multi-hop access):0
1781:RULE 5 (multi-hop access):0
1873:RULE 5 (multi-hop access):0
1942:networkServiceInfo(app_4,tomcat,tcp,8080,root):1
1943:vulExists(app_4,vul_app,tomcat,remoteExploit,privE
1896:RULE 2 (remote exploit of a server program):0
1894:hacl(app_4,db_45,tcp,5432):1
1895:execCode(app_4,root):0
1893:RULE 5 (multi-hop access):0

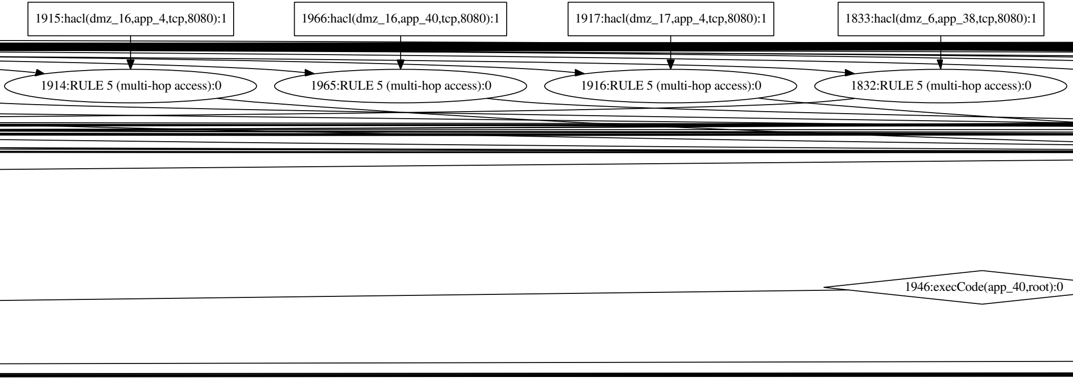

1829:hacl(dmz_4,app_38,tcp,8080):1
1905:hacl(dmz_11,app_4,tcp,8080):1
1950:hacl(dmz_0,app_40,tcp,8080):1
1907:hacl(dmz_12,app_4,tcp,8080):1
1828:RULE 5 (multi-hop access):0
1904:RULE 5 (multi-hop access):0
1949:RULE 5 (multi-hop access):0
1906:RULE 5 (multi-hop access):0
Escalation):1
1945:hacl(
1944:RU

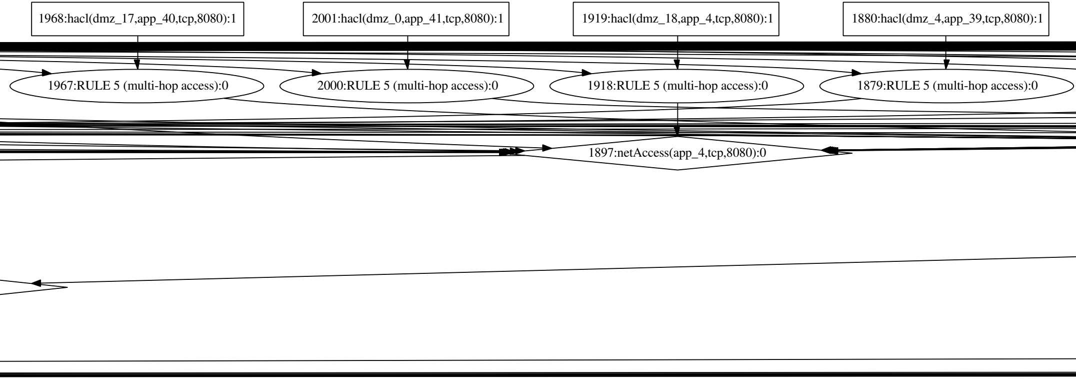

1952:hacl(dmz_1,app_40,tcp,8080):1
1737:hacl(dmz_9,app_36,tcp,8080):1
1954:hacl(dmz_10,app_40,tcp,8080):1
1784:hacl(dmz_7,app_37,tcp,8080):1
1951:RULE 5 (multi-hop access):0
1736:RULE 5 (multi-hop access):0
1953:RULE 5 (multi-hop access):0
1783:RULE 5 (multi-hop access):0
app_40,db_45,tcp,5432):1
LE 5 (multi-hop access):0

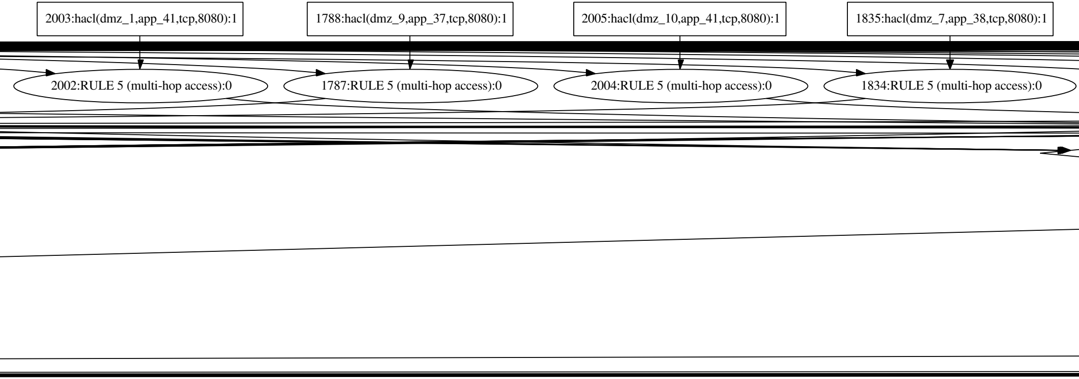

1876:hacl(dmz_21,app_39,tcp,8080):1
1956:hacl(dmz_11,app_40,tcp,8080):1
1958:hacl(dmz_12,app_40,tcp,8080):1
1909:hacl(dmz_13,app_4,tcp,8080):1
1875:RULE 5 (multi-hop access):0
1955:RULE 5 (multi-hop access):0
1957:RULE 5 (multi-hop access):0
1908:RULE 5 (multi-hop access):0

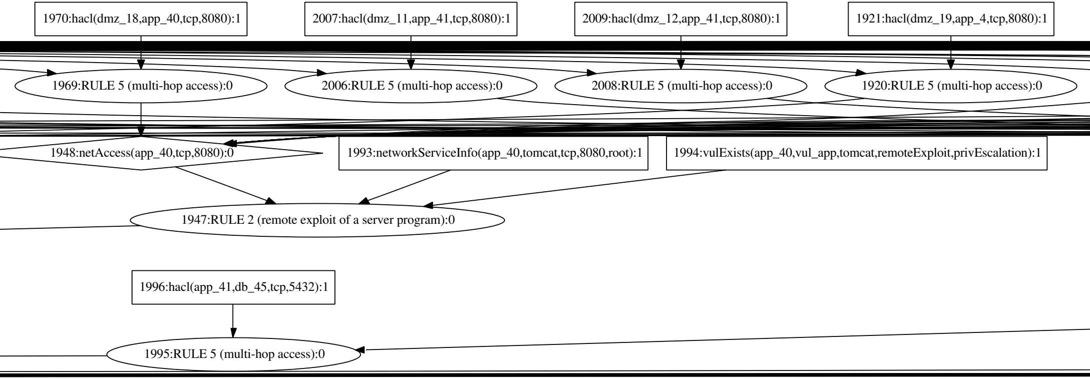

1831:hacl(dmz_5,app_38,tcp,8080):1
1960:hacl(dmz_13,app_40,tcp,8080):1
1911:hacl(dmz_14,app_4,tcp,8080):1
1786:hacl(dmz_8,app_37,tcp,8080):1
1830:RULE 5 (multi-hop access):0
1959:RULE 5 (multi-hop access):0
1910:RULE 5 (multi-hop access):0
1785:RULE 5 (multi-hop access):0

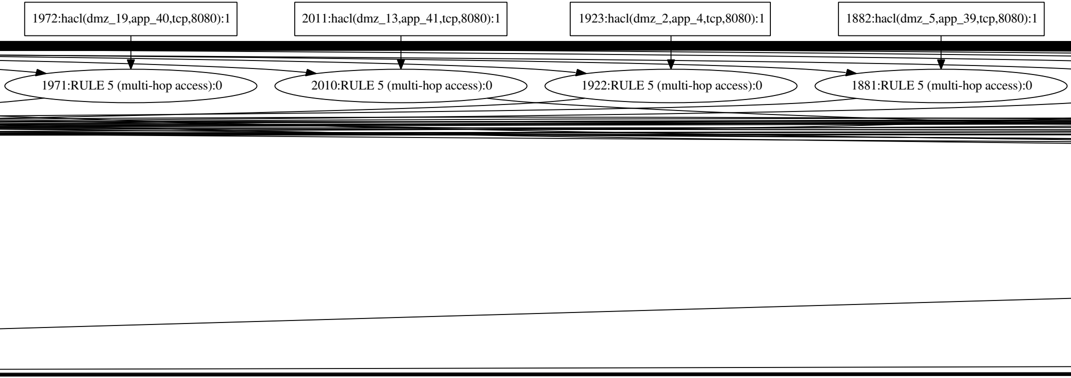

1962:hacl(dmz_14,app_40,tcp,8080):1
1913:hacl(dmz_15,app_4,tcp,8080):1
1878:hacl(dmz_3,app_39,tcp,8080):1
1964:hacl(dmz_15,app_40,tcp,8080):1
1961:RULE 5 (multi-hop access):0
1912:RULE 5 (multi-hop access):0
1877:RULE 5 (multi-hop access):0
1963:RULE 5 (multi-hop access):0

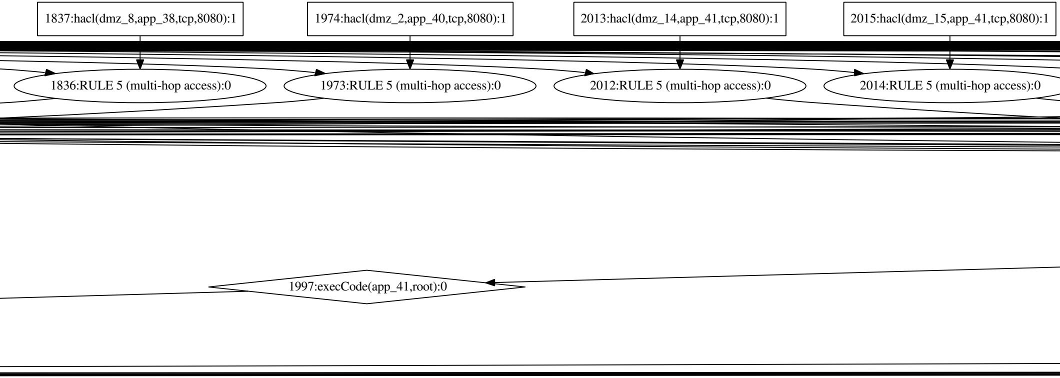

1915:hacl(dmz_16,app_4,tcp,8080):1
1966:hacl(dmz_16,app_40,tcp,8080):1
1917:hacl(dmz_17,app_4,tcp,8080):1
1833:hacl(dmz_6,app_38,tcp,8080):1
1914:RULE 5 (multi-hop access):0
1965:RULE 5 (multi-hop access):0
1916:RULE 5 (multi-hop access):0
1832:RULE 5 (multi-hop access):0
1946:execCode(app_40,root):0

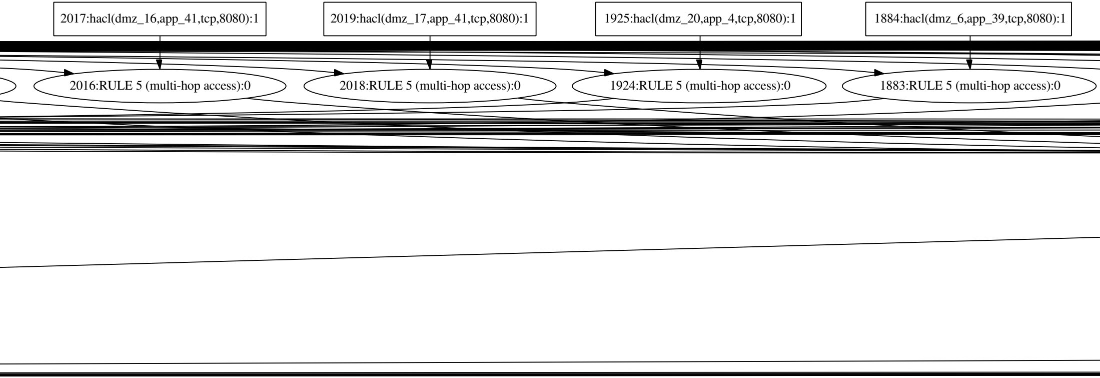

1968:hacl(dmz_17,app_40,tcp,8080):1
2001:hacl(dmz_0,app_41,tcp,8080):1
1919:hacl(dmz_18,app_4,tcp,8080):1
1880:hacl(dmz_4,app_39,tcp,8080):1
1967:RULE 5 (multi-hop access):0
2000:RULE 5 (multi-hop access):0
1918:RULE 5 (multi-hop access):0
1879:RULE 5 (multi-hop access):0
1897:netAccess(app_4,tcp,8080):0

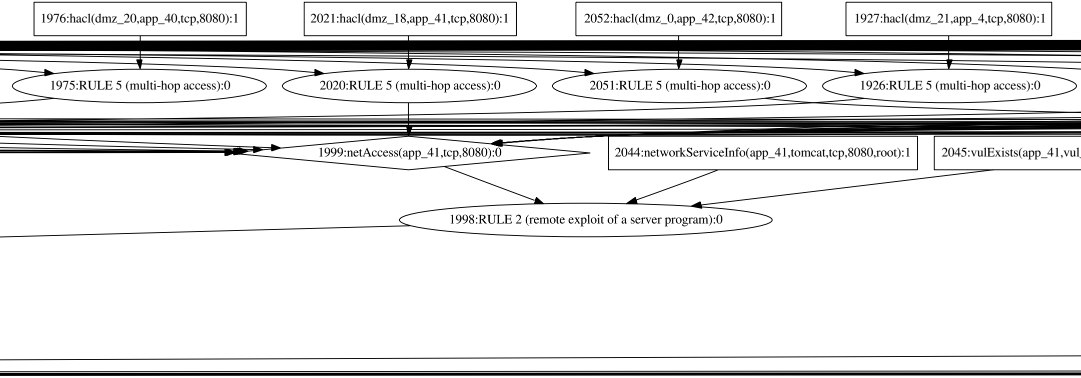

2003:hacl(dmz_1,app_41,tcp,8080):1
1788:hacl(dmz_9,app_37,tcp,8080):1
2005:hacl(dmz_10,app_41,tcp,8080):1
1835:hacl(dmz_7,app_38,tcp,8080):1
2002:RULE 5 (multi-hop access):0
1787:RULE 5 (multi-hop access):0
2004:RULE 5 (multi-hop access):0
1834:RULE 5 (multi-hop access):0

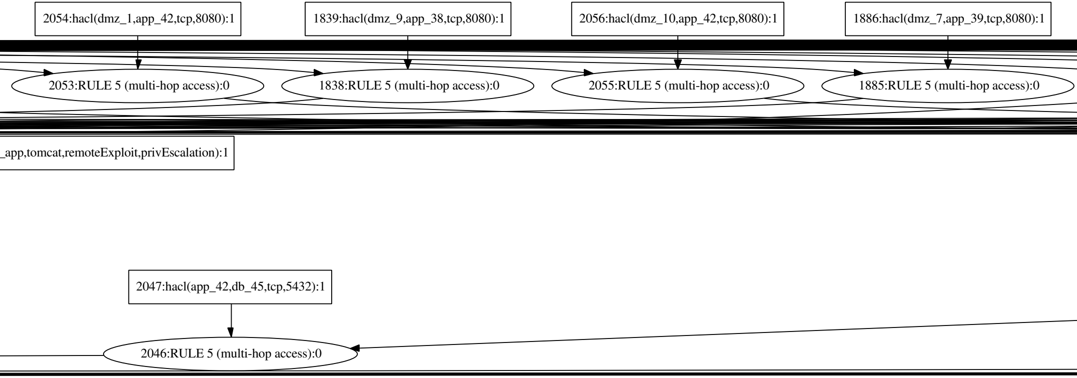

1970:hacl(dmz_18,app_40,tcp,8080):1
2007:hacl(dmz_11,app_41,tcp,8080):1
2009:hacl(dmz_12,app_41,tcp,8080):1
1921:hacl(dmz_19,app_4,tcp,8080):1
1969:RULE 5 (multi-hop access):0
2006:RULE 5 (multi-hop access):0
2008:RULE 5 (multi-hop access):0
1920:RULE 5 (multi-hop access):0
1948:netAccess(app_40,tcp,8080):0
1993:networkServiceInfo(app_40,tomcat,tcp,8080,root):1
1994:vulExists(app_40,vul_app,tomcat,remoteExploit,privEscalation):1
1947:RULE 2 (remote exploit of a server program):0
1996:hacl(app_41,db_45,tcp,5432):1
1995:RULE 5 (multi-hop access):0

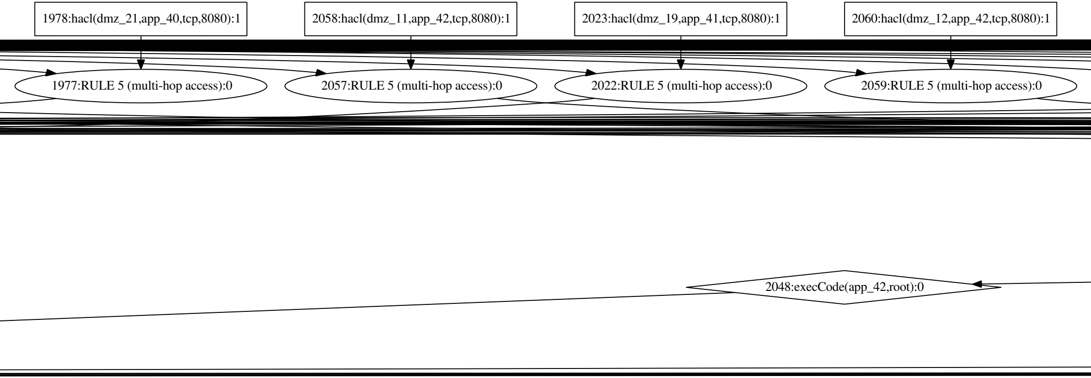

1972:hacl(dmz_19,app_40,tcp,8080):1
2011:hacl(dmz_13,app_41,tcp,8080):1
1923:hacl(dmz_2,app_4,tcp,8080):1
1882:hacl(dmz_5,app_39,tcp,8080):1
1971:RULE 5 (multi-hop access):0
2010:RULE 5 (multi-hop access):0
1922:RULE 5 (multi-hop access):0
1881:RULE 5 (multi-hop access):0

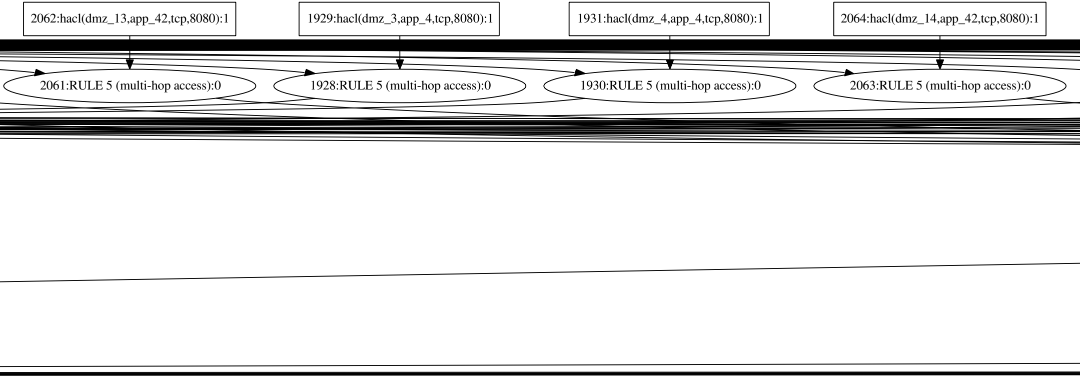

1837:hacl(dmz_8,app_38,tcp,8080):1
1974:hacl(dmz_2,app_40,tcp,8080):1
2013:hacl(dmz_14,app_41,tcp,8080):1
2015:hacl(dmz_15,app_41,tcp,8080):1
1836:RULE 5 (multi-hop access):0
1973:RULE 5 (multi-hop access):0
2012:RULE 5 (multi-hop access):0
2014:RULE 5 (multi-hop access):0
1997:execCode(app_41,root):0

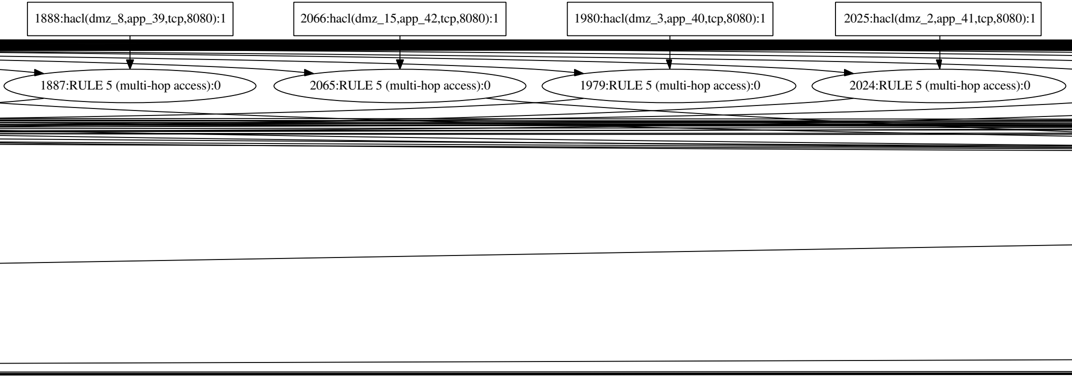

2017:hacl(dmz_16,app_41,tcp,8080):1
2019:hacl(dmz_17,app_41,tcp,8080):1
1925:hacl(dmz_20,app_4,tcp,8080):1
1884:hacl(dmz_6,app_39,tcp,8080):1
2016:RULE 5 (multi-hop access):0
2018:RULE 5 (multi-hop access):0
1924:RULE 5 (multi-hop access):0
1883:RULE 5 (multi-hop access):0

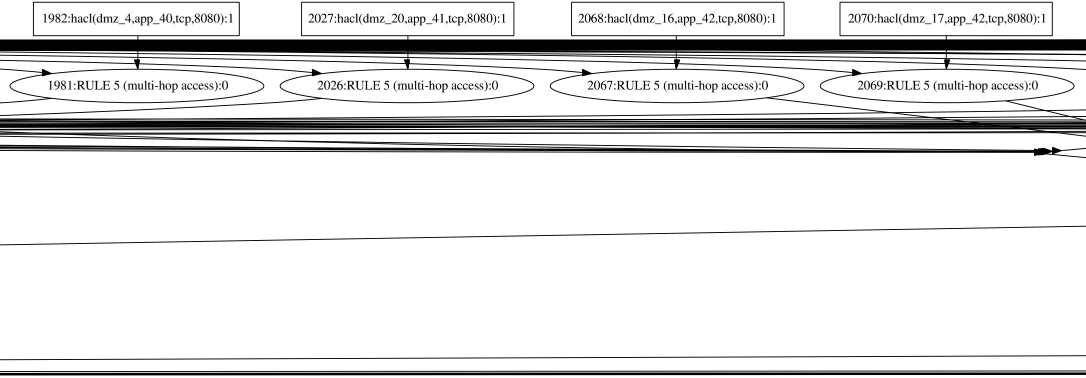

1976:hacl(dmz_20,app_40,tcp,8080):1
2021:hacl(dmz_18,app_41,tcp,8080):1
2052:hacl(dmz_0,app_42,tcp,8080):1
1927:hacl(dmz_21,app_4,tcp,8080):1
1975:RULE 5 (multi-hop access):0
2020:RULE 5 (multi-hop access):0
2051:RULE 5 (multi-hop access):0
1926:RULE 5 (multi-hop access):0
1999:netAccess(app_41,tcp,8080):0
2044:networkServiceInfo(app_41,tomcat,tcp,8080,root):1
2045:vulExists(app_41,vul
1998:RULE 2 (remote exploit of a server program):0

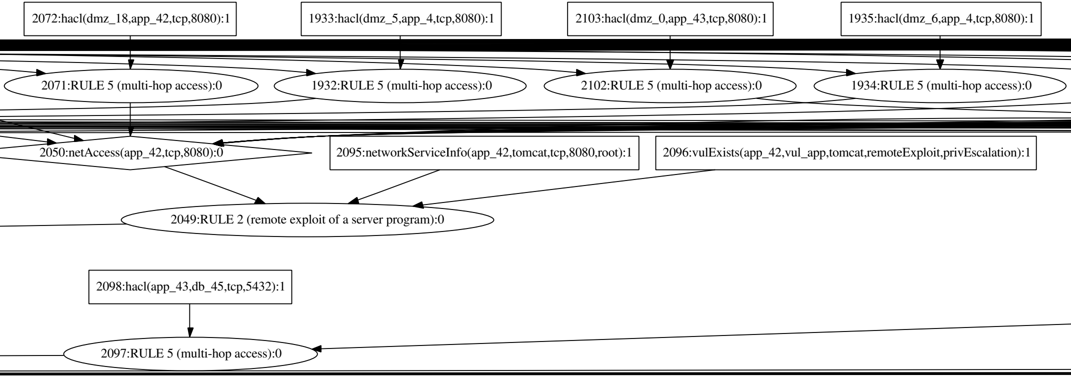

2054:hacl(dmz_1,app_42,tcp,8080):1
1839:hacl(dmz_9,app_38,tcp,8080):1
2056:hacl(dmz_10,app_42,tcp,8080):1
1886:hacl(dmz_7,app_39,tcp,8080):1
2053:RULE 5 (multi-hop access):0
1838:RULE 5 (multi-hop access):0
2055:RULE 5 (multi-hop access):0
1885:RULE 5 (multi-hop access):0
_app,tomcat,remoteExploit,privEscalation):1
2047:hacl(app_42,db_45,tcp,5432):1
2046:RULE 5 (multi-hop access):0

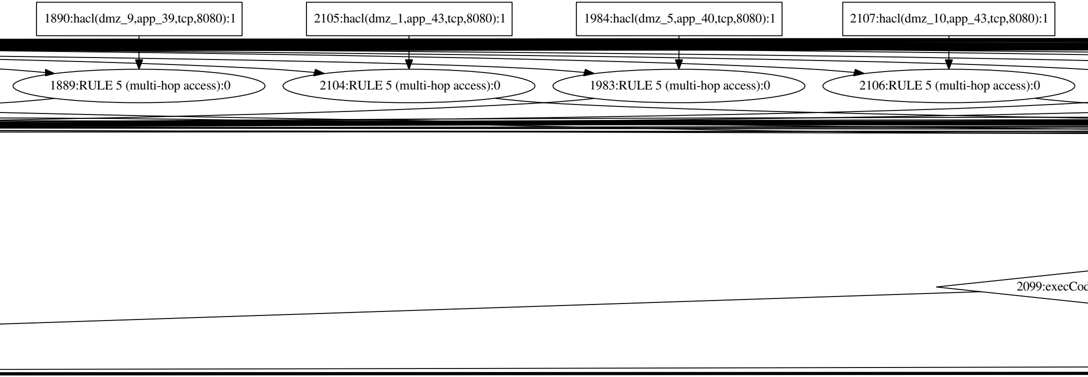

1978:hacl(dmz_21,app_40,tcp,8080):1
2058:hacl(dmz_11,app_42,tcp,8080):1
2023:hacl(dmz_19,app_41,tcp,8080):1
2060:hacl(dmz_12,app_42,tcp,8080):1
1977:RULE 5 (multi-hop access):0
2057:RULE 5 (multi-hop access):0
2022:RULE 5 (multi-hop access):0
2059:RULE 5 (multi-hop access):0
2048:execCode(app_42,root):0

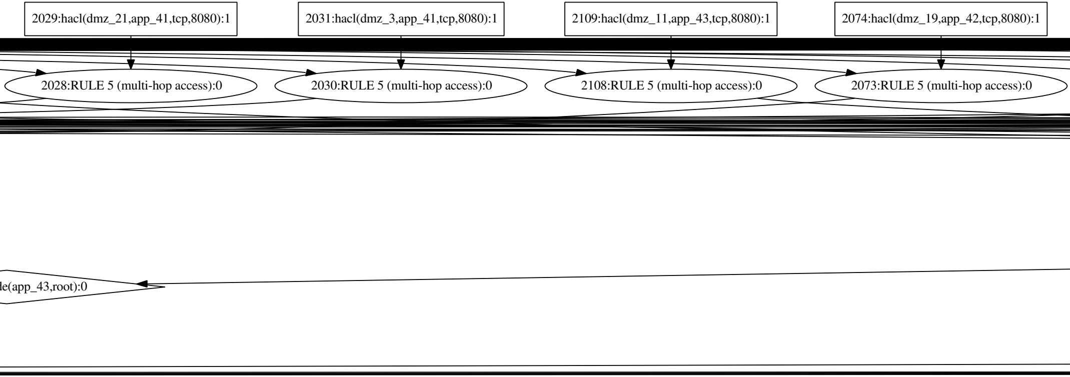

2062:hacl(dmz_13,app_42,tcp,8080):1
1929:hacl(dmz_3,app_4,tcp,8080):1
1931:hacl(dmz_4,app_4,tcp,8080):1
2064:hacl(dmz_14,app_42,tcp,8080):1
2061:RULE 5 (multi-hop access):0
1928:RULE 5 (multi-hop access):0
1930:RULE 5 (multi-hop access):0
2063:RULE 5 (multi-hop access):0

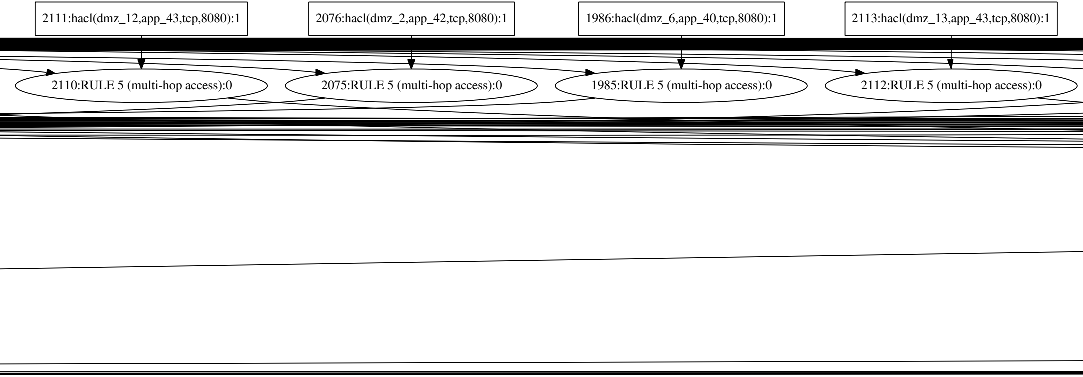

1888:hacl(dmz_8,app_39,tcp,8080):1
2066:hacl(dmz_15,app_42,tcp,8080):1
1980:hacl(dmz_3,app_40,tcp,8080):1
2025:hacl(dmz_2,app_41,tcp,8080):1
1887:RULE 5 (multi-hop access):0
2065:RULE 5 (multi-hop access):0
1979:RULE 5 (multi-hop access):0
2024:RULE 5 (multi-hop access):0

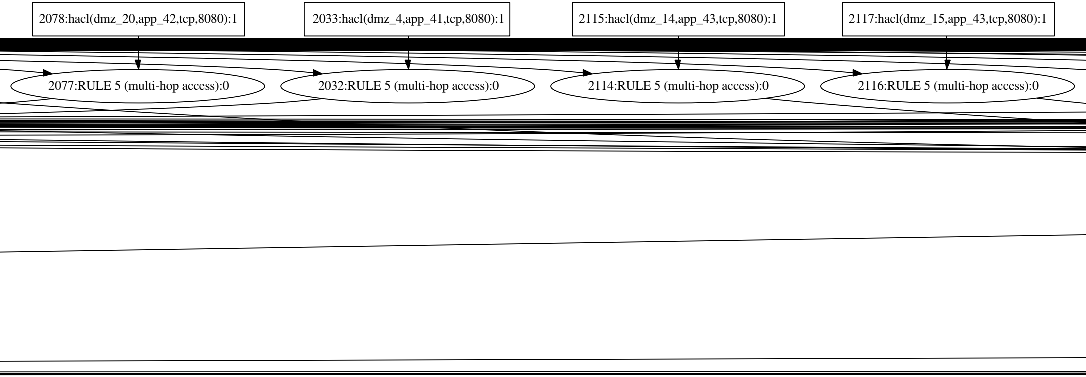

1982:hacl(dmz_4,app_40,tcp,8080):1
2027:hacl(dmz_20,app_41,tcp,8080):1
2068:hacl(dmz_16,app_42,tcp,8080):1
2070:hacl(dmz_17,app_42,tcp,8080):1
1981:RULE 5 (multi-hop access):0
2026:RULE 5 (multi-hop access):0
2067:RULE 5 (multi-hop access):0
2069:RULE 5 (multi-hop access):0

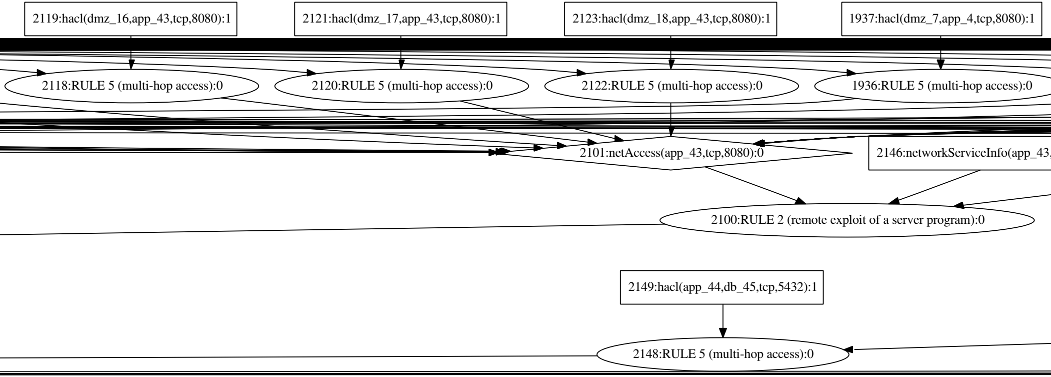

2072:hacl(dmz_18,app_42,tcp,8080):1
1933:hacl(dmz_5,app_4,tcp,8080):1
2103:hacl(dmz_0,app_43,tcp,8080):1
1935:hacl(dmz_6,app_4,tcp,8080):1
2071:RULE 5 (multi-hop access):0
1932:RULE 5 (multi-hop access):0
2102:RULE 5 (multi-hop access):0
1934:RULE 5 (multi-hop access):0
2050:netAccess(app_42,tcp,8080):0
2095:networkServiceInfo(app_42,tomcat,tcp,8080,root):1
2096:vulExists(app_42,vul_app,tomcat,remoteExploit,privEscalation):1
2049:RULE 2 (remote exploit of a server program):0
2098:hacl(app_43,db_45,tcp,5432):1
2097:RULE 5 (multi-hop access):0

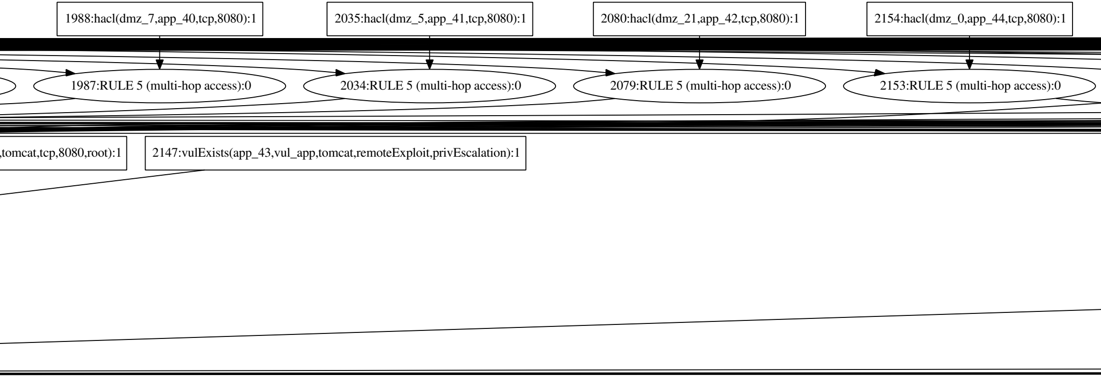

1890:hacl(dmz_9,app_39,tcp,8080):1
2105:hacl(dmz_1,app_43,tcp,8080):1
1984:hacl(dmz_5,app_40,tcp,8080):1
2107:hacl(dmz_10,app_43,tcp,8080):1
1889:RULE 5 (multi-hop access):0
2104:RULE 5 (multi-hop access):0
1983:RULE 5 (multi-hop access):0
2106:RULE 5 (multi-hop access):0
2099:execCod

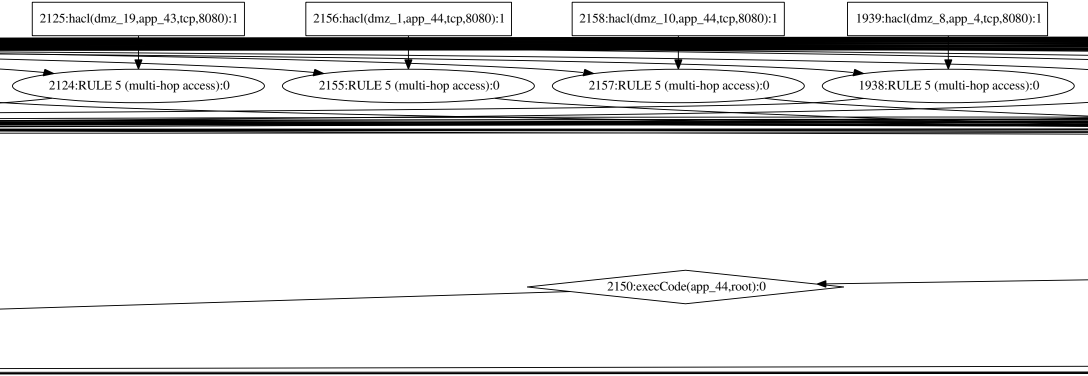

2029:hacl(dmz_21,app_41,tcp,8080):1
2031:hacl(dmz_3,app_41,tcp,8080):1
2109:hacl(dmz_11,app_43,tcp,8080):1
2074:hacl(dmz_19,app_42,tcp,8080):1
2028:RULE 5 (multi-hop access):0
2030:RULE 5 (multi-hop access):0
2108:RULE 5 (multi-hop access):0
2073:RULE 5 (multi-hop access):0
le(app_43,root):0

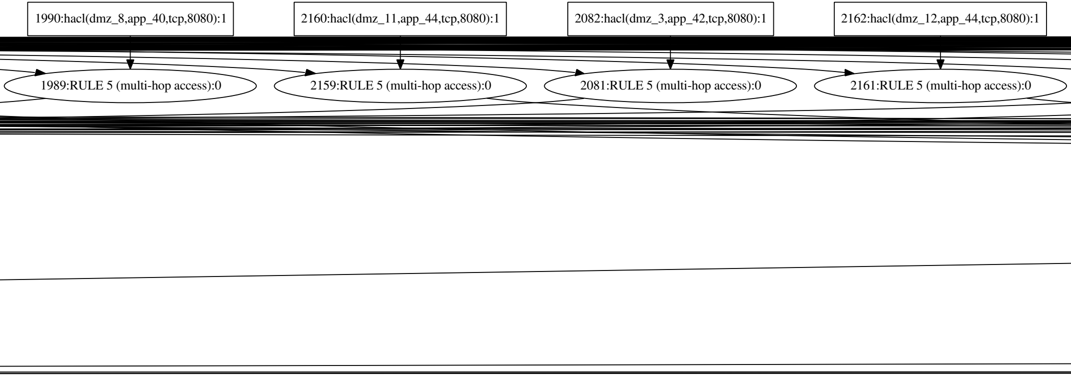

2111:hacl(dmz_12,app_43,tcp,8080):1
2076:hacl(dmz_2,app_42,tcp,8080):1
1986:hacl(dmz_6,app_40,tcp,8080):1
2113:hacl(dmz_13,app_43,tcp,8080):1
2110:RULE 5 (multi-hop access):0
2075:RULE 5 (multi-hop access):0
1985:RULE 5 (multi-hop access):0
2112:RULE 5 (multi-hop access):0

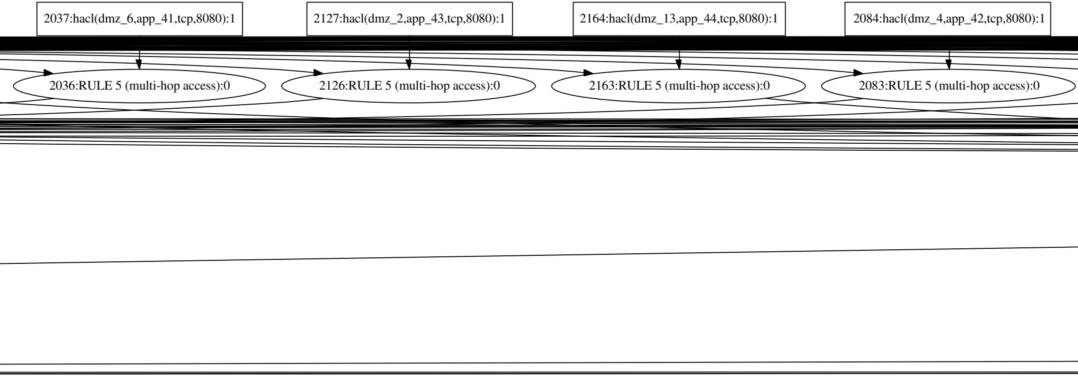

2078:hacl(dmz_20,app_42,tcp,8080):1
2033:hacl(dmz_4,app_41,tcp,8080):1
2115:hacl(dmz_14,app_43,tcp,8080):1
2117:hacl(dmz_15,app_43,tcp,8080):1
2077:RULE 5 (multi-hop access):0
2032:RULE 5 (multi-hop access):0
2114:RULE 5 (multi-hop access):0
2116:RULE 5 (multi-hop access):0

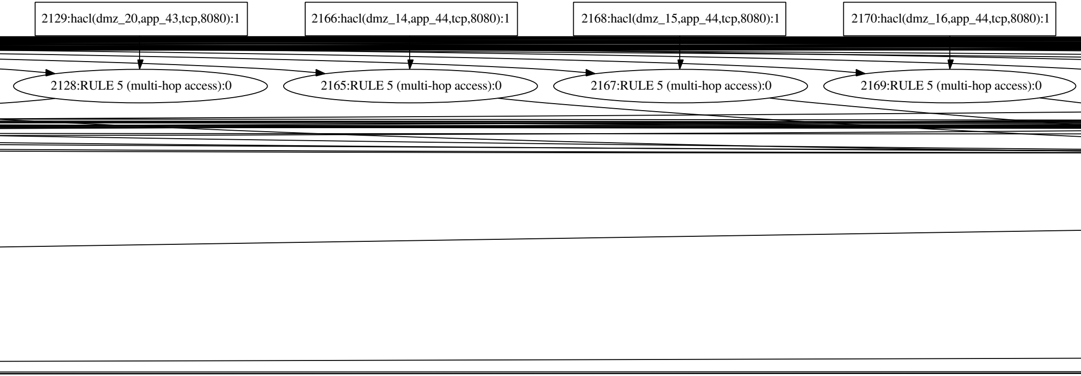

2119:hacl(dmz_16,app_43,tcp,8080):1
2121:hacl(dmz_17,app_43,tcp,8080):1
2123:hacl(dmz_18,app_43,tcp,8080):1
1937:hacl(dmz_7,app_4,tcp,8080):1
2118:RULE 5 (multi-hop access):0
2120:RULE 5 (multi-hop access):0
2122:RULE 5 (multi-hop access):0
1936:RULE 5 (multi-hop access):0
2101:netAccess(app_43,tcp,8080):0
2146:networkServiceInfo(app_43,
2100:RULE 2 (remote exploit of a server program):0
2149:hacl(app_44,db_45,tcp,5432):1
2148:RULE 5 (multi-hop access):0

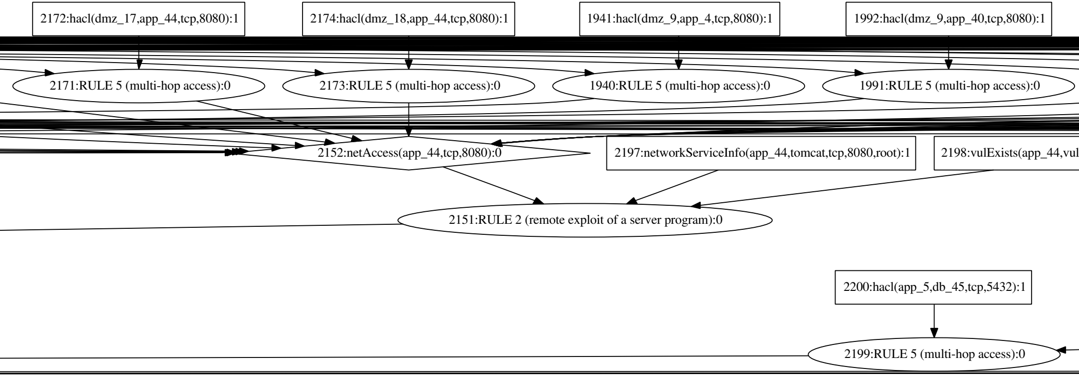

1988:hacl(dmz_7,app_40,tcp,8080):1
2035:hacl(dmz_5,app_41,tcp,8080):1
2080:hacl(dmz_21,app_42,tcp,8080):1
2154:hacl(dmz_0,app_44,tcp,8080):1
1987:RULE 5 (multi-hop access):0
2034:RULE 5 (multi-hop access):0
2079:RULE 5 (multi-hop access):0
2153:RULE 5 (multi-hop access):0
,tomcat,tcp,8080,root):1
2147:vulExists(app_43,vul_app,tomcat,remoteExploit,privEscalation):1

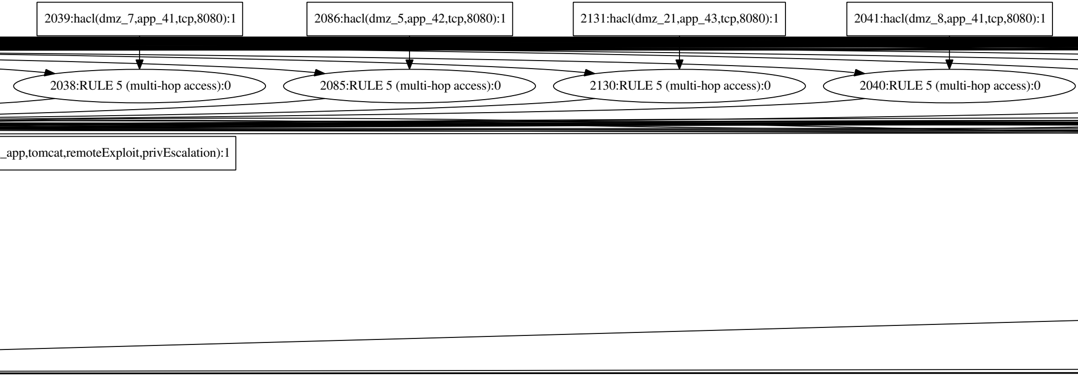

2125:hacl(dmz_19,app_43,tcp,8080):1
2156:hacl(dmz_1,app_44,tcp,8080):1
2158:hacl(dmz_10,app_44,tcp,8080):1
1939:hacl(dmz_8,app_4,tcp,8080):1
2124:RULE 5 (multi-hop access):0
2155:RULE 5 (multi-hop access):0
2157:RULE 5 (multi-hop access):0
1938:RULE 5 (multi-hop access):0
2150:execCode(app_44,root):0

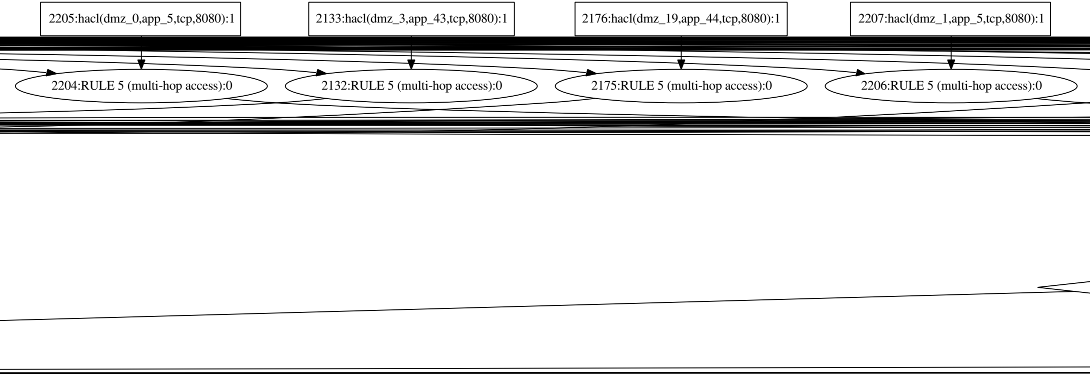

1990:hacl(dmz_8,app_40,tcp,8080):1
2160:hacl(dmz_11,app_44,tcp,8080):1
2082:hacl(dmz_3,app_42,tcp,8080):1
2162:hacl(dmz_12,app_44,tcp,8080):1
1989:RULE 5 (multi-hop access):0
2159:RULE 5 (multi-hop access):0
2081:RULE 5 (multi-hop access):0
2161:RULE 5 (multi-hop access):0

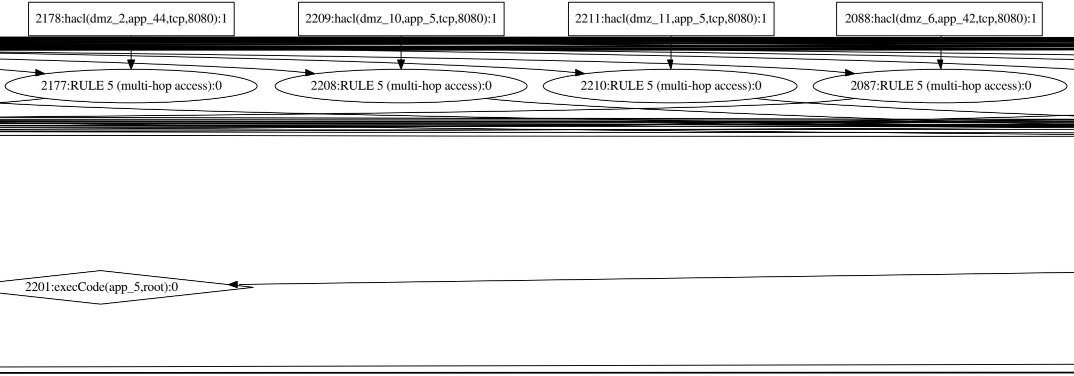

2037:hacl(dmz_6,app_41,tcp,8080):1
2127:hacl(dmz_2,app_43,tcp,8080):1
2164:hacl(dmz_13,app_44,tcp,8080):1
2084:hacl(dmz_4,app_42,tcp,8080):1
2036:RULE 5 (multi-hop access):0
2126:RULE 5 (multi-hop access):0
2163:RULE 5 (multi-hop access):0
2083:RULE 5 (multi-hop access):0

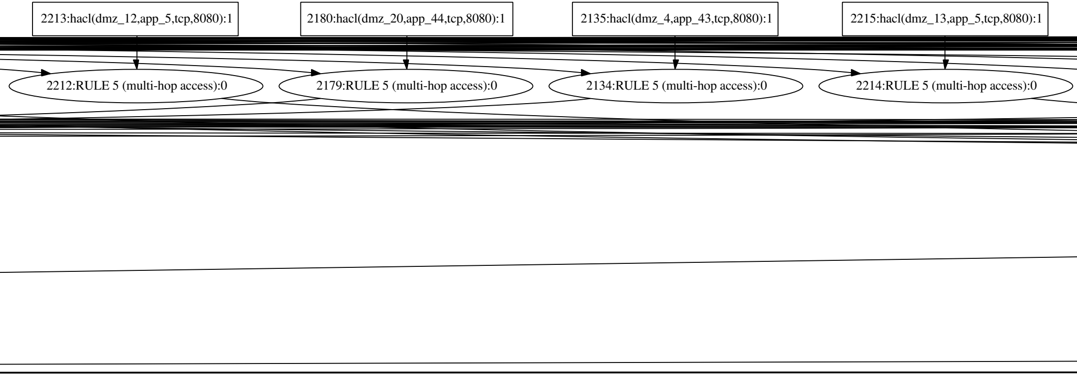

2129:hacl(dmz_20,app_43,tcp,8080):1
2166:hacl(dmz_14,app_44,tcp,8080):1
2168:hacl(dmz_15,app_44,tcp,8080):1
2170:hacl(dmz_16,app_44,tcp,8080):1
2128:RULE 5 (multi-hop access):0
2165:RULE 5 (multi-hop access):0
2167:RULE 5 (multi-hop access):0
2169:RULE 5 (multi-hop access):0

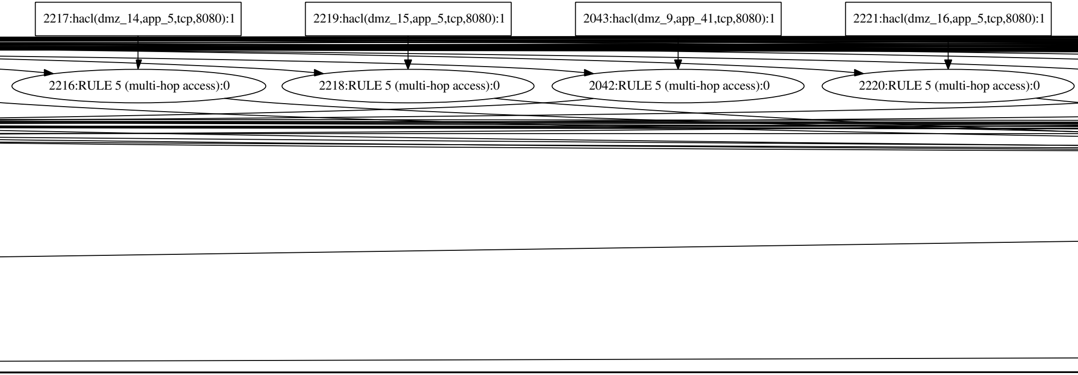

2172:hacl(dmz_17,app_44,tcp,8080):1
2174:hacl(dmz_18,app_44,tcp,8080):1
1941:hacl(dmz_9,app_4,tcp,8080):1
1992:hacl(dmz_9,app_40,tcp,8080):1
2171:RULE 5 (multi-hop access):0
2173:RULE 5 (multi-hop access):0
1940:RULE 5 (multi-hop access):0
1991:RULE 5 (multi-hop access):0
2152:netAccess(app_44,tcp,8080):0
2197:networkServiceInfo(app_44,tomcat,tcp,8080,root):1
2198:vulExists(app_44,vul
2151:RULE 2 (remote exploit of a server program):0
2200:hacl(app_5,db_45,tcp,5432):1
2199:RULE 5 (multi-hop access):0

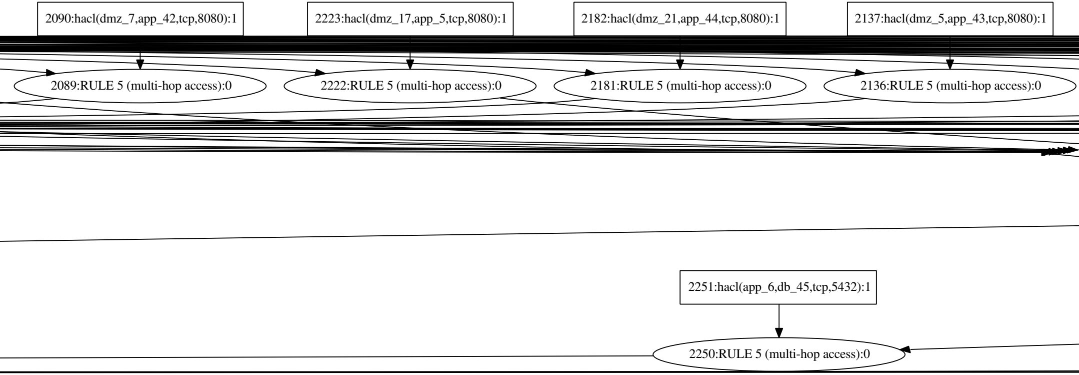

2039:hacl(dmz_7,app_41,tcp,8080):1
2086:hacl(dmz_5,app_42,tcp,8080):1
2131:hacl(dmz_21,app_43,tcp,8080):1
2041:hacl(dmz_8,app_41,tcp,8080):1
2038:RULE 5 (multi-hop access):0
2085:RULE 5 (multi-hop access):0
2130:RULE 5 (multi-hop access):0
2040:RULE 5 (multi-hop access):0
_app,tomcat,remoteExploit,privEscalation):1

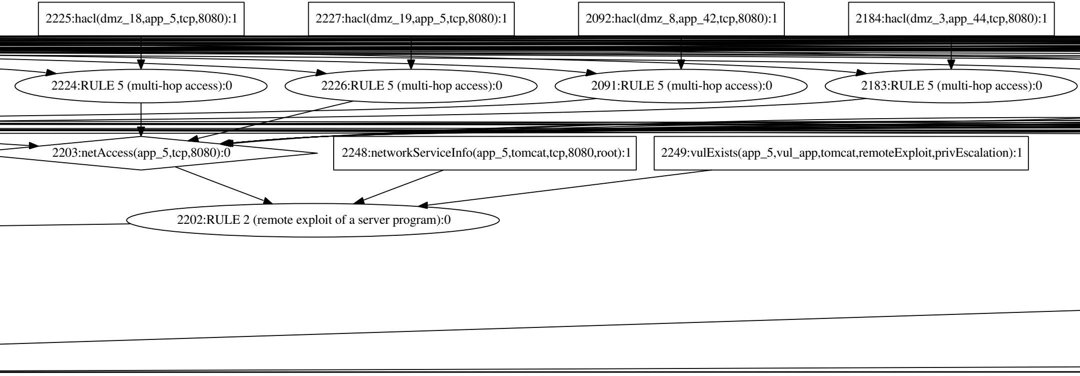

2205:hacl(dmz_0,app_5,tcp,8080):1
2133:hacl(dmz_3,app_43,tcp,8080):1
2176:hacl(dmz_19,app_44,tcp,8080):1
2207:hacl(dmz_1,app_5,tcp,8080):1
2204:RULE 5 (multi-hop access):0
2132:RULE 5 (multi-hop access):0
2175:RULE 5 (multi-hop access):0
2206:RULE 5 (multi-hop access):0

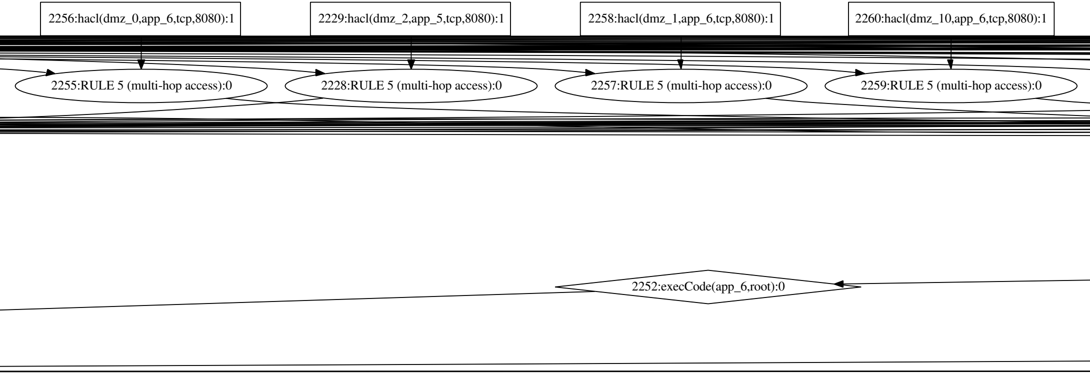

2178:hacl(dmz_2,app_44,tcp,8080):1
2209:hacl(dmz_10,app_5,tcp,8080):1
2211:hacl(dmz_11,app_5,tcp,8080):1
2088:hacl(dmz_6,app_42,tcp,8080):1
2177:RULE 5 (multi-hop access):0
2208:RULE 5 (multi-hop access):0
2210:RULE 5 (multi-hop access):0
2087:RULE 5 (multi-hop access):0
2201:execCode(app_5,root):0

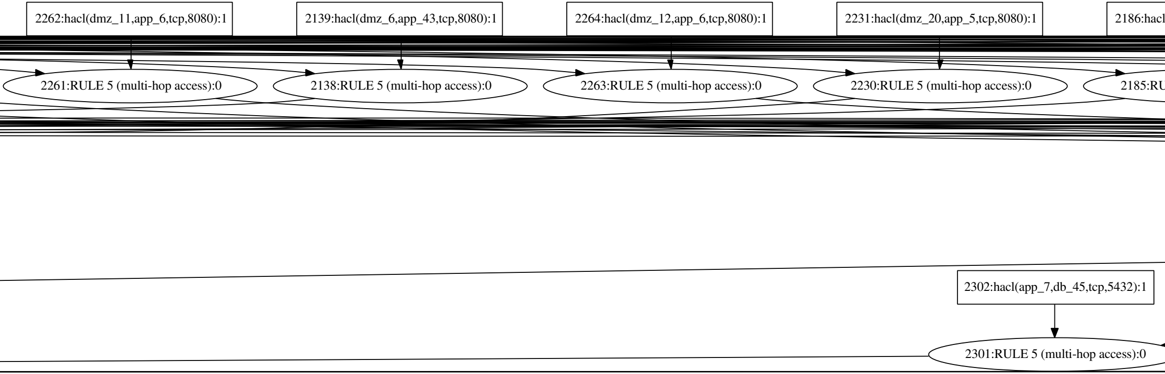

2213:hacl(dmz_12,app_5,tcp,8080):1
2180:hacl(dmz_20,app_44,tcp,8080):1
2135:hacl(dmz_4,app_43,tcp,8080):1
2215:hacl(dmz_13,app_5,tcp,8080):1
2212:RULE 5 (multi-hop access):0
2179:RULE 5 (multi-hop access):0
2134:RULE 5 (multi-hop access):0
2214:RULE 5 (multi-hop access):0

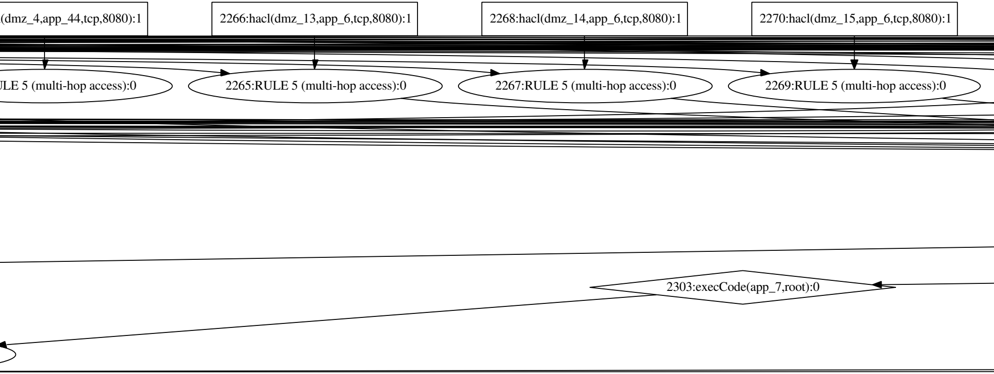

2217:hacl(dmz_14,app_5,tcp,8080):1
2219:hacl(dmz_15,app_5,tcp,8080):1
2043:hacl(dmz_9,app_41,tcp,8080):1
2221:hacl(dmz_16,app_5,tcp,8080):1
2216:RULE 5 (multi-hop access):0
2218:RULE 5 (multi-hop access):0
2042:RULE 5 (multi-hop access):0
2220:RULE 5 (multi-hop access):0

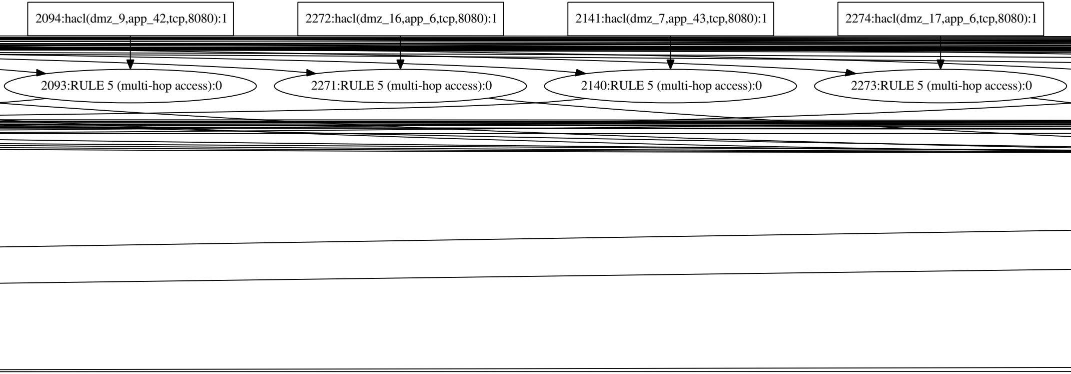

2090:hacl(dmz_7,app_42,tcp,8080):1
2223:hacl(dmz_17,app_5,tcp,8080):1
2182:hacl(dmz_21,app_44,tcp,8080):1
2137:hacl(dmz_5,app_43,tcp,8080):1
2089:RULE 5 (multi-hop access):0
2222:RULE 5 (multi-hop access):0
2181:RULE 5 (multi-hop access):0
2136:RULE 5 (multi-hop access):0
2251:hacl(app_6,db_45,tcp,5432):1
2250:RULE 5 (multi-hop access):0

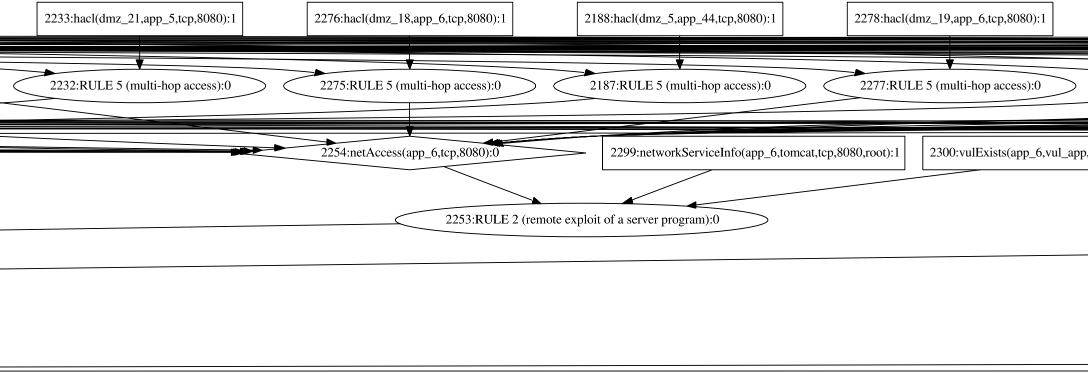

2225:hacl(dmz_18,app_5,tcp,8080):1
2227:hacl(dmz_19,app_5,tcp,8080):1
2092:hacl(dmz_8,app_42,tcp,8080):1
2184:hacl(dmz_3,app_44,tcp,8080):1
2224:RULE 5 (multi-hop access):0
2226:RULE 5 (multi-hop access):0
2091:RULE 5 (multi-hop access):0
2183:RULE 5 (multi-hop access):0
2203:netAccess(app_5,tcp,8080):0
2248:networkServiceInfo(app_5,tomcat,tcp,8080,root):1
2249:vulExists(app_5,vul_app,tomcat,remoteExploit,privEscalation):1
2202:RULE 2 (remote exploit of a server program):0

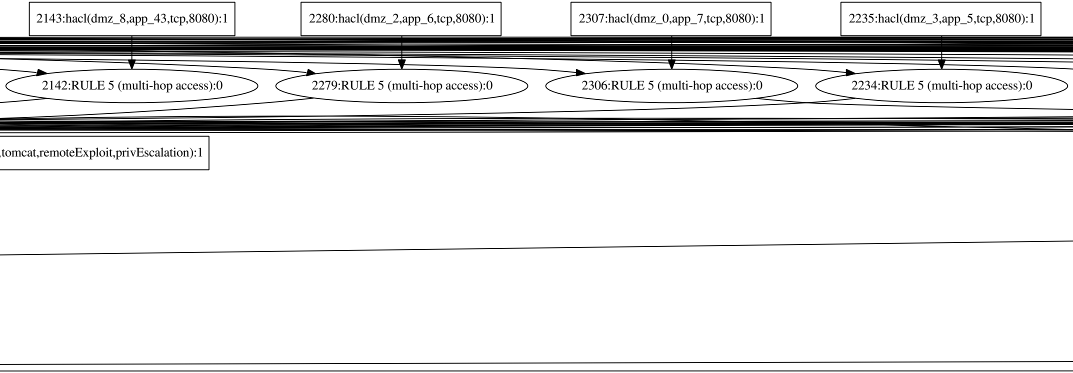

2256:hacl(dmz_0,app_6,tcp,8080):1
2229:hacl(dmz_2,app_5,tcp,8080):1
2258:hacl(dmz_1,app_6,tcp,8080):1
2260:hacl(dmz_10,app_6,tcp,8080):1
2255:RULE 5 (multi-hop access):0
2228:RULE 5 (multi-hop access):0
2257:RULE 5 (multi-hop access):0
2259:RULE 5 (multi-hop access):0
2252:execCode(app_6,root):0

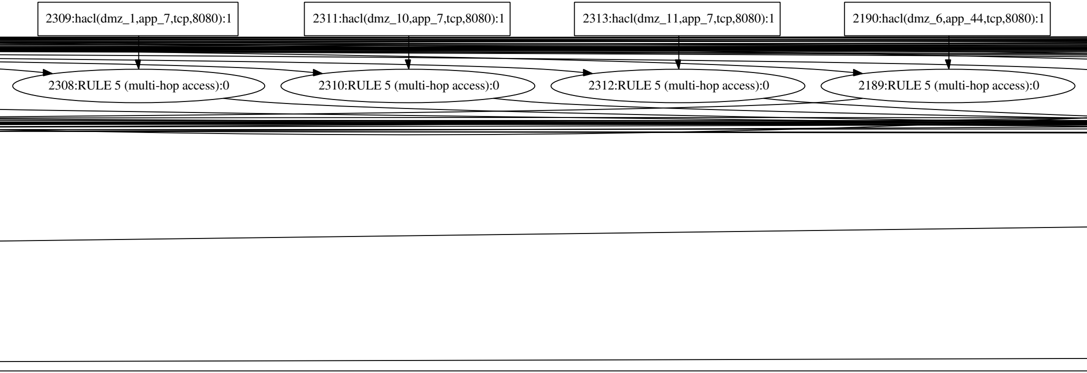

2262:hacl(dmz_11,app_6,tcp,8080):1
2139:hacl(dmz_6,app_43,tcp,8080):1
2264:hacl(dmz_12,app_6,tcp,8080):1
2231:hacl(dmz_20,app_5,tcp,8080):1
2186:hacl
2261:RULE 5 (multi-hop access):0
2138:RULE 5 (multi-hop access):0
2263:RULE 5 (multi-hop access):0
2230:RULE 5 (multi-hop access):0
2185:RU
2302:hacl(app_7,db_45,tcp,5432):1
2301:RULE 5 (multi-hop access):0

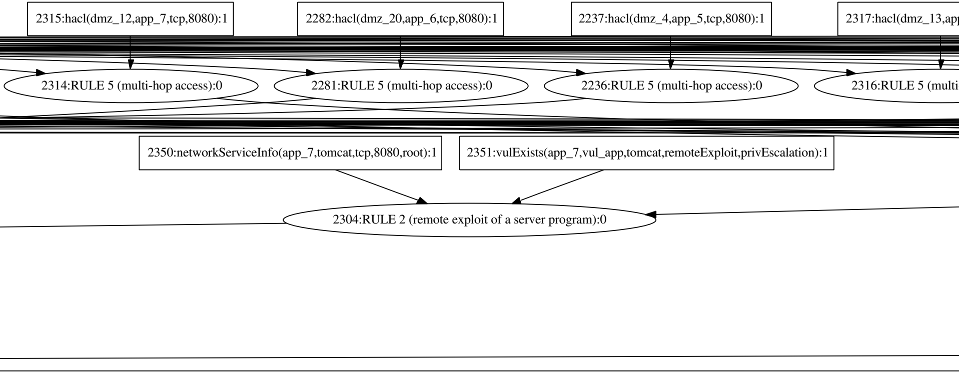

(dmz_4,app_44,tcp,8080):1
2266:hacl(dmz_13,app_6,tcp,8080):1
2268:hacl(dmz_14,app_6,tcp,8080):1
2270:hacl(dmz_15,app_6,tcp,8080):1
LE 5 (multi-hop access):0
2265:RULE 5 (multi-hop access):0
2267:RULE 5 (multi-hop access):0
2269:RULE 5 (multi-hop access):0
2303:execCode(app_7,root):0

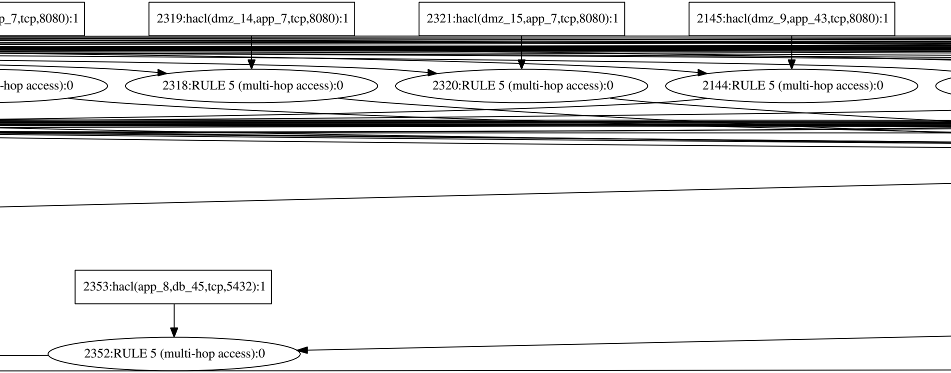

2094:hacl(dmz_9,app_42,tcp,8080):1
2272:hacl(dmz_16,app_6,tcp,8080):1
2141:hacl(dmz_7,app_43,tcp,8080):1
2274:hacl(dmz_17,app_6,tcp,8080):1
2093:RULE 5 (multi-hop access):0
2271:RULE 5 (multi-hop access):0
2140:RULE 5 (multi-hop access):0
2273:RULE 5 (multi-hop access):0

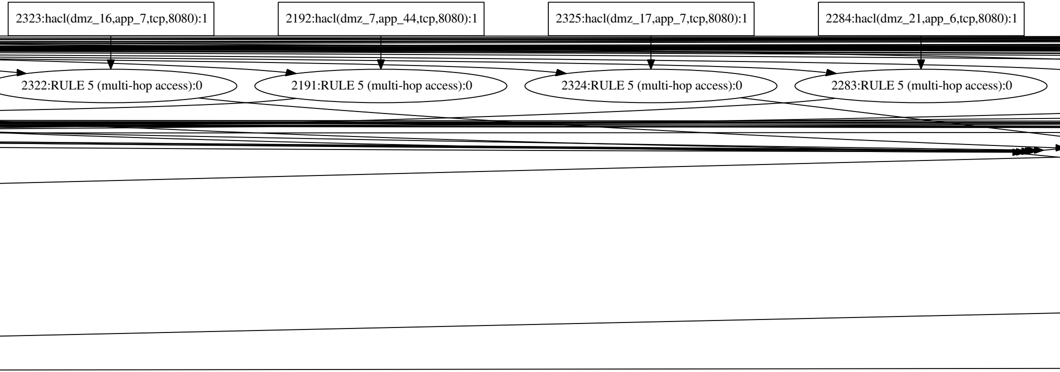

2233:hacl(dmz_21,app_5,tcp,8080):1
2276:hacl(dmz_18,app_6,tcp,8080):1
2188:hacl(dmz_5,app_44,tcp,8080):1
2278:hacl(dmz_19,app_6,tcp,8080):1
2232:RULE 5 (multi-hop access):0
2275:RULE 5 (multi-hop access):0
2187:RULE 5 (multi-hop access):0
2277:RULE 5 (multi-hop access):0
2254:netAccess(app_6,tcp,8080):0
2299:networkServiceInfo(app_6,tomcat,tcp,8080,root):1
2300:vulExists(app_6,vul_app,
2253:RULE 2 (remote exploit of a server program):0

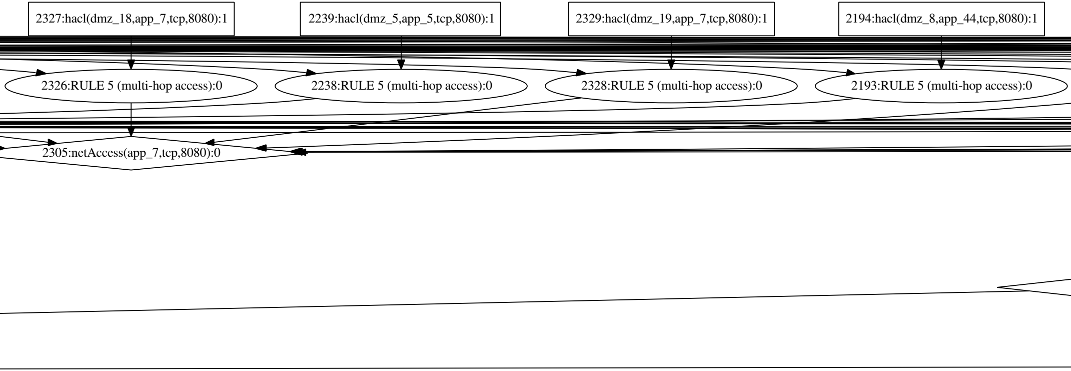

2143:hacl(dmz_8,app_43,tcp,8080):1
2280:hacl(dmz_2,app_6,tcp,8080):1
2307:hacl(dmz_0,app_7,tcp,8080):1
2235:hacl(dmz_3,app_5,tcp,8080):1
2142:RULE 5 (multi-hop access):0
2279:RULE 5 (multi-hop access):0
2306:RULE 5 (multi-hop access):0
2234:RULE 5 (multi-hop access):0
,tomcat,remoteExploit,privEscalation):1

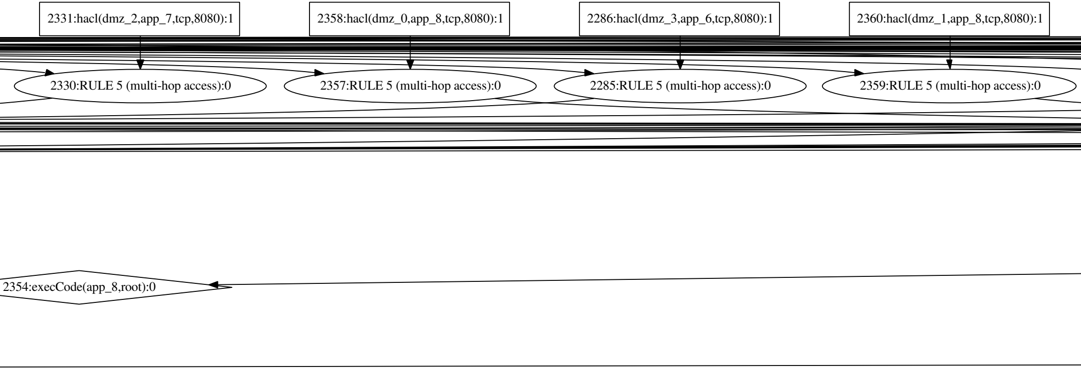

2309:hacl(dmz_1,app_7,tcp,8080):1
2311:hacl(dmz_10,app_7,tcp,8080):1
2313:hacl(dmz_11,app_7,tcp,8080):1
2190:hacl(dmz_6,app_44,tcp,8080):1
2308:RULE 5 (multi-hop access):0
2310:RULE 5 (multi-hop access):0
2312:RULE 5 (multi-hop access):0
2189:RULE 5 (multi-hop access):0

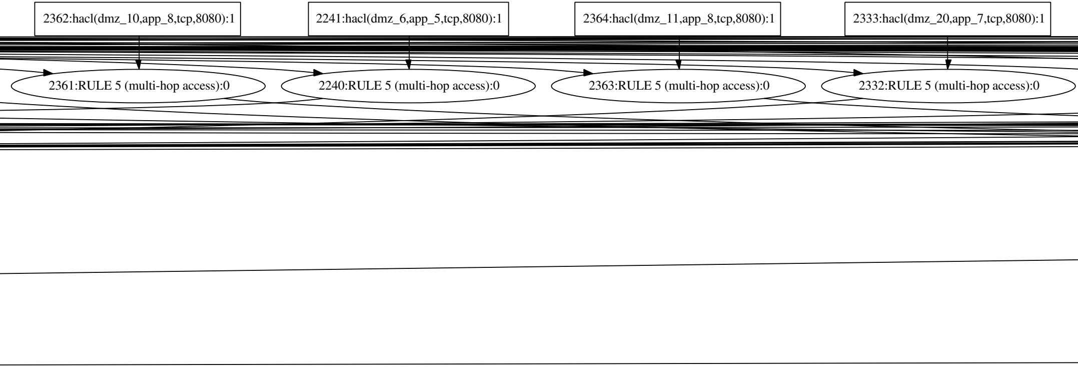

2315:hacl(dmz_12,app_7,tcp,8080):1
2282:hacl(dmz_20,app_6,tcp,8080):1
2237:hacl(dmz_4,app_5,tcp,8080):1
2317:hacl(dmz_13,ap
2314:RULE 5 (multi-hop access):0
2281:RULE 5 (multi-hop access):0
2236:RULE 5 (multi-hop access):0
2316:RULE 5 (multi
2350:networkServiceInfo(app_7,tomcat,tcp,8080,root):1
2351:vulExists(app_7,vul_app,tomcat,remoteExploit,privEscalation):1
2304:RULE 2 (remote exploit of a server program):0

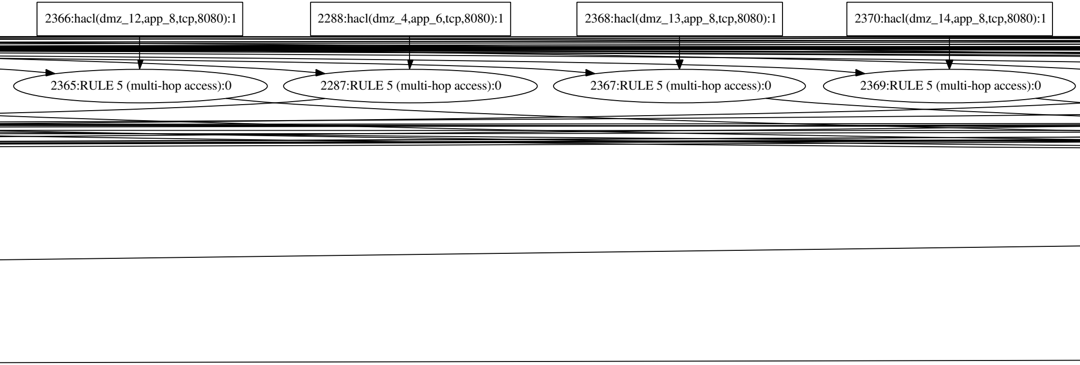

p_7,tcp,8080):1
2319:hacl(dmz_14,app_7,tcp,8080):1
2321:hacl(dmz_15,app_7,tcp,8080):1
2145:hacl(dmz_9,app_43,tcp,8080):1
-hop access):0
2318:RULE 5 (multi-hop access):0
2320:RULE 5 (multi-hop access):0
2144:RULE 5 (multi-hop access):0
2353:hacl(app_8,db_45,tcp,5432):1
2352:RULE 5 (multi-hop access):0

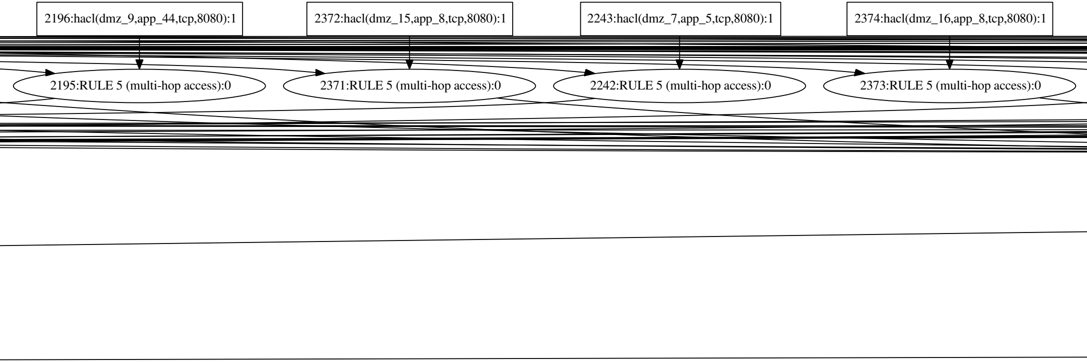

2323:hacl(dmz_16,app_7,tcp,8080):1
2192:hacl(dmz_7,app_44,tcp,8080):1
2325:hacl(dmz_17,app_7,tcp,8080):1
2284:hacl(dmz_21,app_6,tcp,8080):1
2322:RULE 5 (multi-hop access):0
2191:RULE 5 (multi-hop access):0
2324:RULE 5 (multi-hop access):0
2283:RULE 5 (multi-hop access):0

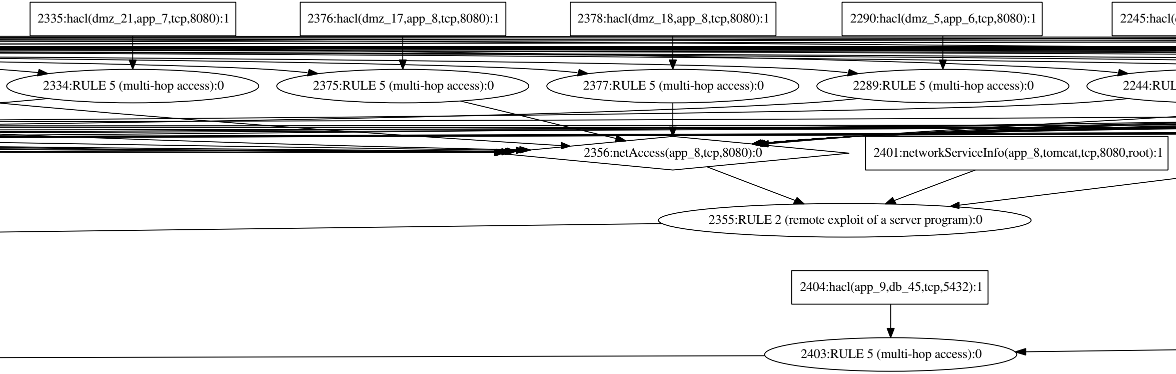

2327:hacl(dmz_18,app_7,tcp,8080):1
2239:hacl(dmz_5,app_5,tcp,8080):1
2329:hacl(dmz_19,app_7,tcp,8080):1
2194:hacl(dmz_8,app_44,tcp,8080):1
2326:RULE 5 (multi-hop access):0
2238:RULE 5 (multi-hop access):0
2328:RULE 5 (multi-hop access):0
2193:RULE 5 (multi-hop access):0
2305:netAccess(app_7,tcp,8080):0

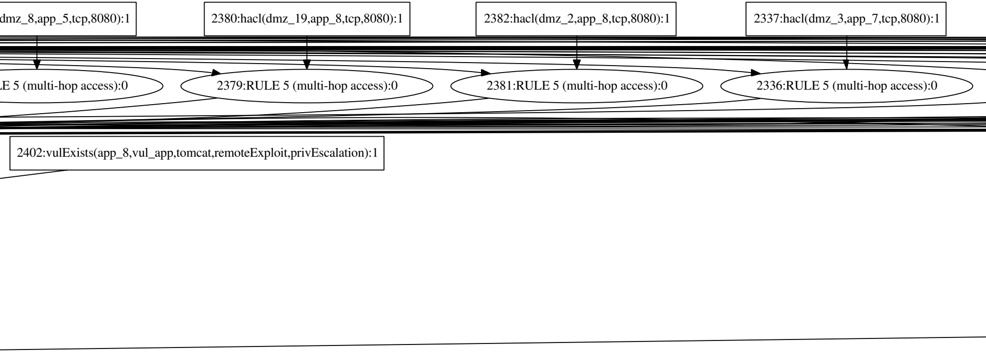

2331:hacl(dmz_2,app_7,tcp,8080):1
2358:hacl(dmz_0,app_8,tcp,8080):1
2286:hacl(dmz_3,app_6,tcp,8080):1
2360:hacl(dmz_1,app_8,tcp,8080):1
2330:RULE 5 (multi-hop access):0
2357:RULE 5 (multi-hop access):0
2285:RULE 5 (multi-hop access):0
2359:RULE 5 (multi-hop access):0
2354:execCode(app_8,root):0

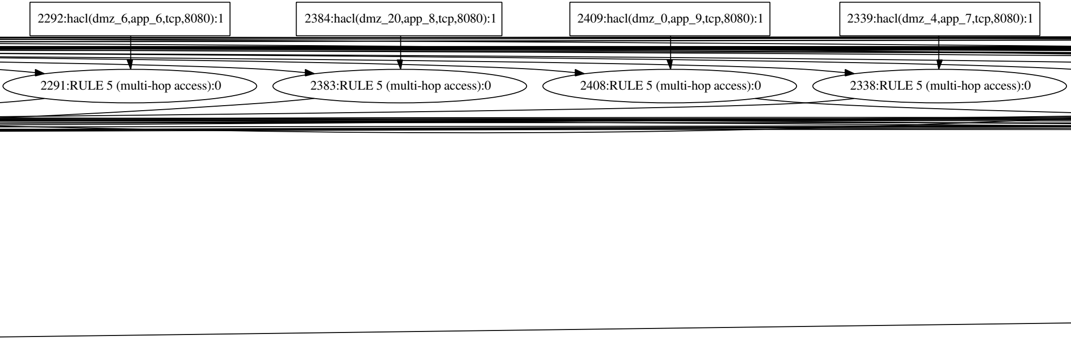

2362:hacl(dmz_10,app_8,tcp,8080):1
2241:hacl(dmz_6,app_5,tcp,8080):1
2364:hacl(dmz_11,app_8,tcp,8080):1
2333:hacl(dmz_20,app_7,tcp,8080):1
2361:RULE 5 (multi-hop access):0
2240:RULE 5 (multi-hop access):0
2363:RULE 5 (multi-hop access):0
2332:RULE 5 (multi-hop access):0

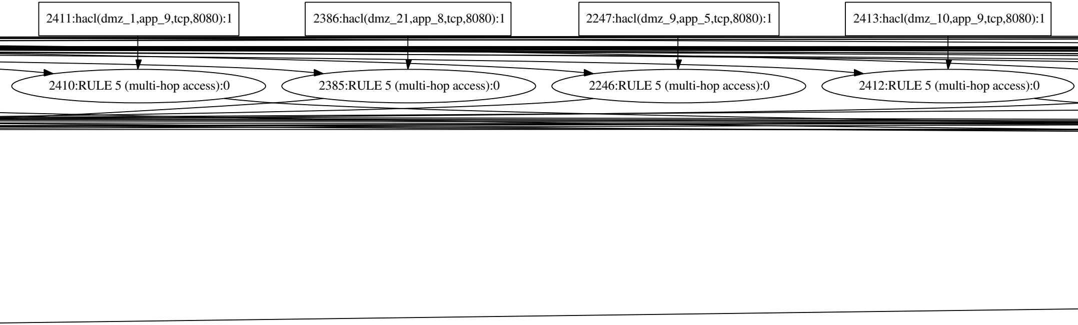

2366:hacl(dmz_12,app_8,tcp,8080):1
2288:hacl(dmz_4,app_6,tcp,8080):1
2368:hacl(dmz_13,app_8,tcp,8080):1
2370:hacl(dmz_14,app_8,tcp,8080):1
2365:RULE 5 (multi-hop access):0
2287:RULE 5 (multi-hop access):0
2367:RULE 5 (multi-hop access):0
2369:RULE 5 (multi-hop access):0

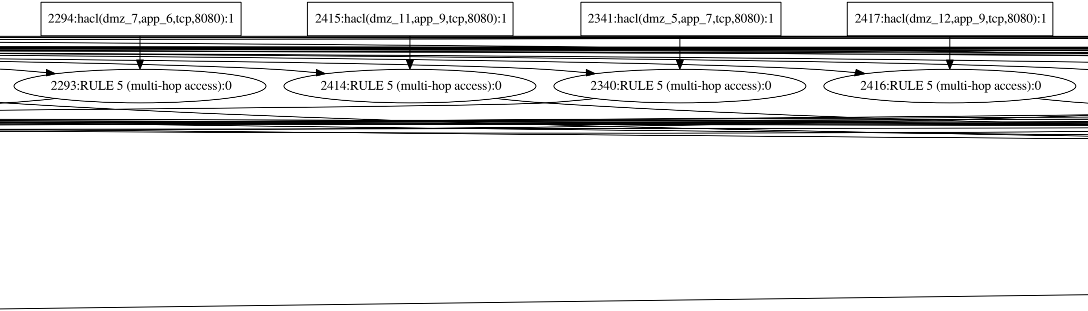

2196:hacl(dmz_9,app_44,tcp,8080):1
2372:hacl(dmz_15,app_8,tcp,8080):1
2243:hacl(dmz_7,app_5,tcp,8080):1
2374:hacl(dmz_16,app_8,tcp,8080):1
2195:RULE 5 (multi-hop access):0
2371:RULE 5 (multi-hop access):0
2242:RULE 5 (multi-hop access):0
2373:RULE 5 (multi-hop access):0

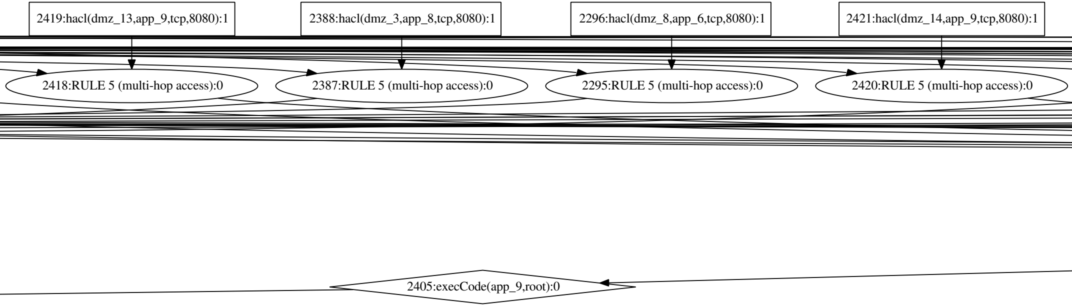

2335:hacl(dmz_21,app_7,tcp,8080):1
2376:hacl(dmz_17,app_8,tcp,8080):1
2378:hacl(dmz_18,app_8,tcp,8080):1
2290:hacl(dmz_5,app_6,tcp,8080):1
2245:hacl(
2334:RULE 5 (multi-hop access):0
2375:RULE 5 (multi-hop access):0
2377:RULE 5 (multi-hop access):0
2289:RULE 5 (multi-hop access):0
2244:RUL
2356:netAccess(app_8,tcp,8080):0
2401:networkServiceInfo(app_8,tomcat,tcp,8080,root):1
2355:RULE 2 (remote exploit of a server program):0
2404:hacl(app_9,db_45,tcp,5432):1
2403:RULE 5 (multi-hop access):0

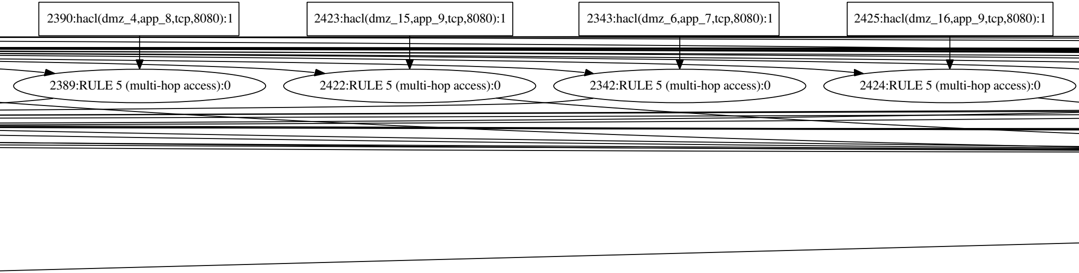

dmz_8,app_5,tcp,8080):1
2380:hacl(dmz_19,app_8,tcp,8080):1
2382:hacl(dmz_2,app_8,tcp,8080):1
2337:hacl(dmz_3,app_7,tcp,8080):1
LE 5 (multi-hop access):0
2379:RULE 5 (multi-hop access):0
2381:RULE 5 (multi-hop access):0
2336:RULE 5 (multi-hop access):0
2402:vulExists(app_8,vul_app,tomcat,remoteExploit,privEscalation):1

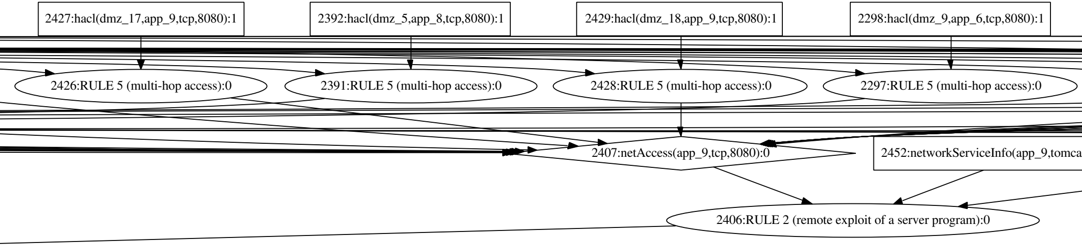

2292:hacl(dmz_6,app_6,tcp,8080):1
2384:hacl(dmz_20,app_8,tcp,8080):1
2409:hacl(dmz_0,app_9,tcp,8080):1
2339:hacl(dmz_4,app_7,tcp,8080):1
2291:RULE 5 (multi-hop access):0
2383:RULE 5 (multi-hop access):0
2408:RULE 5 (multi-hop access):0
2338:RULE 5 (multi-hop access):0

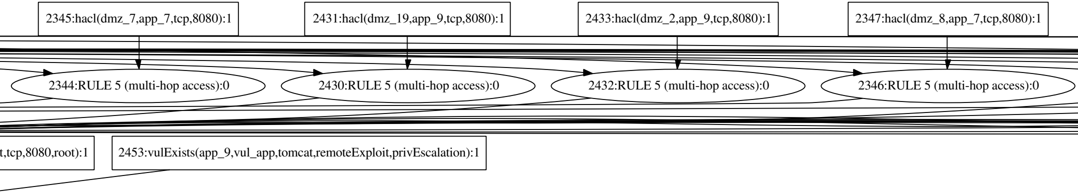

2411:hacl(dmz_1,app_9,tcp,8080):1
2386:hacl(dmz_21,app_8,tcp,8080):1
2247:hacl(dmz_9,app_5,tcp,8080):1
2413:hacl(dmz_10,app_9,tcp,8080):1
2410:RULE 5 (multi-hop access):0
2385:RULE 5 (multi-hop access):0
2246:RULE 5 (multi-hop access):0
2412:RULE 5 (multi-hop access):0

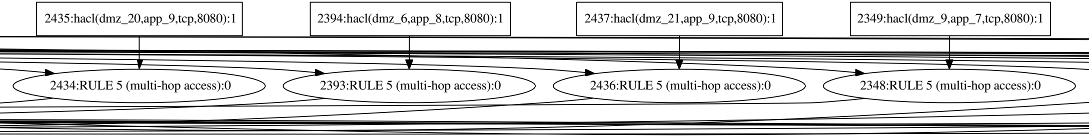

2294:hacl(dmz_7,app_6,tcp,8080):1
2415:hacl(dmz_11,app_9,tcp,8080):1
2341:hacl(dmz_5,app_7,tcp,8080):1
2417:hacl(dmz_12,app_9,tcp,8080):1
2293:RULE 5 (multi-hop access):0
2414:RULE 5 (multi-hop access):0
2340:RULE 5 (multi-hop access):0
2416:RULE 5 (multi-hop access):0

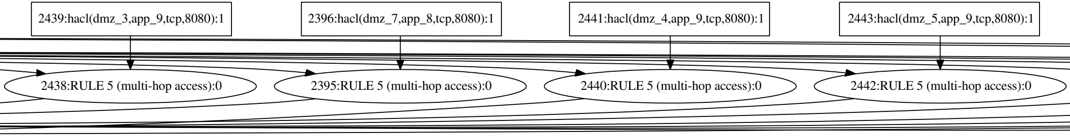

2419:hacl(dmz_13,app_9,tcp,8080):1
2388:hacl(dmz_3,app_8,tcp,8080):1
2296:hacl(dmz_8,app_6,tcp,8080):1
2421:hacl(dmz_14,app_9,tcp,8080):1
2418:RULE 5 (multi-hop access):0
2387:RULE 5 (multi-hop access):0
2295:RULE 5 (multi-hop access):0
2420:RULE 5 (multi-hop access):0
2405:execCode(app_9,root):0

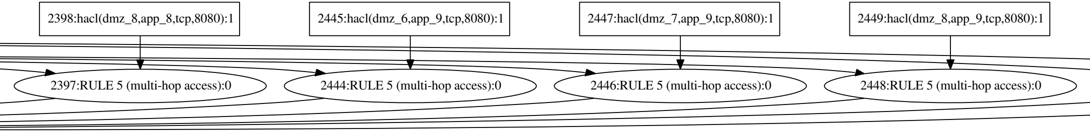

2390:hacl(dmz_4,app_8,tcp,8080):1
2423:hacl(dmz_15,app_9,tcp,8080):1
2343:hacl(dmz_6,app_7,tcp,8080):1
2425:hacl(dmz_16,app_9,tcp,8080):1
2389:RULE 5 (multi-hop access):0
2422:RULE 5 (multi-hop access):0
2342:RULE 5 (multi-hop access):0
2424:RULE 5 (multi-hop access):0

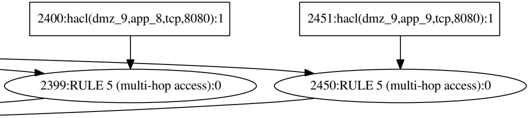

2427:hacl(dmz_17,app_9,tcp,8080):1
2392:hacl(dmz_5,app_8,tcp,8080):1
2429:hacl(dmz_18,app_9,tcp,8080):1
2298:hacl(dmz_9,app_6,tcp,8080):1
2426:RULE 5 (multi-hop access):0
2391:RULE 5 (multi-hop access):0
2428:RULE 5 (multi-hop access):0
2297:RULE 5 (multi-hop access):0
2407:netAccess(app_9,tcp,8080):0
2452:networkServiceInfo(app_9,tomca
2406:RULE 2 (remote exploit of a server program):0


2345:hacl(dmz_7,app_7,tcp,8080):1
2431:hacl(dmz_19,app_9,tcp,8080):1
2433:hacl(dmz_2,app_9,tcp,8080):1
2347:hacl(dmz_8,app_7,tcp,8080):1
2344:RULE 5 (multi-hop access):0
2430:RULE 5 (multi-hop access):0
2432:RULE 5 (multi-hop access):0
2346:RULE 5 (multi-hop access):0
t,tcp,8080,root):1
2453:vulExists(app_9,vul_app,tomcat,remoteExploit,privEscalation):1


2435:hacl(dmz_20,app_9,tcp,8080):1
2394:hacl(dmz_6,app_8,tcp,8080):1
2437:hacl(dmz_21,app_9,tcp,8080):1
2349:hacl(dmz_9,app_7,tcp,8080):1
2434:RULE 5 (multi-hop access):0
2393:RULE 5 (multi-hop access):0
2436:RULE 5 (multi-hop access):0
2348:RULE 5 (multi-hop access):0


2439:hacl(dmz_3,app_9,tcp,8080):1
2396:hacl(dmz_7,app_8,tcp,8080):1
2441:hacl(dmz_4,app_9,tcp,8080):1
2443:hacl(dmz_5,app_9,tcp,8080):1
2438:RULE 5 (multi-hop access):0
2395:RULE 5 (multi-hop access):0
2440:RULE 5 (multi-hop access):0
2442:RULE 5 (multi-hop access):0


2398:hacl(dmz_8,app_8,tcp,8080):1
2445:hacl(dmz_6,app_9,tcp,8080):1
2447:hacl(dmz_7,app_9,tcp,8080):1
2449:hacl(dmz_8,app_9,tcp,8080):1
2397:RULE 5 (multi-hop access):0
2444:RULE 5 (multi-hop access):0
2446:RULE 5 (multi-hop access):0
2448:RULE 5 (multi-hop access):0


2400:hacl(dmz_9,app_8,tcp,8080):1
2451:hacl(dmz_9,app_9,tcp,8080):1
2399:RULE 5 (multi-hop access):0
2450:RULE 5 (multi-hop access):0